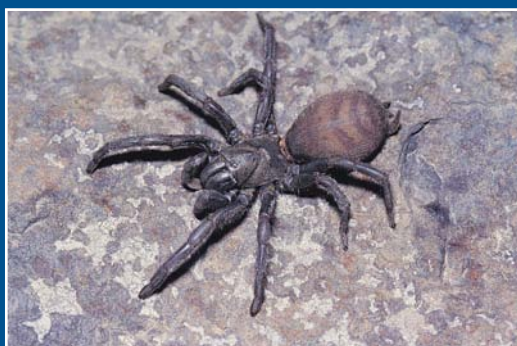
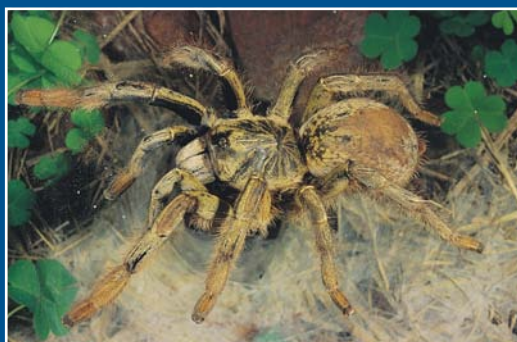


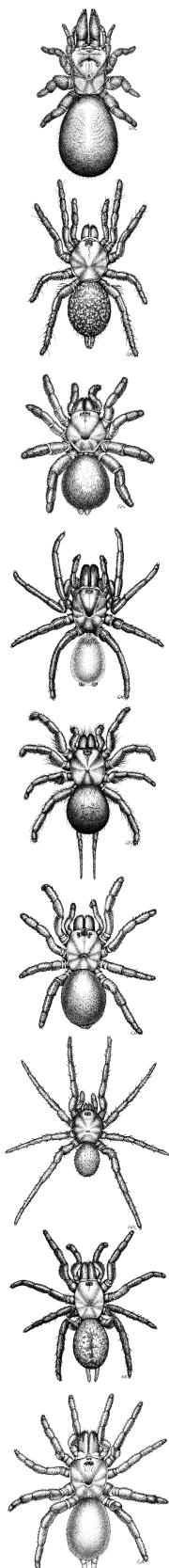
Plant Protection Research Institute Handbook No. 13

Baboon and Trapdoor Spiders of Southern Africa: An Identification Manual

A.S. Dippenaar-Schoeman



Plant Protection Research Institute
Agricultural Research Council



Baboon and Trapdoor Spiders of Southern Africa: An Identification Manual

A.S. Dippenaar-Schoeman

ARC-Plant Protection Research Institute



Agricultural Research Council, Pretoria

2002

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Cover images

Front cover (clockwise from top left)

Ctenizidae: *Stasimopus rufidens*

Theraphosidae: *Pterinochilus nigrovulvus*

Cyrtoucheniidae: *Ancylotrypa pretoriae*

Migidae: *Moddrigea peringueyi* (photographer: N. Larsen)

Back cover

Left, Idiopidae: *Gorgyrella schreineri minor*

Right, Nemesiidae: *Hermacha* sp. (N. Larsen)

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A

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INTRODUCTION

The Mygalomorphae is a suborder that includes some of the largest and most fascinating spiders — the baboon and trapdoor spiders. Among the arachnids they are regarded as more primitive than most. The oldest fossil records date back to the Triassic Period (Selden & Gall, 1992). They are long-lived animals that are able to survive for up to 20 years in captivity.

Most mygalomorph families are terrestrial and live in silk-lined retreats, either in burrows of various shapes made in the soil or in sac-like chambers made under rocks or on tree trunks. The entrances to these retreats are either open or closed with a trapdoor made out of silk and soil particles. Most species are nocturnal and hide during the day in the retreats. At night they wait at the entrance for passing prey or they wander around in search of food. They prey on a variety of insects and small animals and form an important part of the ecological food web.

Except for scientific descriptions, checklists and short notes in popular field guides, little information is available on the mygalomorph spider fauna of Southern Africa. In this manual, information compiled from published records, with some additional observations, provides the reader with tools for identifying and understanding the vast mygalomorph spider fauna of the subregion. It forms part of a series of practical identification manuals for the families, subfamilies, genera and the more common species of spiders found in Southern Africa.

This manual contains the following information:

- illustrated keys, diagnostic and descriptive characters, taxonomic notes, and notes on the natural history and distribution of the mygalomorph families, subfamilies and genera;
- illustrated keys to species (where possible), and taxonomic notes and distribution records for each species;
- a comprehensive bibliography;

- appendixes with the following:
 - * alphabetical list of families, genera and species;
 - * list of generic synonyms;
 - * list of abbreviations.

Currently, 15 families of mygalomorph spiders are recognized worldwide, 11 of which are found in the Afrotropical Region and 10 in Southern Africa. They are represented by the following numbers of genera and species:

FAMILY	GENERA	SPECIES
Atypidae	1	1
Barychelidae	3	5
Ctenizidae	1	40
Cyrtacheeniidae	2	37
Dipluridae	2	5
Idiopidae	6	65
Microstigmatidae	1	6
Migidae	2	26
Nemesiidae	5	49
Theraphosidae	5	47
TOTAL	28	281

The keys apply only to the Southern African members of these families and may not always be applicable to other mygalomorphs of the world. They should be used with reference to the diagnostic characters given for each family, subfamily, genus or species. The keys are artificial in that they do not reflect relationships or natural classifications.

Geographical coverage: The area covered is the Southern African subregion here defined as the area south of the Cunene and Zambezi Rivers. It includes the following seven countries: Botswana, Lesotho, Mozambique (southern half), Namibia, South Africa, Swaziland and Zimbabwe (fig. 1a).

Special attention is given to the South African

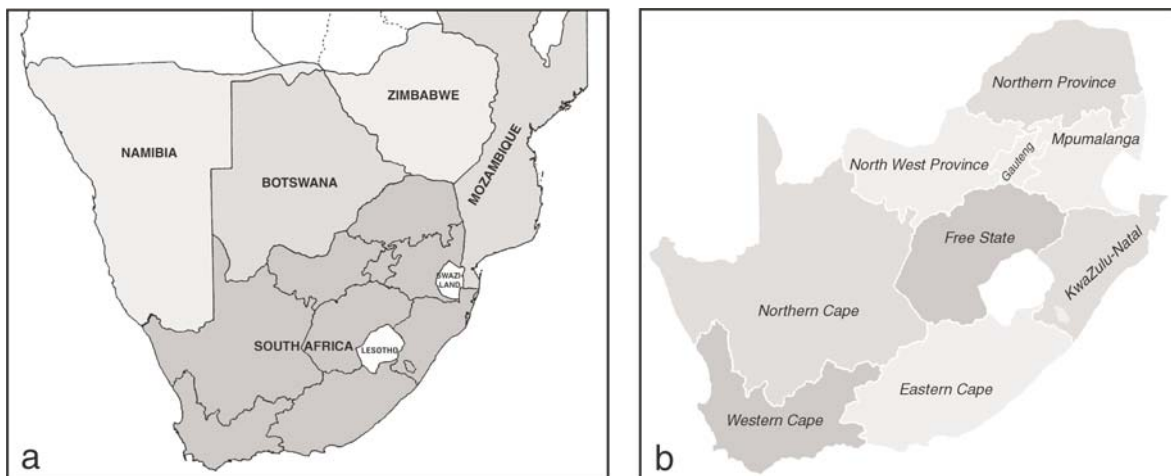


Fig. 1. **a:** Map of Southern Africa showing countries referred to in the text; **b:** South Africa showing the nine provinces.

fauna and the distributions of species are grouped according to the nine provinces (fig. 1b): Eastern Cape, Free State, Gauteng, KwaZulu-Natal, Mpumalanga, Northern Cape, Northern Province, North West Province and Western Cape. Distribution data for the families

are based on published records while new records refer to unpublished records from the National Collection of Arachnida (NCA) at the Plant Protection Research Institute, Agricultural Research Council, Pretoria.

HISTORICAL BACKGROUND

South African spiders were first mentioned by Petiver in 1702. The first mygalomorph collected and described from Southern Africa was the theraphosid *Mygale atra* Latreille, 1832. Only in 1871 did Ausserer establish the first Southern African genus, *Harpactira*, with *M. atra* as the type species. Owing to colonial expansion, hundreds of specimens from Africa were dispatched to museums in France, Germany and the United Kingdom. During the Second Anglo-Boer War (1899–1902) large consignments of mygalomorph spiders were dispatched from South Africa to the United Kingdom (Smith, 1990a). Some of the most important research on Southern African mygalomorphs was carried out by:

- * Simon (1889–1907), who established 38% of the presently recognized Southern African genera and 4.3% of the species;
- * Pocock (1889–1903), who made important contributions by describing 30 mygalomorph species, 10% of the presently known fauna;
- * Purcell (1902–1908), stationed at the South African Museum for ten years, was responsible for the establishment of 21% of the mygalomorph genera and 75 species that represent 26% of the known species;
- * Hewitt (1910–1935) worked both at the Transvaal and Albany Museums and described 41% of the presently known species, a total of 117;
- * Tucker (1917–1920) described one genus and 12 species of African mygalomorphs, while Strand (1906, 1907) described 5 species and Lawrence (1927–1952) 9 species.

It was only in the 1980s that Griswold, while working at the Natal Museum for a short period, undertook the first revisionary studies of the families Microstigmatidae and Migidae, while Coyle (1984, 1995) revised the genera of the Dipluridae. The only other revisions are those of De Wet & Dippenaar-Schoeman (1991), who revised the genus *Ceratogyrus*, and Smith (1990a), who presented a detailed account of the theraphosids of Africa. Some African theraphosid genera are currently being revised in the UK by Gallon (2001, in press). However, most (86%) of the African mygalomorph genera still require revision.

HIGHER CLASSIFICATION

At present, about 35 000 spider species are recognized, grouped into 108 families. Arrangement into a 'natural' system is still a matter of controversy as can be seen by the more than 20 different spider classification systems that have been proposed since 1900 (Foelix, 1996).

The spider order Araneae is usually divided into the Mesothelae and Opisthothelae. The Mesothelae are represented by a single family (Liphistiidae), two genera and 40 species. They have several primitive characters such as a segmented abdomen and four pairs of spinnerets. They are not known from the Afrotropical Region. The Opisthothelae are represented by two suborders, the Mygalomorphae and the Araneomorphae.

- **Mygalomorphae:** Represented by 15 families, 260 genera and about 2200 species, including, e.g., the baboon spiders (known as tarantulas in the New World) and trapdoor spiders. They have unsegmented abdomens, four booklungs, usually four spinnerets (lacking anterior median spinnerets) and their fangs are directed paraxially (fig. 4h).
- **Araneomorphae:** Represented by 93 families, about 2700 genera and 32 800 species (known as the 'true' or less primitive spiders), representing about 94% of the known species. They usually have two booklungs and/or tracheae, six spinnerets and their fangs are directed diaxially (fig. 4g).

Raven (1985) and Coddington & Levi (1991) presented a classification (fig. 2) for all mygalomorphs based on 39 characters. Two main microorders, the Tuberculotae and Fornicephalae, are recognized.

CLASS ARACHNIDA

ORDER ARANEAE

OPISTHOTHELA

SUBORDER MYGALOMORPHAE

MICROORDER TUBERCULOTAE

[*sloping thoracic region; serrula present; distinct eye tubercle*]

• Mecicobothrioidina

[*loss of tarsal spines; absence of palpal conductor; sloping thorax*]

Mecicobothriidae

Microstigmatidae

• Quadrithelina

[*with corrugiform trichobothria*]

Dipluridae

Nemesiidae

• Theraphosoidina

[*trichobothria on tibiae, metatarsi and tarsi of all legs and palps*]

Barychelidae

Paratropidae

Theraphosidae

MICROORDER FORNICEPHALAE

[*arched head region; stout tarsi; slender front legs*]

• Atypoidina

[*reduction of tarsal trichobothria; rastellum absent*]

Atypidae

Antrodiaetidae

• Rastelloidina

[*elevated cephalic region, broad procurved fovea; rastellum present*]

Cyrtaucheniidae

Idiopidae

• Ctenizoidina

[*strong spines laterally on tibiae, metatarsi and tarsi I and II*]

Ctenizidae

Actinopodidae

Migidae

Fig. 2. Classification of the mygalomorph families of the world (after Raven, 1985, and Coddington & Levi, 1991). Families in boldface are discussed in the text.

MORPHOLOGY

The body of a spider is divided into two major regions: the **cephalothorax** (prosoma) and **abdomen** (opisthosoma) connected by a narrow pedicel. The following morphological details are considered in the accounts on the families: cephalothorax: carapace, sternum, eyes, chelicerae, mouthparts; appendages (legs and palps); and abdomen (dorsum and venter of abdomen, spinnerets and genitalia).

Mygalomorph spiders are sexually dimorphic. Males differ from females not only in the shape of the genitalia, size and colour, but the dimorphism extends to numerous other characters, e.g. the presence or absence of setae and scopulae on the legs, differences in the shape of the teeth on the tarsal claws, the shape of the carapace, modification or reduction of characters, e.g. preening comb and rastellum. Sexual dimorphism renders the identification of conspecifics or congeners problematic. Frequently, the characters used to unite groups are limited to either males or females. Descriptions of species are often based on one sex only.

Cephalothorax

Carapace (fig. 3a): The carapace is divided into two regions, cephalic and thoracic. In most species the division is clearly demarcated by the cervical groove. Behind the cervical groove a depression, known as the fovea, is present in all mygalomorphs. It serves as an attachment site for the dorsal muscles of the sucking stomach and the muscles to each chelicera. The shape of the fovea is an important generic character in being straight, pro- or recurved. In *Ceratogyrus*, a genus of the Theraphosidae, the fovea has a horn-like extension (fig. 69e). Several furrows radiate from the fovea, and are very distinct in, e.g., Theraphosidae. The width of the clypeus, the area between the anterior eyes and the carapace edge, is another important generic character.

The integument in mygalomorphs varies from very hairy (Theraphosidae) to almost smooth, without hairs (Ctenizidae).

Sternum (fig. 3b): The undivided sternal plate (sternum) lies on the ventral side of the carapace. The sternal plate is usually covered with short setae. In the Mygalomorphae the sternum bears sigilla, small circular impressions devoid of setae. The sigilla correspond with the internal thoracic attachment sites for the muscles that actuate the legs. The sigilla play an important role in the identification of the genera. They vary in number from 2 to 6 and their shapes vary from circular to pear-shaped. Sigilla are either positioned close to the edge of the sternum or more centrally. Anteriorly the sternum is marked by a distinct groove, the labiosternal junction.

Eyes (fig. 3c): Mygalomorph spiders have eight simple eyes arranged in two or three rows. They are frequently grouped together on an eye tubercle. The most common arrangement is in two rows that can be straight, procurved or recurved. The eyes are named according to their position on the carapace, namely anterior median eyes (AME), posterior median eyes (PME) and posterior lateral eyes (PLE). The median ocular quadrangle (MOQ) is the area included by the four median eyes.

Chelicerae (fig. 4a, b): Each chelicera consists of a stout basal section (paturon) and a smaller, movable distal section, the fang. The fang usually rests in a groove, the chelicer furrow. One or both sides of the furrow may be armed with teeth (promarginal and retromarginal teeth) that are used to masticate prey. The chelicerae in some families, such as the Theraphosidae, bear dense scopulae and/or stridulating structures on the outside. In the mygalomorphs the fangs are usually stout and long. Movement of the chelicerae is

paraxial (fig. 4h) in all families except the Migidae where it is directed more obliquely (fig. 52c). The distal edge of each chelicera is usually provided with strong bristles or spines that are collectively known as a rastellum (fig. 4b). These spines are used to dig and smoothen the walls of burrows. They are usually present in the Rastelloidina group of burrowing mygalomorphs. The rastellum is usually less developed in males.

Mouthparts (fig. 4a): The basal segment (coxa) of the palp is enlarged to form chewing mouthparts, the endites (gnathocoxa). In the Mygalomorphae the endites are broadened laterally and frequently the anterior part of each endite has an anterior lobe (fig. 7g). In the Tuberculotae group the rim of the endite bears a cuticular, serrated ridge known as the serrula. This is used in a saw-like fashion to cut prey. The promargins of the endites are fringed with scopulae, dense coverings of setae used to filter the liquefied food. Between the endites is the labium, which is free of the sternum. In the Mygalomorphae, most genera bear cuspules on the endites and labium (fig. 3b). Cuspules are small, socketed spines that can be cylindrical or expanded at the tips.

Appendages

Legs (fig. 3a): Each of the eight legs has seven segments. The legs usually bear setae, spines, various sensory setae and receptors. Some of the sensory setae are fine and hair-like, set vertically in conspicuous sockets and are known as trichobothria. In the Barychelidae and Theraphosidae these tarsal trichobothria are short, thick and clavate (fig. 7c). In some families the tarsi and sometimes the metatarsi have dense, short, stiff setae covering the ventral surface. This brush of setae is known as scopulae, and improves the spider's grip on the substrate or prey. Terminally, the tarsi can have two (fig. 7a) or three claws (fig. 7b). The two-clawed spiders (Barychelidae and Theraphosidae) have a thick pad of iridescent scopulae that surrounds and obscures the paired claws (fig. 7a). In the males of some genera, tibia I is provided with a mating spur that is variable in shape (fig. 4c). A preening comb,

consisting of a distal cluster of 2–6 setae, with bases touching, is sometimes present on metatarsi III and/or IV in some genera (fig. 23f).

Palps (fig. 3a): The palps are leg-like appendages consisting of six segments (compared to seven leg segments). The palpal metatarsus is lacking. In females the palp is simple and usually bears a single tarsal claw. In adult males, the last segment of the palp is modified into a secondary copulatory organ (fig. 4e). In the mygalomorphs, the male copulatory organ is very simple. The tarsus of the palp (the cymbium) carries an extension in the form of a pear-shaped bulb, or palpal organ. A blind duct spirals through the bulb and opens at the tip. The narrow portion of the tip is called the embolus. The bulb acts as a reservoir for sperm, functioning like a pipette, and can take up a droplet of sperm. The sperm is then stored inside the duct until mating occurs.

Abdomen

The abdomen (fig. 3a,b) is joined to the cephalothorax by a thin pedicel through which the circulation and feeding systems are canalized. The exoskeleton of the abdomen is much thinner than that of the cephalothorax and this allows great expansion of the abdomen when prey is being fed on, or when a large number of eggs is being formed in the female. The abdomen is variable in size and configuration, but is usually elliptical, oval or globose in most mygalomorphs, and usually hirsute. The heart is sometimes visible through the integument as a longitudinal mark. The dorsum can be decorated with patterns consisting of, for example, spots, bands, chevrons or a folium.

Spinnerets (fig. 4d): In all Southern African mygalomorph spiders (except the Atypidae with three pairs and some barychelids with one pair), two pairs of spinnerets are situated ventrally in front of the anal opening. The spinnerets have great mobility and are well provided with muscles. The position, thickness and number of spinneret segments are characters used at the generic level.

MORPHOLOGY

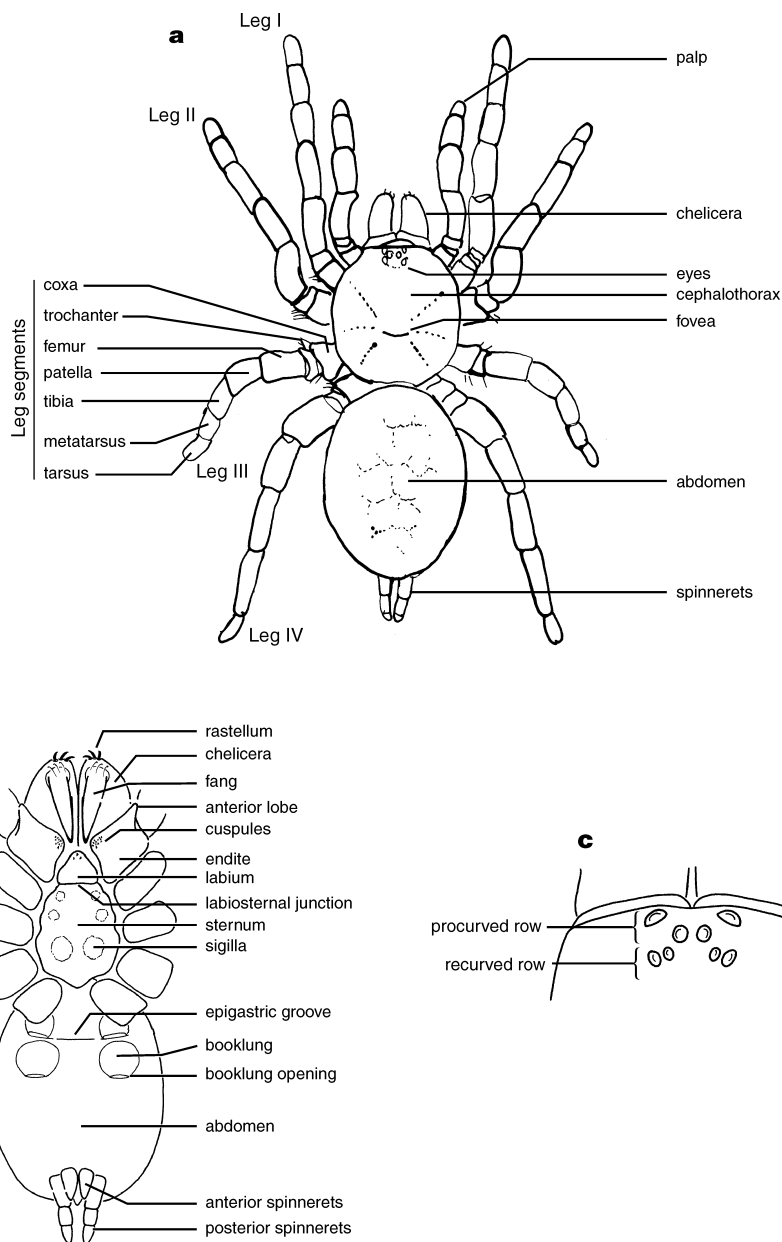


Fig.3. Mygalomorphae external morphology. **a:** dorsal view of body; **b:** ventral view of body; **c:** eye pattern, dorsal view.

Respiratory system (fig. 3b): Mygalomorph spiders possess two pairs of booklungs, one pair situated above the epigynal furrow and the second pair just below. The external openings of the booklungs are present on either side of the epigastric region. The openings are slit-like, except in the Microstigmatidae in which they are oval (fig. 7f). The branchial opercula are

formed by two pairs of pale or cream-coloured external plates visible on the ventral side of the abdomen, just in front of and below the epigastric furrow.

Genitalia (fig. 4f): In contrast to most araneomorph spiders, the genitalia of mygalomorph spiders are simple in both sexes. In the female,

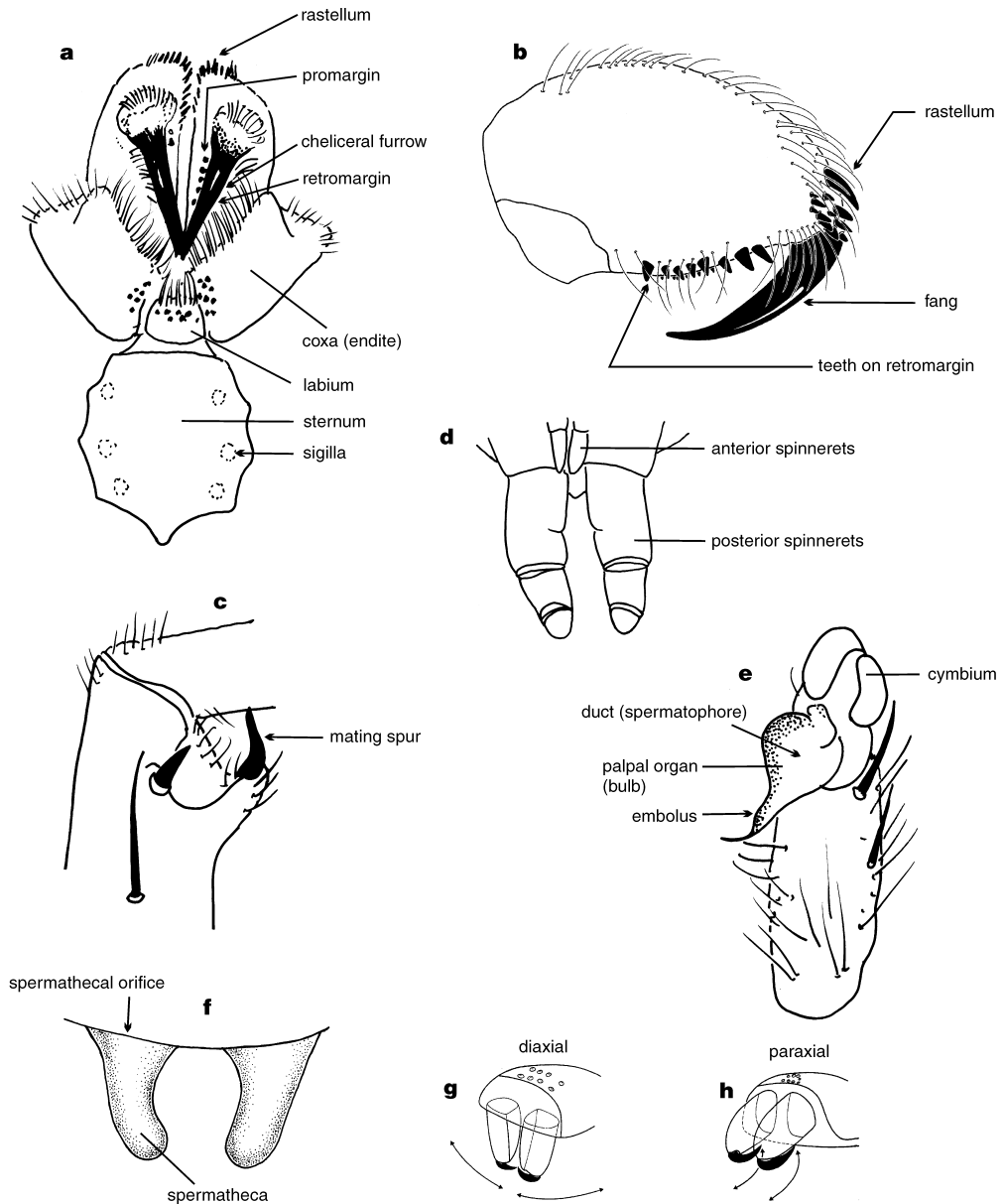


Fig. 4. Mygalomorphae external morphology. **a:** mouthparts and sternum, ventral view; **b:** chelicera, lateral view, showing rastellum; **c:** mating spur on front leg of male; **d:** spinnerets, ventral view; **e:** male palp, showing the secondary sex organ; **f:** spermathecae of female; **g:** diaxial cheliceral pattern; **h:** paraxial cheliceral pattern.

the terminal part of the oviduct is known as the *uterus externus*. In the Mygalomorphae it ends in the primary genital opening (gonopore) situated between the anterior booklungs in the epigynal groove. The pair of sac-like spermathecae or seminal receptacles connects directly to the *uterus externus*, which is also

the site where fertilization takes place. The spermathecae are variable in shape and open through the spermathecal orifices to the outside. The orifices usually have a wide diameter (fig. 4f). There is no sclerotized external epigynum, so it is difficult to determine when a female is mature.

NATURAL HISTORY

The Mygalomorphae are a diverse group of spiders and most species (except the Microstigmatidae) live in silk-lined retreats. The retreats can either be vertical burrows or chambers made under rocks or under bark on trees. The retreats are left open or can be closed by a trapdoor. Extensions to the entrance frequently include lids, signal threads, collars, turrets or catch webs. These structures extend the range at which the substrate vibration receptors, located on the palps and legs of the spiders, can detect prey (Coyle, 1986). Most prey is captured at or close to the entrance of the retreat. The construction of trapdoors and other structures around the retreats have evolved independently many times (Coyle *et al.*, 1992). The Microstigmatidae are the only free-running mygalomorph spiders in Southern Africa that do not live in a burrow or web. Mygalomorphs are usually nocturnal and hide during the

day in a retreat. At night some species wander around in search of food, while most of the burrow-living spiders lie-and-wait for prey at the entrance to the retreat.

Mygalomorph spiders produce very thin and dry silk threads. Several hundred threads are produced simultaneously to form ribbons, and with movement of the spinnerets and abdomen, a silk carpet is laid down on the substrate. Slow perpendicular movements of the body and spinnerets are used to enlarge this carpet. Silk is used in various ways, such as lining of burrows, and construction of retreats, webs, trapdoors and egg sacs.

Burrows

Burrow shapes

Burrows are made in a variety of microhabitats. The shape of the burrow and the microhabitat in

Table 1. Types of retreats of Southern African Mygalomorphae spiders.

FAMILY	TYPE OF RETREAT/BURROW
Atypidae	silk-lined burrow entrance, an excavated ambush chamber lined with a silk layer that is used to trap prey (fig. 6e,f)
Barychelidae	variable silk-lined burrows with one or more entrances (fig. 15a–c) or entrance with leaf-and/or grass-covered turret, or shallow retreat under stones with one or two trapdoors.
Ctenizidae	silk-lined burrows usually with rigid, cork-like trapdoors that are either circular or D-shaped (fig. 19a–c)
Cyrtacheniidae	simple silk-lined burrows or burrows with side passages; frequently Y-shaped with a flexible wafer trapdoor or closed with mud pellets (fig. 26a–c)
Dipluridae	tubular or funnel-shaped silk retreat made in crevices with entrance extending outwards to form irregular, interconnected funnel- or sheet-like webs (fig. 31f)
Idiopidae	silk-lined burrows or chambers closed with wafer- or cork-like trapdoors (fig. 35e)
Microstigmatidae	free-running wanderers, hiding under debris on the forest floor
Migidae	bag- or sac-like arboreal retreats or terrestrial silk-lined burrows closed with flap-like trapdoors (fig. 52h)
Nemesiidae	silk-lined burrows that are either simple or Y-shaped, or silk-lined tunnels and chambers made under rocks (fig. 60a,b)
Theraphosidae	silk-lined burrows or silk-lined chambers made under rocks, usually without a trapdoor but entrance covered with a thin layer of silk when not active (fig. 5a,i)

which it is made, differ between genera and families. Burrows are frequently found on open grassy plains, excavated vertically in the ground. A thick layer of silk covers the inside walls of the burrow. The silk layer prevents the soil from caving in and renders a well-balanced microclimate. The eggs are usually deposited at the bottom of the burrow and are covered with a layer of silk to protect them from predators, parasites and microbial infections. The spider normally digs only one hole during its life-time and enlarges it as it grows older. The depth of the burrows varies depending on obstacles in the ground, the size of the spider, hardness or softness of the substrate, soil type and slope of the ground. Although many burrows are classified as having a simple shape, the shape is often variable, depending on obstacles such as roots or pebbles that the spider encounters in the soil (Decae, 1996).

Some of the variations encountered are:

- burrows consisting of a single shaft that can be uniform in width (fig. 5a) or with a chamber at the bottom (fig. 5f) or a shaft of varying width (fig. 5g);
- burrows with side passages or shafts (fig. 5c,e);
- burrows of varying shape, e.g. Y-shaped (fig. 5d) or U-shaped (fig. 5h), or
- sac-shaped burrows made, for example, under rocks (fig. 5i).

Burrow construction

The spiders use a variety of excavating methods. The fangs and rastellum on the chelicerae are used by most trapdoor spiders to loosen soil (Coyle *et al.*, 1992) while spiders without rastellums (e.g. Theraphosidae) use their chelicerae and fangs. Spiders with rastellums are able to initiate and excavate new burrows while spiders without rastellums usually adapt existing holes in the ground as their burrows. Smith (1990a) suggested that most African theraphosids are opportunistic burrowers, extending, for example, insect, mole and lizard holes.

The construction of burrows of trapdoor spiders takes place in different stages (Coyle, 1981). During the initial excavation stage a burrow is

made 1.5–2.0 times the spider's length. Then follows the door-construction stage. During the third excavation stage the burrow is lengthened under cover of the trapdoor.

Different methods are used to remove the soil from the burrow. Loose soil is ejected from the burrow by some idiopids, using their legs or palps (Coyle *et al.*, 1992). Others push the clumps of soil together with the front legs, bind them together with silk threads and carry them in their chelicerae to the outside. Solid clumps of soil of a manageable size are picked up with the fangs and removed (De Wet & Dippenaar-Schoeman, 1991). Migids possibly use the keels on the fangs to loosen bark or soil when a burrow is constructed (Griswold, 1987a).

Why a burrow?

Burrows provide spiders with the following protection:

- against predators and parasites;
- for the eggs and developing spiderlings in the brood chamber;
- during the moulting process;
- during mating;
- while intercepting or ambushing prey;
- during inactive periods, especially in winter;
- against flooding, as the silk is waterproof;
- against veld fires when spiders withdraw deep into the burrow;
- against thermal stress as temperature and humidity are relatively stable in the burrow;
- against fungal and bacterial attack owing to antibiotic and antifungal properties of the silk (the macromolecular structure of silk is inert and most enzymes cannot decompose it).

Burrow entrances

A variety of structures is used to cover the entrances of burrows. In the trapdoor families the burrows are closed with a trapdoor that usually fits perfectly into the burrow entrance. The outer part of the trapdoor is always well camouflaged to blend in with the surrounding substrate. The thickness and shape of trapdoors vary between families, genera and species. Frequently, the

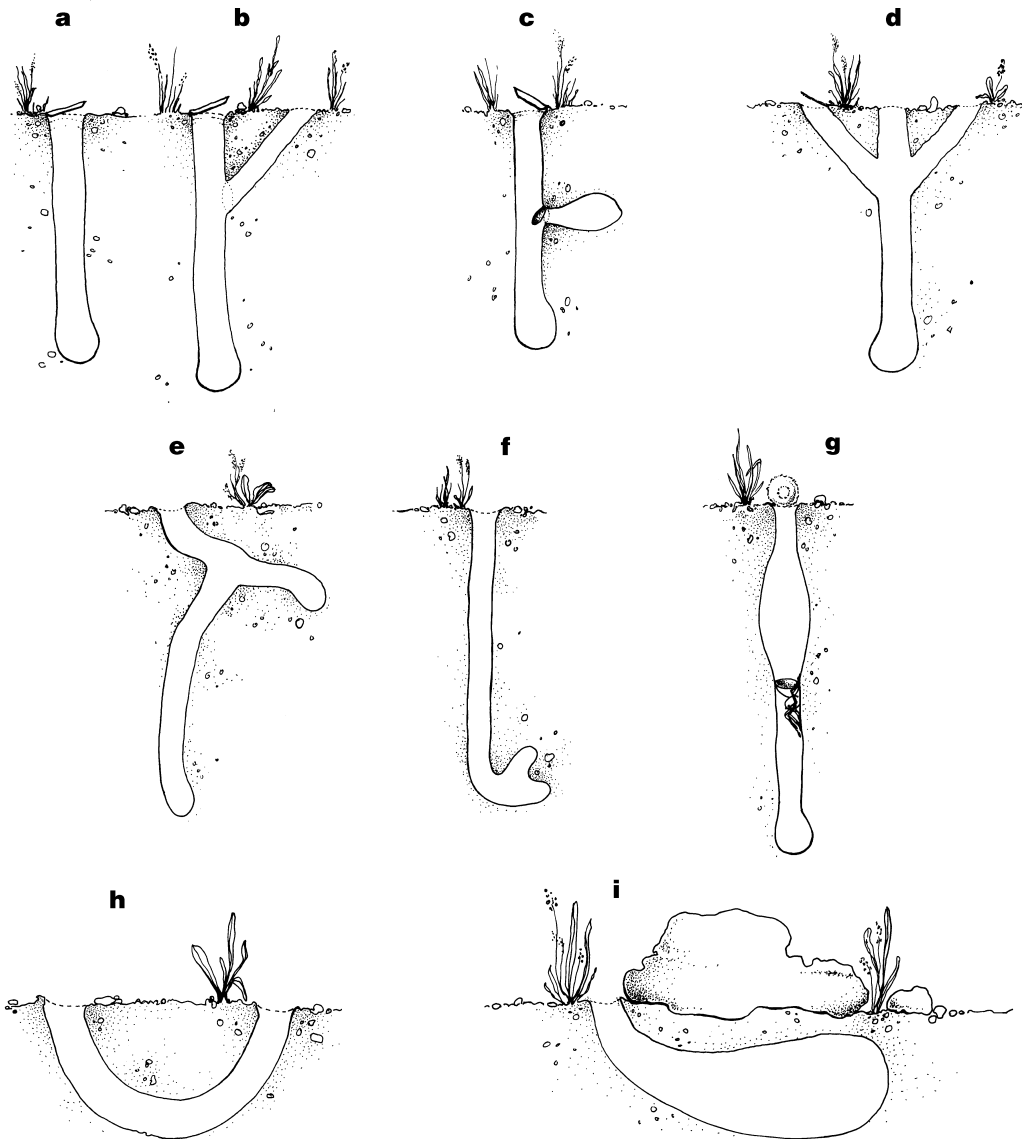


Fig. 5. Burrows of mygalomorph spiders. **a:** simple vertical burrow with wafer-lid; **b:** Y-shaped burrow; **c:** burrow with side shaft closed with a trapdoor; **d:** burrow with two side shafts; **e:** burrow with chamber to the side; **f:** burrow with chamber at bottom; **g:** burrow of *Galeosoma* sp.; **h:** curved burrow with two entrances; **i:** retreat chamber below a rock.

entrances to the burrows are provided with various structures that can serve as an early prey detection system.

Cork-lid trapdoors: One type of trapdoor resembles the cork of a bottle and is known as a cork-lid trapdoor (fig. 6b). It fits snugly into the entrance. The shape of the lid varies between species, from round to D-shaped. The underside is usually

unevenly convex with round edges. It is usually provided with a circle of small pits (fig. 6c). These pits provide holding spaces that enable the spider to pull the lid closed by numerous strong setae on the front legs and palps. When the spider closes the trapdoor it is very difficult to prise it open. The cork-lid trapdoor probably provides sufficient protection, as the use of side tunnels or other structures is not common.

Folding collars: The lining of the burrow sometimes extends slightly onto the soil surface. This short, flexible collar collapses inwards to close off the burrow and to camouflage the entrance. At night it is opened when the spider takes up position in the entrance. Debris is sometimes added to the silk extension as camouflage.

Pellet or stone lid: A pellet made of silk and soil particles or small stones is used by some spiders to close the burrow entrance.

Silk layers: Entrances sealed with a silk layer can serve as protection in, e.g., the theraphosids where entrances are frequently covered with a thin layer of silk when the spiders are not active. Thicker layers are used as a type of web to catch prey, as found in the Atypidae. In the African atypids the entrance chamber is closed with a thick layer of silk and it is part of the spider's prey-detection system (fig. 6e,f). Prey landing on the silk is impaled through the silk.

Wafer-lid trapdoors: Wafer-lid trapdoors consist of a flexible, limp flap that is usually merely a continuation of the burrow's wall lining. Spiders that close their burrows with this type of lid usually have longer legs and usually rush farther out of their burrows to catch their prey than other spiders. Wafer-lid trapdoors remain open more readily while the spider is out, allowing rapid return into the burrow. Members of, for example, the Cyrtaucheniidae, Idiopidae and Migidae construct this type of trapdoor. In some idiopids the thin wafer lid is gradually transformed over a period of weeks from being thin, flat and rather smooth-edged, to one that is thicker, more concave with silk tabs along the edge (Coyle *et al.*, 1992). These silk tabs may increase the prey-sensing effectiveness of the door and allow the door to be pulled further into the burrow and held more securely.

Turrets: The burrows sometimes extend above the soil surface in the form of an aerial tube or turret. The turret is usually formed by the incorporation of plant material and soil particles. It can be rigid and with or without a trapdoor. The turret protects

the burrow against flooding and can also serve as an early prey-detection device.

Trapdoor construction

Trapdoors are made of soil, often clay, molded into shape and reinforced with silk (Decae *et al.*, 1982). Different methods are used to construct trapdoors (Coyle *et al.*, 1992).

Door-moulding: In some families the door is moulded from pellets of excavated soil particles, often clay reinforced with silk. This method has been observed in ctenizids (Coyle, 1981), migids (Todd, 1945) and idiopids (Coyle *et al.*, 1992).

Door-cutting: In some nemesiids and idiopids the door is cut out of one end of the sealed retreat (Todd, 1945).

Trapdoor camouflage

The outside of the lid is always well camouflaged and resembles the immediate surroundings (fig. 20c). Van Dam & Roberts (1917) observed that plant material found in the vicinity is usually incorporated into the outside of the door, such as bundles of short straws placed upright or across the lid, or twigs stuck upright onto the lid. Pebbles are incorporated into the door by e.g. *Galeosoma pilosum* (Van Dam & Roberts, 1917).

Additional defence mechanisms

Although the burrows with trapdoors provide protection against most natural enemies, e.g. wasps and centipedes, some predators are still able to invade the burrows and additional defence mechanisms are used to overcome this.

- **body plugs:** in *Galeosoma* (Idiopidae) the hardened posterior part of the abdomen is used as plug to provide a false bottom to closed-off parts of the burrow (fig. 37g);
- **emergency exits:** side shafts are provided with emergency exits through which the spiders can escape (fig. 5b,d);
- **folding collars:** built into sections lower down the burrow, these are pulled closed to seal off the bottom part;
- **side shafts:** are closed with pebbles or trapdoors (fig. 5c);

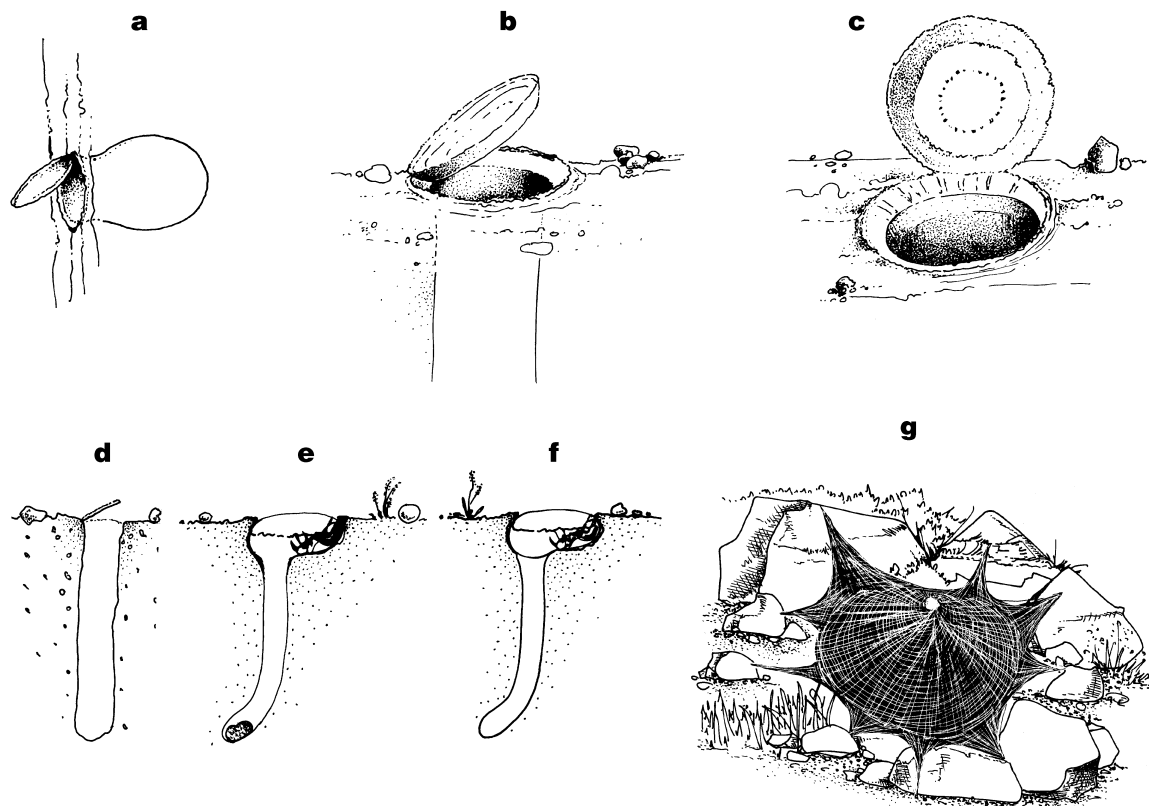


Fig. 6. Retreats, trapdoors and webs of mygalomorph species. **a:** sac-like retreat of a migid in a tree; **b:** cork-lid trapdoor; **c:** underside of a trapdoor; **d:** burrow with a wafer-lid; **e, f:** burrow of *Calommata* with (e) and without (f) eggs; **g:** web of a diplurid.

- **pebbles and stones:** pebbles and stones are positioned halfway down the main burrow. When in danger, the spider pulls the stone down to seal the lower part off from the top part.

Prey-capture methods

The prey-capture behaviour of mygalomorphs has not been as well studied as that of araneomorphs, and information is to a large extent scattered in the literature. Buchli (1969) published a review of the literature on prey capture while Coyle (1986) discussed the role that silk plays in the capture process in mygalomorphs.

Burrow-living mygalomorphs

Most burrow-living mygalomorphs are sit-and-wait predators. Prey is usually detected by substrate vibration receptors on the palps and legs of

the spiders. The receptors may consist of trichobothria that detect prey-generated air currents, or silk- or soil-vibration detectors such as the slit sensilla or club-shaped trichobothria (Coyle, 1986). The prey usually consists of small animals such as insects that wander within range of the burrow, in which case the spiders cannot be too selective. Different methods of prey capture have been reported by Decae *et al.* (1982).

Method I: regarded as more primitive (Decae *et al.*, 1986). Here the spider sits in the entrance of the burrow and prey passing close-by is pounced on. The spider usually only catches prey that wander close to the burrow. This method has been observed in the Barychelidae (Raven, 1994), the more primitive Nemesiidae (Buchli, 1969) and some Theraphosidae (De Wet & Dippenaar-Schoeman, 1991).

Method II: this method is more commonly found in the true trapdoor spiders. The spider waits behind a slightly open door for potential prey to pass by. The spider then leaps onto it, flinging the door open in the process. These spiders never leave the burrow completely, as the claws of the fourth legs keep a firm grip on the rim. The hunting area is restricted to the immediate surroundings of the burrow that can be reached by the spider. Most members of Ctenizidae use this method and it has also been observed in the Barychelidae (Raven, 1994), Migidae and Theraphosidae.

Method III: here spiders make use of a early-detection system. The spider increases the prey detection area by assemblages of silk threads, twigs, grass or debris around the burrow (Main, 1978). The spider waits in the entrance, with legs resting on the plant material or silk threads. Vibrations are detected when prey comes into contact with this material. Members of the Atypidae, Cyrtachenidae, Migidae, Nemesiidae and Theraphosidae use this method.

Method IV: has been observed in a ctenizid trapdoor spider in Greece. The web extends from the burrow entrance in such a way that it not only warns the spider of prey, but also prevents prey from escaping (Decae *et al.*, 1982).

Web-living mygalomorphs

Although some diplurids excavate burrows for retreats, most do not. They construct tubular or funnel-shaped silk retreats in crevices, under stones or under logs while some make them above-ground in shady places in tree trunks and holes in riverbeds (Coyle, 1986). A sheetweb is connected to the burrow, rendering a funnel-and-sheetweb prey-capture strategy (fig. 6g).

Free-living mygalomorphs

The microstigmatids are the only Southern African mygalomorphs that do not construct a burrow or retreat. They appear to make minimal use of silk and are characteristic members of forest cryptofauna (Griswold, 1985a; Van der Merwe, 1994).

Natural history

Mating

Mating usually takes place in spring and summer. Prior to mating, the male transfers sperm from the genital opening under the abdomen to the secondary sexual organs on the palps. This is achieved by depositing sperm onto a small sperm web. The sperm is then absorbed by the palpal organ, in which it is stored until mating. Adult males now usually change their life-style completely to become wanderers in search of a female.

Sexual dimorphism is evident in many species with the male being smaller than the female, and often looking very different. To ensure recognition by the female, the male uses various techniques to approach her. In some species, the male has an intricate courtship ritual that involves waving and raising of the palps. In some of the burrowing spiders, the male carefully approaches the female. He reveals his presence by tapping rhythmically against the sides of the burrow. Courtship in most mygalomorphs is usually of short duration (Petrunkévitch, 1911; M. Paulsen, pers. comm.). The mating spur on the front leg of the male is used to force open the jaws of the female. This prevents her from attacking him during mating.

Egg-laying

Eggs are usually laid during summer. Egg-laying does not necessarily take place directly after mating. In *Ceratogyrus bechuanicus* the female lays eggs only five months after mating (De Wet, 1991). The number of eggs produced by the different spider genera varies greatly. Eggs are usually deposited in an egg sac made of silk. The number of eggs per sac and the construction of the sac differ greatly between species, as do also its size, shape and colour. In most families the egg sac is deposited at the bottom of the retreat.

The female tends the eggs and young. Newly-hatched spiderlings are not always mobile after hatching. In *Ceratogyrus darlingi* (Smith, 1990a) and *C. bechuanicus* (De Wet, 1991) of the family Theraphosidae the young start moving about

approximately 50 days after hatching. In some genera the young spiders stay with the mother for some time (De Wet, 1991).

Ecdysis

As young spiders grow, they undergo a number of moults. The first moult takes place in the egg sac. A few days before moulting commences, the spider stops eating. First the skin under the carapace parts just above the coxae. The carapace then lifts off like a lid but remains attached at the pedicel. The skin of the abdomen then tears at the side and the abdomen breaks free. The legs, palps and chelicerae are freed from the skin with rhythmic movements, and finally the spider pulls free. At this stage the spider is soft and defenceless against predators and it takes a while before the new skin hardens.

In young spiders, moulting is completed within a few minutes but as the spider matures the process may last for an hour or more. Males usually moult fewer times than females. Araneomorph spiders moult only until they reach maturity, while a female mygalomorph spider which lives much longer, may also moult after reaching maturity. If a leg is lost between moults, the spider is capable of regenerating a new one, which appears after the next moult. Initially, the new leg is shorter and thinner than the others.

Dispersal

As a large number of spiderlings emerge from the nest simultaneously, local overpopulation may quickly lead to competition for available food and even cannibalism. Most mygalomorphs are more or less gregarious. Not much information is available on their dispersal. Cutler & Guarisco (1995) summarized the literature on juvenile dispersal. In most families they disperse by walking away from the burrow. If a favourable patch of ground is found near the burrow of the female, the small spiderling will settle there. They aggregate in such a way that many burrows of juveniles are frequently found grouped around the burrow of the adult female (she is known as the matriarch of the cluster).

This differs between families and genera, however. In *Ceratogyrus bechuanicus* and

C. brachycephalus, small colonies are found scattered over a wide area while in some *Pterinochilus* species large numbers of burrows are usually grouped together in colonies of up to 106 burrows per 80 m² (De Wet, 1991). Reichling (2000) reported on group dispersal by a *Brachypelma* sp., while ballooning has been observed in ctenizid spiderlings (Coyle, 1983, 1985).

Longevity

Most araneomorph spiders in temperate regions live for only one year or sometimes two. However, the mygalomorphs are renowned for their longevity and the atypids can reach seven years and the theraphosids over 20 years of age (Canard, 1986). As a rule, only the females have a high life expectancy. Because males have no role to play after the mating season, they frequently do not live as long as the females and die a few weeks after having mated (Perrett, 1974b; De Wet, 1991; Paulsen, 1998).

Apart from death due to age, there are various other factors that can influence the longevity of a spider, e.g. shortage of food and water, cannibalism, unfavourable habitats, adverse climatic conditions, fires, predation and destruction of their natural habitats by man.

Prey

Mygalomorphs prey on a variety of small animals such as:

- **insects:** Ants, beetles (e.g. Tenebrionidae), cicadas, cockroaches, Orthoptera (e.g. grasshoppers, locusts, crickets), Isoptera (termites), Lepidoptera (mostly Saturniidae and Sphingidae), Hymenoptera (driver ants of the family Dorylidae) (De Wet, 1991; Coyle, 1995; Paulsen, 1999a);
- **arachnids:** Spiders, solifugids (Paulsen, 1999a) and scorpions (Paulsen, pers. comm.); millipedes (Coyle, 1995);
- **small reptiles, amphibians and snails:** Frogs and lizards (Paulsen, pers. comm.), snails (Coyle, 1995).

Natural enemies

Spiders of all stages are attacked by a wide

variety of predators, parasitoids and parasites. They are a food source for a number of animals such as birds, centipedes, reptiles (lizards, chameleons), insectivorous mammals [honey badger, *Mellivora capensis* (Smithers, 1983), shrews, bats, mice and baboons] and other arachnids such as scorpions, solifugids and spiders. Members of the spider family Palpimanidae are frequently found associated with trapdoor spiders and they may prey on them (Van Dam & Roberts, 1917). Mygalomorphs are also attacked by various fungi (Rong & Grobbelaar, 1998).

A number of insects and mites are specialized predators or parasites of spiders in general: Hymenoptera (Sphecidae, Pompilidae, Ichneumonidae) (Ledger, 1979; Scholtz & Holm, 1985; Harris, 1987); dipterous parasitoids (Drosophilidae, Phoridae, Chloropidae); predators of eggs (Sarcophagidae) and endoparasites (Acroceridae). Spiders also have endoparasites such as parasitic nematodes of the family Mermithidae.

Defensive behaviour

Mygalomorphs use different mechanisms to defend themselves against their enemies.

Active defence: Use is made of their ability to produce venom to defend themselves against predators. Mygalomorphs have fairly large fangs and are able to deliver a nasty bite. When threatened, most mygalomorph spiders react by adopting an aggressive posture in which they raise their front legs and throw the front part of their bodies backwards, exposing their large fangs.

Urticating hairs: The release of urticating hairs

from the abdomen is commonly found in Theraphosidae of the New World. The hairs can be shed or inserted by direct contact with potential predators. According to Bertani & Marques (1996), hair-flicking is restricted to burrow-inhabiting spiders of the Aviculariinae and all members of the Theraphosinae, whereas contact urticating hairs are used only by arboreal spiders of the subfamily Aviculariinae. Urticating hairs are absent in theraphosids of the Afro-tropical Region.

Stridulation: When alarmed, some members of the Theraphosidae produce a hissing sound, similar to that of snakes, by rubbing the setae on the chelicerae and palp together.

Toxicity to man

Little is known about the effect of the venom of mygalomorphs on man or animals. A species of *Pterinochilus* (Theraphosidae) from East Africa produces a venom with neurotoxic properties based on experimental work with mice and guineapigs. However, compared to that of, for example, the black button spider, the venom is less toxic (Perrett, 1974a). In Southern Africa, painful bites have been reported from *Harpactirella lightfooti*, a theraphosid species known from the Western Cape Province (Finlayson, 1939; Smithers, 1939).

The spider most venomous to man is the male of an Australian mygalomorph, *Atrax robustus*, of the family Hexathelidae. It is commonly known as Sydney's funnelweb spider and 14 known deaths have been attributed to it (R. Raven, pers. comm.). Hexathelidae do not occur in Southern Africa.

Pterinochilus sp. (Theraphosidae): female defending her burrow.



COLLECTING & CONSERVATION

Most mygalomorph species live permanently in burrows and the usual collecting methods such as sweep-netting cannot be used to sample them. However, the males are more agile, wandering around in search of a mate and are frequently collected in pit traps. Some may land in swimming pools or even wander into houses.

Collecting methods

Burrow-living spiders

Burrow entrances are usually well camouflaged and often found beneath grass tufts, rocks or stones. In trapdoor spiders the lid is usually well camouflaged with soil, twigs, leaves, grass or pebbles and extremely difficult to locate. Potential collecting sites need to be scrutinized very closely. Frequently, an old trapdoor or dis-used burrow will serve as an indicator of their presence. Pit traps are also useful for revealing their presence at a particular site.

Most mygalomorphs are nocturnal, and males wandering around at night could lead you to a female. Van Dam & Roberts (1917) found that trapdoor spiders occur in almost any habitat and they provided advice about where to look for spiders:

Soil surface: Look for faintly marked circles in more or less bare patches of ground. Trapdoor spiders are frequently found in clayey rather than sandy or stony soil. The soil surface can be swept with a small broom or the topsoil can be scraped away with a spade or trowel to reveal a burrow entrance.

Webbing at the base of grass tufts: When moving grass tufts to the side, webbing adhering to the base frequently indicates the presence of spiders. Van Dam & Roberts (1917) found their first atypid spider using this method. Baboon spiders' burrows are also frequently found at the base of grass tufts.

Turning stones: Van Dam & Roberts (1917) found that in areas prone to heavy rainfall, a large percentage of trapdoor spiders are found sheltering under stones or rocks not inhabited by scorpions or centipedes. They frequently found individuals of more than one genus living together under a stone.

Moss-covered banks: Burrows are frequently made in moss-covered banks. Pieces of moss are used to camouflage the entrances. A brush or tweezers can be used to pull bits of the moss apart to locate the entrance.

Leaf litter: Burrows are also made under leaves and debris that collect around the roots of plants. Moving the debris and searching for silk threads may assist in locating the burrows.

Soil patches between roots or in crevices: Burrows are sometimes made in the loose earth, lodged between the roots of aloes and other plants. A long pair of tweezers can be used to locate the burrows.

Under or on bark: Members of the Migidae are found under bark with a rough surface. Small pieces of bark and moss are used to camouflage the burrow and entrance. Use a pair of tweezers to pull bits of the moss apart to locate the entrance.

Web-living mygalomorphs

In the web-building mygalomorphs the webs are usually spread out over the soil surface or bark. The spider can be collected by hand after spraying alcohol onto the web to slow it down.

Free-living mygalomorphs

The wandering mygalomorphs include members of the family Microstigmatidae and males of other families. The microstigmatids are forest dwellers and they have been collected by hand

from under debris on the forest floor. Both sexes have also been collected in pit traps (Van der Merwe, 1994).

Conservation

Of the mygalomorphs, it is mainly the larger Theraphosidae that are in great demand as pets and are consequently regarded as commercially threatened by the International Union for the Conservation of Nature (IUCN) (De Wet & Schoonbee, 1991).

It is suspected that the demand for South African theraphosid spiders has increased since the Mexican red-kneed tarantula was placed in Appendix II of the Conservation of International

Trade in Endangered Species (CITES) (De Wet & Schoonbee, 1991).

Some genera of the African theraphosids, e.g. *Ceratogyrus*, are especially popular with pet owners and collectors because of their unique horn-shaped foveal tubercle. In February 1987 three theraphosid genera, *Ceratogyrus*, *Harpac-tira* and *Pterinochilus*, were added to Schedule VII of the Transvaal Provincial Nature Conservation Ordinance of 1983 as Protected Invertebrate Animals. At present, all Provinces in South Africa follow this recommendation as a guideline and permits are needed to collect and transport the above genera in South Africa.



Entrance of a burrow with a grass turret.



Horned baboon spider (Theraphosidae: *Ceratogyrus bechuanicus*), one of the protected species in South Africa.

KEY TO THE SOUTHERN AFRICAN MYGALOMORPH FAMILIES

1. Claw tufts and scopulae present, scopulae on metatarsi and tarsi usually forming thick pads of iridescent hair surrounding and obscuring the paired tarsal claws (fig. 7a); third tarsal claw absent; body hairy 2
- Claw tufts absent; scopulae, if present, never forming thick, iridescent pads; third tarsal claw usually present (fig. 7b); body hairy or smooth 3
2. Tarsi with clavate trichobothria (>6) dorsally over most of segment (fig. 7c); apical segment of posterior spinnerets long and finger-like (fig. 7d); anterior lobe of endites well developed; clypeus wide (Harpactirinae) **THERAPHOSIDAE** (p. 102)
- Tarsi with 4–6 clavate trichobothria restricted basally (absent in some species of *Cyphonisia*); apical segment of posterior spinnerets short and dome-shaped (fig. 7e); anterior lobe of endites not well developed (fig. 14c); clypeus not wide **BARYCHELIDAE** (p. 24)
3. Booklung openings small, oval (fig. 7f); body covered with blunt-tipped or clavate setae (fig. 48b) **MICROSTIGMATIDAE** (p. 76)
- Booklung openings slit-like (fig. 3b); body without blunt-tipped or clavate setae 4
4. Endites strongly elongated and curved (fig. 7g); cephalic region strongly elevated; chelicerae well developed, almost same length as carapace (fig. 7h); fangs long **ATYPIDAE** (p. 20)
- Anterior lobes not strongly elongated and curved; cephalic region and chelicerae and fangs not as strongly elevated or developed 5
5. Fangs short, directed obliquely with two distinct longitudinal keels on outer surface (fig. 8a) **MIGIDAE** (p. 81)
- Fangs longer, paraxially directed, without keels 6
6. Rastellum present, distinct (fig. 8b) 7
- Rastellum absent or, if present, comprising only a few weak spines 9
7. Anterior lateral eyes in front of other eyes, close to clypeal edge (fig. 8c) **IDIOPIDAE** (p. 56)
- Anterior lateral eyes not in front of other eyes, all eyes grouped closely together (fig. 8d) 8
8. Front legs of female with lateral bands of short, thorn-like setae distally (fig. 8e); paired tarsal claws of female with a single row of teeth comprising one long and two smaller teeth **CTENIZIDAE** (p. 29)
- Front legs of female without short, thorn-like spines; paired tarsal claws of front legs of female with two rows of teeth (fig. 8f) or with one large bicuspid tooth, similar in males but teeth in some genera arranged in an S-shaped row (fig. 8g) **CYRTAUCHENIIDAE** (p. 39)
9. Tarsi without scopulae; paired claws with one row of teeth; posterior spinnerets very long and widely spaced (fig. 8h) **DIPLURIDAE** (p. 49)
- Tarsi with scopulae; paired claw with two rows of teeth; posterior spinnerets shorter and closer together **NEMESIIDAE** (p. 91)

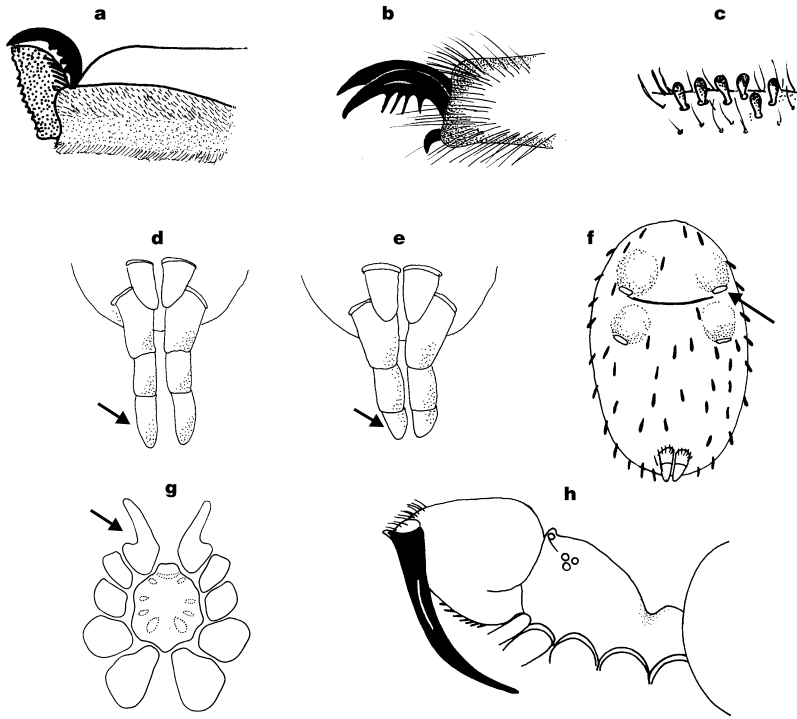


Fig. 7. External morphology of mygalomorphs. **a:** tarsus with two claws and thick scopulae; **b:** tarsus with three claws; **c:** clavate trichobothria; **d:** spinnerets of Theraphosidae; **e:** spinnerets of Barychelidae; **f:** oval booklung openings; **g:** anterior lobe; **h:** lateral view of carapace and chelicera of Atypidae.

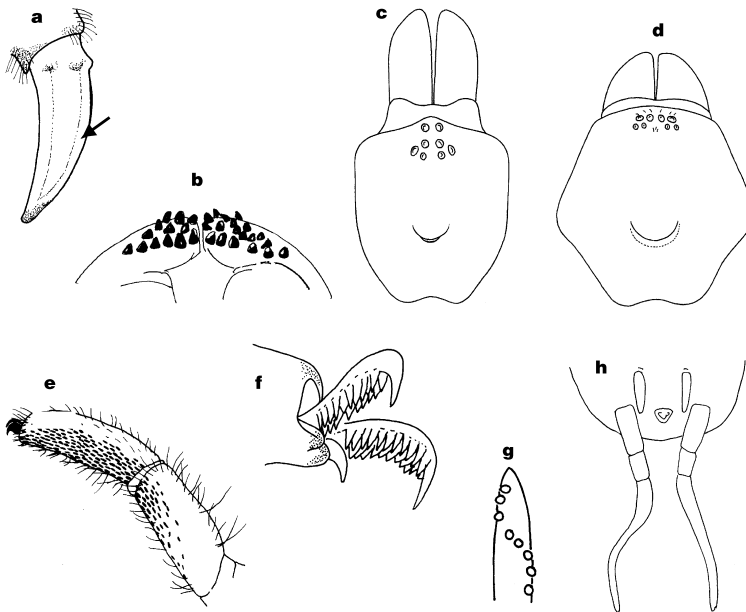


Fig. 8. External morphology of mygalomorphs. **a:** fang, lateral view, showing longitudinal keels; **b:** rastellum on chelicerae; **c:** eye pattern of Idiopinae; **d:** eye pattern of Ctenizidae; **e:** leg I with thorn-like setae; **f:** paired claws on tarsus with two rows of teeth; **g:** S-shaped row of teeth on paired tarsal claw; **h:** diplurid spinnerets.

FAMILY ATYPIDAE

purseweb spiders

(figs 9–12)

The Atypidae are a small family of mygalomorph spiders represented by three genera in Africa as well as in China, Japan and East Asia. Only one species is known from Southern Africa.

Genus recorded from Southern Africa

Calommata Lucas, 1837.

Diagnostic characters

Medium-sized to large (9–30 mm) spiders with the following synapomorphies:

- elongated anterior lobe on endite (fig. 10c),
- truncated median spinnerets (fig. 10e),
- rotating nature of endites (fig. 10b),
- teeth on paired and unpaired tarsal claws raised on a common process.

Descriptive characters

- **carapace:** glabrous; cephalic region more strongly arched than thoracic region (fig. 10b);
- **sternum:** four pairs of sigilla (fig. 10c);
- **eyes:** on a compact transverse tubercle (fig. 10a);
- **chelicerae:** large, dorsally expanded without rastellum (fig. 10b); fangs long, nearly as long as paturon (fig. 10c);
- **mouthparts:** endites on prolateral side with anterior lobe (fig. 10c); labium fused to sternum, or labiosternal junction a narrow groove (fig. 10c); serrula absent;
- **legs:** three claws; legs weakly spinose, without claw tufts, tarsal scopulae or trichobothria; tarsal claws raised on a common process;
- **female palp:** tarsi with dentate claw (fig. 10c);
- **abdomen:** suboval (fig. 10a); with single large tergite or irregularly shaped dorsal scutum in males;
- **spinnerets:** six; anterior spinnerets small, wide apart; median spinnerets truncated; posterior spinnerets with three subequal segments, apical segment finger-like (fig. 10e);
- **genitalia:** female with four (2 + 2) spermathecae each bearing several closely packed terminal receptacula (*Calommata*); male bulbus with distinct

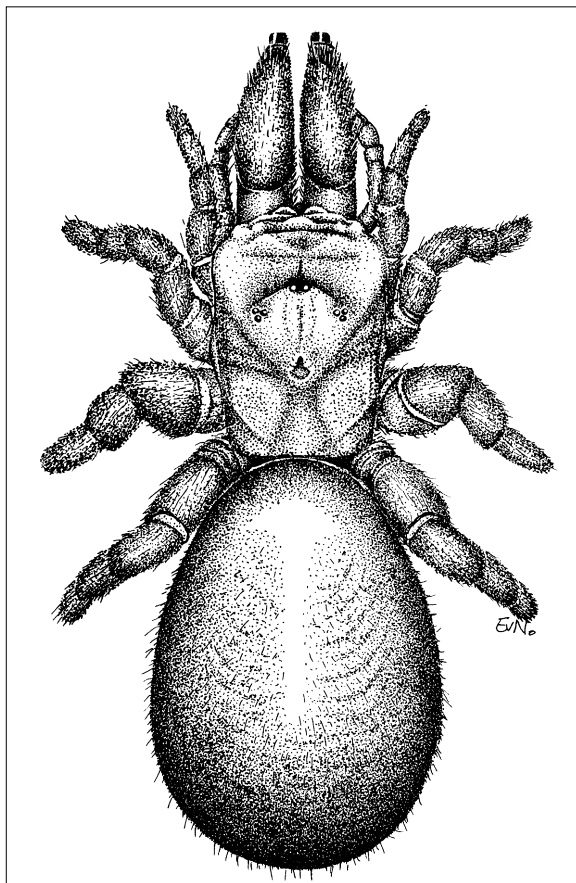


Fig. 9. Atypidae — *Calommata simoni*.

distal haematodocha; embolus short and straight (fig. 10g);

- **body size:** 9–30 mm;
- **colour:** carapace testaceous with darker stains on cephalic region; legs pale testaceous; abdomen dull greyish brown to yellowish brown.

Higher classification

Raven (1985), in his cladistic analysis of the family, placed the atypids in the microorder Fornicephalae under the Atypoidina with the Antrodiaetidae (fig. 2). The Atypoidina are characterized by the reduction or absence of tarsal trichobothria.

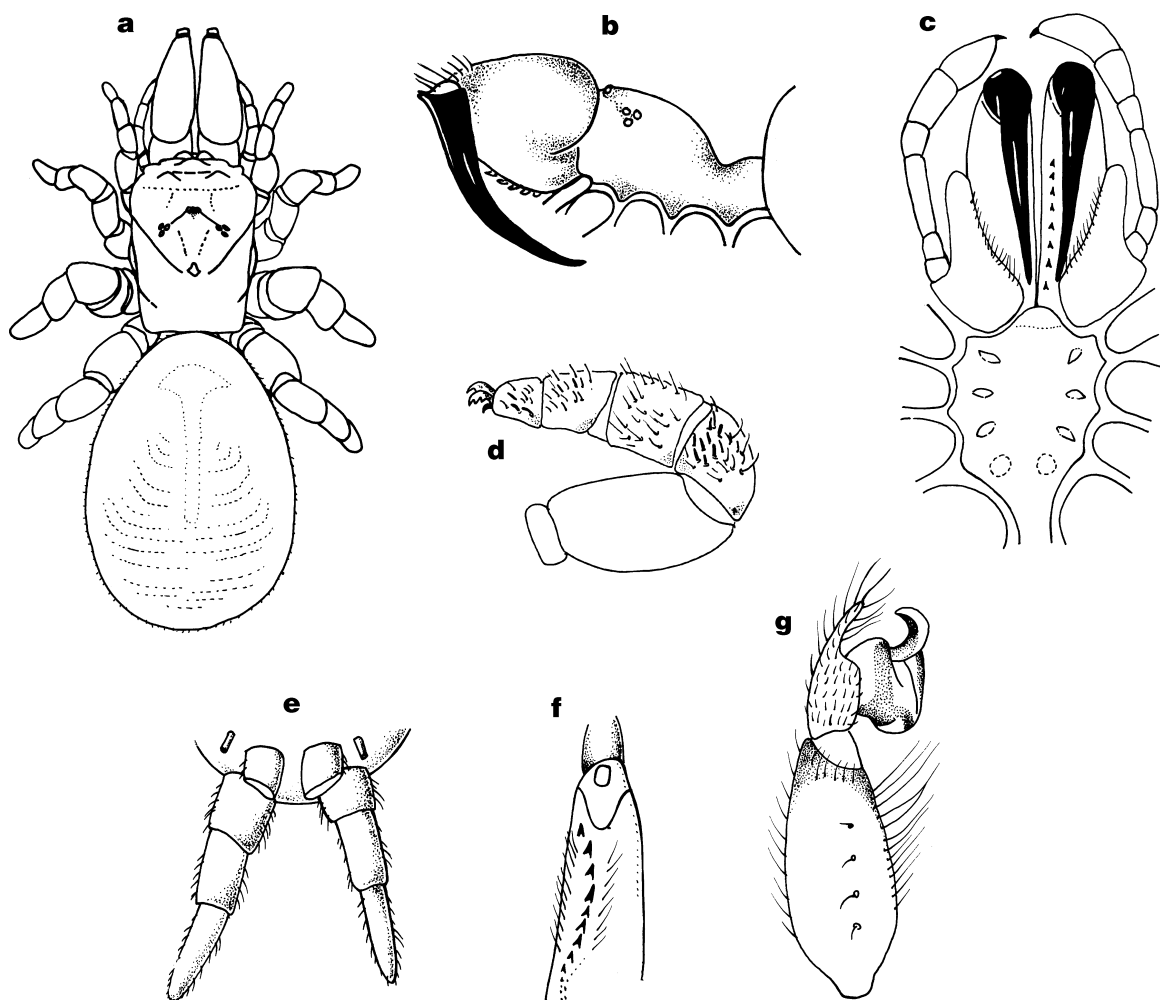


Fig. 10. Atypidae — *Calommata simoni*. **a:** female, dorsal view; **b:** carapace and chelicera, lateral view; **c:** sternum and chelicerae, ventral view; **d:** leg III, lateral view; **e:** spinnerets, ventral view; **f:** chelicera with teeth; **g:** male palp, lateral view.

Natural history

Most atypids live permanently in silk-lined burrows. They use specialized strategies to capture their prey. In some genera the burrows extend into a tough, tubular, prolonged aerial segment while in others the top of the burrow is covered with a tough layer of silk, covered with sand particles. An insect walking over the aerial part generates vibrations, which are transmitted to the spider below. The spider strikes with its long fangs through the silk to impale the prey. Once the prey has been immobilized the spider cuts the silk with one

fang and drags the prey through it. The remains of the prey and liquid droppings are subsequently ejected through an opening at the top of the tube. The females spend their entire lives in these burrows, which are enlarged in size and toughness as they age. They can live for seven years or more.

Genus *CALOMMATA* Lucas, 1837

Calommata Lucas, 1837: 378; Kraus & Baur, 1974: 88; Gertsch & Platnick, 1980: 2; Raven, 1985: 122; Dippenaar-Schoeman & Jocqué, 1997: 52.

Type species: *Actinopus fulvipes* Lucas, 1837.

Diagnostic characters

Eyes situated on a compact transverse tubercle near fovea (fig. 9); fovea bipartite and longitudinal; labiosternal junction a narrow groove (fig. 10c); legs reduced in size, especially leg I (fig. 10a), leg I without spines or spinules, other legs without spines but covered with small spinules (fig. 10d); palpal tibia and tarsus of female flattened (fig. 10c); male palp with short embolus (fig. 10g).

Taxonomic notes

Kraus & Baur (1974) suggested that *Calommata* should not be included in the Atypidae. However, Gertsch & Platnick (1980) recognized two synapomorphies uniting the three atypid genera, namely the elongated anterior lobe of the endites and the structure of the median spinnerets, which are wide with obliquely triangular tips.

Natural history

The burrow and 'catching web' of *Calommata* differ from those of other atypid genera in having no large aerial portion. Charpentier (1995) studied the burrows of *C. simoni* from large colonies in West Africa. The burrow is silk-lined, 25–30 cm deep, with the top part excavated to form a small ambush-chamber, crater-like in shape. The burrow is lined with silk and encloses the surface chamber (fig. 11a,b). The silk on the outer surface of the chamber is adhesive and usually covered with soil particles. The female seals herself in the ambush-chamber when not breeding. The spider lies on its back in the large ambush-chamber, waiting for prey. She bites through the silk layer when prey lands on it. The egg sac is deposited at the bottom of the burrow. While eggs and spiderlings are in the burrow it is not sealed off from the ambush-chamber.

Observations made by Van Dam & Roberts (1917) on *C. simoni* in the Pretoria area differ slightly from those above. They discovered two burrows, one made in bare ground and the other under an upturned grass tuft, in June (winter) and they reported that 'the entrance was open without a lid'. The top of the burrow is raised slightly above the ground level and from the inner rim; it is neatly rounded off, sloping gradually outwards

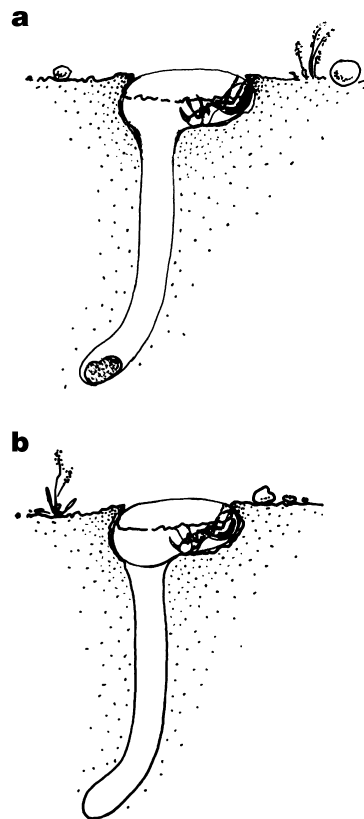


Fig. 11. Burrow of *Calommata simoni* (a) with and (b) without eggs.

and downwards to the soil surface. The outer surface is covered with silk that is covered with earth, resembling the surroundings. The interior of the tube is lined with loose, highly adhesive webbing. This webbing is pinched inwards a few centimetres from the top. The adhesiveness of the webbing probably affords protection against intrusion by enemies. The burrows are vertical for the greater part of their depth of 18–20 cm.

Hewitt (1916b) reported that *C. simoni* has a very pronounced and objectionable odour similar to decomposing stable manure. This may attract insects such as flies to the burrow. An immature female collected by Hewitt (1916b) was found in grassland, occupying a burrow about 18–20 cm deep, lined with thick webbing, but without the protection of a cover.

Distribution

Calommata is a small genus with seven known

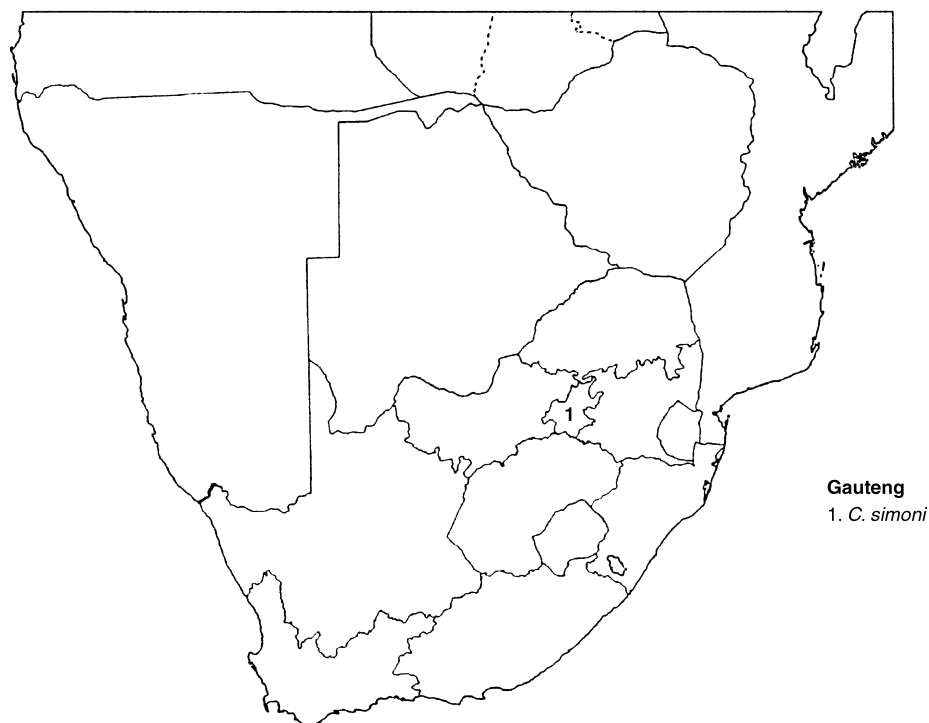


Fig. 12. Distribution of *Calommata simoni* in Southern Africa.

species of which only one has been recorded from Africa (Cameroon, Democratic Republic of Congo, Uganda and Tanzania) and the rest from South East Asia. In Southern Africa it is known only from Gauteng (fig. 12).

Conservation status

Urban development has probably had an adverse effect on *C. simoni* as it is at present regarded as extremely rare. The last live specimens were collected in South Africa in the late 1920s.

Species recorded from Southern Africa

1. *Calommata simoni* Pocock, 1903a (African purseweb spider)

Calommata simoni Pocock, 1903a: 259; Lessert, 1930: 617; Roewer, 1942: 212; Benoît, 1967: 283; Blandin, 1971: 48; Kraus, 1978: 245; Platnick, 1989: 115; 1993: 79.

Calommata transvaalicus Hewitt, 1916b: 180; Roewer, 1942: 212; Benoît, 1967: 283 (synonym); Van Dam & Roberts, 1917: 221.

Type locality: female lectotype and seven female paralectotypes of *C. simoni*, Efuleni, Cameroon, West Africa (BMNH); female holotype (immature) of *C. transvaalicus*, Roodeplaat (17 miles NE Pretoria), Gauteng Province (TM).

Distribution: South Africa (Gauteng: Pretoria area at Roodeplaat, Hatfield, Derdepoort, Mayville, Villieria).

FAMILY BARYCHELIDAE

trapdoor baboon spiders

(figs 13–16)

The family Barychelidae is represented by 41 genera and occurs worldwide in warmer tropical regions but shows a far higher diversity in the Pacific Region than elsewhere (Raven, 1994). Two subfamilies are known from the Afrotropical Region where they are represented by 10 genera and 55 species. From Southern Africa only one subfamily, the Barychelinae, has been recorded and is represented by three genera and five species.

Genera recorded from Southern Africa

Barychelinae: *Cyphonisia* Simon, 1889; *Pisenor* Simon, 1889; *Sipalolasma* Simon, 1892.

Diagnostic characters

Barychelids are medium-sized to large (9–32 mm) spiders with the following synapomorphies:

- absence of third claw (fig. 14b),
- biserially dentate tarsal claws in males (fig. 8f),
- well-developed scopulae on tarsi I and II (fig. 14b).

Descriptive characters

- **carapace:** cephalic region gently sloping, as high in front of fovea as behind; hairiness varies from almost smooth to dense;
- **sternum:** usually as long as wide; sigilla usually small and marginal with anterior pair indistinct (fig. 14c);
- **eyes:** arranged in two or three rows (fig. 14d–f); ocular area usually at least as long as wide or wider; eyes situated on tubercle; anterior lateral eyes usually close to anterior edge of carapace (fig. 14a);
- **chelicerae:** rastellum present or absent, if present consisting of weak spines, less developed in males;
- **mouthparts:** anterior lobe of endite small; cuspules reduced on endites and labium; labium always wider than long (fig. 14c); serrula present;
- **legs:** two claws; claw tufts well developed (fig. 14b); scopulae present on metatarsi and tarsi I & II; tibia I of male with (fig. 14g) or without spur; preening comb absent; tarsi with 4–6 clavate trichobothria (fig. 7c)

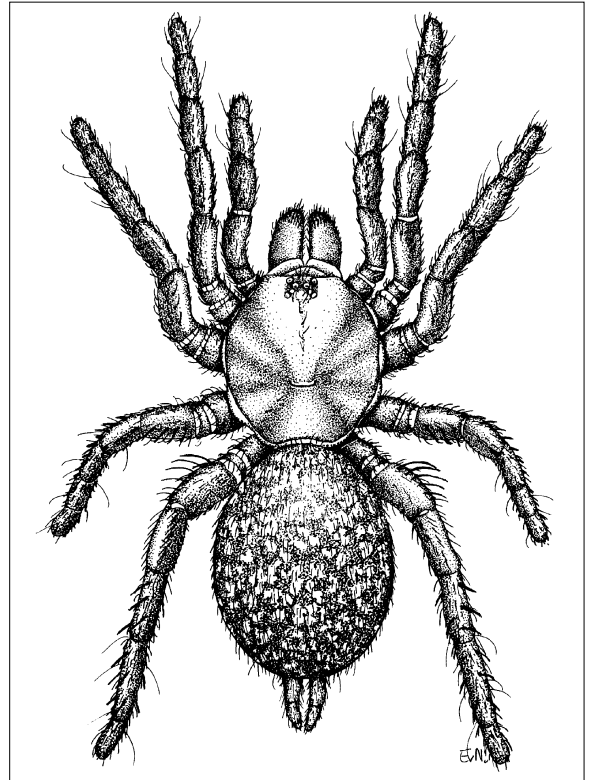


Fig. 13. Barychelidae — *Cyphonisia* sp.

confined basally (absent in some *Cyphonisia* spp.); tibiae, metatarsi and tarsi with long filiform trichobothria; paired tarsal claws in males biserially dentate, in females without teeth or teeth arranged in one row; leg formula usually 4123;

- **female palp:** tarsal claw without teeth;
- **abdomen:** oval; uniformly hairy;
- **spinnerets:** two or four spinnerets; posterior spinnerets with apical segment short and domed (fig. 14h);
- **genitalia:** female genitalia with two entire or divided spermathecae; male palp with cymbium bilobate or one very long lobe (*Pisenor*, *Sipalolasma*); bulbus with small second haematodocha and coniform distal sclerite lacking a conductor (fig. 14i);
- **body size:** 9–32 mm;
- **colour:** females usually 'earth'-coloured, varying from yellowish grey to reddish brown to greyish black; abdomen with pale spots or patches.

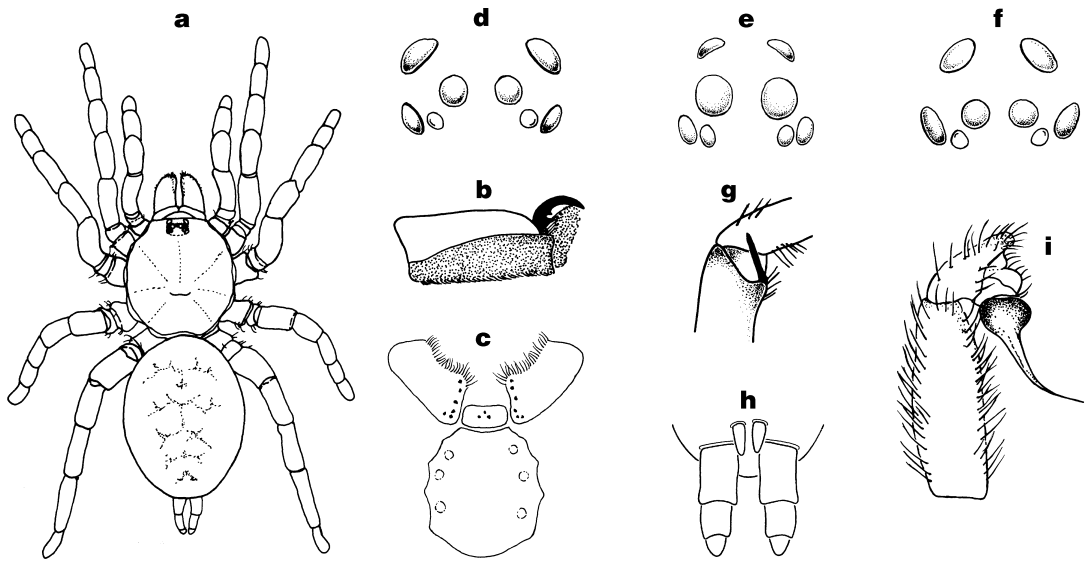


Fig. 14. Barychelidae. **a:** female, dorsal view; **b:** tarsus I showing claws and scopula; **c:** sternum and mouthparts, ventral view; **d:** eye pattern, *Sipalolasma* sp.; **e:** eye pattern, *Pisenor* sp.; **f:** eye pattern, *Cyphonisia* sp.; **g:** tibial spur on leg I; **h:** spinnerets of barychelids; **i:** male palp.

Higher classification

Raven (1985) placed the barychelids in the microorder Tuberculotae (fig. 2) based on their sloping thoracic region, the presence of serrula on the endites and a distinct eye tubercle. They form part of the superfamily Barycheloidea and together with the Theraphosidae and Paratropidae are placed in the Theraphosoidina. In the Theraphosoidina, trichobothria are found on the tibiae, metatarsi and tarsi of all legs and the palps.

Natural history

Barychelids are mainly burrowing spiders and their burrows vary from temporary silk retreats to complex silk-lined burrows that are frequently covered with concealed trapdoors (fig. 15a–c)

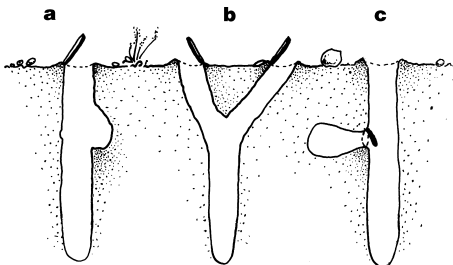


Fig. 15. Burrows of Barychelidae. **a:** single burrow; **b:** Y-shaped burrow; **c:** burrow with side passage.

and often with more than one entrance per burrow (Raven, 1994). A few species are arboreal and make tubular retreats similar to the migids. Some burrows may be found in leaf litter attached to the underside of rocks or fallen trees (Coyle, 1986; Raven, 1994).

Subfamily BARYCHELINAE Simon, 1889

Barycheleae Simon, 1889b: 192.

Barychelinae Raven, 1985: 111; 1994: 336.

Diagnostic characters

Eyes situated on an eye tubercle; anterior eye row strongly procurved with anterior lateral eyes close to clypeal edge (fig. 14a); cuspules absent or reduced on endites and labium (fig. 14c); labium wider than long; rastellum present or absent; apical segment of posterior spinnerets domed (fig. 14h); colour brown to dark brown; body and legs hairy; body size 9–26 mm.

Taxonomic notes

Although Raven (1985) listed some genera such as *Sipalolasma* as *incertae sedis* they are included here, as the taxonomy and placement of species are still in need of attention. No genera of the Barychelidae of the Afrotropical Region have been revised.

Distribution

The genera of the Barychelinae are known from West, Central and Southern Africa. The most common genus, *Cyphonisia*, is represented by 24 species that are widely distribution in the eastern and central parts of Africa. Roewer (1942) erroneously listed two species, *Ciphonisia arctura* and *C. macequece*, from South Africa — they were in fact collected in Zimbabwe and Mozambique, respectively. The first published record of a barychelid from South Africa was reported by Whitmore *et al.* (2001).

KEY TO THE SOUTHERN AFRICAN GENERA OF BARYCHELINAE

1. Ocular area as wide as long or slightly wider behind (fig. 14d); fovea a deep circular pit or procurved ***Sipalolasma***
- Ocular area usually narrower in front than behind (fig. 14f); fovea straight or slightly recurved (fig. 13) **2**
2. Two spinnerets; fovea recurved; tibia I as long as metatarsus I; scopulae on legs I and II scant ***Pisenor***
- Two or four spinnerets; fovea straight, sometimes slightly procurved; tibia I and metatarsus I subequal in size; scopula on leg I dense; legs I and II swollen . . . ***Cyphonisia***

Genus *CYPHONISIA* Simon, 1889

Cyphonisia Simon, 1889c: 409; 1892a: 120; 1903c: 912; Benoit, 1966: 217; Raven, 1985: 157.

Pisenorina Benoit, 1966: 214; Raven, 1985: 157 (synonym of *Cyphonisia*).

Pisenorodes Pocock, 1898c: 504; Raven, 1985: 157 (synonym of *Cyphonisia*).

Type species: *Cyphonisia obesa* Simon, 1889.

Diagnostic characters

Eye tubercle distinct; ocular area as wide as long or wider behind (fig. 14f); anterior lateral eyes close to clypeal edge; fovea straight or slightly procurved (fig. 14a); rastellum absent; labium and endites with cuspules, more than 10 cuspules on endites; two or four spinnerets; scopulae on metatarsi and tarsi I and II long, thin and divided by setae; legs I and II swollen; paired claws of males with two rows of teeth; tibial spur present (fig. 14g); claw tufts present on female palp; body size 15–18 mm.

Taxonomic notes

Cyphonisia is considered a senior synonym of *Pisenorodes* Pocock, 1898, and *Pisenorina* Benoit, 1966, by Raven (1985). He also considered it the correct position of all described species of *Pisenor* Simon, 1889, except for the type species *P. notius* Simon, 1889 (Platnick, 1989). (See also taxonomic notes on p. 27.)

Natural history

Little is known about the behaviour of *Cyphonisia*. In *Cyphonisia obesa*, a spider from tropical West Africa, the burrow is provided with a double-trapdoor entrance, primarily to assist the spider to escape from predators (Blandin & Célérier, 1977). It consists of a shallow chamber with the two trapdoors opening in exactly opposite directions, only a short distance apart. This double-door feature has also been recorded in West Australian barychelids (Main, 1957) inhabiting forested areas but with a range extension into savanna regions.

Distribution

Twenty-one species are known from the Afro-tropical Region, including three from Southern Africa that are mainly found in Mozambique and Zimbabwe (fig. 16).

KEY TO THE SOUTHERN AFRICAN SPECIES OF *CYPHONISIA*

1. Labium with >4 cuspules; endites with compact group of 15–20 cuspules . ***C. arcturus***
- Labium with <4 cuspules; endites with a maximum of 10–12 cuspules **2**
2. Labium with 1–2 cuspules; endites with 8–12 cuspules ***C. macequece***
- Labium with three cuspules; endites with 7–8 cuspules ***C. selindanus***

Species recorded from Southern Africa

1. *Cyphonisia arcturus* (Tucker, 1917)

Diplothele arcturus Tucker, 1917: 118; Roewer, 1942: 217.

Urothele arcturus: Benoit, 1965c: 35.

Cyphonisia arcturus: Raven, 1985: 112; see taxonomic notes p. 27.

Type locality: female holotype (SAM B2189), Arcturus, Salisbury, Zimbabwe.

Distribution: Zimbabwe.

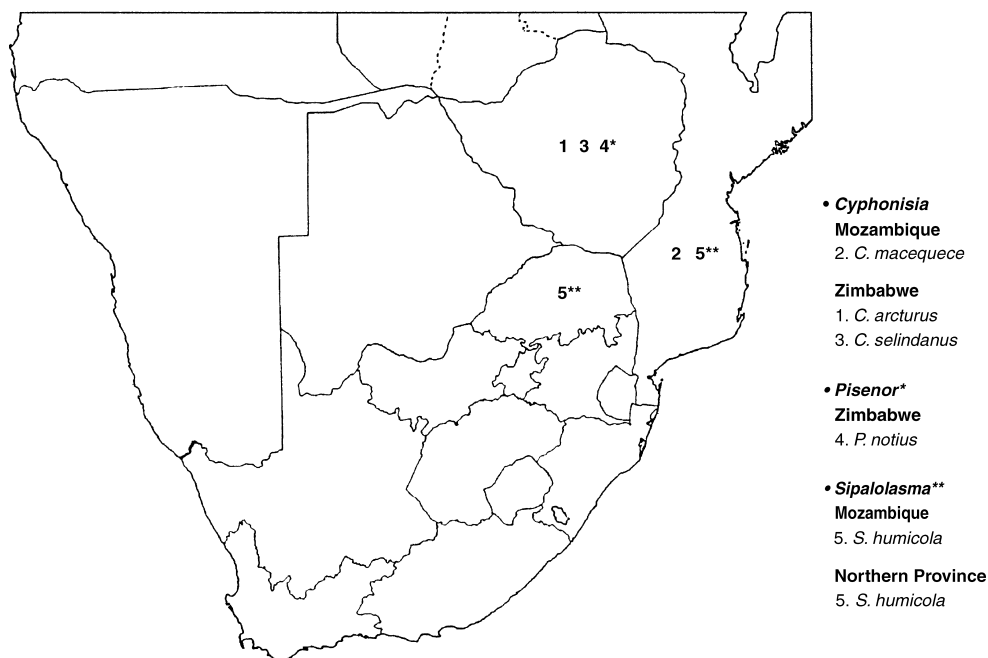


Fig. 16. Distribution of Barychelidae species in Southern Africa.

2. *Cyphonisia macequece* (Tucker, 1920)

Diplothele macequece Tucker, 1920: 440; Roewer, 1942: 217.

Urothele macequece: Benoit, 1965c: 37.

Cyphonisia macequece: Raven, 1985: 160; see taxonomic notes p. 27.

Type locality: female holotype (SAM B2648), Mount Vengo, Macequece, Mozambique.

Distribution: Mozambique.

3. *Cyphonisia selindanus* (Benoit, 1965c)

Urothele selindana Benoit, 1965c: 37; Brignoli, 1983: 131.

Pisenor selindana: Raven, 1985: 160.

Cyphonisia selindana: Raven, 1985: 160; see taxonomic notes p. 27.

Type locality: female holotype (MRAC 116.120), Mount Selinda, Chirinda, Zimbabwe.

Distribution: Zimbabwe.

Taxonomic notes

Raven (1985) considered *Pisenor* a senior synonym of *Urothele* Tullgren, 1910, but he transferred all species attributed to *Pisenor* (except *P. notius*) to *Cyphonisia* without individual citations (Platnick, 1989).

Natural history

Very little is known about the behaviour of *Pisenor*. The only published information is a note by Benoit (1966) who collected *P. notius* Simon from a burrow with a trapdoor covered by bits of leaves.

Distribution

Ten species are known from the Afrotropical Region, distributed from Zimbabwe northwards to the southern parts of Ethiopia, with one known from Southern Africa (fig. 16).

Species recorded from Southern Africa

1. *Pisenor notius* Simon, 1889c

Pisenor notius Simon, 1889c: 411; Roewer, 1942: 216; Benoit, 1966: 230; Raven, 1985: 112; Platnick, 1993: 98.

Type locality: female holotype (MNHN), Zambezi, Zimbabwe.

Genus *PISENOR* Simon, 1889

Pisenor Simon, 1889c: 411; 1892a: 120; 1903c: 912; Benoit, 1966: 228; Raven, 1985: 114; Platnick, 1989: 94.

Urothele Tullgren, 1910: 87; Raven, 1985: 160 (synonym).

Type species: *Pisenor notius* Simon, 1889.

Diagnostic characters

Anterior lateral eyes close to clypeal edge; eyes arranged in a group, wider behind than in front (fig. 14e); eye tubercle distinct; rastellum absent; fovea recurved; two spinnerets; metatarsal preening comb absent; male with tibial spur (fig. 14g); female with one row of teeth on paired claws, male with two rows; body size 15–25 mm; colour reddish brown to dark brown, abdomen frequently with spots.

Distribution: Zimbabwe northwards to the southern parts of Ethiopia.

from *Sipalolasma humicola* having been collected in pit traps in South Africa.

Genus *SIPALOLASMA* Simon, 1892

Sipalolasma Simon, 1892a: 123; Pocock, 1900b: 176; Benoit, 1966: 236; Raven, 1985: 113; Platnick, 1989: 94.

Cyclopelma Benoit, 1965a: 302; Raven, 1985: 151 (synonym).

Type species: *Sipalolasma ellioti* Simon, 1892.

Diagnostic characters

Anterior lateral eyes not close to clypeal edge; ocular area as wide as long or slightly wider behind (fig. 14d); fovea a deep circular pit or procurved; four spinnerets; tibial spur present in male; labium and endites with cuspules; claw tufts absent from female palp; body size 15–29 mm.

Taxonomic notes

Sipalolasma is considered a senior synonym of *Cyclopelma* Benoit, 1965, but was listed as *incertae sedis* in the Barychelidae by Raven (1985).

Natural history

Little is known about their natural history apart

Distribution

Four species of *Sipalolasma* are known from the Afrotropical Region, from Mozambique, Ethiopia and the Democratic Republic of Congo. *Sipalolasma humicola* is reported here for the first time from South Africa, having been collected in the Northern Province and in pit traps during a survey of the Makelali Nature Reserve, Northern Province (Whitmore *et al.*, 2001) (fig. 16).

Species recorded from Southern Africa

1. *Sipalolasma humicola* (Benoit, 1965a)

Cyclopelma humicola Benoit, 1965a: 303; Brignoli, 1983: 130.

Sipalolasma humicola Raven, 1985: 151.

Type locality: female holotype, Mozambique.

Distribution: Mozambique, Ethiopia and Democratic Republic of Congo, South Africa (*Northern Province*: Makelali Nature Reserve; Nylsvley Nature Reserve, Broederstroom).

FAMILY CTENIZIDAE

cork-lid trapdoor spiders

(figs 17–21)

The family Ctenizidae is represented by ten genera and about 96 species worldwide. They inhabit most of the tropical and subtropical areas of the world and are represented by two subfamilies, the Ctenizinae and Pachylomerinae. The seven genera of the Ctenizinae are known from Africa, North America, the Oriental Region, southern Europe and Eurasia, while the three genera of Pachylomerinae occur in north and central America, India, Australia and on the western Pacific islands. From Southern Africa only one subfamily is known, represented by one genus and 40 species.

Genus recorded from Southern Africa

Ctenizinae: *Stasimopus* Simon, 1892.

Diagnostic characters

The ctenizids are medium-sized to very large (15–43 mm) spiders with the following synapomorphy:

- stout, curved spinules present on lateral faces of anterior pairs of legs of the female (fig. 18e).

Descriptive characters

- **carapace:** glabrous, cephalic region domed; fovea procurved (fig. 18a);
- **sternum:** posteriorly truncated and narrowed anteriorly; posterior pair of sigilla shallow (fig. 18d);
- **eyes:** arranged in two rows, anterior row usually slightly procurved (fig. 18b);
- **chelicerae:** rastellum distinct, consisting of thick spines (fig. 18c); cheliceral furrow with two rows of strong teeth;
- **mouthparts:** labium usually wider than long, with few cusps; endites with anterior lobe a long cone with few to numerous cusps (fig. 18d); serrula absent;
- **legs:** three claws; legs short, strong (fig. 17) and densely spinose; female with distal segments of legs I and II with lateral bands of short thorn-like spines (fig. 18e); female with one long and two smaller teeth on paired claw, third claw curved and bare;

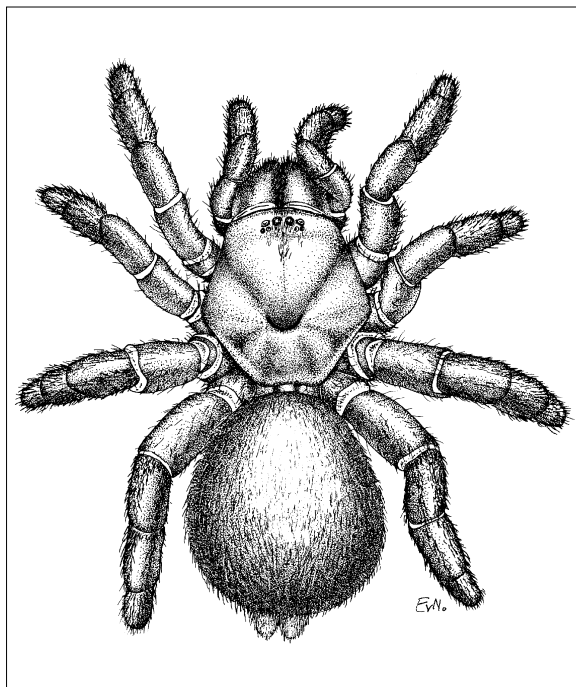


Fig. 17. Ctenizidae — *Stasimopus* sp.

- **abdomen:** oval; covered with a thin layer of short hair;
- **spinnerets:** four; apical segment of posterior spinnerets domed;
- **genitalia:** spermathecae multilobular; male palp simple, bulbous with pyriform conical distal sclerite; second haematodocha small (fig. 18f);
- **body size:** 15–43 mm;
- **colour:** cephalothorax varies from brown to reddish black with legs yellowish brown or reddish brown; abdomen usually a pallid or dull colour.

Higher classification

Raven (1985) listed the Ctenizidae in the microorder Fornicephalae and placed it together with the Migidae and the Actinopodidae in the Ctenizoidina with the Idiopidae as sister group (fig. 2). Raven (1985) transferred several genera previously included in the Ctenizidae to other families.

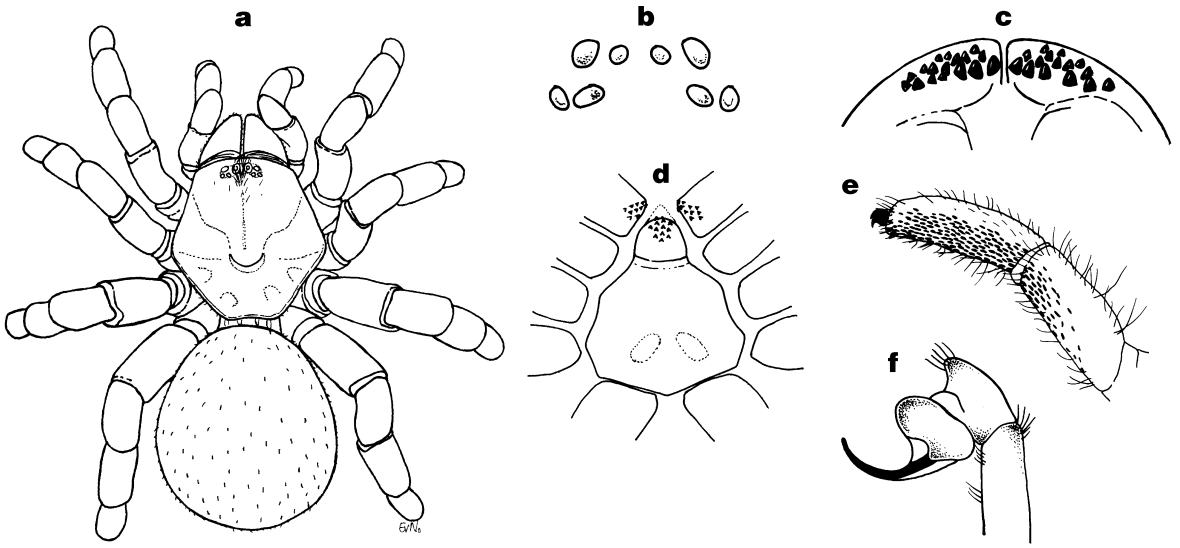


Fig. 18. Ctenizidae — *Stasimopus* sp. **a:** female, dorsal view; **b:** eye pattern, dorsal view; **c:** rastellum, anterior view; **d:** sternum and mouthparts; **e:** leg I with spinules; **f:** male palp, lateral view.

Natural history

Ctenizids are trapdoor spiders that live in silk-lined burrows of various shapes and depths. The rastellum is used during burrow excavation. The burrow is closed with a well-fitting, hinged trapdoor of variable thickness. The trapdoor is made of soil, often clay, molded into shape and reinforced with silk. It usually resembles a cork bottle-stopper, hence the common name of these spiders. The lid is attached to the rim of the burrow by a tough silk hinge (fig. 6b,c). The outside of the trapdoor is usually very well camouflaged by incorporating soil from the immediate vicinity of the burrow.

When disturbed, the spider pulls the lid closed and holds it down with the fangs, claws and setae on the front legs, as well as with the palps that fit into a circle of small holes on the inside of the lid (fig. 19b). Usually during the day, when the spider is not active, the lid is kept closed. It is also kept closed during harsh weather, egg-laying and when the spider sheds its skin. Most trapdoor spiders are nocturnal and open the trapdoor slightly, waiting for prey to pass close-by. They then rush out, capture it and return to the burrow.

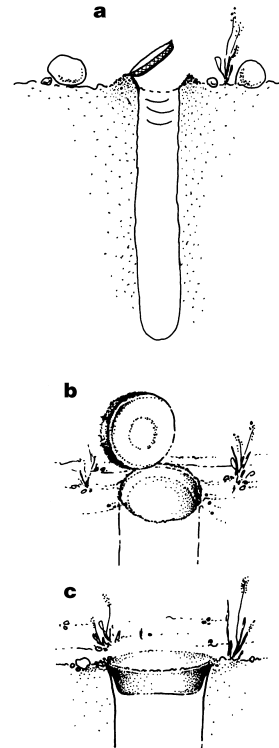


Fig. 19. Ctenizidae — *Stasimopus* sp. **a:** burrow; **b:** lid with circle of claw and tooth marks; **c:** cork-lid, lateral view.

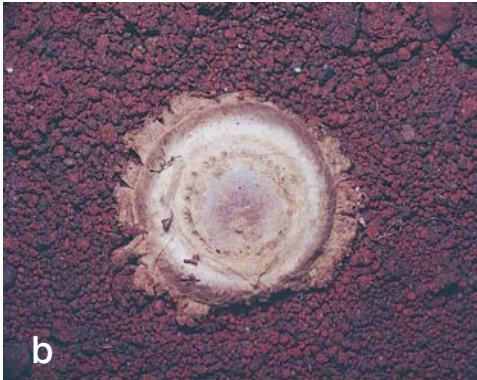


Fig. 20. Ctenizidae — *Stasimopus* sp. **a:** burrow entrance; **b:** underside of trapdoor; **c:** trapdoor from above.

Distribution

Seven genera, known from Africa, North America, southern Europe, Eurasia and the Oriental Region, are included in the Ctenizinae, but only one, *Stasimopus*, is known from Africa (Raven, 1985).

Genus *STASIMOPUS* Simon, 1892

African corklid trapdoor spider

Stasimopus Simon, 1892a: 82; Hewitt, 1915a: 75; Roewer, 1942: 159.

Type species: *Actinopus cafrus* C.L. Koch, 1842.

Diagnostic characters

Eyes arranged in two rows, posterior row wider than anterior row (fig. 18b); fovea strongly procurved (fig. 18a); cheliceral furrow with two rows of strong teeth; posterior sternal sigilla shallow (fig. 18d); rastellum with thick spines (fig. 18c); body stout, thick-set and covered with very short hairs or with large, quite smooth and shiny areas (fig. 17); legs short, strong and densely spined; body size 22–42 mm.

Taxonomic notes

Most of the original descriptions of the species are based on single or a few specimens, frequently males. Variation has hardly been taken into account and a revision of this genus might result in the recognition of fewer species. Purcell (1903c) and Hewitt (1915a) provided keys to some species.

Natural history

Stasimopus species live in silk-lined burrows (fig. 19a) that are usually made in flat areas. The burrows descend vertically to varying depths of between 14 and 22 cm depending on the nature of the soil (Table 2). They are thickly lined to form a thick wall of felt-like silk (fig. 20a). The upper part of the burrow is frequently widened all round, especially at the hinge. The entrance to the burrow is covered with a trapdoor, which is easily recognized by its comparative thickness and the way it fits tightly into the entrance. The shape of the lid varies from circular to D-shaped with a long hinge. The underside of the lid is unevenly convex and rounded at the edges. Tooth or claw marks, which form a distinct circle of small pits (fig. 19b, 20b), provide the spider with a place to grip. In some species the circle of pits is almost entirely obliterated. The external surface of the lid is well

Subfamily CTENIZINAE Thorell, 1887

Ctenizoidae Thorell, 1887: 19.

Halonoproctidae Pocock, 1901c: 207, 209.

Cyclocosmieae Simon, 1903c: 884.

Ctenizinae Thorell; Raven, 1985: 141.

Diagnostic characters

The Ctenizinae are distinguished by tibia III being cylindrical, without a dorsal saddle-shaped depression.

Table 2. Burrows and trapdoors of some *Stasimopus* species.

SPECIES	BURROW AND TRAPDOOR	REFERENCES
<i>S. brevipalpis</i>	lid width 22 mm; hinge width 15 mm; average thickness of lid 4 mm; width of burrow lower down 16.8 mm	Purcell, 1903b
<i>S. erythrognaethus</i>	lid width 30 mm; hinge width 6 mm; burrow width 19 mm; average thickness of lid 4.5 mm; upper part of burrow more funnel-shaped	Purcell, 1903b
<i>S. kentanicus</i>	lid thick; underside unevenly convex, strongly rounded at edges; circle of pits almost entirely obliterated	Purcell, 1903c
<i>S. kolbei</i>	lid thick and cork-shaped; hinge broad; peripheral surface strongly marked; lid edge scarcely bevelled, except on hinge side; underside flattened; pits almost obliterated	Purcell, 1903c
<i>S. oculatus</i>	lid thick, D-shaped with bevelled edge, burrow width 30–60 mm at entrance, lower down 23–50 mm; burrow thickly lined with silk; upper surface of lid coated with mud	Pocock, 1897
<i>S. patersonae</i>	burrow 8–9 cm deep; lid D-shaped; hinge long	Hewitt, 1913a
<i>S. quadratimaculatus</i>	lid very thick (5.5–9 mm), cork-like; not strongly bevelled at edge; lower edge more angular; burrow width at entrance 23–28 mm; burrow width lower down 16–18 mm; burrow depth 18–19 cm	Purcell, 1903c
<i>S. robertsi</i>	lid thickness 6 mm; lid width 25.3 mm; burrow vertical with width at top 25 mm, width at bottom 28 mm; distinct pits on lid; burrows made in hard, bare ground	Van Dam & Roberts, 1917
<i>S. schreineri</i>	lid thick, nearly circular except at hinge; upper surface irregular, concave or nearly flat, coated with mud; underside flat or more convex; circle of pits absent or reduced	Purcell, 1903b
<i>S. suffuscus</i>	similar to <i>S. robertsi</i> but burrow larger	Van Dam & Roberts, 1917
<i>S. unispinosus</i>	lid very thick, cork-like, not strongly bevelled at edge, lower edge more rounded; lid width 6–8 mm; lid thickness 2.3–4.0 mm	Purcell, 1903c

camouflaged with sand and debris (fig. 20c). It can be irregular, concave or nearly flat, or the edge of the lid can be strongly bevelled and extending gradually onto the soil surface.

Females and juveniles only leave their burrows to capture prey that are within reach of the burrow entrance. Adult males are more slender with longer legs and usually wander around in search of a female. However, two males of *S. robertsi* were collected from burrows with lids that closely resembled those of the female, but smaller

(Hewitt, 1916b). During egg-laying and hatching the lid is firmly fastened down with silk attached to the lining of the burrow all round the lower edge (Purcell, 1903c). The egg sac of *S. quadratimaculatus* is a white, soft-textured and sub-spherical sac, except on one side, which is flat with a narrow, free flap. Purcell (1903c) recorded about 50 eggs at the bottom of the burrow described above. Published information on the burrows and trapdoors of some *Stasimopus* species is summarized in Table 2.

Distribution

Stasimopus is a fairly large genus represented by 44 species in Africa, 40 of which (including two subspecies) are known from Southern Africa. It is abundant throughout the area south of the Limpopo. According to Hewitt (1915a), most of the species are confined to the Cape provinces and the Free State. The more northern provinces have not yet been well collected. In general, more than one species are only occasionally encountered at the same locality (fig. 21).

KEY TO THE FEMALES OF SOUTHERN AFRICAN SPECIES OF *STASIMOPUS*

(adapted from Purcell, 1903c, and Hewitt, 1915a)

1. Tarsus of palp with group of short spinules at base extending over half or more the length of segment 2
 - Tarsus of palp with few short spinules at base 4
 - Tarsus of palp without distinct spinules at base (KwaZulu-Natal) *S. rufidens*
2. Tibia I shorter than metatarsus I; armed with 5–9 spines on inner surface (Worcester) *S. erythrognathus*
 - Tibia I slightly longer than metatarsus I; armed with 8–11 spines on inner surface (Willowmore) *S. bimaculatus*
 - Tibia I subequal in length to metatarsus I; armed with >11 spines on inner surface 3
3. Tibia I with <16 spines on inner surface; tibia of palp with large distal group of spinules dorsally extending over at least a third the length of segment; metatarsus III with 10–12 spines on anterior surface (Steinkopf) *S. schultzei*
 - Tibia I with 20–23 spines on inner surface; tibia of palp with 8 or 9 minute apical spinules dorsally; metatarsus III with 22–24 spines on anterior surface (Little Namaqualand) *S. obscurus*
 - Tibia I with about 24 spines on inner surface; tibia of palp with few spinules distally (few or none are stout); metatarsus III with 24 or more spines on anterior surface (Pretoria, Potchefstroom) *S. robertsi*
4. Metatarsus I with long band of spinules on dorsal surface covering a third to half or more the length of segment 5

- Metatarsus I with moderate band of spinules on dorsal surface covering a quarter to third the length of segment 11
- Metatarsus I with short band of spinules on dorsal surface covering a sixth or less of the length of segment 17
- 5. Apex of metatarsus III with spines 6
- Apex of metatarsus III without spines 8
- 6. Metatarsus IV with apical tuft consisting of a single large stout spine, flanked on one or both sides by 1–3 spiniform setae (De Aar, Hanover) *S. unispinosus*
 - Not as above 7
- 7. Tibia of palp with some stout spinules dorsally (Free State, Gauteng, Northern Cape Province) *S. oculatus*
 - Tibia of palp without stout spinules dorsally (Middledrift) *S. spinosus*
- 8. Tarsus of palp with basal patch of 9–20 spinules dorsally 9
 - Tarsus of palp without basal patch of spinules (Kimberley) *S. poweri*
- 9. Tibia of palp dorsally with some short, stout spinules (Kroonstad, Mafikeng) *S. coronatus*
 - Tibia of palp without some short, stout spinules dorsally 10
- 10. Posterior lateral eyes small, subequal to or only slightly larger than posterior median eyes (Swellendam) *S. brevipalpis*
 - Posterior lateral eyes much larger than posterior median eyes (Montagu) *S. quadrimaculatus*
- 11. Tibia of palp with some spinules at apex 12
 - Tibia of palp without spinules at apex 15
- 12. Tarsus of palp with eight distinct spinules at base and band of spinules down each side of tarsus (Smithfield) *S. nanus*
 - Not as above 13
- 13. Tarsus of palp with band of spinules (16–18) on inner side extending to base 14
 - Tarsus of palp with short basal patch extending about a seventh the length of segment (Venterskroon, Potchefstroom) *S. nigellus*
- 14. Eyes unusually small; posterior median eyes rounded (Heidelberg) *S. suffuscus*

- Eyes not as small; posterior median eyes oval (Kroonstad) ***S. dreyeri***
- 15. Metatarsus III with stout spines (Port Elizabeth) ***S. patersonae***
- Metatarsus III without spines 16
- 16. Distance between anterior lateral eyes and posterior lateral eyes about equal to diameter of posterior lateral eye (Pearston) ***S. astutus***
- Distance between anterior lateral eyes and posterior lateral eyes about twice the diameter of posterior lateral eye (Eastern Cape Province) ***S. schoenlandi***
- 17. Ocular area as wide as length of metatarsus I and a third of tarsus I; posterior lateral eyes always much smaller than posterior median eyes (Victoria West) ***S. maraisi***
- Not as above 18
- 18. Ocular area only two-and-a-half times as wide as long (Kentani) ***S. kolbei***
- Ocular area very wide; posterior width equal to length of metatarsus I and half or more of tarsus I; posterior eye row recurved 19
- Ocular area about as wide as or slightly narrower than length of metatarsus I; posterior lateral eyes usually as large as or larger than posterior median eyes 20
- 19. Patella III with a number of slender dorsal spines at apex as well as stout ones along anterior surface (Hanover, De Aar, Port Elizabeth) ***S. schreineri***
- Patella III without slender dorsal spines at apex (Clanwilliam) ***S. leipoldti***
- 20. Tibia of palp without spinules 24
- Tibia of palp with spinules 21
- 21. Patch of spinules at apex of tibia I dorsally only about half the length of patch at base of metatarsus (Venterskroon) ***S. gigas***
- Not as above 22
- 22. Metatarsus IV with spines medioventrally . . . 23
- Metatarsus IV without spines medioventrally (Qumbu) ***S. qumbu***
- 23. Band of spinules dorsally on metatarsus I strongly developed (Port Elizabeth) ***S. castaneus***
- Band of spinules dorsally on metatarsus I less developed (Port Alfred) ***S. tysoni***
- 24. Metatarsus III without spines at apex 25
- Metatarsus III with cluster of spines at apex . . 26
- 25. Patella III dorsally with a number of short, stout, red spinules at apex (Umtata) ***S. umtalicus***
- Patella III with distal dorsal spines black and finely pointed (Kentani) . . . ***S. kentanticus***
- 26. Diameter of anterior lateral eyes less than distance between anterior lateral eyes and anterior median eyes (King William's Town) ***S. insculptus***
- Diameter of anterior lateral eyes wider than distance between anterior lateral eyes and anterior median eyes (Bathurst) ***S. artifex***

Species recorded from Southern Africa

1. *Stasimopus artifex* Pocock, 1902a

Stasimopus artifex Pocock, 1902a: 10; Hewitt, 1913a: 405; Roewer, 1942: 159.

Type locality: female holotype (AM), Bathurst district (33.50S; 26.49E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Bathurst, Rokeby Park, Seaview, Kleinmond).

2. *Stasimopus astutus* Pocock, 1902a

Stasimopus astutus Pocock, 1902a: 11; Purcell, 1903c: 86; Hewitt, 1914b: 28; 1917a: 697; Tucker, 1917: 82; Roewer, 1942: 159.

Type locality: female holotype (AM), Pearston (32.35S; 25.09E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Pearston, Bedford, Jansenville).

3. *Stasimopus bimaculatus* Purcell, 1903c

Stasimopus bimaculatus Purcell, 1903c: 75, 86; Hewitt, 1915a: 78; Roewer, 1942: 159.

Type locality: female holotype (SAM 9942), Willowmore (33.10S; 23.37E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Willowmore).

New records: *Western Cape*: Karoo National Park; *Free State*: Kroonstad.

4. *Stasimopus brevipalpis* Purcell, 1903c

Stasimopus brevipalpis Purcell, 1903c: 75, 86; Hewitt, 1915a: 80, 81; Tucker, 1917: 81, 85; Roewer, 1942: 159.

Type locality: one male and female syntypes (SAM 8846 & 8895), farm 'Bonnie Vale', Bushman's Drift, Breede River, Swellendam Division, near Ashton (33.50S; 20.05E), Western Cape Province.

Namibia

21. *S. obscurus*

Botswana

10. *S. fordii*

North West Province

7. *S. coronatus*

20. *S. nigellus*

29. *S. robertsi*

Northern Cape

16. *S. longipalpis*

17. *S. maraisi*

21. *S. obscurus*

22. *S. oculatus*

25. *S. poweri*

32. *S. schreineri*

40. *S. unispinosus*

Western Cape

3. *S. bimaculatus*

4. *S. brevipalpis*

9. *S. erythrognathus*

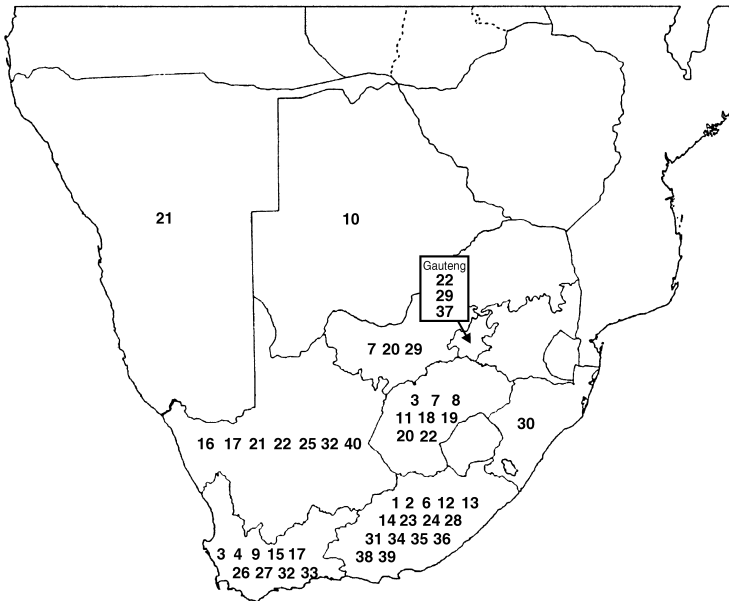
15. *S. leipoldti*

17. *S. maraisi*

26. *S. purcelli*

27. *S. quadratimaculatus*

33. *S. schultzei*



Gauteng

22. *S. oculatus*

29. *S. robertsi*

37. *S. suffusus*

Free State

3. *S. bimaculatus*

7. *S. coronatus*

8. *S. dreyeri*

11. *S. gigas*

18. *S. minor*

19. *S. nanus*

20. *S. nigellus*

22. *S. oculatus*

KwaZulu-Natal

30. *S. rufidens*

Eastern Cape

1. *S. artifex*

2. *S. astutus*

6. *S. castaneus*

12. *S. insculptus*

13. *S. kentanicus*

14. *S. kolbei*

23. *S. palpiger*

24. *S. patersonae*

28. *S. qumbu*

31. *S. schoenlandi*

32. *S. schreineri*

34. *S. spinipes*

35. *S. spinosus*

36. *S. steyburgensis*

38. *S. tysoni*

39. *S. umtalicus*

Fig. 21. Distribution of *Stasimopus* species in Southern Africa.

Distribution: South Africa (*Western Cape*: Swellendam, Robertson).

5. *Stasimopus caffrus* (C.L. Koch, 1842)

Actinopus caffrus C.L. Koch, 1842: 98.

Stasimopus caffrus: Simon, 1892a: 82; Roewer, 1942: 159.

Type locality: female holotype, South Africa (no exact locality).

Distribution: South Africa.

6. *Stasimopus castaneus* Purcell, 1903c

Stasimopus castaneus Purcell, 1903c: 80, 86; Roewer, 1942: 159.

Type locality: female holotype (SAM 5378), Port Elizabeth (33.58S; 25.36E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Port Elizabeth).

7. *Stasimopus coronatus* Hewitt, 1915a

Stasimopus coronatus Hewitt, 1915a: 80, 87; Roewer, 1942: 159.

Type locality: female syntypes (AM), Kroonstad (27.40S; 27.15E), Free State Province.

Distribution: South Africa (*Free State*: Kroonstad; *North West Province*: Mafikeng).

8. *Stasimopus dreyeri* Hewitt, 1915a

Stasimopus dreyeri Hewitt, 1915a: 81, 86; 1916b: 203; Roewer, 1942: 159.

Type locality: female holotype (AM), Kroonstad (27.40S; 27.15E), Free State Province.

9. *Stasimopus erythrognathus* Purcell, 1903c

Stasimopus erythrognathus Purcell, 1903c: 73, 86; Hewitt, 1914b: 32; 1915a: 78; 1917a: 696; Roewer, 1942: 159.

Type locality: six female syntypes (SAM 2857), Worcester (33.39S; 19.26E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Worcester).

10. *Stasimopus fordii* Hewitt, 1927a

Stasimopus fordii Hewitt, 1927: 422; Roewer, 1942: 159.

Type locality: male holotype (AM), Masite, Botswana.

Distribution: Botswana.

11. *Stasimopus gigas* Hewitt, 1915c

Stasimopus gigas Hewitt, 1915c: 318; 1915a: 82, 89; 1917a: 698; Roewer, 1942: 159.

Type locality: male holotype, Vrededorf road, Venterskroon (26.53S; 27.16E), Free State Province.

Distribution: South Africa (*Free State*: Venterskroon).

12a. *Stasimopus insculptus* Pocock, 1901a

Stasimopus insculptus Pocock, 1901a: 285; Purcell, 1903c: 85; Hewitt, 1913a: 404; 1915a: 83; 1917a: 696; Roewer, 1942: 160.

Type locality: male holotype, King William's Town, (32.50S; 27.17E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: King William's Town, also from Pirie Forest near King William's Town).

12b. *Stasimopus insculptus peddiensis* Hewitt, 1917a

Stasimopus insculptus peddiensis Hewitt, 1917a: 690, 696; Roewer, 1942: 160.

Type locality: four male syntypes, Peddie (33.12S; 27.01E), *Eastern Cape* Province.

Distribution: South Africa (*Eastern Cape*: Peddie).

13. *Stasimopus kentanicus* Purcell, 1903c

Stasimopus kentanicus Purcell, 1903c: 82, 86; Hewitt, 1915a: 82, 84; Tucker, 1917: 85; Roewer, 1942: 160.

Type locality: 85 male (SAM 14,685, B 782) and two female syntypes (SAM 12411), Kentani (32.38S; 28.25E), *Eastern Cape* Province.

Distribution: South Africa (*Eastern Cape*: Kentani).

14. *Stasimopus kolbei* Purcell, 1903c

Stasimopus kolbei Purcell, 1903c: 83, 86; Hewitt, 1915a: 83; Roewer, 1942: 160.

Type locality: two female syntypes (SAM 5317, 4542), Qoloro River mouth, Kentani (32.38S; 28.25E), *Eastern Cape* Province.

Distribution: South Africa (*Eastern Cape*: Kentani).

15. *Stasimopus leipoldti* Purcell, 1902b

Stasimopus leipoldti Purcell, 1902b: 348; 1903c: 85; Hewitt, 1915a: 81; Roewer, 1942: 160.

Type locality: female holotype (SAM 2909), Clanwilliam (32.13S; 18.59E), *Western Cape* Province.

Distribution: South Africa (*Western Cape*: Clanwilliam).

16. *Stasimopus longipalpis* Hewitt, 1917a

Stasimopus longipalpis Hewitt, 1917a: 689; Roewer, 1942: 160.

Type locality: three male syntypes (NM), Kimberley (28.42S; 24.59E), *Northern Cape* Province.

Distribution: South Africa (*Northern Cape*: Kimberley).

17. *Stasimopus maraisi* Hewitt, 1914b

Stasimopus maraisi Hewitt, 1914b: 24; 1915a: 81; 1927: 425; Roewer, 1942: 160.

Type locality: series of female syntypes (AM), farm Driefontein, 12 miles from Victoria West (31.24S; 23.07E), *Northern Cape* Province.

Distribution: South Africa (*Northern Cape*: Victoria West; *Western Cape*: Karoo National Park, Beaufort West).

18. *Stasimopus minor* Hewitt, 1915c

Stasimopus minor Hewitt, 1915c: 320; 1917a: 698; Roewer, 1942: 160.

Type locality: male holotype, Bloemfontein (29.07S; 26.14E), *Free State* Province.

Distribution: South Africa (*Free State*: Bloemfontein).

19. *Stasimopus nanus* Tucker, 1917

Stasimopus nanus Tucker, 1917: 87; Roewer, 1942: 160.

Type locality: female holotype (SAM 89), Smithfield (30.13S; 26.32E), *Free State* Province.

Distribution: South Africa (*Free State*: Smithfield).

20. *Stasimopus nigellus* Pocock, 1902b

Stasimopus nigellus Pocock, 1902b: 319; Hewitt, 1917a: 697; 1919b: 93; 1925: 286; Roewer, 1942: 160.

Type locality: male holotype (TM), Vredeford road, Venterskroon (26.53S; 27.16E), *Free State* Province.

Distribution: South Africa (*Free State*: Venterskroon; *North West* Province: Potchefstroom).

21. *Stasimopus obscurus* Purcell, 1908

Stasimopus obscurus Purcell, 1908: 208; Griffin & Dippenaar-Schoeman, 1991: 160.

Type locality: female holotype, Little Namaqualand (no exact locality), *Northern Cape* Province.

Distribution: South Africa (*Northern Cape*: Little Namaqualand) and Namibia.

22. *Stasimopus oculatus* Pocock, 1897

Stasimopus oculatus Pocock, 1897: 728; Hewitt, 1915a: 80; 1915c: 321; Roewer, 1942: 160.

Type locality: female holotype, Bloemfontein (29.07S; 26.14E), *Free State* Province.

Distribution: South Africa (*Free State*: Bloemfontein, Venterskroon, Reddersburg, Kroonstad, Jagersfontein, Winburg, Ladybrand; *Gauteng*: Modderfontein; *Northern Cape*: Kimberley).

23. *Stasimopus palpiger* Pocock, 1902a

Stasimopus palpiger Pocock, 1902a: 9; Purcell, 1903c: 85; Hewitt, 1917a: 698; Roewer, 1942: 160.

Type locality: male holotype, Graaff Reinet (32.18S; 24.37E), *Eastern Cape* Province.

Distribution: South Africa (*Eastern Cape*: Graaff Reinet).

24. *Stasimopus patersonae* Hewitt, 1913a

Stasimopus patersonae Hewitt, 1913a: 408; 1914b: 30 (male & female); 1915a: 82; 1917a: 697; Roewer, 1942: 160.

Type locality: male holotype (AM), Perseverance Uitenhage road, near Port Elizabeth (33.58S; 25.36E), *Eastern Cape* Province.

Distribution: South Africa (*Eastern Cape*: Port Elizabeth, Redhouse, Alicedale).

25. *Stasimopus poweri* Hewitt, 1915a

Stasimopus poweri Hewitt, 1915a: 85; Roewer, 1942: 160.

Type locality: five female syntypes (AM), Modder River near Kimberley (24.01S; 27.11E), *Northern Cape* Province.

Distribution: South Africa (*Northern Cape*: Kimberley).

26. *Stasimopus purcelli* Tucker, 1917*Stasimopus purcelli* Tucker, 1917: 84; Roewer, 1942: 161.**Type locality:** male holotype (SAM 150.432), Caledon (34.13S; 19.25E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Caledon).**27. *Stasimopus quadratimaculatus* Purcell, 1903c***Stasimopus quadratimaculatus* Purcell, 1903c: 78; Roewer, 1942: 161.**Type locality:** six female syntypes (SAM 12397–12402), lowveld at the Montagu Baths (33.47S; 20.07E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Montagu).**28. *Stasimopus qumbu* Hewitt, 1913a***Stasimopus qumbu* Hewitt, 1913a: 407; 1914b: 30.**Type locality:** female holotype (AM), farm Shawbury, Qumbu (33.10S; 28.51E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Qumbu).**29. *Stasimopus robertsi* Hewitt, 1910***Stasimopus robertsi* Hewitt, 1910: 75; 1913a: 412; Van Dam & Roberts, 1917: 230; Roewer, 1942: 161.*Stasimopus dubius* Hewitt, 1913a: 410; 1916b: 204 (synonym).**Types locality:** male holotype (TM), Pretoria (25.35S; 28.11E), Gauteng Province.**Distribution:** South Africa (*Gauteng*: widespread throughout the Pretoria and Centurion area, Wonderboom Poort, Mayville, Pretoria North, Witfontein, Skinner Court, Lyttelton Junction, Roodeplaat, Seekoegat; *North West Province*: Potchefstroom).**30. *Stasimopus rufidens* (Ausserer, 1871)***Cyrtocarenum rufidens* Ausserer, 1871: 160.*Pachylomerus natalensis* O. P.-Cambridge, 1889: 35.*Stasimopus rufidens* Pocock, 1898a: 199; Strand, 1907c: 174; Hewitt, 1915d: 132; Roewer, 1942: 161.**Type locality:** female holotype, Estcourt (29.01S; 29.53E), KwaZulu-Natal.**Distribution:** South Africa (*KwaZulu-Natal*: Estcourt, Mooi River, Durban).**31. *Stasimopus schoenlandi* Pocock, 1900a***Stasimopus schönlandi* Pocock, 1900a: 319; Strand, 1907c: 176; Hewitt, 1913a: 406; 1914b: 27; Roewer, 1942: 161.*Stasimopus schönlandi rufitarsis* Strand, 1907c: 178.**Type locality:** female holotype, Grahamstown (33.19S; 26.22E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Grahamstown, Jansenville, Brak Kloof, Atherstone Station, Somerset East, Middleton, Kamacks road near Uitenhage, Port Elizabeth, Debe Nek).**32. *Stasimopus schreineri* Purcell, 1903b***Stasimopus schreineri* Purcell, 1903b: 19; Roewer, 1942: 161.**Type locality:** series of female and male syntypes (SAM 9513, 9516, 9551, 9558, 9159), Hanover (31.55S; 24.27E) and Poortjiesfontein, De Aar (30.40S; 24.1E), Northern Cape Province.**Distribution:** South Africa (*Northern Cape*: Hanover, De Aar; *Eastern Cape*: Perseverance (near Port Elizabeth), Schurffeberg, Somerset East).**33. *Stasimopus schultzei* Purcell, 1908***Stasimopus schultzei* Purcell, 1908: 207; Griffin & Dippenaar-Schoeman, 1991: 156; Roewer, 1942: 161.**Type locality:** two female syntypes, Steinkopf (29.15S; 17.44E), Little Namaqualand, Western Cape Province.**Distribution:** South Africa (*Western Cape*: Steinkopf).**34. *Stasimopus spinipes* Hewitt, 1917a***Stasimopus spinipes* Hewitt, 1917a: 692; 1927: 422; Roewer, 1942: 161.**Type locality:** series of male and female syntypes (AM), East London (33.1S; 27.58E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: East London).**35. *Stasimopus spinosus* (Hewitt, 1914b)***Stasimopus spinosus* Hewitt, 1914b: 26 (originally a subspecies of *schoenlandi*); Hewitt, 1927: 424; Roewer, 1942: 161.**Type locality:** female holotype (AM), Annshaw, Middledrift (32.19S; 27.00E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Middledrift, Debe Nek, Alice).**36. *Stasimopus steynsburgensis* Hewitt, 1915c***Stasimopus steynsburgensis* Hewitt, 1915c: 317; Roewer, 1942: 161.**Type locality:** male holotype (BMNH), Steynsburg (31.18S; 25.48E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Steynsburg).**37. *Stasimopus suffuscus* Hewitt, 1916b***Stasimopus suffuscus* Hewitt, 1916b: 203; Roewer, 1942: 161.**Type locality:** female holotype (TM), Beerlaagte, Heidelberg district (26.30S; 28.22E), Gauteng Province.**Distribution:** South Africa (*Gauteng*: Heidelberg).**38. *Stasimopus tysoni* Hewitt, 1919b***Stasimopus tysoni* Hewitt, 1919b: 93; Roewer, 1942: 161.**Type locality:** series of female and male syntypes (TM), Port Alfred (25.46S; 27.46E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Port Alfred).**39a. *Stasimopus umtalicus* Purcell, 1903c***Stasimopus umtalicus* Purcell, 1903c: 81; Hewitt, 1927: 422; Roewer, 1942: 161.**Type locality:** female holotype (SAM 8848), Umtata (32.38S; 28.49E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Umtata).

39b. *Stasimopus umtalicus rangeri* Hewitt, 1927

Stasimopus umtalicus rangeri Hewitt, 1927: 421; Roewer, 1942: 161.

Type locality: male holotype (AM) from farm Gleniffer (29.41 S; 25.32 E), Kei Road Station, Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Kei Road).

40. *Stasimopus unispinosus* Purcell, 1903b

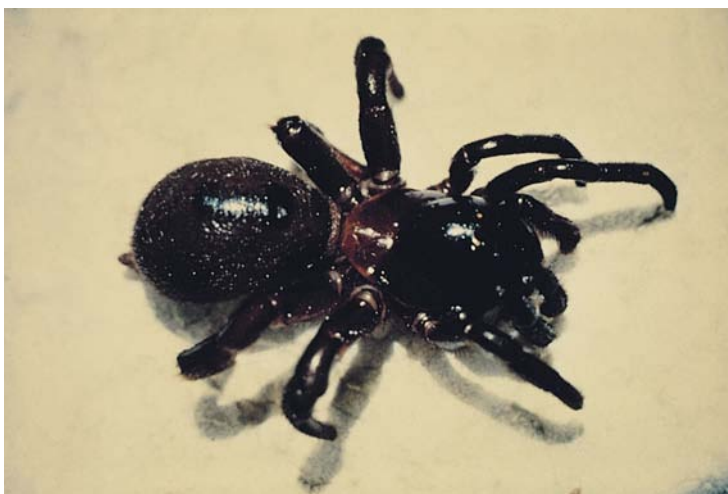
Stasimopus unispinosus Purcell, 1903b: 22; Hewitt, 1913a: 412 (male); Roewer, 1942: 161.

Type locality: female holotype and juveniles (SAM 9426, 9523, 9579), Hanover (31.05 S; 24.27 E) and neighbouring farm Poortjesfontein, De Aar (30.40 S; 24.1 E), Northern Cape Province.

Distribution: South Africa (Northern Cape: Hanover, De Aar).



Female trapdoor spider
(Ctenizidae: *Stasimopus rufidens*).



Female trapdoor spider
(Ctenizidae: *Stasimopus robertsi*).

wafer-lid trapdoor spiders

(figs 22–27)

The Cyrtoucheniidae are represented by 18 genera and 110 species worldwide. They are fairly widely distributed and are represented by three subfamilies, two of which, the Cyrtoucheniinae and Aporoptychinae, are known from Southern Africa where they are represented by two genera and 37 species.

Genera recorded from Southern Africa

Cyrtacheniinae: *Homostola* Simon, 1892;

Aporoptychinae: *Ancylotrypa* Simon, 1889.

Diagnostic characters

Cyrtacheniids are medium-sized to large (9–32 mm) spiders with the following synapomorphies:

- presence of scopulae (sometimes weak) on tarsi I and II,
- multilobular spermathecae,
- spination of tarsi I and II.

Descriptive characters

- **carapace:** cephalic region raised; fovea broad and pro- or recurved (fig. 22);
- **sternum:** with three pairs of sigilla (fig. 24b);
- **eyes:** in two rows forming a rectangular group that is slightly wider behind than in front (figs 23b, 24e, 25e);
- **chelicerae:** rastellum consisting of several short, blunt spines on a low mound on the inner surface (fig. 23c); cheliceral furrow with teeth on one (fig. 25c) or both margins (fig. 24c);
- **mouthparts:** endites rectangular or broad with few cusps; labium either wider than long or as long as wide, with (fig. 24b) or without (fig. 25b) cusps; serrula absent;
- **legs:** three claws; tarsus I scopulate in female; all tarsi scopulate in males or only sparsely on tarsi I and II; anterior paired tarsal claws usually with two rows of teeth in female (fig. 23d); teeth arranged in an S-shaped row in males (fig. 23e); preening comb present or absent (fig. 23f); front legs often shorter and thicker than leg IV;

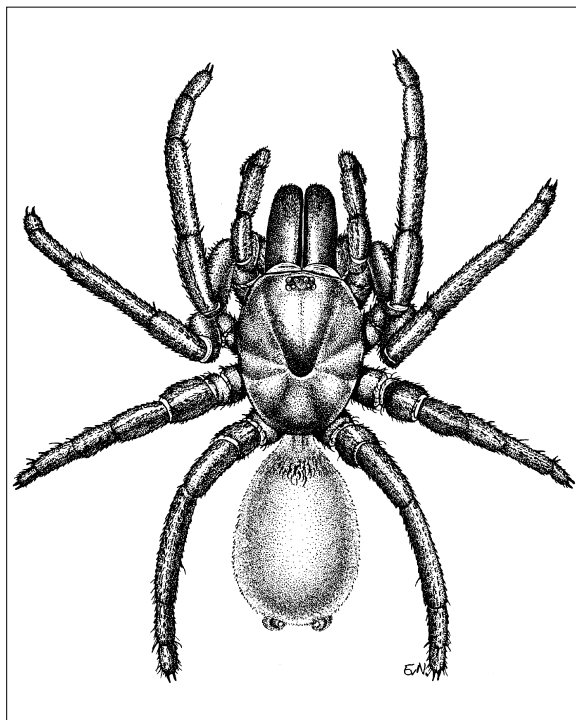


Fig. 22. Cyrtaucheniidae — *Homostola* sp.

- **abdomen:** oval; dorsum with pattern in some species;
- **spinnerets:** four; apical segment of posterior spinnerets domed (fig. 24a) or triangular (fig. 25a);
- **genitalia:** female genitalia with spermathecae multilobular; male palp simple, bulbous pyriform, with a small haematodocha, without conductor; cymbium simple (fig. 23g);
- **body size:** 9–32 mm;
- **colour:** varies from dark chestnut-brown to reddish yellow.

Higher classification

The Cyrtaucheniidae are placed in the micro-order Fornicephalae and belong to the Rastelloidina (fig. 2). According to Raven (1985) they are the presumed sister group of the Ctenizoidina (Ctenizidae, Actinopodidae and Migidae) as they share an elevated cephalic region, a broad, deep, procurved fovea and a rastellum.

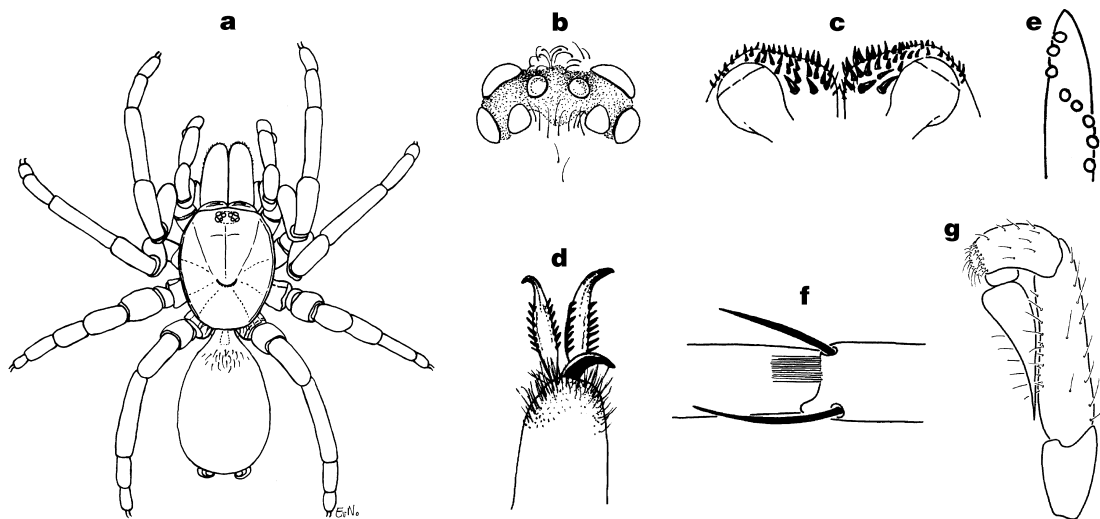


Fig. 23. Cyртаuchenіidae. **a:** female, dorsal view; **b:** eye pattern; **c:** rastellum, anterior view; **d:** tarsal claws of female with two rows of teeth; **e:** tarsal claw of male with S-shaped row of teeth; **f:** preening comb on metatarsus; **g:** male palp.

Taxonomic notes

The family Cyртаuchenіidae was elevated to family rank and redelimited by Raven (1985). It includes genera previously placed in the Ctenizidae. According to Goloboff (1993) the family is possibly paraphyletic.

Natural history

The cyртаuchenіids are burrowing spiders that line their burrows with silk. The shape of the burrow varies between species from a simple burrow to burrows with side tunnels (fig. 26a–c). The lid is usually a wafer-type consisting of a flexible limp flap which merely is a continuation of the web tubing. Spiders making this type of lid frequently rush out of their burrows to catch their prey. They usually have long legs. The wafer-type doors remain open while the spider is out, allowing a rapid return to the burrow. The lining of the burrow is sometimes extended above the soil surface to form a turret (Coyle, 1986).

KEY TO THE SOUTHERN AFRICAN SUBFAMILIES OF CYRTAUCHENIIDAE

1. Preening comb present on apex of metatarsus IV (fig. 23f), reduced in males; chelicerae with two rows of teeth along fang furrow (fig. 24c) **Cyртаuchenіinae**
- Preening comb absent; chelicerae with one row of teeth (fig. 25c) . . . **Aporoptychinae**

Subfamily CYRTAUCHENIINAE Simon, 1889

Cyртаuchenіeae Simon, 1889b: 179.
Cyртаuchenіinae Simon; Raven, 1985: 126.

Diagnostic characters

Apical segment of posterior spinnerets short and dome-shaped, distal segment not longer than half the length of penultimate segment (fig. 24a); labium wider than long, with cusps (fig. 24b); preening comb present on apex of metatarsus IV (fig. 23f), reduced in males; chelicerae with two rows of teeth on fang furrow (fig. 24c).

Taxonomic notes

This subfamily is represented by two genera, *Homostola* and *Cyртаuchenіus*. Both genera are in need of revision.

Distribution

Only *Homostola* occurs in Southern Africa while *Cyртаuchenіus* is known from the northern parts of Africa and the southern Mediterranean Region.

Genus *HOMOSTOLA* Simon, 1892

African leaf-litter trapdoor spider

Homostola Simon, 1892b: 271; Roewer, 1942: 167; Raven, 1985: 127; Platnick, 1989: 62.

Stictogaster Purcell, 1902b: 362; Roewer, 1942: 165; Raven, 1985: 159 (synonym).

Paromostola Purcell, 1903c: 94; Roewer, 1942: 168; Raven, 1985: 157 (synonym).

Type species: *Homostola vulpecula* Simon, 1892.

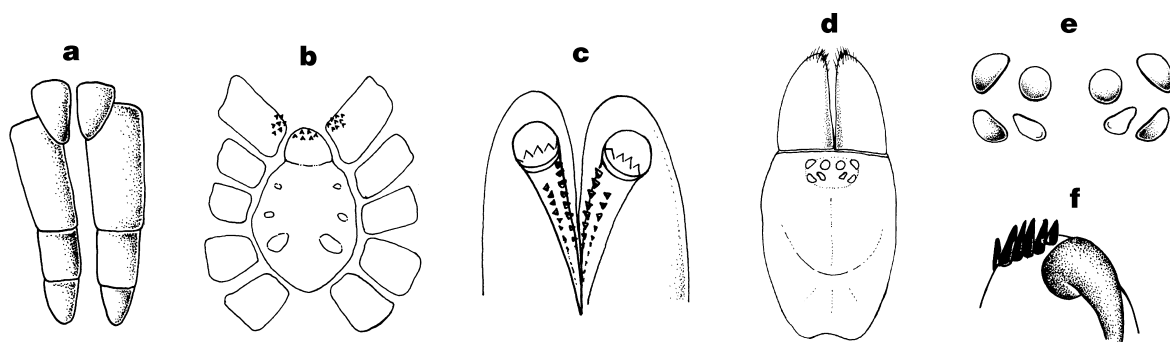


Fig. 24. Cyртаuchenіidae, Cyртаuchenііnae — *Homostola* sp. **a:** spinnerets; **b:** sternum; **c:** cheliceral furrows with two rows of teeth; **d:** carapace, dorsal view; **e:** eye pattern; **f:** rastellum.

Diagnostic characters

Carapace lightly hirsute to glabrous; cephalic region usually strongly arched; fovea broad, transverse to procurved (fig. 24d); clypeus narrow to absent; eye group rectangular, twice as wide as long (fig. 24e), on distinct eye tubercle; rastellum usually comprises several short, blunt spines (fig. 24f); sternum with six sigilla that are variable in shape (fig. 24b); colour varies from dark brown to pale yellowish, carapace frequently with darker median areas; abdomen sometimes with spots; body size 13.5–23.0 mm.

Taxonomic notes

Homostola was transferred by Raven (1985) from the Ctenizidae and he considered it a senior synonym of *Stictogaster* Purcell, 1902, and *Paramostola* Purcell, 1904.

Natural history

Few published records exist on their behaviour. Burrows of *Homostola zebrina* have been found mainly under rocks and stones (Van Dam & Roberts, 1917). These sheltered habitats seem to increase the survival rate in areas with heavy rains. Burrows containing females and their offspring usually have only one entrance while most burrows without young have two distinct entrances. The latter burrows are T-shaped with the entrances above the ends of the cross-piece. In some instances the trapdoor lids are covered with bits of straw (Van Dam & Roberts, 1917). Males have also been collected from burrows.

During a survey of ground fauna from five habitat types that included open and closed humus-

rich forested areas in the Ngome State Forest, *H. zebrina* was more abundant in grassy areas and pine plantations with penetrating forest than in the forested areas (Van der Merwe, 1994). Individuals were found in leaf litter and males were readily collected in pit traps. The females live in fairly shallow burrows (16–20 cm deep) made under leaf litter and covered with loose-fitting lids.

Distribution

Nine species of *Homostola* are known from the Afrotropical Region with five recorded from Southern Africa (fig. 27a).

KEY TO THE FEMALES OF SOUTHERN AFRICAN SPECIES OF HOMOSTOLA

- | | | |
|----|--|----------------------|
| 1. | Labium with >10 cuspules | 2 |
| — | Labium with <10 cuspules | 3 |
| 2. | Labium with >50 cuspules; total body length about 13 mm | <i>H. pardalina</i> |
| — | Labium with about 25 cuspules; total body length about 20 mm | <i>H. reticulata</i> |
| 3. | Tarsal claw of leg IV with 3–10 teeth on paired claws | 4 |
| — | Tarsal claw of leg IV with reduced number of teeth on paired claws | <i>H. abernethyi</i> |
| 4. | Tarsal claw of leg IV with >6 teeth on paired claws; body dark in colour | <i>H. vulpecula</i> |
| — | Tarsal claw of leg IV with three teeth on paired claws; body pale yellow | <i>H. zebrina</i> |

Species recorded from Southern Africa

1. *Homostola abernethyi* (Purcell, 1903c)

Paramostola abernethyi Purcell, 1903c: 94; Roewer, 1942: 168.
Homostola abernethyi: Raven, 1985: 157 (transfer to *Homostola*).

Type locality: female holotype (juvenile?) (SAM 12704), Kentani (32.30S; 28.21E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Kentani).

2. *Homostola pardalina* (Hewitt, 1913a)

Paramostola pardalina Hewitt, 1913a: 424; Roewer, 1942: 168.
Homostola pardalina: Raven, 1985: 157.

Type locality: female holotype (TM), Barberton (25.48S; 31.3E), Mpumalanga Province.

Distribution: South Africa (*Mpumalanga*: Barberton).

New record: *Gauteng*: Suikerbosrand Nature Reserve.

3. *Homostola reticulata* (Purcell, 1902b)

Stictogaster reticulatus Purcell, 1902b: 362; Simon, 1903c: 897;
 Tucker, 1917: 132; Roewer, 1942: 165.
Homostola reticulata: Raven, 1985: 159.

Types locality: two female syntypes, Bonnie Vale Farm at Bushman's Drift on the Breede River, Swellendam (34.1S; 20.26E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Bushman's Drift, Swellendam).

4. *Homostola vulpecula* Simon, 1892b

Homostola vulpecula Simon, 1892b: 271; 1892a: 108; Roewer, 1942: 167; Raven, 1985: 27; Platnick, 1989: 62.

Type locality: female holotype, Zululand (no exact locality), KwaZulu-Natal Province.

Distribution: South Africa (*KwaZulu-Natal*: Zululand).

New records: *KwaZulu-Natal*: Empangeni, Ngome State Forest.

5. *Homostola zebrina* Purcell, 1902b

Homostola zebrina Purcell, 1902b: 359; 1903c: 94; Hewitt, 1915d: 130; 1916b: 202; Tucker, 1917: 97; Roewer, 1942: 167.

Types locality: female syntypes (SAM 8445), Pietermaritzburg (29.28S; 30.28E), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Pietermaritzburg, Durban, Lower Umkomaas, Dumisa, Makowe, Umbilo, Clairmont, Howick, Eshowe, Ngome State Forest; *Mpumalanga*: Lake Chrissie, Lochiel, Steynsdorp, Oshoek, Carolina district; *Gauteng*: Pretoria district: Bon Accord) and Swaziland.

Diagnostic characters

Eye area wider than long; anterior eye row pro-curved and posterior row straight (fig. 25e); apical segment of posterior spinnerets digitiform, distal segment longer than or equal to penultimate segment (fig. 25a); labium as wide as long or longer than wide without cuspules or with reduced number (fig. 25b); preening comb absent; chelicerae with single row of large teeth (fig. 25c).

Taxonomic notes

This subfamily is represented by seven genera with two, *Ancylotrypa* and *Acontius*, known from the Afrotropical Region. Both genera are in need of revision.

Distribution

Acontius is known from tropical West Africa, while *Ancylotrypa* is widely distributed in Southern Africa.

Genus *ANCYLOTRYPA* Simon, 1889

African wafer-lid trapdoor spiders

Ancylotrypa Simon, 1889c: 406; Roewer, 1942: 162; Raven, 1985: 157; Platnick, 1989: 61.

Pelmatorycter Pocock, 1902a: 13; Roewer, 1942: 168; Raven, 1985: 157 (synonym).

Type species: *Ancylotrypa fossor* Simon, 1889.

Diagnostic characters

Carapace usually glabrous; cephalic region strongly arched, posteriorly narrowed (fig. 25d); fovea broad, transverse or procurved; clypeus narrow or absent (fig. 25d), eye group rectangular, twice as wide as long (fig. 25e), eye tubercle low or absent; chelicerae broad, rastellum with several short blunt spines on low mound (fig. 25f); endites broadly rectangular, anterior lobe indistinct (fig. 25b); endites in female with few cuspules, absent in male; sternum posteriorly broad; posterior sigilla large, either pear-shaped or oval (fig. 25b); colour varies from yellowish brown to almost blackish brown; abdomen sometimes with bands or spots; body size 9–32 mm.

Taxonomic notes

Raven (1985) transferred *Ancylotrypa* from Ctenizidae to the Cyrtachenidae and considered it a senior synonym of *Pelmatorycter* Pocock, 1902.

Natural history

Species of *Ancylotrypa* live in vertical, silk-lined

Subfamily APOROPTYCHINAE Simon 1889

Aporoptychae Simon, 1889b: 179.

Rhytidicoleae Simon, 1903c: 885.

Aporoptychinae, Raven, 1985: 129.

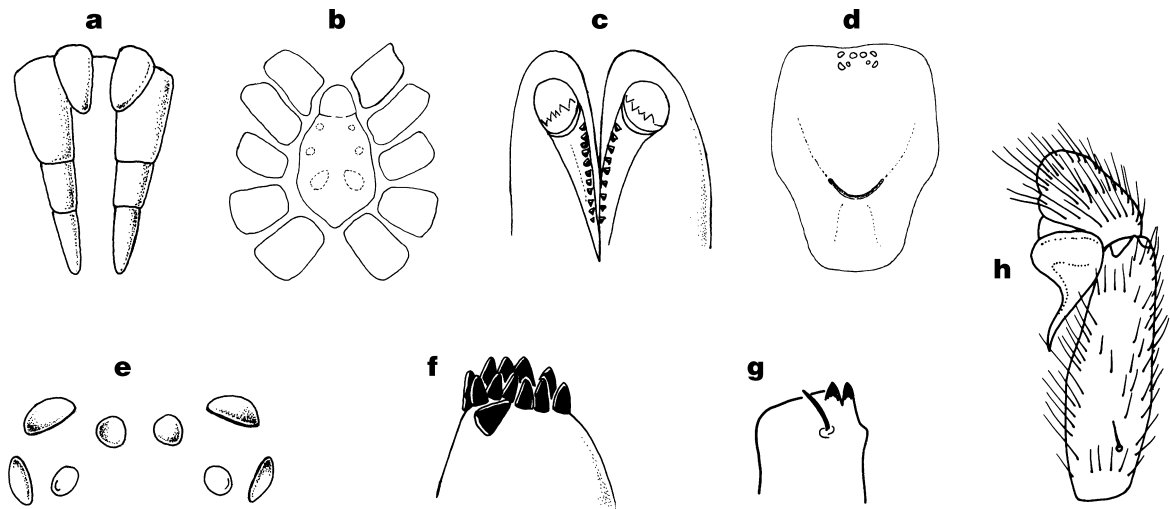


Fig. 25. Cyртаuchenіidae, Аporotуchіnae — *Ancylotrypa* sp. **a:** spinnerets; **b:** sternum; **c:** cheliceral furrows with one row of teeth; **d:** carapace; **e:** eye pattern; **f:** rastellum; **g:** femur of male palp showing apophysis; **h:** male palp, *A. zuluensis*.

burrows made in habitats ranging from grassy areas to open, barren ground and are frequently found under logs, stones or rock overhangs, which afford shelter. The depth of the burrow varies between species with the main portion being as deep as 32 cm (Table 3). Burrow shapes vary from simple to Y- or U-shaped (fig. 26 a–c). In some species side chambers are made with or without lids. The burrows are closed with different types of soft lids. During the day most of spiders retire to the lower portion of the burrow. Males are more active and are easily collected in pit traps. In built-up areas they are often found in swimming pools.

Species of *Ancylotrypa pretoriae* have frequently been recorded from areas heavily

infested with the harvester termite *Hodotermes mossambicus* in South Africa. Field observations indicated that they prey on these termites and are considered an important predator of them (Van den Berg & Dippenaar-Schoeman, 1991). *Ancylotrypa vryheidensis* was more abundant in indigenous forest with low undergrowth and grassy areas than in forest with dense undergrowth during a year-long survey at Ngome State Forest (Van der Merwe, 1994).

Distribution

Ancylotrypa is a fairly large genus of trapdoor spiders represented by 48 species known from Ethiopia in the north to South Africa in the south. From Southern Africa, 32 species are known, having been recorded from throughout South Africa, Botswana and Namibia (fig. 27b).

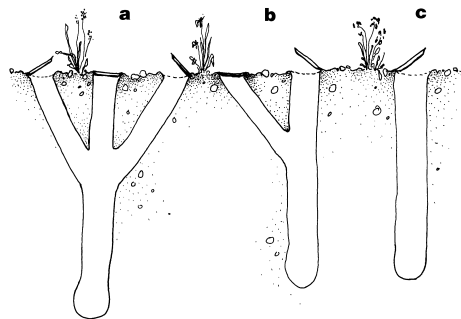


Fig. 26. Cyртаuchenіid burrows. **a:** burrow with three entrances; **b:** Y-shaped burrow; **c:** simple burrow with wafer-lid.

KEY TO MALES OF SOUTHERN AFRICAN SPECIES OF ANCYLOTRYPA

(adapted from Hewitt, 1916b)

- Femora of palp with apophysis (fig. 25g) at apex 2

— Femora of palp without apophysis at apex 6
- Femora of palp with single short, horn-like apophysis; two sigilla (Dunbrody) **A. cornuta**

- Femora of palp with two short apophyses (fig. 25g) at apex 3
- 3. Palp, when pressed forward, reaches at least one- to two-fifths of length of tibia I; cheliceral fang furrow with seven or more teeth 4
- Palp shorter, when pressed forward, extends about four-fifths of distance along patella I; cheliceral fang furrow with six teeth (Venterskroon) **A. brevicornis**
- 4. Palp, when pressed forward, only extends as far as basal fifth of tibia I; cheliceral fang furrow with seven teeth; body size 11 mm (Pretoria North, Rustenburg) .. **A. nuda**
- Palp, when pressed forward, extends about two-fifths of tibia I or longer; cheliceral fang furrow with more than seven teeth; body size >11 mm 5
- 5. Embolus of palpal organ long and curved (fig. 25h); posterior sigilla more or less oval, own diameter apart; body size 12 mm (Hluhluwe) **A. zuluensis**
- Embolus of palpal organ shorter and less curved; posterior sigilla pear-shaped, one-and-a-half times own diameter apart; body size 13.7 mm (Klipspruit) **A. breyeri**
- 6. Metatarsi of all legs with scopula below 7
- Metatarsi of some legs without scopula 9
- 7. All metatarsi distally with dense scopula below 8
- Metatarsi III and IV with weak scopula near apex; metatarsi I and II with weak scopula in apical third (Peddie) **A. tookei**
- 8. Colouration distinct, legs and abdomen black and carapace and upper chelicerae bright red; palpi and legs hirsute (Ngqeleni) **A. bulcocki**
- Dark olive-brown, carapace nearly black, abdomen dull purple; palpi and legs less hairy but bearing short, stiff setae (Alicedale) **A. parva**
- 9. Metatarsi I and usually also II with distinct scopula near apex 10
- Metatarsi I and usually also II without scopula below 13
- 10. Tarsus II with four small spines; tarsus IV with a distal group of small spines internally and double series of longer spines externally (South Africa, no exact locality) **A. nigriceps**
- Tarsus II without spines or with only two on outer side; tarsus IV with no external spines or only 1–4 small ones and none or 1–2 internally 11
- 11. Coxa III with tuft of stiff setae postero-ventrally; abdomen without long hairs above 12
- Coxa III without a tuft of stiff setae postero-ventrally; abdomen dorsum with long hairs (Pretoria) **A. pretoriae**
- 12. Patella of palp without spines (Bloemfontein) **A. dreyeri**
- Patella of palp with two spines anteriorly near apex (Barberton) **A. barbertoni**
- 13. Tibia of palp elongate 14
- Tibia of palp short 16
- 14. Posterior sternal sigilla very large, pear-shaped, touching at mid-line (Steinkopf) **A. namaquensis**
- Posterior sigilla moderately large, not touching at mid-line 15
- 15. Posterior sigilla their own length apart (Matjiesfontein) **A. pallidipes**
- Posterior sigilla twice their own length apart (De Aar) **A. pusilla**
- 16. Coxa III with a tuft of stiff setae on postero-ventral border (Pretoria, Magaliesberg) **A. brevipalpis**
- Coxa III without a tuft of stiff setae on posteroventral border (Bedford) .. **A. sororum**

List of species from Southern Africa

1. *Ancylotrypa barbertoni* (Hewitt, 1913a)

Pelmatorycter barbertoni Hewitt, 1913a: 430; 1916b: 198; Roewer, 1942: 168.

Ancylotrypa barbertoni: Raven, 1985: 157.

Type locality: male and female syntypes (TM), Barberton (25.48S; 31.03E), Mpumalanga Province.

Distribution: South Africa (Mpumalanga: Barberton).

2. *Ancylotrypa bicornuta* Strand, 1906b

Ancylotrypa bicornuta Strand, 1906b: 3; Roewer, 1942: 163.

Type locality: male holotype, Cape Town (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (Western Cape: Cape Town).

3. *Ancylotrypa brevicornis* (Hewitt, 1919a)

Pelmatorycter brevicornis Hewitt, 1919a: 209; Roewer, 1942: 168.

Ancylotrypa brevicornis: Raven, 1985: 157.

Type locality: series of male and female syntypes (TM),

Table 3. Burrows and trapdoors of some *Ancylotrypa* species.

SPECIES	BURROW AND TRAPDOOR	REFERENCES
<i>A. brevicornis</i>	burrow Y-shaped with both tunnels projecting upwards, opening on the soil surface	Hewitt, 1919a
<i>A. crudeni</i>	burrow bifurcated with one tunnel ending blindly below the surface and other projecting above the ground	Hewitt, 1915a
<i>A. dentatus</i>	burrow (25 cm in depth) well lined with silk, with lid a flexible limp flap that is merely a continuation of burrow tubing, edge slightly overlaps burrow entrance while upper surface lies flush with the ground; double entrance present; trapdoor about 9 mm in diameter and sub-circular in outline, with hinge occupying about half the circumference	Purcell, 1903b
<i>A. dreyeri</i>	burrow with two long, blind side tunnels leading into the central burrow, one on each side in the upper half of its length; silk lining of upper part of burrow projects a little above the surface, distal end folds inwards, thus closing entrance to burrow	Hewitt, 1915c
<i>A. nuda</i>	burrow made in soft earth mixed with pebbles; trapdoor in form of hood without distinct hinge; lid joined to rim on all sides except in front, over which it folds double; burrow when open U-shaped, openings flush with the ground, curve representing top of the main 12-cm deep burrow	Van Dam & Roberts, 1917
<i>A. pretoriae</i>	burrow sometimes found in hard open ground or in grassy areas; lid wafer-like, flat and thin, lying level with ground; burrow penetrates ground obliquely just below surface for nearly 7 cm before it drops vertically to a depth of 32 cm; sometimes provided with a short chamber with a lid at side near bottom in which spider takes refuge	Van Dam & Roberts, 1917
<i>A. lateralis</i>	burrow bifurcated with both tunnels projecting above ground	Hewitt, 1915a

Venterskroon (26.53S; 27.16E), southeast of Potchefstroom, North West Province.

Distribution: South Africa (North West Province: Venterskroon).

4. *Ancylotrypa brevipalpis* (Hewitt, 1916b)

Pelmatorycter brevipalpis Hewitt, 1916b: 196; Roewer, 1942: 168.

Ancylotrypa brevipalpis: Raven, 1985: 157.

Type locality: male holotype (TM), NE Pretoria (23.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Roodeplaat, Pretoria (Hatfield, Koedoespoort), Centurion (Lyttelton Junction); North West Province: Schoeman's Rust, near Crocodile River bridge).

5. *Ancylotrypa breyeri* (Hewitt, 1919b)

Pelmatorycter breyeri Hewitt, 1919b: 91; Roewer, 1942: 168.

Ancylotrypa breyeri: Raven, 1985: 157.

Types locality: male and female syntypes (TM), Klipspruit (24.3S; 29.35E), KwaZulu-Natal.

Distribution: South Africa (KwaZulu-Natal: Klipspruit).

6. *Ancylotrypa bulcocki* (Hewitt, 1916b)

Pelmatorycter bulcocki Hewitt, 1916b: 200; Roewer, 1942: 168.

Ancylotrypa bulcocki: Raven, 1985: 157.

Type locality: male and female syntypes (AM).

Ngqeleni (31.13S; 29.13E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Ngqeleni).

7. *Ancylotrypa coloniae* (Pocock, 1902a)

Pelmatorycter coloniae Pocock, 1902a: 13; Roewer, 1942: 168.

Ancylotrypa coloniae: Raven, 1985: 157.

Type locality: holotype immature specimen, Jansenville (32.56S; 24.40E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Jansenville).

8. *Ancylotrypa cornuta* Purcell, 1904

Ancylotrypa cornuta Purcell, 1904: 119; Tucker, 1917: 130; Roewer, 1942: 163.

Type locality: male holotype, Dunbrody (33.28S; 25.33E), Uitenhage district, Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Dunbrody).

9. *Ancylotrypa crudeni* (Hewitt, 1915a)

Pelmatorycter crudeni Hewitt, 1915a: 72; 1923: 62 (male); Roewer, 1942: 169.

Ancylotrypa crudeni: Raven, 1985: 157.

Type locality: series of female syntypes (AM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Alicedale).

10. *Ancylotrypa dentata* (Purcell, 1903b)

Cyrtachenius dentatus Purcell, 1903b: 29.

Ancylotrypa dentata: Roewer, 1942: 169.

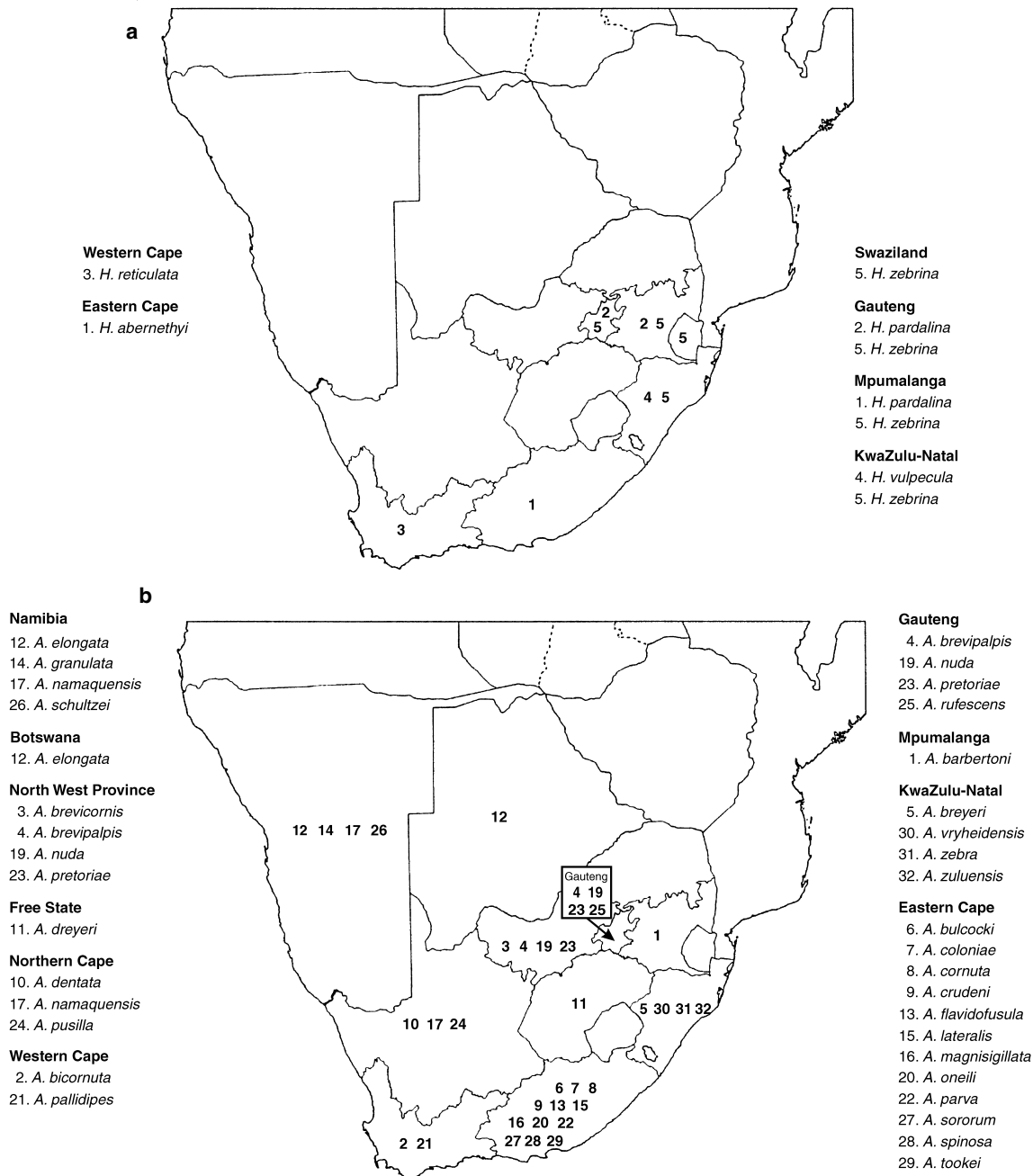


Fig. 27. Distribution of (a) *Homostola* and (b) *Ancylotrypa* species in Southern Africa.

Type locality: four female syntypes (SAM), Hanover (31.5S; 24.27E), Northern Cape Province.

Distribution: South Africa (Northern Cape: Hanover).

11. *Ancylotrypa dreyeri* (Hewitt, 1915c)

Pelmatorycter dreyeri Hewitt, 1915c: 299; Roewer, 1942: 169.
Ancylotrypa dreyeri: Raven, 1985: 157.

Type locality: male and female syntypes (AM), Bloemfontein (30.19S; 26.48E), Free State Province.

Distribution: South Africa (Free State: Bloemfontein).

12. *Ancylotrypa elongata* Purcell, 1908

Ancylotrypa elongata Purcell, 1908: 211; Roewer, 1942: 163;
Eagle, 1985: 131; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype, Kooa (24.51S; 24.28E), Kalahari, Botswana.

Distribution: Botswana and Namibia.

13. *Ancylotrypa flavidofusula* (Hewitt, 1915a)

Pelmatorycter flavidofusulus Hewitt, 1915a: 104; Roewer, 1942: 169.

Ancylotrypa flavidofusula: Raven, 1985: 157.

Type locality: female holotype (AM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Alicedale).

14. *Ancylotrypa granulata* (Hewitt, 1935)

Pelmatorycter granulata Hewitt, 1935: 461; Roewer, 1942: 169.

Ancylotrypa granulata: Raven, 1985: 157; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: two female syntypes (TM), Kuke Pan (23.18S; 24.22E), Namibia.

Distribution: Namibia.

15. *Ancylotrypa lateralis* (Purcell, 1902b)

Cyrtauchenius lateralis Purcell, 1902b: 357; Roewer, 1942: 169.

Pelmatorycter lateralis: Simon, 1903d: 89; Raven, 1985: 157.

Type locality: female type (SAM 4232), Dunbrody (33.28S; 25.33E), Sundays River, Uitenhage, Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Dunbrody, Uitenhage).

16. *Ancylotrypa magnisigillata* (Hewitt, 1914b)

Pelmatorycter magnisigillata Hewitt, 1914b: 33; Roewer, 1942: 169.

Ancylotrypa magnisigillata: Raven, 1985: 157.

Type locality: female holotype (AM), Kokstad (30.33S; 29.23E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Kokstad).

17. *Ancylotrypa namaquensis* (Purcell, 1908)

Pelmatorycter namaquensis Purcell, 1908: 209; Roewer, 1942: 169.

Ancylotrypa namaquensis: Raven, 1985: 157; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: two male syntypes, Steinkopf (29.15S; 17.44E), Little Namaqualand, Northern Cape Province.

Distribution: South Africa (Northern Cape: Steinkopf), Namibia.

18. *Ancylotrypa nigriceps* (Purcell, 1902)

Cyrtauchenius nigriceps Purcell, 1902b: 358.

Pelmatorycter nigriceps: Simon, 1903c: 899.

Ancylotrypa nigriceps: Roewer, 1942: 169; Raven, 1985: 157.

Type locality: male holotype (SAM 4204), Johannesburg (26.12S; 28.05E), Gauteng Province.

Distribution: South Africa (Gauteng: Johannesburg).

19. *Ancylotrypa nuda* (Hewitt, 1916b)

Pelmatorycter nudus Hewitt, 1916b: 192; Roewer, 1942: 169.

Ancylotrypa nuda: Raven, 1985: 157.

Type locality: male holotype (TM), Little Wonderboom (25.36S; 28.12E), Magaliesberg, Gauteng Province.

Distribution: South Africa (Gauteng: Little Wonderboom, Wonderboom Poort; North West Province: Magaliesberg, Wolhuterskop, Rustenburg).

20. *Ancylotrypa oneili* (Purcell, 1902b)

Cyrtauchenius oneili Purcell, 1902b: 355.

Pelmatorycter oneili: Simon, 1903c: 899; Roewer, 1942: 169.

Ancylotrypa oneili: Raven, 1985: 157.

Type locality: female holotype (SAM 8506), Dunbrody (33.28S; 25.33E), Sundays River, Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Dunbrody).

21. *Ancylotrypa pallidipes* (Purcell, 1904)

Pelmatorycter pallidipes Purcell, 1904: 120; Tucker, 1917: 130; Roewer, 1942: 169.

Ancylotrypa pallidipes: Raven, 1985: 157.

Type locality: male holotype (SAM 12780), Matjiesfontein (33.14S; 20.35E), Western Cape Province.

Distribution: South Africa (Western Cape: Matjiesfontein).

22. *Ancylotrypa parva* (Hewitt, 1916b)

Pelmatorycter parvus Hewitt, 1916b: 198; Roewer, 1942: 169.

Ancylotrypa parva: Raven, 1985: 157.

Type locality: male holotype (TM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Alicedale).

23. *Ancylotrypa pretoriae* (Hewitt, 1913a)

Pelmatorycter pretoriae Hewitt, 1913a: 426; 1916b: 194; Roewer, 1942: 169.

Ancylotrypa pretoriae: Raven, 1985: 157.

Type locality: male holotype (TM), Garstfontein, Pretoria (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Pretoria (Rietondale, Garstfontein, Mayville, Pretoria North); Centurion (Lyttelton Junction, Irene), Zeekoegat; North West Province: Schoemansrust near Crocodile River).

24. *Ancylotrypa pusilla* Purcell, 1903b

Ancylotrypa pusilla Purcell, 1903b: 27; Tucker, 1917: 94 (female); Roewer, 1942: 163.

Types locality: male holotype (SAM 9455), Hanover (31.5S; 24.27E), near De Aar, Northern Cape Province, females De Aar (SAM 2608, 2609), Vlagkop 8 km N Hanover.

Distribution: South Africa (Northern Cape: De Aar, Hanover).

25. *Ancylotrypa rufescens* (Hewitt, 1916b)

Pelmatorycter pretoriae rufescens Hewitt, 1916b: 195.

Pelmatorycter rufescens: Roewer, 1942: 169.

Ancylotrypa rufescens: Roewer, 1985: 157.

Type locality: male and female syntypes (TM), Roodeplaat (25.37S; 28.22E), NE Pretoria, Gauteng

Province.

Distribution: South Africa (*Gauteng*: Roodeplaat).

26. *Ancylotrypa schultzei* (Purcell, 1908)

Pelmatorycter schultzei Purcell, 1908: 210; Roewer, 1942: 170.
Ancylotrypa schultzei: Raven, 1985: 157; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype, Kubub, Namibia.

Distribution: Namibia.

27. *Ancylotrypa sororum* (Hewitt, 1916b)

Pelmatorycter sororum Hewitt, 1916b: 199; Roewer, 1942: 170.
Ancylotrypa sororum: Raven, 1985: 157.

Type locality: male holotype (AM), Bedford (25.48S; 31.3E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Bedford).

28. *Ancylotrypa spinosa* Simon, 1889b

Ancylotrypa spinosa Simon, 1889c: 407; Pocock, 1897: 732;
 Tucker, 1917: 96 (female); Roewer, 1942: 163.
Bolostromus spinosa: Simon, 1892a: 100.

Type locality: male holotype, Port Elizabeth (33.58S; 25.36E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Port Elizabeth).

29. *Ancylotrypa tookei* (Hewitt, 1919b)

Pelmatorycter tookei Hewitt, 1919b: 92; Roewer, 1942: 170.
Ancylotrypa tookei: Raven, 1985: 157.

Type locality: five male syntypes (AM), Peddie (33.12S;

27.1E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Peddie).

30. *Ancylotrypa vryheidensis* (Hewitt, 1915d)

Pelmatorycter vryheidensis Hewitt, 1915d: 128; Roewer, 1942: 170.
Ancylotrypa vryheidensis: Raven, 1985: 157.

Type locality: female holotype, Vryheid (27.46S; 30.48E), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Vryheid, Ngome State Forest).

31. *Ancylotrypa zebra* (Simon, 1892b)

Cyrtauchenius zebra Simon, 1892b: 272.
Pelmatorycter zebra: Roewer, 1942: 170.
Ancylotrypa zebra: Raven, 1985: 157.

Type locality: female holotype, Zululand (no exact locality), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Zululand).

32. *Ancylotrypa zuluensis* (Lawrence, 1937)

Pelmatorycter zuluensis Lawrence, 1937: 214; Roewer, 1942: 170.
Ancylotrypa zuluensis: Raven, 1985: 157.

Type locality: two male syntypes (NM), Hluhluwe Game Reserve (28.2S; 32.17E), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Hluhluwe Game Reserve).



Female wafer-lid trapdoor spider
 (Cyrtaucheniiidae: *Ancylotrypa pretoriae*).

FAMILY DIPLURIDAE

sheetweb mygalomorphs

(figs 28–32)

The Dipluridae occur worldwide and are represented by 20 genera and about 275 species in four subfamilies, the Diplurinae, Masteriinae, Euagrinae and Ischnothelinae. Only the last two subfamilies, represented by two genera and 5 species, are known from Southern Africa.

Genera recorded from Southern Africa

Ischnothelinae: *Thelechoris* Karsch, 1881;

Euagrinae: *Allothele* Tucker, 1920.

Diagnostic characters

Diplurids are small to medium-sized (5–22 mm) spiders with the following synapomorphies:

- long posterior spinnerets (fig. 29c),
- widely spaced posterior median spinnerets (fig. 30a),
- cephalic region lower than elevated thoracic region.

Descriptive characters

- **carapace:** cephalic region low, thoracic region elevated; overall hirsute; fovea small, pit-like (fig. 28); clypeus usually narrow;
- **sternum:** heart-shaped, narrow or wide; sigilla oval and marginal; labiosternal groove distinct;
- **eyes:** in compact group on eye tubercle, twice as wide as long (fig. 29d);
- **chelicerae:** porrect (fig. 29a); rastellum absent; cheliceral furrow with teeth; fangs long;
- **mouthparts:** endites short, cuspules present or absent; labium wider than long without cuspules (fig. 29b); serrula distinct;
- **legs:** three claws; paired claws with numerous teeth in one row; scopulae absent; tarsi long and slender; legs I or II (male) usually with mating spur consisting of apophyses on tibia and/or metatarsus (fig. 29e);
- **abdomen:** oval; hirsute; frequently with spots or chevron markings (figs 28, 29h);
- **spinnerets:** posterior spinnerets very long, widely spaced, longer than carapace; median spinnerets short and widely spaced (fig. 30a);

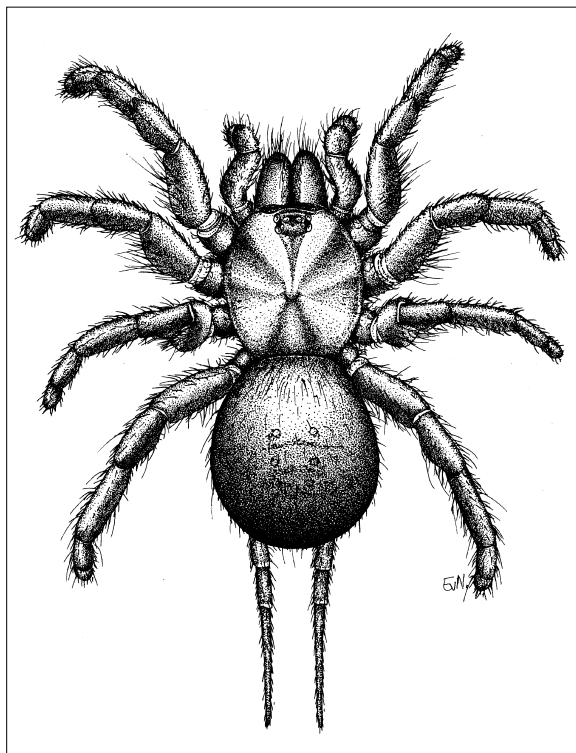


Fig. 28. Dipluridae — *Allothele* sp.

- **genitalia:** spermathecae either one or two on each side (fig. 29f); bulbous of male palp pyriform with elongated embolus; cymbium bilobate and spinose (Euagrinae) (fig. 29g) or an elongated lobe (Ischnothelinae) (fig. 30i);
- **body size:** 5–22 mm;
- **colour:** varies from pale tan to orange-brown to purple-brown or blackish brown, with spots or chevron markings on dorsum (fig. 29h).

Higher classification

The Dipluridae belong to the microorder Tuberculotae, in having a serrula on the endites, the thoracic region sloping and the eyes on a tubercle. The Dipluridae are the sister group of the Crassitarsae (spiders with tarsal scopulae and reduced median tarsal claws) and together with the Nemesiidae they form the Quadrithelina

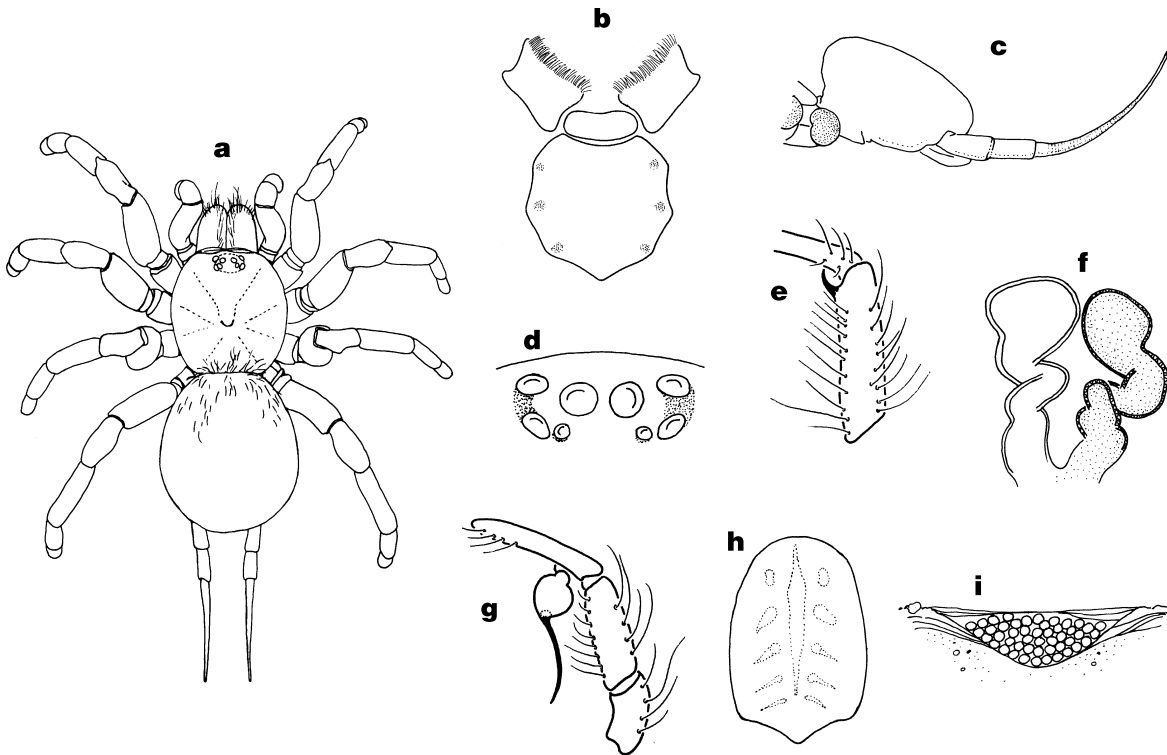


Fig. 29. Dipluridae. **a:** female, dorsal view; **b:** sternum; **c:** abdomen with spinnerets, lateral view, showing pseudosegments; **d:** eye pattern; **e:** mating spur on tibia I; **f:** spermathecae; **g:** male palp, retrolateral view; **h:** shape of abdomen; **i:** egg sac. (After Coyle, 1995.)

group (spiders with corrugiform trichobothria) (Raven, 1985).

Taxonomic notes

Raven (1985) discussed the family, subfamilies and genera and provided a key to the genera.

Natural history

Diplurids use ill-defined sheetwebs with a funnel-like retreat to detect and capture prey in trees or from under stones, soil crevices, logs or moss mats or above-ground in protected spots at the base of plants. The spider hides in the retreat in one corner of the web. According to Coyle (1986), some diplurids excavate burrows for a retreat but many do not. The web of *Lathrothele grabensis* Benoit, a species found throughout central Africa, has the funnel part of the web made in crevices or deep depressions in the soil. The funnel retreat opens either directly onto a

sheet-like capture web or by two or three short access tunnels. The capture web consists primarily of an irregular, hammock-shaped sheet (Coyle 1995). Coyle & Kether (1990) reported on the prey-capture behaviour and prey of the Dipluridae.

**KEY TO THE SOUTHERN AFRICA
SUBFAMILIES OF DIPLURIDAE**

1. Endites with cuspsules; sternum as wide as long; cymbium of male palp with an elongated apophysis (fig. 29g); posterior spinnerets with long, tapering, pseudo-segmented terminal segment (fig. 29c) **Ischnothelinae**
- Endites without cuspsules; sternum longer than wide; cymbium of male palp bilobed and spinose (fig. 30i); posterior spinnerets long and tapering but terminal segment not pseudo-segmented (fig. 30a) . . . **Euagrinae**

Subfamily ISCHNOTHELINAE F.O. P.-Cambridge

curtain-web spiders

Ischnothelinae F.O. P.-Cambridge, 1897: 35; Raven, 1985: 76; Coyle, 1995: 24.

Diagnostic characters

Carapace very flat, hirsute with pit-like or short fovea; eye group wider than long (fig. 29d), close to clypeal edge; chelicerae porrect; serrula broad; legs long, spinose; leg formula 4321; tibia I of male with elongated distal process and thorn-like spine (fig. 29e); metatarsus I with proximal process; paired claws with one S-shaped row of teeth; scopulae absent; posterior spinnerets elongate, apical segment pseudosegmented (fig. 29c).

Taxonomic notes

Coyle (1995) revised the subfamily Ischnothelinae and found the monophyly of this subfamily supported by the following synapomorphies: elongated cymbial apophysis (fig. 29g); cheliceral furrow with two rows of teeth; fovea procurved; trichobothrial bases collariform; silk spigots fused; posterior spinnerets with long, tapering, pseudosegmented terminal segment (fig. 29c).

Distribution

The subfamily is represented by two genera, *Thelechoris* and *Lathrothele*, in the Afrotropical Region. Only *Thelechoris* is known from Southern Africa.

Genus THELECHORIS Karsch, 1881

African curtain-web spiders

Thelechoris Karsch, 1881: 196; Benoit, 1964b: 422; Brignoli, 1983: 127; Raven, 1983a: 347; 1985: 77; Platnick, 1993: 91; 1998: 82; Coyle, 1995: 115.

Type species: *Thelechoris rutenbergi* Karsch, 1881.

Diagnostic characters

Carapace with soft, downy hairs; 1–3 foveal bristles; cephalic region moderately elevated above thoracic region; sternum broad, as wide as long; abdomen shown in fig. 29h; male palpal organ with bulb roughly heart-shaped, abrupt, constricted at embolus base (fig. 29g); embolus proportionally long; male tibia I mating apophysis small and terminal (fig. 29e); metatarsus I straight and smooth; spermathecal stalks spiralled (fig. 29f); medium- to dark brown, sometimes with purplish tint; body size 11–17 mm.

Taxonomic notes

Simon (1892a) first regarded *Thelechoris* as a synonym of *Ischnothele* but Benoit (1965a) restored it as a valid genus. According to Raven (1985), *Ischnothele* is not known to occur in Africa. *Thelechoris* therefore includes all the African species previously listed under *Ischnothele*.

Natural history

Thelechoris striatipes (then *karschi*), is the best known African curtain-web spider. The species occurs from Kenya southwards to Namibia. It is a moderately large diplurid that builds a conspicuous, perennial, three-dimensional capture web, up to 1.5 m in width (Coyle & Meigs, 1992). The capture web consists of interconnected sheets and passageways funneling into a protected tubular silk retreat. The webs are found in a wide variety of habitats from rock piles and road embankments to tree trunks and shrubs. They provide living space for many co-inhabitants. In total, 59–61 species of spiders and insects have been found living on the webs. Spiders include mysmenids, scytodids, palpimanids, prodidomids, pisaurids, mimetids, ctenids, oonopids, theridiids, salticids and sparassids (Coyle & Meigs, 1992). The most common co-inhabitant is the mysmenid *Kilifia inquilina*, a host-specific kleptoparasite. The second-most common species is the salticid *Portia schultzi*, which may be an important predator of the host spiderlings. In coastal Kenya, *P. schultzi* appears to prefer the webs of *T. striatipes* (Murphy & Murphy, 1983) and even oviposits in them (Forster & Murphy, 1986). Based on the drawings of Strand (1908), *T. rutenbergi* makes funnel-shaped nests on the trunks of trees. The courtship and mating behaviour of *T. striatipes* was studied by Coyle & O'Shields (1990). Courtship consists of an early non-contact phase of vibratory signalling. Then follows a contact phase involving leg-fencing and sometimes lunging. The male clasps the female palp with the tibial apophysis, tilts the female upwards and backwards. The palpal organ is then inserted alternately. The egg sac is made in the wall of the female's retreat. It is elongated and hammock-shaped (fig. 29i) and contains between 56 and 277 eggs. The female

appears to spend most of her time in the retreat positioned over the egg sac (Coyle, 1995). Prey includes ants, beetles, cicadas, flies, grasshoppers, hemipterans, isopods, millipedes, snails, spiders, wasps and winged termites (Coyle, 1995).

Distribution

One species is known from Southern Africa and Madagascar (fig. 32).

Species recorded from Southern Africa

1. *Thelechoris striatipes* (Simon, 1889a)

Entomothela striatipes Simon, 1889a: 236; 1891: 329.

Thelechoris rutenbergi (misidentification); Lenz, 1891: 153; Strand, 1916: 54.

Thelechoris karschi Bösenberg & Lenz, 1894: 27; Benoit, 1964b: 424; 1971: 147; Raven, 1983b: 553; Platnick, 1989: 82; 1993: 91; Coyle & O'Shields, 1990: 281; Coyle & Meigs, 1992: 289; Coyle, 1995: 117 (synonym).

Ischnothela masonica Pocock, 1901b: 337; Roewer, 1942: 204; Benoit, 1964b: 425 (synonym).

Ischnothela karschi: Strand, 1907c: 263; Berland, 1914: 51; Lessert, 1936: 207; Roewer, 1942: 204; Holm, 1954: 199; Coyle, 1986: 279; Forster & Murphy, 1986: 29; Baert & Murphy, 1987: 194.

Ischnothela rutenbergi (misidentification); Strand, 1907c: 263 (in part).

Ischnothela gracilis Tucker, 1917: 123; Lessert, 1936: 207; Benoit, 1964b: 425.

Ischnothela cassetti Tucker, 1920: 444; Benoit, 1964b: 425.

Ischnothela catamita Roewer, 1953: 56 (misidentification).

Thelechoris striatipes, Willey & Coyle, 1992: 151; Coyle, 1995: 117; Platnick, 1998: 121.

Type locality: male lectotype and female paralectotype (MNHN 7008), Nossi Bé, NE coast of Madagascar.

Distribution: Madagascar, Tanzania, Kenya, Malawi, Zambia, Zimbabwe, Mozambique, Angola, Namibia, Botswana and South Africa (Northern Province).

Subfamily EUAGRINAE Raven, 1979

Euagrini Raven, 1979: 635; 1985: 78.

Diagnostic characters

Carapace very flat and hirsute; fovea small and pit-like; eye group twice as wide as long; labium and endites without cuspules; serrula broad; leg formula 4321; paired tarsal claw with one row of S-shaped teeth; unpaired claw elongate, curved and finely dentate; tarsi III and IV at least with spines, scopulae absent; apical segment of posterior spinnerets digitiform (fig. 30a).

Distribution

This subfamily occurs worldwide and is repre-

sented by nine genera with only one genus, *Allothela*, known from Africa.

Genus *ALLOTHELE* Tucker, 1920

Allothela Tucker, 1920: 441; Coyle, 1984: 4; Raven, 1985: 78.

Type species: *Allothela teretis* Tucker, 1920.

Diagnostic characters

Carapace with dense hair cover consisting of thin recumbent setae (fig. 28); eyes in compact quadrangle, wider than long; fovea a deep transverse groove, usually recurved, with two erect setae side-by-side in front of fovea; sigilla small, round and subequal in size (fig. 29b); endites and labium lack cuspules; cheliceral furrow with 9–14 medium-sized to large teeth on promargin and 6–50 along proximal one-third on retrolateral side; leg III usually longer than I or II; males with a well-developed non-terminal mating apophysis on tibia II with stout tooth-like apical and subapical spines (fig. 30b); palpal bulb of male simple, pyriform with elongated, ridged embolus (fig. 30c); females with setae-lined spermathecae; median spinnerets short, unsegmented (fig. 30a) with distinct hirsute sclerite just anterior to base; posterior spinnerets longer than carapace with terminal segment longer than basal or middle segment; medium- to dark brown, carapace with radiating stripes, abdomen with markings; body size 8–12 mm.

Taxonomic notes

Allothela was synonymized with *Euagrus* Ausserer, 1875, by Benoit (1964b). When Coyle (1984) revised the African genus *Allothela* he removed it from the synonymy of *Euagrus* (sometimes erroneously called *Evagrus*).

Natural history

Very little is known about the behaviour and ecology of *Allothela* species (Coyle, 1984). They seem to be adapted to savanna and forest habitats with dry winters and rainy summer seasons. They make sheetwebs with funnel-retreats partly (or wholly) sheltered in subterranean cavities, under rocks, in rotten logs, in leaf litter or under bark (fig. 31f) (Coyle, 1984). The males abandon their webs in search of mates during the wet summer months. *Allothela teretis* typically builds a sheet curtain-web in cool, shady places such as on tree trunks and in or across holes on stream

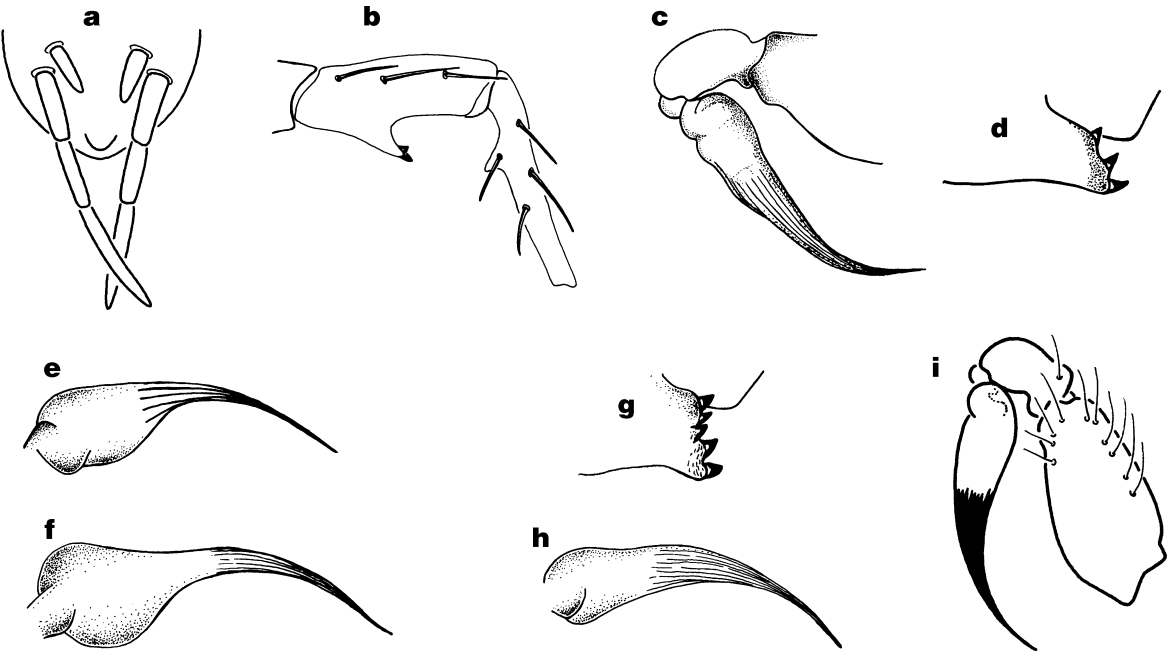


Fig. 30. Dipluridae, Euagrinae — *Allothele* spp. **a:** spinnerets, ventral view; **b:** mating spur, *A. teretis*; **c:** bulb, *A. teretis*; **d:** mating spur, *A. caffer*; **e:** bulb, *A. caffer*; **f:** bulb, *A. australis*; **g:** mating spur, *A. malawi*; **h:** bulb, *A. malawi*; **i:** male palp. (b–h after Coyle, 1984.)

KEY TO THE SOUTHERN AFRICAN SPECIES OF ALLOTHELE (adapted from Coyle, 1984)	
1. Males.....	2
— Females.....	5
2. Mating apophysis on tibia II distally slender, with two stout spines on tip (fig. 30b); embolus (fig. 30c) (KwaZulu-Natal) ..	<i>A. teretis</i>
— Mating apophysis on tibia II relatively short, stout, rarely with fewer than three spines (fig. 30 d,g).....	3
3. Basal three-fifths of embolus broad but tapers abruptly to become thin and straight in distal two-fifths, curved only at tip (fig. 30e) in retrolateral view (KwaZulu-Natal) ..	<i>A. caffer</i>
— Embolus tapers gradually and curves gently over entire length (fig. 30h).....	4
4. Palpal bulb proportionally wide in retrolateral view; embolus strongly curved near tip (fig. 30f) (Eastern Cape Province) ..	<i>A. australis</i>
— Palpal bulb not especially wide in retrolateral view; embolus not strongly curved near tip (fig. 30h) (Kruger National Park) ..	<i>A. malawi</i>
5. Single, large, roughly oval spermathecal bulb (fig. 31a).....	<i>A. australis</i>
— Each spermathecal trunk terminates in two distinct bulbs, a median and a lateral bulb (which is sometimes bifurcate) ..	6
6. Spermathecal trunks short and broad at base; median bulb irregular, sclerotized, with one or two forward-projecting lobes (fig. 31b) ..	<i>A. caffer</i>
— Spermathecal trunks not as short and broad at base; median bulb lacks forward-projecting lobes (fig. 31d) ..	7
7. Anterior genital lip prolonged posteriorly well past edge of anterior booklung (fig. 31e) ..	<i>A. teretis</i>
— Anterior genital lip prolonged very little or not at all past posterior edge of anterior booklung ..	<i>A. malawi</i>

banks. *Isela okuncana*, a mysmenid, lives as a kleptoparasite on the web of *A. terestis* (Griswold, 1985b). *Allothele australis* lives in a sheetweb with a funnel retreat, located beneath stones and in rock crevices, usually provided with two entrances. *A. caffer* constructs similar sheetwebs under stones, as does *A. malawi* under rotten logs near streams.

Distribution

Allothele represents a distinct group of diplurids found only in the southern parts of Africa where they are represented by four species (Coyle, 1984) (fig. 32).

List of species from Southern Africa

1. *Allothele australis* (Purcell, 1903c)

Thelechoris australis Purcell, 1903c: 106; Hewitt, 1915d: 132; Tucker, 1917: 120 (misidentification).

Allothele australis: Tucker, 1920: 442; Coyle, 1984: 8; Platnick, 1989: 78.

Type locality: female holotype (SAM 8899), Dunbrody (33.28S; 25.33E), Sundays River near Kirkwood, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Dunbrody, Cookhouse, Grahamstown, Somerset West, Line Drift, Peddie).

2. *Allothele caffer* (Pocock 1902b)

Euagrus caffer Pocock, 1902b: 318; Hewitt, 1915d: 132; 1919b: 109; Benoit, 1971: 148; Raven, 1983a: 348.

Euagrus caffer australis: Hewitt, 1919b: 95.

Euagrus caffer: Benoit, 1964b: 419 (in part).

Allothele caffer: Tucker, 1920: 442 (in part).

Allothele caffer var. *australis*: Bonnet, 1955: 231.

Allothele caffer: Tucker, 1920: 442 (in part); Lessert, 1933: 85; Coyle, 1984: 12; Platnick, 1989: 78.

Type locality: two male and two female syntypes (BMNH), Durban (29.57S; 30.59E), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Durban, Burman Bush, The Bluff, Umhlali, Umhlanga) and Malawi.

3. *Allothele malawi* Coyle, 1984

Allothele malawi Coyle, 1984: 14; Platnick, 1989: 78.

Type locality: male holotype and five male paratypes (MRAC 155.230), Lichenya Plateau, Linje River, Mt Mulanje, Malawi.

Distribution: Malawi and South Africa (*Mpumalanga*: Kruger National Park (Punda Miliq)).

4. *Allothele teretis* Tucker, 1920

Allothele teretis Tucker, 1920: 441; Bonnet, 1955: 231; Roewer, 1942: 203; Coyle, 1984: 17; Platnick, 1989: 78.

Evagrus teretis: Benoit, 1964b: 421.

Type locality: male and two female syntypes (SAM B4021), Mfongosi, near Ubombo, Zululand, KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Mfongosi, Estcourt, Griffin Hill about 100 km SE Estcourt, Kranskop, Middledrift at Tugela River, Mpofana near Dundee, Weenen, Spienkop, Winterton, Muden).

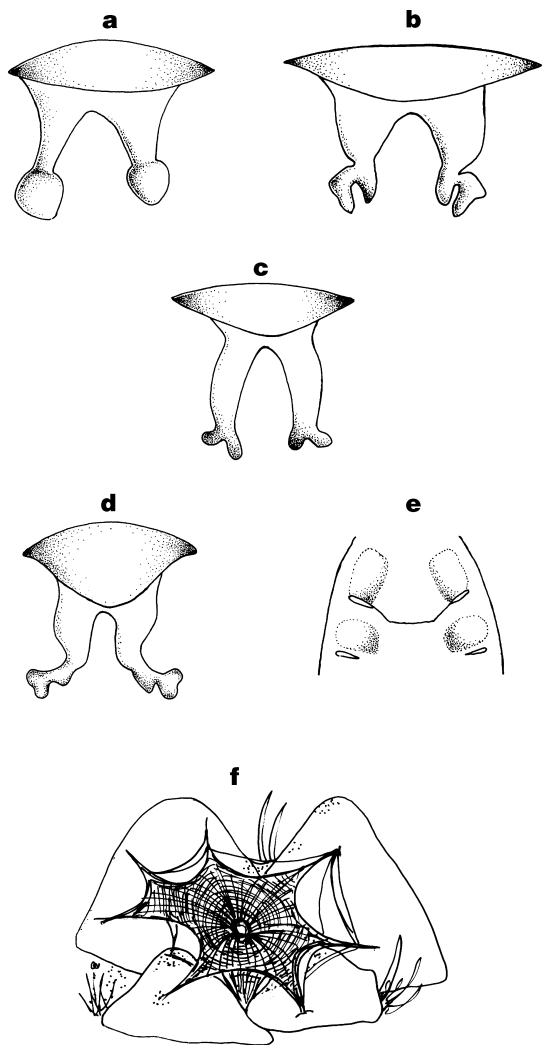


Fig. 31. Dipluridae, Euagrinae — *Allothele* spp. **a:** spermathecae, *A. australis*; **b:** spermathecae, *A. caffer*; **c:** spermathecae, *A. malawi*; **d:** spermathecae, *A. terestis*; **e:** epigynal opening, *A. terestis*; **f:** sheetweb of a diplurid. (a–e after Coyle, 1984.)

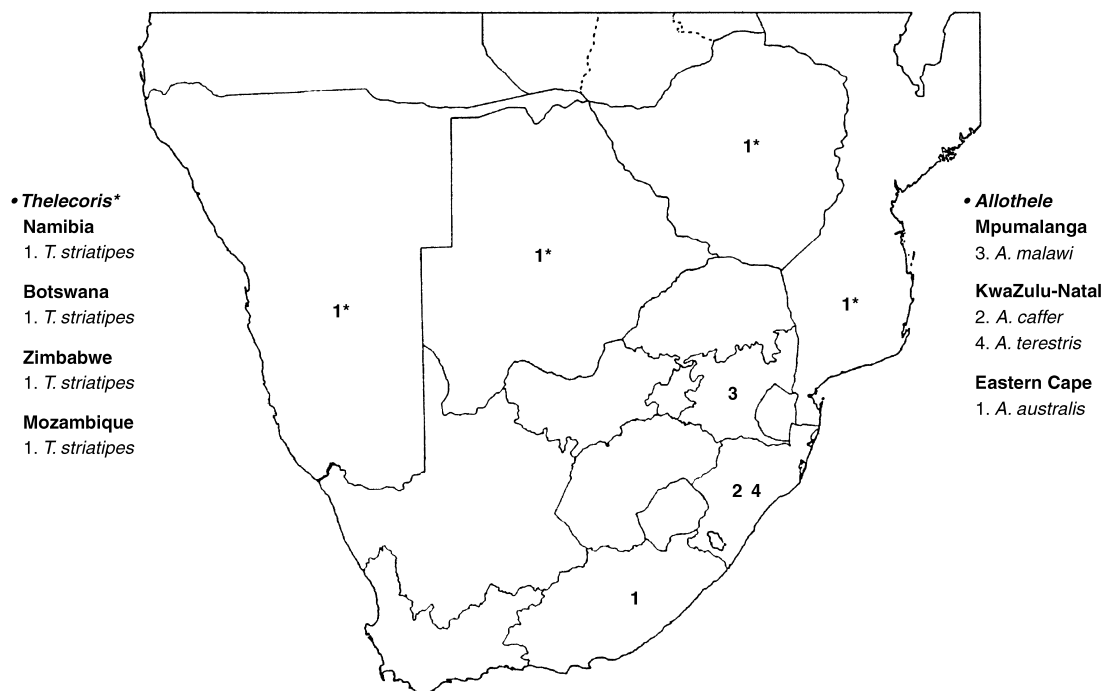


Fig. 32. Distribution of *Thelechoris** and *Allothele* species in Southern Africa.



Female sheetweb mygalomorph
 (Dipluridae: *Allothele australis*).

FAMILY IDIOPIDAE

front-eyed trapdoor spiders

(figs 33–46)

Idiopids are known from India, Australia, New Zealand, South and Central America, Madagascar and Africa. They are represented by 19 genera and about 200 species in three subfamilies, the Arbanitinae, Genysinae and Idiopinae. The last two subfamilies are known from the Afrotropical Region where they are represented by eight genera and 96 species. From Southern Africa, only the subfamily Idiopinae is known, represented by six genera and 65 species.

Genera recorded from Southern Africa

Idiopinae: *Ctenolophus* Purcell, 1904; *Galeosoma* Purcell, 1903; *Gorgyrella* Purcell, 1902; *Heligmomerus* Simon, 1892; *Idiops* Perty, 1833; *Segregara* Tucker, 1917.

Diagnostic characters

Idiopids are medium-sized to large (8–33 mm) spiders with the following synapomorphies:

- distal sclerite of male palp open along one side with second haematodocha extending almost to tip of embolus (fig. 34b),
- cymbium of male palp with one lobe rounded, other acutely pointed,
- palpal tibia of males swollen and usually with an excavation prolaterally, bearing short, thorn-like spines usually arranged in a half-circle (fig. 34b).

Descriptive characters

- **carapace:** glabrous in females and usually granulate in males; cephalic region arched; fovea broad and procurved (fig. 33);
- **sternum:** four or six sigilla; labiosternal groove shallow;
- **eyes:** anterior lateral eyes close together, on eye tubercles on or near clypeal edge (Idiopinae) (fig. 34c); arrangement of other eyes varies between genera;
- **chelicerae:** rastellum on distinct mound (fig. 34d);

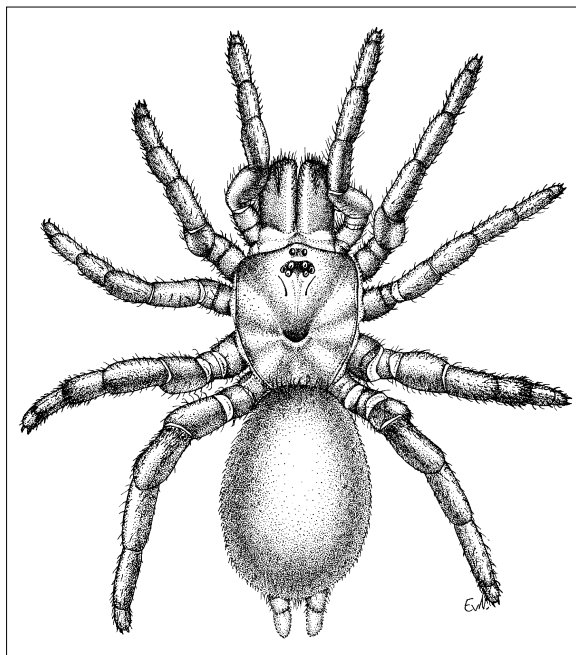


Fig. 33. Idiopidae — *Ctenolophus* sp.

cheliceral furrow usually with strong cheliceral teeth on inner row with fewer and sometimes smaller teeth or denticles on outer row (fig. 35b);

- **mouthparts:** labium usually wider than long, cuspsules present in females (fig. 35c) absent in males; endites rectangular, anterior lobe small; serrula absent;
- **legs:** three claws; front legs of females with strong rows of setae arranged on lateral edge of tarsi, metatarsi and sometimes tibia; coxae of legs with patches of dense spinules or stiff setae in some genera; patellae III and IV dorsally with rows of spinules in some genera; female with teeth on paired claws, similar in size or number; male with one row of a few long teeth on paired claws; male usually with scopulae on tarsi I–IV;
- **abdomen:** oval, except in *Galeosoma* in which the apical part is domed, covered by a shield in females (fig. 37a);
- **spinnerets:** four; posterior spinnerets with apical segment domed; anterior spinnerets very small;
- **genitalia:** male palp with distal sclerite open along one side with second haematodocha extending down bulb almost to embolus tip; cymbium with one lobe

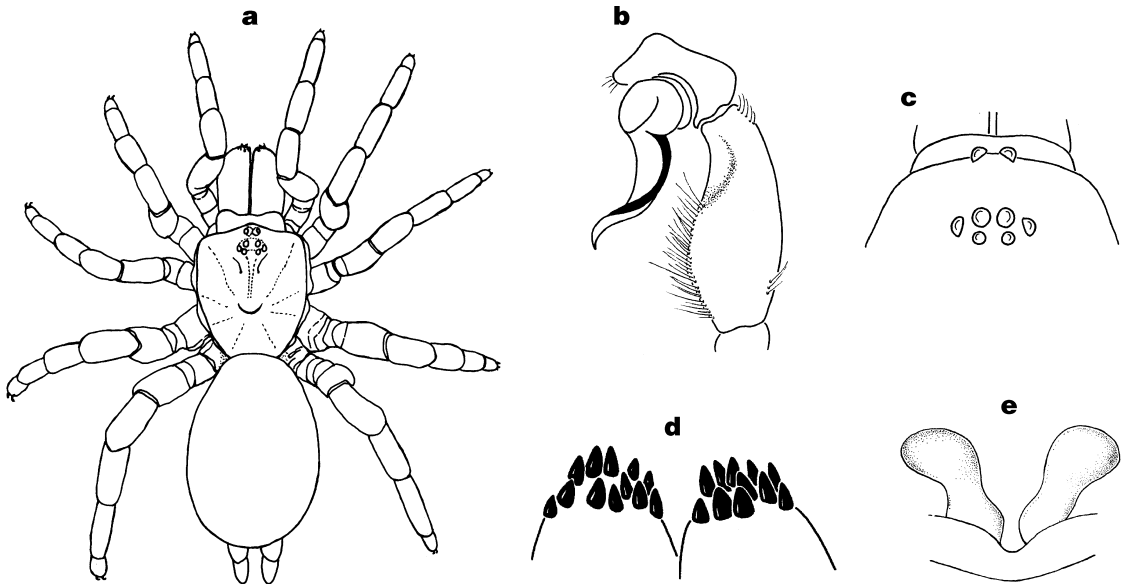


Fig. 34. Idiopidae, Idiopinae. **a:** female, dorsal view; **b:** male palp; **c:** eye pattern; **d:** rastellum; **e:** spermathecae.

rounded and other acutely pointed; palpal tibia of males swollen and usually with an excavation prolaterally, bearing short, thorn-like spines (fig. 34b), usually arranged in a half-circle; spermathecae sac-like, spermathecal ducts short and broad (fig. 34e);

- **body size:** 8–33 mm;
- **colour:** various shades of brown, yellow, red, olive to purplish.

Higher classification

Raven (1985) placed the Idiopidae in the microorder Fornicephalae (fig. 2) in the Rastelloidina with the Migidae, Actinopodidae and Ctenizidae. The Cyrtachenidae constitutes the sister group of the Idiopidae.

Taxonomic note

Raven (1985) reviewed and redelimited the Idiopidae and provided a key to the genera. He transferred several genera previously placed in the Ctenizidae to the Idiopidae.

Natural history

Idiopids are trapdoor spiders that use their rastellums to excavate burrows. They make their burrows in a variety of soil types, often on open grassy plains where the soil is soft during summer, enabling them to dig or enlarge their burrows

before the soil hardens during the winter months. The females live permanently in silk-lined burrows while adult males usually wander around in search of a mate.

Subfamily IDIOPINAE Simon, 1889

Idiopinae Simon, 1889b: 178.

Idiopinae, Raven, 1985: 59, 138.

Diagnostic characters

Carapace glabrous in females and granulate in males, with cephalic region arched and fovea strongly procurved (fig. 34a); anterior lateral eyes set far in front of other eyes close to clypeal edge (fig. 34c); labium wider than long with few cuspules (fig. 35c), cuspules absent in males; rastellum consisting of a distinct process with strong setae (fig. 34d); posterior pair of sigilla absent (except in *Segregara* and *Gorgyrella*); distal segments of front legs with numerous lateral spines; scopulae absent from tarsi of female but usually present on all tarsi of male; tibia I of male with a single, distal spur with two apophyses distally; palpal tibia of males with an excavation prolaterally, bearing short, thorn-like spines usually arranged in a half-circle (fig. 35d).

Taxonomic notes

Although the Idiopinae is a very uniform and distinctive group of spiders, problems still surround

generic relationships. According to Raven (1985) questions regarding the number of genera in Southern Africa can only be addressed after revisions of all genera. To date, no generic revisions of the Idiopidae of Southern Africa have been undertaken.

For the Southern African genera, Tucker (1917) is followed here. He recognized the number, size and position of sigilla as being distinct for each genus. Therefore several genera with three pairs of small sigilla previously listed in *Idiops* are here moved to *Segregara*. However, all genera are in need of revision as the validity of several species and even genera is in doubt.

Natural history

Members of the Idiopinae close their burrows with a trapdoor hinged at one side with silk. The thickness of the lids varies from wafer-thin to thick and cork-like in appearance. The spiders are nocturnal and sit at the entrance of the burrows with the door slightly open, waiting for prey to pass by. If disturbed, the spider retreats into the burrow, closing the door tightly behind it. During the day they are usually found at the bottom of the burrow. Idiopines are more or less gregarious and several individuals are usually found in the same area, often making their burrows in open grassland. The males do not live permanently in burrows, but move around in search of females.

Although most burrows are made in the ground, Coyle *et al.* (1992) collected an Idiopinae species from East Africa that makes arboreal, tubular, silk-lined retreats in clayey soils lodged in deep bark crevices 1.3–2.0 m above ground level. It constructs the trapdoor using the door-moulding method. The trapdoors is well camouflaged, with the door edges bevelled and extending outwards to numerous closely spaced tabs made of silk. According to Coyle *et al.* (1992), these tabs may increase prey-sensing effectiveness or may allow the door to be pulled further into the retreat and held more securely.

Distribution

The subfamily Idiopinae is known from South and Central America, Africa, India and West Australia. It is represented in Southern Africa by six genera and 62 species.

KEY TO THE SOUTHERN AFRICA GENERA OF IDIOPINAE

(adapted from Griswold, 1984, and Raven, 1985)

- 1. Abdomen truncated, with apical part domed with chitinated shield in females (fig. 37a); soft-bodied in males . . . **Galeosoma**
- Abdomen soft-skinned, evenly rounded (fig. 33) **2**
- 2. Tibia III excavated dorsally at base, saddle-shaped with bare patch posteriorly among rows of spinules (fig. 41a); two pairs of sigilla (fig. 41b) **Heligmomerus**
- Tibia III convex; two or three pairs of sigilla . . **3**
- 3. Sternum with three pairs of sigilla (fig. 39b) . . **4**
- Sternum with two pairs of sigilla (fig. 35c) . . . **5**
- 4. Posterior sigilla on sternum enlarged (fig. 39b) **Gorgyrella**
- Posterior sigilla on sternum small (fig. 45a) **Segregara**
- 5. Cheliceral furrow with row of strong teeth on inner row, teeth in outer row reduced, or only small denticles posteriorly (fig. 35b) **Ctenolophus**
- Cheliceral furrow with two rows of equally strong teeth (fig. 43c) **Idiops**

Genus CTENOLOPHUS Purcell, 1904

Ctenolophus Purcell, 1904: 117; Roewer, 1942: 149; Raven, 1985: 138.

Type species: *Acanthodon kolbei* Purcell, 1902.

Diagnostic characters

Median ocular quadrangle usually wider posteriorly than anteriorly (fig. 35a); anterior lateral eyes large, near clypeal edge, usually on tubercles; posterior row of eyes procurved; posterior lateral eyes usually large and reniform or pear-shaped (fig. 35a); cheliceral furrow with a row of large teeth on inner row, outer row of teeth reduced or only a few small denticles posteriorly (fig. 35 b); labium broader than long with cuspules (fig. 35c); sternum with two pairs of small marginal sigilla (fig. 35c); coxa III with strips of rather slender setae on posteroventral border; male palp as in fig. 35d; body size 8–18 mm.

Taxonomic notes

Purcell (1904) moved four species that he described in *Acanthodon* to *Ctenolophus*, a genus created by him. Hewitt (1913a) regarded

Table 4. Somatic characters used to distinguish Southern African Idiopinae genera.

Genus/size	Tibia II saddle-shaped	Abdominal shield	Posterior sigilla	Size of posterior sigilla	Shape of chelicerae	Teeth on cheliceral furrow
<i>Ctenolophus</i> (8–18 mm)	absent	absent	absent	—	normal	one row with small denticles in a second row
<i>Galeosoma</i> (16–21 mm)	absent	present	absent	—	normal	one row
<i>Gorgyrella</i> (19–33 mm)	absent	absent	present	large	narrowed	two rows with fewer teeth in outer row
<i>Heligmomerus</i> (18–33 mm)	present	absent	absent	—	narrowed	two rows
<i>Idiops</i> (9–35 mm)	absent	absent	absent	—	normal	two rows with equal number of teeth
<i>Segregara</i> (12–22 mm)	absent	absent	present	small	normal	two rows with unequal number of teeth

Ctenolophus a synonym of *Acanthodon* but *A. petittii*, the type species, was moved to *Idiops*, equivalent to considering *Ctenolophus* congeneric with *Idiops*. However, according to Tucker (1917), *Idiops* and *Ctenolophus* differ in one respect, namely the row of teeth on the cheliceral groove, which was also recognized by Raven (1985).

Natural history

Little is know about the behaviour of *Ctenolophus* species except for the following few observations. *Ctenolophus spiricola* from Kentani in the Eastern Cape Province constructs a silk-lined burrow that descends to a depth of 6–8 cm (fig. 35e) (Purcell, 1903c). At the bottom, the burrow gradually widens, while at the entrance it is more compressed with an oval opening. The edges of the tube-lining are spread out horizontally onto the surface of the ground, forming a flat rim about 1 mm wide except at the side of the hinge. The hinge is broad and situated along one of the longer edges of the oval rim. The lid, which is D-shaped in outline, is strongly but irregularly convex on its upper surface and covered with earth and minute green vegetable growth, often including pieces of moss. The lower surface is

smooth and concave right up to the edge, which is quite sharp and fits against the outer edge of the outspread rim of the tube. The burrows of *C. spiricola* are found sheltered under roots in forested areas.

Ctenolophus oomi has been collected from various localities in the Carolina district (Van Dam & Roberts, 1917). Some burrows were found on the upper face of an embankment of an old road. The burrows were not very deep, about 7.5 cm, with the silk lining extending above the soil surface. All the lids were decorated with bits of grass. The burrows of *C. kolbei* are tubular and about 10 mm in diameter for the greater part except about 10 mm from the top where they widen gradually to about 15 mm at the opening. The opening has a broad rim of silk, about 4 mm wide. The lid is flat and not thickened, and it merely closes against the rim. The hinge is almost as broad as the greatest diameter of the lid (Purcell, 1902b).

Distribution

Ctenolophus is an African genus with eight known species, five of which occur in Southern Africa (fig. 36).

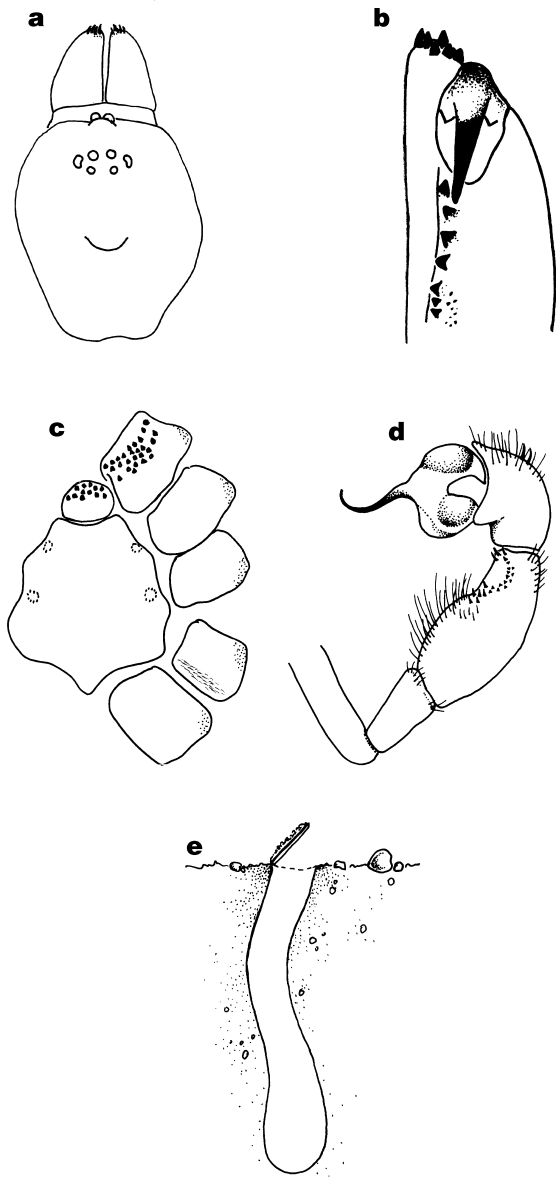


Fig. 35. *Ctenolophus* sp. **a:** carapace, dorsal view; **b:** cheliceral fang furrow; **c:** sternum; **d:** male palp; **e:** burrow.

2. Ocular area short, length less than or equal to one-third the distance between anterior edge of carapace and centre of fovea 3
- Ocular area longer, length almost equal to two-fifths the distance between anterior edge of carapace and centre of fovea 4
3. Abdomen anteriorly with setigerous tubercles (Kentani) ***C. kolbei***
- Abdomen anteriorly without setigerous tubercles (Kentani) ***C. spiricola***
4. Coxa III with strongly developed, narrow patch of slender, bristly setae; posterior median eyes closer to posterior lateral eyes than to each other (Newington) ***C. fenoulheti***
- Coxa III with weakly developed, thin band of setae; posterior eyes subequal in size (Paulpietersburg) ***C. oomi***

Species recorded from Southern Africa

1. ***Ctenolophus cregoei* (Purcell, 1902b)**
Acanthodon cregoei Purcell, 1902b: 352; Hewitt, 1919b: 75.
Ctenolophus cregoei: Purcell, 1904: 119; Roewer, 1942: 149.
Type locality: male holotype (SAM 981), Durban (29.57S; 30.59E), KwaZulu-Natal.
Distribution: South Africa (*KwaZulu-Natal*: Durban).
2. ***Ctenolophus fenoulheti* Hewitt, 1913a**
Ctenolophus fenoulheti Hewitt, 1913a: 414; Roewer, 1942: 149.
Type locality: female holotype (AM), Newington, Soutpansberg district, Northern Province (Note: according to an atlas 'Newington' is close to Pilgrim's Rest (24.51S; 31.25E) in Mpumalanga Province).
Distribution: South Africa (*Mpumalanga*: Newington).

3. ***Ctenolophus kolbei* (Purcell, 1902b)**
Acanthodon kolbei Purcell, 1902b: 353.
Ctenolophus kolbei: Purcell, 1904: 118; Roewer, 1942: 150.
Type locality: female syntypes (SAM 4543), Kentani district (32.30S; 28.21E), Eastern Cape Province
Distribution: South Africa (*Eastern Cape*: Kentani).

4. ***Ctenolophus oomi* Hewitt, 1913a**
Ctenolophus oomi Hewitt, 1913a: 415.
Acanthodon oomi: Hewitt, 1916b: 187; Roewer, 1942: 150.
Type locality: female holotype (NM), Lüneburg, near Paulpietersburg (27.25S; 30.50E), KwaZulu-Natal.
Distribution: South Africa (*KwaZulu-Natal*: Paulpietersburg; *Mpumalanga*: Lake Chrissie, Oshoek, Carolina).

KEY TO THE SOUTHERN AFRICAN SPECIES OF CTENOLOPHUS (adapted from Hewitt, 1916b, 1919b)

1. Males; metatarsus I distinctly curved (seen from side); excavation of palpal tibia with broad, semicircular band of short, dense spinules (fig. 35d); (Durban) ***C. cregoei***
- Females 2

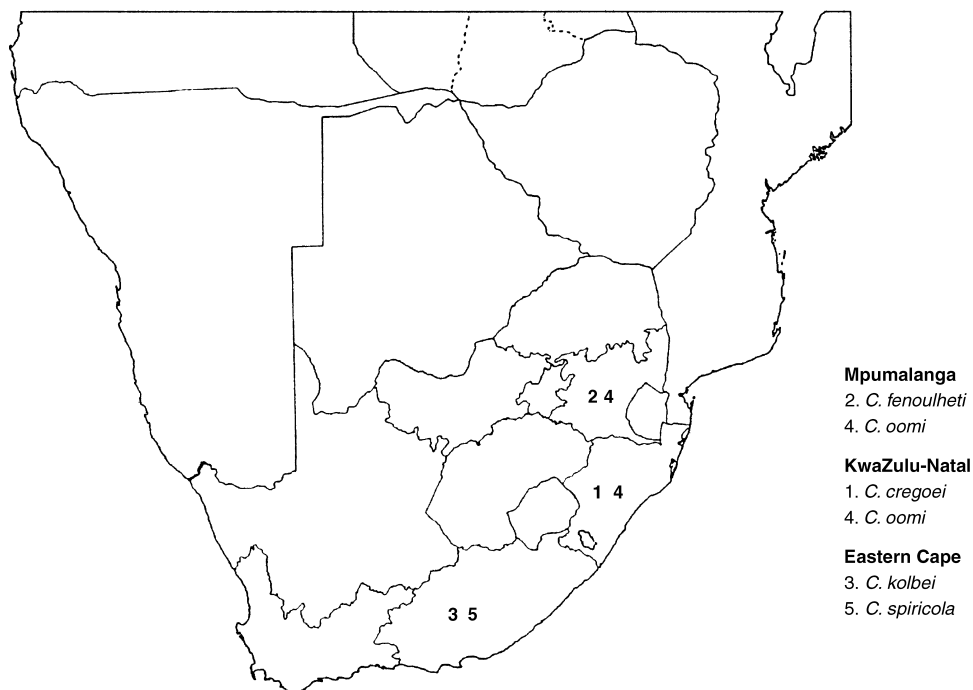


Fig. 36. Distribution of *Ctenolophus* species in Southern Africa.

5. *Ctenolophus spiricola* (Purcell, 1903c)

Acanthodon spiricola Purcell, 1903c: 88; Hewitt, 1914b: 17; 1919b: 77.

Ctenolophus spiricola: Purcell, 1904: 118; Roewer, 1942: 150.

Type locality: two female syntypes (SAM 12413, 12414), village near Kentani (32.30S; 28.21E), Eastern Cape Province. Hewitt (1914b) described a male from the same locality.

Distribution: South Africa (Eastern Cape: Kentani).

Taxonomic notes

According to Hewitt (1919b), the shape of the shield is fairly constant in adults collected from the same area. However, in some species, such as *G. coronotum*, *G. hirsutum*, *G. pallidum* and *G. pilosum*, there is a gradation in shape, with intermediate forms that are difficult to identify. The shape of the shield in juveniles may vary considerably and to distinguish between species, only adult female material can be used. The males of only two species are known and their abdomens are evenly contoured with no raised surfaces. A key to some of the species of *Galeosoma* was provided by Hewitt (1935).

Natural history

Members of the genus *Galeosoma* live in silk-lined burrows made in the soil, with the entrance sealed by a wafer-type trapdoor (fig. 37g). The female spider uses the hardened posterior part of her abdomen as a false bottom to close off and protect the lower part of the burrow. According to Hewitt (1913a), the abdominal shield prevents predators of *Galeosoma*, such as centipedes and pompilid wasps, from entering

Genus *GALEOSOMA* Purcell, 1903

shield-bum trapdoor spider

Galeosoma Purcell, 1903c: 92; Raven, 1985: 138.

Type species: *Galeosoma scutatum* Purcell, 1903.

Diagnostic characters

Cuticula of upper part of abdomen very thick, hard and inflexible in female (fig. 37a), forming a symmetrical oval piece of armour encasing dorsal and upper surface, completely covering spinnerets from above, underside soft-skinned; cheliceral furrow with one row of teeth; labium and endites with few cuspules; two pairs of small sigilla situated close to sternal edge; coxae III without cuspules; male palp as in fig. 37b; body size 16–21 mm, shield width 8–11 mm.

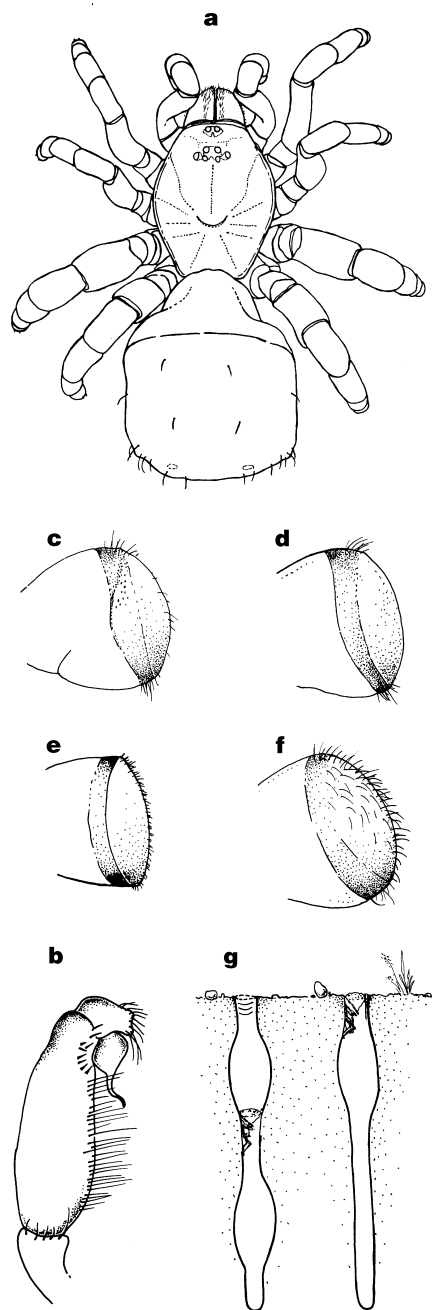


Fig. 37. *Galeosoma* spp. **a:** female, dorsal view; **b:** male palp; **c–f:** lateral view of shields in (c) *G. mossambicus*, (d) *G. vandami*, (e) *G. planiscutatum*, and (f) *G. coronatum*; **g:** burrows.

the burrow from above.

The entrance to the burrow of *G. schreineri* is raised slightly above the level of the ground (Hewitt, 1913a). The burrow descends vertically,

expanding below into one or two chambers. The shield of the spider very closely fits the narrow passage and the spider can completely block the way against intruders by stationing itself with the shield upwards. The burrow is closed with a thin wafer-like lid. The spider uses the wider portions of the burrow as 'turning chambers' (fig. 37g). The burrow entrances of *G. hirsutum* and *G. pilosum* are raised well above ground level (sometimes as much as 1.5 cm) and are decorated with bits of grass or small pebbles, particularly the lids. This remarkable disguise resembles worn-off tufts of dead grass. The lid is usually concave above and fits onto the rim rather than into it. The depth of the burrow is about 12.5 cm with the narrowest part being 7 mm and the widest part 16 mm (Van Dam & Roberts, 1917). In *G. robertsi* the entrance is level with the ground and is not decorated with grass. A turning chamber is situated halfway down the burrow. Young of *G. robertsi* have been found from March to April in these burrows.

The undeveloped shield of the male is marked by distinct colouration and hairiness, and the area corresponding to the shield of the female is sparsely covered with stiff setae, which become longer along the lateral and posterior margins. In newly-hatched young there is no trace of an abdominal shield (Hewitt, 1916b). The only other spider genus with a similarly shaped abdomen belongs to the genus *Cyclocosma*, known from Mexico and China.

Distribution

Galeosoma is represented by 12 species and three subspecies and is known only from the southern parts of Africa. Except for *G. mossambicus*, known from Mozambique, and *G. vernayi* from Botswana and Namibia, the rest of the species have been recorded from South Africa (fig. 38).

Species recorded from Southern Africa

1a. *Galeosoma coronatum* Hewitt, 1915b

Galeosoma coronatum Hewitt, 1915b: 95; 1919a: 206; 1919b: 81; Roewer, 1942: 150.

Type locality: female holotype (AM), Kroonstad (27.40S; 27.15E), Free State Province.

Distribution: South Africa (*Free State:* Kroonstad; *North West Province:* Potchefstroom).

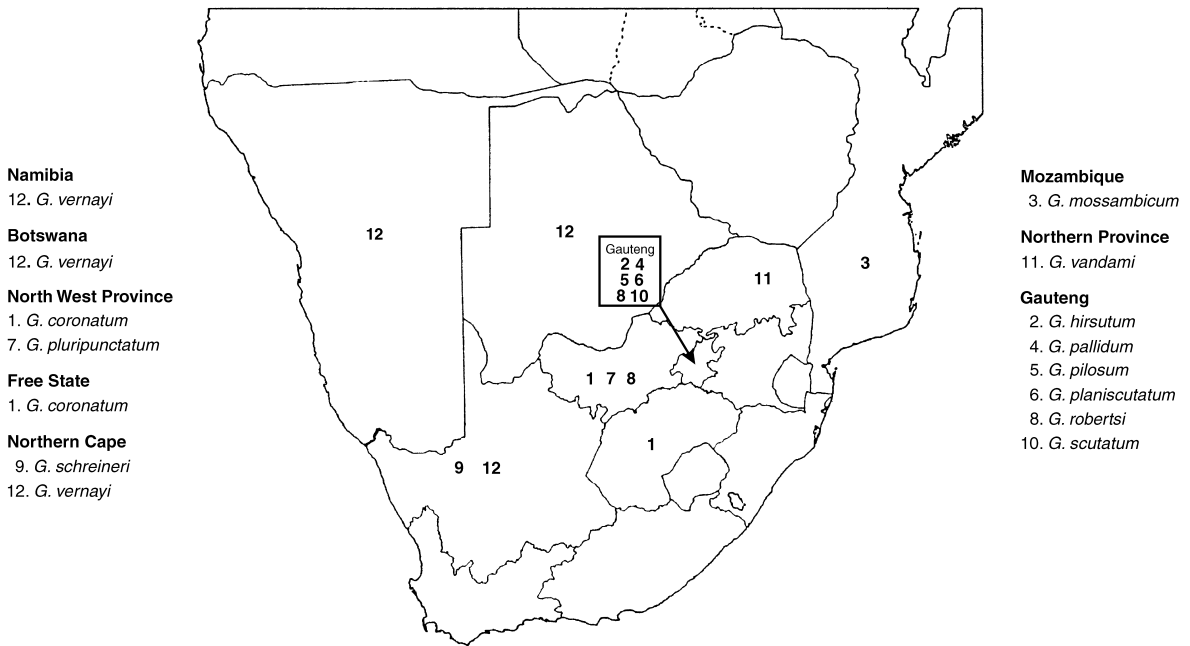


Fig. 38. Distribution of *Galeosoma* species in Southern Africa.

1b. *Galeosoma coronatum sphaeroideum* Hewitt, 1919b

Galeosoma coronatum sphaeroideum Hewitt, 1919b: 82.

Type locality: two female syntypes (TM), Potchefstroom (26.40S; 27.30E), North West Province.

Distribution: South Africa (North West Province: Potchefstroom).

2. *Galeosoma hirsutum* Hewitt, 1916b

Galeosoma hirsutum Hewitt, 1916b: 190; Van Dam & Roberts, 1917: 226; Roewer, 1942: 150.

Type locality: female syntypes (TM), Roodeplaat (25.37S; 28.22E), near Pretoria, and from Zeekoegat, Pretoria, Gauteng Province.

Distribution: South Africa (Gauteng: Roodeplaat, Zeekoegat, Witfontein, Pretoria North, Derdepoort, Rosslyn).

3. *Galeosoma mossambicum* Hewitt, 1919b

Galeosoma mossambicum Hewitt, 1919b: 78; Roewer, 1942: 150.

Type locality: female syntypes, Mazambo and Papai, Mozambique.

Distribution: Mozambique.

4. *Galeosoma pallidum* Hewitt, 1915a

Galeosoma pallidum Hewitt, 1915a: 94; Roewer, 1942: 150.

Type locality: three female syntypes (TM), Saltpan, Pretoria district (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Pretoria).

5. *Galeosoma pilosum* Hewitt, 1916b

Galeosoma pilosum Hewitt, 1916b: 190; 1919a: 211 (male); Roewer, 1942: 150.

Type locality: series of female syntypes (TM) from various localities in and around Pretoria (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Pretoria district: Mayville, Wonderboom Poort, Pretoria College, Koedoespoort, Garstfontein, Centurion: Lyttelton).

6. *Galeosoma planiscutatum* Hewitt, 1919b

Galeosoma planiscutatum Hewitt, 1919b: 79; Roewer, 1942: 150.

Type locality: four female syntypes (TM), Pretoria district (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Buffelsdraai).

7. *Galeosoma pluripunctatum* Hewitt, 1919b

Galeosoma pluripunctatum Hewitt, 1919b: 81; Roewer, 1942: 150.

Type locality: female holotype (TM), Mooivlei Rustenburg district (25.15S; 28.29E), North West Province.

Distribution: South Africa (North West Province: Rustenburg).

8a. *Galeosoma robertsi* Hewitt, 1916b

Galeosoma robertsi Hewitt, 1916b: 186; Van Dam & Roberts, 1917: 225; Roewer, 1942: 150.

Type locality: series of female syntypes (TM) from various localities in and around Pretoria (25.35S; 28.11E), Gauteng Province.

Table 5. Somatic characters of the abdominal shield of the females of some *Galeosoma* species.

SPECIES SIZE DISTRIBUTION	OUTLINE	SHAPE OF SHIELD	SHAPE OF RIDGE	DORSAL SURFACE
<i>G. coronotum</i> 17 mm Kroonstad	oval	strongly convex	ridge partly defined; less distinct posteriorly	bearing numerous long hairs (fig. 37f)
<i>G. hirsutum</i> 20.5 mm Roodeplaat	oval	convex	ridge partly defined anteriorly, upper edge well defined posteriorly	surface rough and pitted with numerous stiff setae giving it a bearded appearance
<i>G. mossambicum</i> 16 mm Mozambique	oval	slightly convex	ridge well-defined but less strongly curved anteriorly; margin consist of two parts	short, fine hair anteriorly, glossy and coarsely pitted posteriorly (fig. 37c)
<i>G. pallidum</i> 21 mm Saltpan	round, disk-like	flat	ridge not well defined but more obtusely inclined anteriorly, junction forms a slightly upturned ridge posteriorly	dorsal surface flat
<i>G. pilosum</i> 19 mm Pretoria	sub-rotund	flat	ill-defined anteriorly, posterior ridge broadly rounded	posteriorly faintly to coarsely pitted; bearing long hairs
<i>G. planiscutatum</i> 7 mm Buffelsdraai	oval	flat	ridge well-defined, upturned all round	shallowly pitted (fig. 37e)
<i>G. pluripunctatum</i> 17 mm Rustenburg	round	moderately convex	ridge partly defined, practically obsolete anteriorly but with sharply defined and slightly upturned ridge posteriorly	closely and finely pitted to smooth and glossy, devoid of long, stiff hairs
<i>G. robertsi</i> 18 mm Pretoria	round to oval	strongly convex	ridge well-defined anteriorly, rounded posteriorly	polished appearance, no long hairs or setae
<i>G. schreineri</i> 20.5 mm De Aar	round to oval	convex	ridge well-defined with slightly raised edge anteriorly and acutely inclined edge posteriorly	shield thick, coarsely pitted and hairy, hair forming a loose fringe around edge
<i>G. scutatum</i> 16.5 mm Krugersdorp	oval	convex	ridge partly defined, posterior margin thickened	coarsely pitted, shiny, glabrous except at margin where clothed with hair
<i>G. vandami</i> 18.5 mm Leydsdorp	oval	convex	ridge well-defined but less prominent anteriorly, with a sharp upturned edge posteriorly	few long, stiff, scattered setae but numerous hairs around margin (fig. 37d)
<i>G. vernayi</i> 18 mm Botswana	oval	convex	ridge narrow but well-defined all round	coarsely and unevenly pitted

Distribution: South Africa (*Gauteng*: Pretoria).

8b. *Galeosoma robertsi crinitum* Hewitt, 1919b

Galeosoma robertsi crinitum Hewitt, 1919b: 82.

Type locality: four female syntypes (TM), Potchefstroom (26.40S; 27.30E), North West Province.

Distribution: South Africa (*North West Province*: Potchefstroom).

9. *Galeosoma schreineri* Hewitt, 1913a

Galeosoma schreineri Hewitt, 1913a: 421; Roewer, 1942: 150.

Type locality: series of female syntypes (AM), De Aar (30.40S; 24.1E), Northern Cape Province.

Distribution: South Africa (*Northern Cape*: De Aar).

10. *Galeosoma scutatum* Purcell, 1903c

Galeosoma scutatum Purcell, 1903c: 92; Roewer, 1942: 150.

Type locality: female holotype (SAM 9733), Luipaardsvlei near Krugersdorp (26.6S; 27.46E), Gauteng Province.

Distribution: South Africa (*Gauteng*: Luipaardsvlei).

11a. *Galeosoma vandami* Hewitt, 1915b

Galeosoma vandami Hewitt, 1915b: 95; 1925: 283; Roewer, 1942: 151.

Type locality: two female syntypes (TM), Griffin Mine, Leydsdorp (23.59S; 30.31E), Northern Province.

Distribution: South Africa (*Northern Province*: Leydsdorp, Wilhelmshöhe near Blaauwberg, Soutpansberg district, Gravelotte).

11b. *Galeosoma vandami circumjunctum*
Hewitt, 1919b

Galeosoma vandami circumjunctum Hewitt, 1919b: 89; Roewer, 1942: 151.

Type locality: two subadults (TM), N'Wenedzi River (22.20S; 30.37E), Soutpansberg district, Northern Province.

Distribution: South Africa (Northern Province: Soutpansberg district).

12. *Galeosoma vernayi* Hewitt, 1935

Galeosoma vernayi Hewitt, 1935: 462; Roewer, 1942: 151; Eagle, 1985: 131; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (TM 5448), Molepolole (24.26S; 25.32E), Kalahari, Botswana.

Distribution: South Africa (Northern Cape), Botswana and Namibia.

Genus *GORGYRELLA* Purcell, 1902

Gorgyrella Purcell, 1902b: 350; Roewer, 1942: 151; Raven, 1985: 138.

Type species: *Gorgyrella namaquensis* Purcell, 1902b.

Diagnostic characters

Posterior median eyes widely spaced and close to posterior lateral eyes (fig. 39a); cheliceral furrow with two rows of teeth, outer row with fewer teeth than inner row, sometimes with one or two strong outer flanking teeth in the middle or with smaller teeth posteriorly; chelicerae anteriorly narrowed; sternum with three pairs of sternal sigilla, posterior pair enlarged (fig. 39b); lateral margin of carapace slightly sinuate above base of leg III (fig. 39a); coxae I–III with large areas densely studded with spinules; male palp as in fig. 39c; body size 19–33 mm in females (males unknown).

Taxonomic note

Gorgyrella is closely related to *Segregara* in having three pairs of sternal sigilla, but they differ in that the posterior pair of sigilla is enlarged in *Gorgyrella*.

Natural history

According to Purcell (1903c), the burrows of *Gorgyrella* can be easily distinguished from those of *Stasimopus*.

The burrowing behaviour of only one of the species is known. The burrows of *G. schreineri* are slanting and widen funnel-like towards the opening (Purcell, 1903c). The opening is oval in

cross-section with the hinge-side flattened. The lid is D-shaped and strongly truncated at the hinge margin. The hinge is very wide, in width exceeding the cylindrical part, with the edge strongly bevelled. There is no circle of conspicuous pits on the lower surface, only a small group of pits in the centre. The egg sac of *G. schreineri* is a white oval capsule, almost flat on one side (22 × 13 mm), but strongly convex on the other. It has a tough, parchment-like texture. The oval edge of the egg sac has a broad, free flap. According to Purcell (1903c), it contains 80–90 eggs, which only occupy half of the inner cavity. Adult females have been collected from September to January, and egg sacs have been found in December and young in January.

Distribution

Gorgyrella is an African genus with four species and one subspecies known only from the southern parts of Africa. One species, *G. hirschhorni*, is found in Zimbabwe, while the others have been recorded from the Cape provinces of South Africa (fig. 40). Roewer (1942) erroneously reported that the type locality of *G. inermis* is in Tanzania — Tucker (1917) described it from Cradock in the Eastern Cape Province, South Africa.

KEY TO FEMALES OF SOUTHERN
AFRICAN SPECIES OF *GORGYRELLA*

- 1. Coxae with numerous spinules (fig. 39b) 2
 - Only coxa III with band of long dense setae (Cradock) *G. inermis*
- 2. Spinules present on coxa I 3
 - Spinules absent from coxa I or with a few confined to narrow band on edge (Van Rhynsdorp) *G. namaquensis*
- 3. Coxa III with few coarse spinules (widely distributed) *G. schreineri*
 - Coxa III with a continuous U-shaped strip of spinules (Harare) *G. hirschhorni*

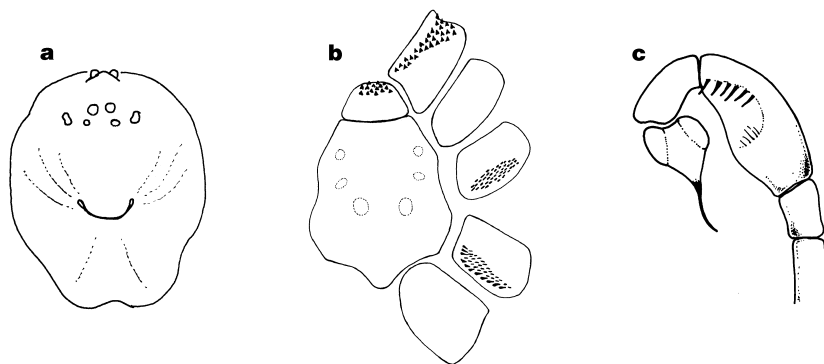


Fig. 39. *Gorgyrella* sp. **a**: carapace, dorsal view; **b**: sternum; **c**: male palp.

Species recorded from Southern Africa

1. *Gorgyrella hirschhorni* (Hewitt, 1919a)

Acanthodon (*Gorgyrella*) *hirschhorni* Hewitt, 1919a: 209;
Fitzpatrick, 2001: 177.
Gorgyrella hirschhorni: Roewer, 1942: 151.

Type locality: female holotype (McGregor Museum, Kimberley), Salisbury [Harare] (17.50S; 31.04E), Zimbabwe.

Distribution: Zimbabwe.

2. *Gorgyrella inermis* Tucker, 1917

Gorgyrella inermis Tucker, 1917: 135; Roewer, 1942: 151.

Type locality: female holotype (SAM B1625), Cradock (32.5S; 25.38E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Cradock).

3. *Gorgyrella namaquensis* Purcell, 1902b

Gorgyrella namaquensis Purcell, 1902b: 350; 1904: 119; Tucker, 1917: 134; Roewer, 1942: 151.

Type locality: female holotype (SAM 8469). Western slope of the Gifftberg, south of Van Rhyn's Dorp (31.36S; 18.45E), Western Cape Province.

Distribution: South Africa (Western Cape: Van Rhyn's Dorp).

4a. *Gorgyrella schreineri* Purcell, 1903b comb. nov.

Gorgyrella schreineri Purcell, 1903b: 25; Tucker, 1917: 134;
Roewer, 1942: 151.

Acanthodon schreineri: Hewitt, 1916b: 184.

Idiops schreineri: Roewer, 1942: 154.

Type locality: two female syntypes (SAM 9518), Hanover (31.5S; 24.27E), Vlagkop and Poortjesfontein, Northern Cape Province.

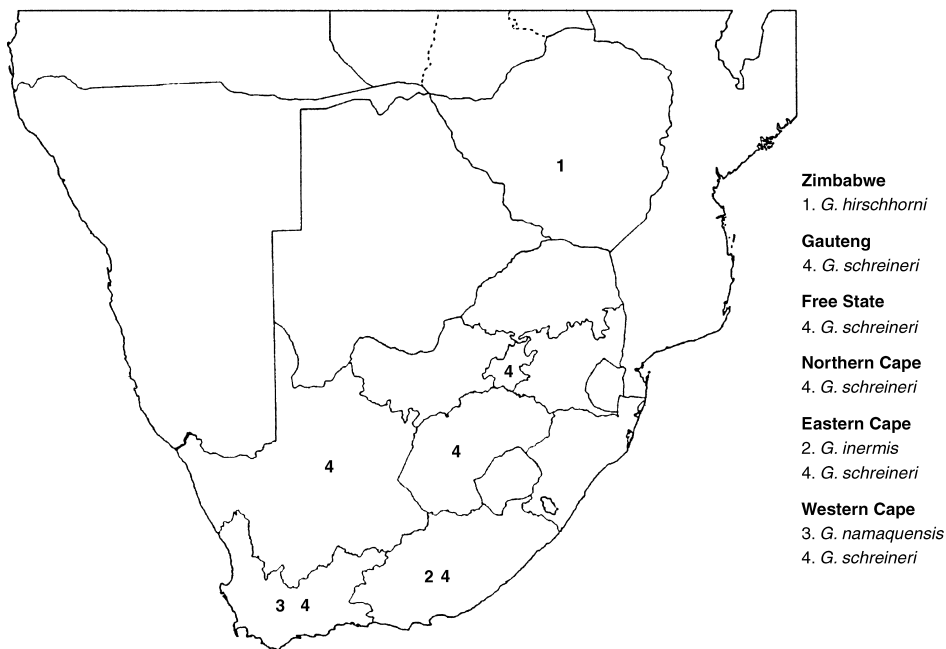


Fig. 40. Distribution of *Gorgyrella* species in Southern Africa.

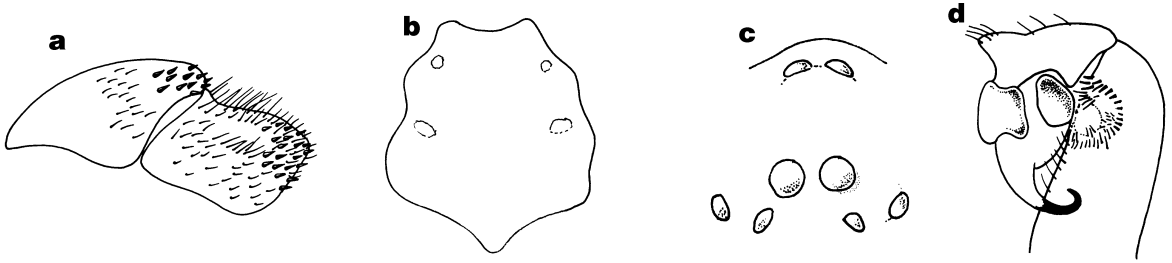


Fig. 41. *Heligmomerus* sp. **a:** tibia III, saddle-shaped; **b:** sternum; **c:** eye pattern; **d:** male palp.

Distribution: South Africa (*Northern Cape*: Hanover, De Aar, Kimberley, Victoria West; *Free State*: Bloemfontein; *Eastern Cape*: Somerset East, Grahamstown, Fort Brown, Adelaide, Whittlesea; *Western Cape*: Worcester, Karoo National Park).

4b. *Gorgyrella schreineri minor* (Hewitt, 1916b) comb. nov.

Acanthodon schreineri minor Hewitt, 1916b: 183; 1919b: 76.

Type locality: male and female syntypes (TM), Roodeplaat (25.37S; 28.22E), 17 km NE Pretoria, Gauteng.

Distribution: South Africa (*Gauteng*: Roodeplaat, Bon Accord, Pretoria: Muckleneuk).

Note: The original species, *Gorgyrella schreineri*, was described by Purcell (1903b). A new variety of *G. schreineri* was described by Hewitt (1916b) as *Acanthodon schreineri minor*, differing from the 'typical *schreineri*' by its smaller size. However, Hewitt did not transfer *schreineri* from *Gorgyrella* to *Acanthodon*. To complicate matters, Roewer (1942) erroneously listed *Acanthodon schreineri* as a species of *Idiops*, crediting Hewitt (1916) for it instead of only for the subspecies '*schreineri minor*'. The subspecies '*schreineri minor*' is here placed in *Gorgyrella*.

Genus *HELIGMOMERUS* Simon, 1892

Heligmomerus Simon, 1892a: 91; Raven, 1985: 138.

Type species: *Heligmomerus prostates* Simon, 1892a.

Diagnostic characters

Eye area short (fig. 41c); fovea broad; chelicerae anteriorly narrowed; cheliceral furrow with two rows of teeth; sternum with four sigilla (fig. 41b); tibia III excavated dorsally at base, saddle-shaped with a bare patch posteriorly among spinules (fig. 41a); male palp as in fig. 41d; body size 18–33 mm.

Natural history

Nothing is known about their behaviour, except that Hewitt (1916b) reported that males of *H. astutus* Hewitt were found hunting insects at an electric light.

Distribution

Seven species of *Heligmomerus* are known, five from Africa, one from India and one from Ceylon. According to Tucker (1917), *Heligmomerus* is not as successfully established in Africa as *Idiops*. The three species recorded from Southern Africa are found in Zimbabwe, the northern parts of South Africa and the Kalahari desert (fig. 42).

Species recorded from Southern Africa

1. *Heligmomerus astutus* (Hewitt, 1915c)

Idiops astutus Hewitt, 1915c: 310; Roewer, 1942: 152; Fitzpatrick, 2001: 177.

Heligmomerus astutus, Hewitt, 1916b: 192.

Type locality: male holotype, Bulawayo, Zimbabwe.

Distribution: Zimbabwe.

Note: Hewitt (1915c) described *Idiops astutus* but transferred it to *Heligmomerus* in 1916b. He suggested that it could be the male of either *H. caffer* or *H. deserti*. He also speculated that the types of both *H. caffer* and *H. deserti* could be immature specimens.

2. *Heligmomerus caffer* Purcell, 1903a

Heligmomerus caffer Purcell, 1903a: 305; Tucker, 1917: 125; Roewer, 1942: 151.

Type locality: female holotype (SAM), Shilovane, (24.05S; 30.14E), near Leydsdorp, Northern Province.

Distribution: South Africa (*Northern Province*: Shilovane, Moorddrift, Waterberg).

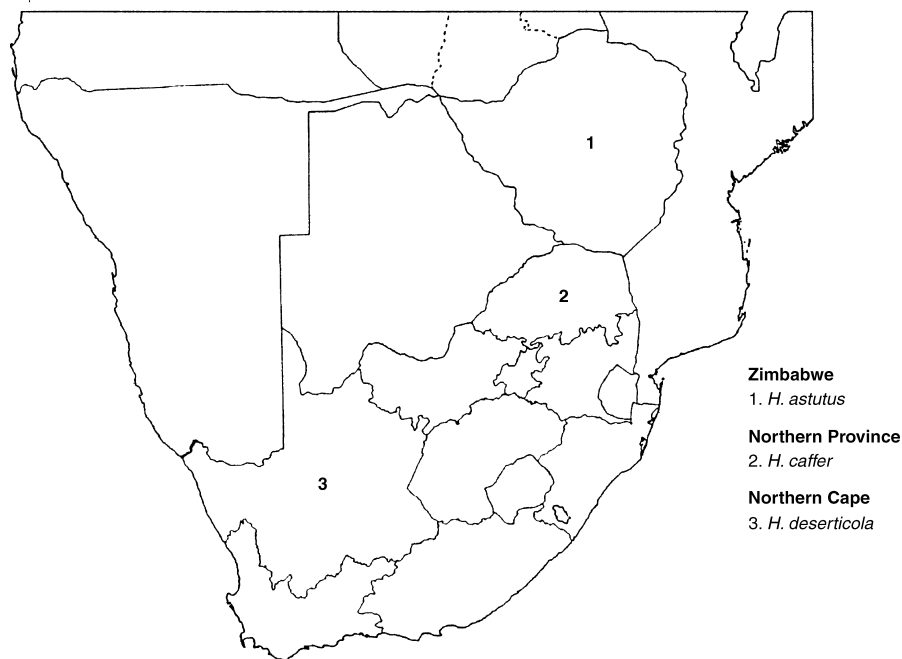


Fig. 42. Distribution of *Heligmomerus* species in Southern Africa.

3. *Heligmomerus deserti* Pocock, 1901a

Heligmomerus deserti Pocock, 1901a: 287; Roewer, 1942: 151.

Type locality: female holotype, Kalahari Desert (no exact locality).

Distribution: Southern Africa (known only from type locality).

Genus *IDIOPS* Perty, 1833

Idiops Perty, 1833: 197; O. P.-Cambridge, 1870a: 101.

Acanthodon Guérin, 1838: 163; Walckenaer, 1837: 434 (synonym of *Idiops*); Pocock, 1897: 731; 1903c: 350.

Titanidiops Simon, 1903c: 890.

Type species: *Idiops fuscus* Perty, 1833.

Diagnostic characters

Posterior eye row procurved (fig. 43a); median ocular area widest posteriorly; carapace narrower posteriorly (fig. 43a); cheliceral fang furrows with equal rows of teeth along inner and outer margins (fig. 43c); sternum with two pairs of sigilla, posterior pair absent (fig. 43b); coxae without spinules; body size 10–35 mm.

Taxonomic notes

Simon (1892a) included both *Acanthodon* and *Idiops* in the Idiopidae, applying *Idiops* only to the American species. Simon (1903c) united *Acanthodon* and *Idiops*. He proposed that *Idiops*

be used for the American species while *Titanidiops* is available for the African species. Purcell (1904) removed all his species from *Acanthodon* and placed *A. fryi* and *A. versicolor* in *Idiops*, the others in a new genus, *Ctenolophus*. However Hewitt (1910, 1913a) ignored this and continued to recognize *Acanthodon*.

Natural history

The burrow of *Idiops* is usually made in soft, stone-free soil during the rainy season, soil which hardens as the seasons progress. The burrow is frequently made in gently sloping, open grassy plains with low scrub, but this varies between species. For example, a burrow of *I. microps* was found on a steeply-sloping roadside cutting. The entrance is usually closed with a trapdoor that varies from cork-like to flat, with the outer surface well camouflaged with, for example, sand grains firmly stuck together so that the trapdoor resembles the surface of the surrounding soil, or with tufts of grass or even dry, black lichen (Van Dam & Roberts, 1917). *Idiops schreineri minor* was collected from a sheltered area at Roodeplaat. It constructs a burrow with a double-door, one above the other. The outer lid is D-shaped and

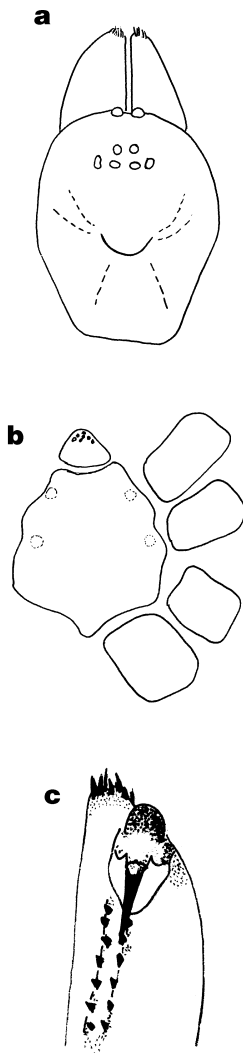


Fig. 43. *Idiops* sp. **a:** carapace, dorsal view; **b:** sternum; **c:** cheliceral fang furrow.

larger than the lid below, and much thicker, overlapping the burrow. The lower lid is wafer-thin and, for the greater part, attached to the top lid with only the front part free (Hewitt, 1919b).

A female of *I. kentanicus* was collected from open ground under an *Acacia* tree. The burrow was only 45 mm deep and perfectly straight, wider below but becoming narrower near the opening, which was slightly oval (Purcell, 1903c). The lid was flat and inflexible, the upper side being very rough and uneven and the underside smooth, flat and lined with white silk. The edge of the lid closed against the outspread rim of

the tube-lining.

Van Dam & Roberts (1917) found the burrows of *I. pretoria* and *I. gunningi* in grassland. The D-shaped lids were covered with attached pieces of dry grass and provided with minute holes on the underside. The hinge was broad. In *I. pretoria* the burrow slants conspicuously.

Members of *Idiops* have on various occasions been collected in areas with high densities of the termite *Hodotermes mossambicus* (pers. obs.). Males are frequently collected in pit traps. A female collected at Steelpoort, was flushed from her burrow by small red ants. She was found sitting at the entrance.

Distribution

Idiops is a large genus, represented by about 89 species in Africa, South America, Central Asia, Syria, Arabia, India and Burma. Fifty-nine species occur in Africa, with 31 species recorded from Southern Africa, including one from Mozambique, three from Zimbabwe, one from Botswana and three from Namibia (fig. 44).

Species recorded from Southern Africa

1. *Idiops arnoldi* Hewitt, 1914b

Idiops arnoldi Hewitt, 1914b: 21; Roewer, 1942: 152; Fitzpatrick, 2001: 177.

Type locality: male holotype (NMB) (no exact locality), probably Bulawayo, Zimbabwe.

Distribution: Zimbabwe.

2. *Idiops bersebaensis* Strand, 1917

Idiops bersebaensis Strand, 1917: 162; Roewer, 1942: 152; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (NMW), Berseba, Namibia.

Distribution: Namibia.

3. *Idiops briodae* (Schenkel, 1937)

Titanidiops briodae Schenkel, 1937: 377.
Idiops briodae, Roewer, 1942: 152; Fitzpatrick, 2001: 177.

Type locality: female holotype, Zimbabwe.

Distribution: Zimbabwe.

4. *Idiops castaneus* Hewitt, 1913a

Idiops castaneus Hewitt, 1913a: 420; 1914b: 23; Roewer, 1942: 152.

Type locality: female holotype (AM), Newington (24.52S; 31.25E), Northern Province.

Note: according to Hewitt (1913b), not Newington in the Soutpansberg district but Newington in Gazankulu,

Namibia

2. *I. bersebaensis*
 6. *I. damarensis*
 16. *I. mafe*
 20. *I. pallidipes*
 28. *I. striatipes*

Botswana

19. *I. palapyi*
 23. *I. pulcher*
 24. *I. pulloides*
 28. *I. striatipes*

North West Province

25. *I. pullus*

Gauteng

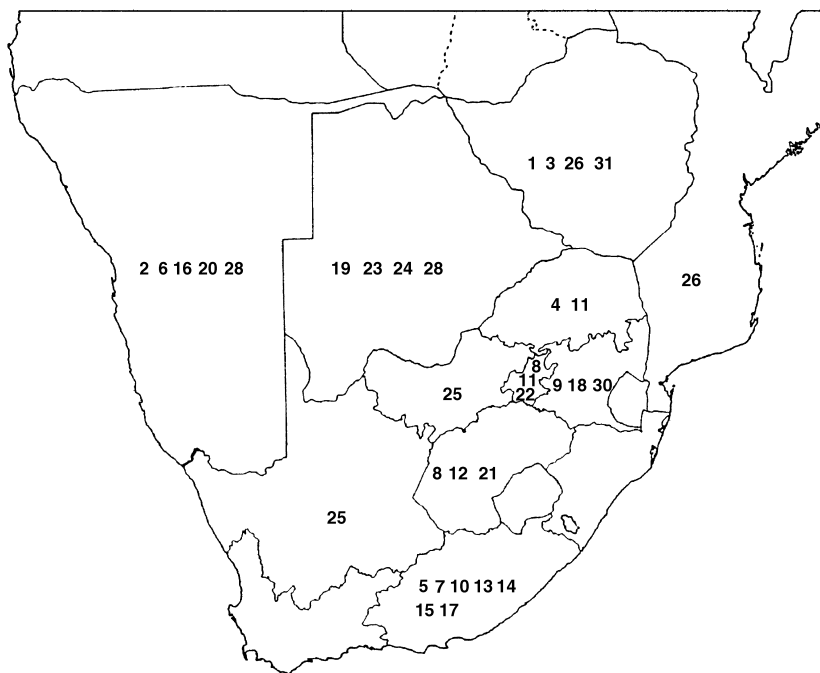
8. *I. fryi*
 11. *I. gunningi*
 22. *I. pretoriae*

Free State

8. *I. fryi*
 12. *I. hamiltoni*
 21. *I. parvus*

Northern Cape

25. *I. pullus*

**Zimbabwe**

1. *I. arnoldi*
 3. *I. briodae*
 26. *I. pungwensis*
 31. *I. versicolor*

Mozambique

26. *I. pungwensis*

Northern Province

4. *I. castaneus*
 11. *I. gunningi*

Mpumalanga

9. *I. gerhardti*
 18. *I. nigropilosus*
 30. *I. vandami*

Eastern Cape

5. *I. crudeni*
 7. *I. flaveolus*
 10. *I. gracilipes*
 13. *I. hepburni*
 14. *I. hirsutus*
 15. *I. kentanicus*
 17. *I. microps*

Fig. 44. Distribution of *Idiops* species in Southern Africa.

Northern Province. According to Hewitt (1915b), based on an immature specimen and should not have been described.

Distribution: South Africa (Northern Province: Newtonton).

5. *Idiops crudeni* (Hewitt, 1914b)

Acanthodon crudeni Hewitt, 1914b: 18; 1919b: 77.
Idiops crudeni, Roewer, 1942: 152.

Type locality: series of female syntypes and three male syntypes (AM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Alicedale).

6. *Idiops damarensis* Hewitt, 1934

Idiops damarensis Hewitt, 1934: 410; Roewer, 1942: 152; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: male and female syntypes (AM), Omatjienne near Otjiwarongo, Namibia.

Distribution: Namibia.

7. *Idiops flaveolus* (Pocock, 1901a)

Acanthodon flaveolum Pocock, 1901a: 286; Hewitt, 1919b: 107.
Idiops flaveolus: Roewer, 1942: 152.

Type locality: female holotype (SAM), near Grahams-town (33.19S; 26.22E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Grahams-town, East London).

8. *Idiops fryi* (Purcell, 1903c)

Acanthodon fryi Purcell, 1903c: 91.
Idiops fryi: Purcell, 1904: 118; Hewitt, 1913a: 421; Tucker, 1917: 125; Roewer, 1942: 152.

Type locality: female holotype (SAM B12756), Johannesburg (26.15S; 28.40E), Gauteng.

Distribution: South Africa (Gauteng: Johannesburg; Free State: Zonderhout at Holfontein).

9. *Idiops gerhardti* Hewitt, 1913a

Idiops gerhardti Hewitt, 1913a: 419; Roewer, 1942: 153.

Type locality: female holotype (AM), Doornkop, Witbooi near Belfast (25.42S; 28.40E), Mpumalanga Province.

Note: According to Hewitt (1915) the type is perhaps not quite mature.

Distribution: South Africa (Mpumalanga: Belfast).

10. *Idiops gracilipes* (Hewitt, 1919b)

Acanthodon gracilipes Hewitt, 1919b: 68, 75.
Idiops gracilipes: Roewer, 1942: 153.

Type locality: male and female syntypes (TM), East London (33.1S; 27.58E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: East London).

11a. *Idiops gunningi* Hewitt, 1913a

Idiops gunningi Hewitt, 1913a: 417; Van Dam & Roberts, 1917: Roewer, 1942: 153.

Type locality: female holotype (AM), Zwartspruit Pretoria, Gauteng.

Distribution: South Africa (*Gauteng*: Pretoria).

11b. *Idiops gunningi elongatus* Hewitt, 1915a

Idiops gunningi elongatus Hewitt, 1915a: 97; Van Dam & Roberts, 1917: 232.

Type locality: four females (TM), Moorddrift (24.17S; 28.58E), Waterberg district, Northern Province.

Distribution: South Africa (*Northern Province*: Moorddrift).

12. *Idiops hamiltoni* (Pocock, 1902b)

Acanthodon hamiltoni Pocock, 1902b: 320; Hewitt, 1914: 58; 1923: 58.

Idiops hamiltoni: Roewer, 1942: 153.

Type locality: subadult female holotype, Vredefort Road (27.01S; 27.22E), Free State Province.

Distribution: South Africa (*Free State*: Vredefort).

13. *Idiops hepburni* (Hewitt, 1919b)

Acanthodon hepburni Hewitt, 1919b: 73.

Idiops hepburni: Roewer, 1942: 153.

Type locality: male and female syntypes (TM), Majuba Nek, Herschel district (30.38S; 27.06E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Majuba Nek, Herschel district).

14. *Idiops hirsutus* (Hewitt, 1919b)

Acanthodon hirsutus Hewitt, 1919b: 69.

Idiops hirsutus: Roewer, 1942: 153.

Type locality: one female and four male syntypes (TM), East London (33.18S; 27.58E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: East London).

15. *Idiops kentanicus* (Purcell, 1903c)

Acanthodon kentanicus Purcell, 1903c: 89.

Ctenolophus kentanicus: Purcell, 1904: 118.

Idiops kentanicus: Tucker, 1917: 92; Roewer, 1942: 153.

Type locality: female holotype (SAM 12412), Kentani, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Kentani).

16. *Idiops mafeae* Lawrence, 1927a

Idiops mafeae Lawrence, 1927a: 2; Roewer, 1942: 153, Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (SAM B5882), Mafa, Namibia.

Distribution: Namibia.

17. *Idiops microps* (Hewitt, 1913c)

Acanthodon microps Hewitt, 1913c: 471; 1919b: 75 (male).

Idiops microps: Roewer, 1942: 153.

Type locality: female holotype (AM), Grey Reservoir, Grahamstown (33.19S; 26.22E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Grahamstown).

18. *Idiops nigropilosus* (Hewitt, 1919b)

Acanthodon nigropilosus Hewitt, 1919b: 70.

Idiops nigropilosus: Roewer, 1942: 153.

Type locality: male and female syntypes (TM), Arnheimburg (26.03S; 30.50E), Carolina district, Mpumalanga Province.

Distribution: South Africa (*Mpumalanga*: Arnheimburg, Carolina).

19. *Idiops palapyi* Tucker, 1917

Idiops palapyi Tucker, 1917: 90; Roewer, 1942: 154; Eagle, 1985: 131; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: male holotype (SAM 14628), Palapye, Botswana.

Distribution: Botswana.

20. *Idiops pallidipes* Purcell, 1908

Idiops pallidipes Purcell, 1908: 207; Roewer, 1942: 154; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype, southern Hereroland, Namibia.

Distribution: Namibia.

21. *Idiops parvus* Hewitt, 1915a

Idiops parvus Hewitt, 1915a: 71; Roewer, 1942: 154.

Type locality: female holotype (AM), Zonderhout, Holfontein (27.53S; 17.16E), Free State Province.

Distribution: South Africa (*Free State*: Zonderhout, Holfontein).

22. *Idiops pretoriae* (Pocock, 1898b)

Acanthodon pretoriae Pocock, 1898b: 319; Hewitt, 1910: 74; 1913a: 418; 1915a: 97; 1915c: 310; Van Dam & Roberts, 1917: 231.

Idiops pretoriae: Roewer, 1942: 154.

Type locality: male holotype (AMNH), Pretoria (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (*Gauteng*: Pretoria district: Saltpan, Skinner's Court, Lyttelton Junction).

23. *Idiops pulcher* Hewitt, 1914b

Idiops pulcher Hewitt, 1914b: 23; Roewer, 1942: 154.

Type locality: female holotype (AM), Thsessebe, Tati, Botswana.

Distribution: Botswana.

24. *Idiops pulloides* Hewitt, 1919a

Idiops pulloides Hewitt, 1919a: 210; Roewer, 1942: 154; Eagle, 1985: 131.

Type locality: male holotype (McGregor Museum, Kimberley), Molepolole (24.26S; 25.32E), Botswana.

Distribution: Botswana.

25. *Idiops pullus* Tucker, 1917

Idiops pullus Tucker, 1917: 88; 1920: 439; Roewer, 1942: 154.

Type locality: male holotype (SAM 14706), west of Mafikeng (25.53S; 25.39E) and north of Vryburg, North West Province.

Distribution: South Africa (*North West Province*: Mafikeng, Vryburg; *Northern Cape*: Kimberley (male SAM B 4185)).

26. *Idiops pungwensis* Purcell, 1904

Idiops pungwensis Purcell, 1904: 116; Hewitt, 1914b: 23; Roewer, 1942: 154.

Type locality: male holotype (SAM 13576), Pungwe River, Mozambique.

Distribution: Mozambique and Zimbabwe.

27. *Idiops spiriferus* Roewer, 1942

Acanthodon spiricola Purcell, 1903; Hewitt, 1914b: 14.

Idiops spiriferus: Roewer, 1942: 154 (name preoccupied Purcell, 1903). Tucker, 1917: 134.

Type locality: male holotype (no exact locality), South Africa.

Distribution: South Africa.

28. *Idiops striatipes* Purcell, 1908

Idiops striatipes Purcell, 1908: 206; Hewitt, 1914b: 23; Roewer, 1942: 154; Eagle, 1985: 131; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (SAM), Sekoma [Sekgoma], Kalahari, Botswana.

Distribution: Botswana and Namibia.

29. *Idiops thorelli* O. P.-Cambridge, 1870b

Idiops thorelli O. P.-Cambridge, 1870b: 156; Pocock, 1897: 731; 1898b: 320; Roewer, 1942: 154.
Acanthodon thorelli: Hewitt, 1919b: 75.

Type locality: male holotype, (no exact locality).

Distribution: South Africa.

30. *Idiops vandami* (Hewitt, 1925)

Acanthodon vandami Hewitt, 1925: 279.
Idiops vandami: Roewer, 1942: 154.

Type locality: male and female syntypes (AM), Barberton (25.48S; 31.3E), Mpumalanga.

Distribution: South Africa (*Mpumalanga*: Barberton).

31. *Idiops versicolor* (Purcell, 1903c)

Acanthodon versicolor Purcell, 1903c: 90.
Idiops versicolor: Purcell, 1904: 118; Hewitt, 1914b: 23; Roewer, 1942: 154; Fitzpatrick, 2001: 177.

Type locality: female holotype (SAM 12001), Baviaan Kopje, three miles East of Umtali (18.59S; 32.40E), Zimbabwe.

Distribution: Zimbabwe.

Genus *SEGREGARA* Tucker, 1917

Segregara Tucker, 1917: 125; Roewer, 1942: 156; Raven, 1985: 138.

Type species: *Gorgyrella abrahami* (Hewitt, 1913).

Diagnostic characters

Posterior eye row procurved, eyes widely spaced (fig. 45b); tibia III cylindrical; sternum with three pairs of small marginal sigilla (fig. 45a); coxae, at least III, with spinules or stiff setae (fig. 45c); size of female varies from 12–22 mm.

Taxonomic notes

Closely related to *Gorgyrella* but differs from it by the shape of the posterior sigilla, which is larger in *Gorgyrella*. *Segregara* was proposed by Tucker (1917) to include two species originally described in *Acanthodon* and *Gorgyrella*. They are included in *Segregara* based on the presence and position of the sternal sigilla. Tucker (1917) provided a detailed discussion of the characteristics of the sigilla.

Natural history

Burrows of *Segregara abrahami* are made in the steeply sloping hillsides or in sloping ground under stones or in areas protected by vegetation (Hewitt, 1913c). The burrows run horizontally for 2 cm or more before descending. The trapdoor hangs almost vertically and the lid is heavy, thick and D-shaped, with the edges strongly bevelled. The hinge of the trapdoor is much longer than the width of the burrow.

The burrows of *S. transvaalensis* are made in bare ground, between tufts of grass and on road embankments. The burrow entrances extend slightly above ground level. The trapdoors are very thin and wafer-like, almost circular and provided beneath with 3–8 minute tooth or claw marks nearer to the hinge than the centre (Van Dam & Roberts, 1917). The burrows are not quite vertical, sometimes slanting backwards for about 2.5 cm, then dropping vertically for about 5 cm and doubling back again for 5 cm. The total length of burrows are 12.5 mm. At Roodeplaat Dam Nature Reserve, and Mosdene Nature Reserve near Naboomspruit, females of *S. transvaalensis* were collected from burrows closed with thin wafer-like lids.

Burrows of *I. monticola* are found on moss-covered banks under stones. The lids are circular, thin with the edge slightly overlapping the rim of the burrow. They are covered on the outside with moss and earth, while the undersides bear small holes. The burrows are shallow, sometimes horizontal or slanting. *I. monticola* males have been collected from burrows.

Distribution

Ten species are known from Africa, all of which are known from Southern Africa (fig. 46).

KEY TO THE SOUTHERN AFRICAN SPECIES OF *SEGREGARA*

(based partly on the key of Hewitt, 1916b)

- 1. Males 2
- Females 8
- 2. Tibia I subequal in length to metatarsus I 3
- Tibia I shorter than metatarsus I 4
- 3. Mating spur on tibia I with distal tubercle strongly flattened and apex obtuse; coxa III without subspiniiform setae in a band of spinules on posteroventral border (Pigg's Peak, Swaziland) ***S. monticoloides***
- Mating spur on tibia I with distal tubercle bearing a long, pointed, black process; coxa III with a band of stiff setae along posteroventral border (Magaliesberg) ***S. monticola***
- 4. Metatarsus I almost straight 5
- Metatarsus I slightly to strongly bowed 6
- 5. Tibia I with two large tubercles, distal one with stout, black, spur-like process, proximal tubercle short with sharp, conical, black apex; ventrally 4–5 external spines (Zululand) ***S. pectinipalpis***
- Tibia I with two large tubercles, distal one with long, black, flattened process with blunt tip; inferiorly four external spines ***S. mossambicus***
- 6. Metatarsus I arcuate in basal half, strongly bent medially; frontal eyes about half of their diameter apart (Jansenville) ***S. ochreolus***
- Metatarsus I slightly bowed, not bent or thickened in any part 7

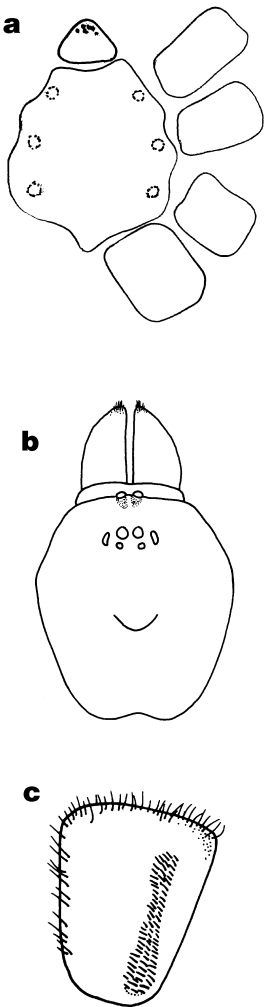


Fig. 45. *Segregara* sp. **a**: sternum; **b**: carapace, dorsal view; **c**: coxa III with spinules.

- 7. Coxa III without spinules (Alicedale) ***S. abrahami***
- Coxa III with well-defined strip of short, stiff, spiniiform setae (Woodbush) · ***S. sylvestris***
- 8. Coxa II without spinules 9
- Coxa II with spinules 11
- 9. Coxa III with patch of stout spinules along entire length of segment (Gravelotte) ***S. paucispinulosus***
- Coxa III with patch of stiff setae only posteroventrally 10
- 10. Patella III with about 30 spines (Zululand) ***S. grandis***

- Patella III with 5–8 spines (Magaliesberg)
..... ***S. monticola***
- 11. Patella III with 2–5 spinules; labium with
4–6 strong cuspules ***S. abrahami***
- Patella III with six spinules anteriorly and five
posteriorly; labium with three apical and six
posterior cuspules ***S. transvaalensis***

Species recorded from Southern Africa

1. *Segregara abrahami* (Hewitt, 1913c)

Gorgyrella abrahami Hewitt, 1913c: 473.
Acanthodon abrahami: Hewitt, 1915c: 305; 1919b: 76.
Segregara abrahami: Tucker, 1917: 134; Roewer, 1942: 156.

Type locality: female syntypes (AM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alicedale, Burgersdorp, Bushmans River, near Dassie Klip).

2. *Segregara grandis* (Hewitt, 1915c) comb. nov.

Acanthodon grandis Hewitt, 1915c: 308.
Idiops grandis: Roewer, 1942: 153.

Type locality: female holotype (NM), Umfolozi (28.27S; 32.10E), Zululand, KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Umfolozi).
Note: Hewitt (1916) collected a species close to *I. grandis* from Forbes Reef, Swaziland, and Lochiel, Ermelo district, Mpumalanga.

3. *Segregara monticola* (Hewitt, 1916b) comb. nov.

Acanthodon monticola Hewitt, 1916b: 185; 1919b: 76; Van Dam & Roberts, 1917: 222.
Idiops monticola: Roewer, 1942: 153.

Type locality: a female and three male syntypes (TM), Magaliesberg, Little Wonderboom (25.33S; 28.09E), Gauteng Province.

Distribution: South Africa (*Gauteng*: Magaliesberg, Daspoort, Pretoria).

4. *Segregara monticoloides* (Hewitt, 1919b) comb. nov.

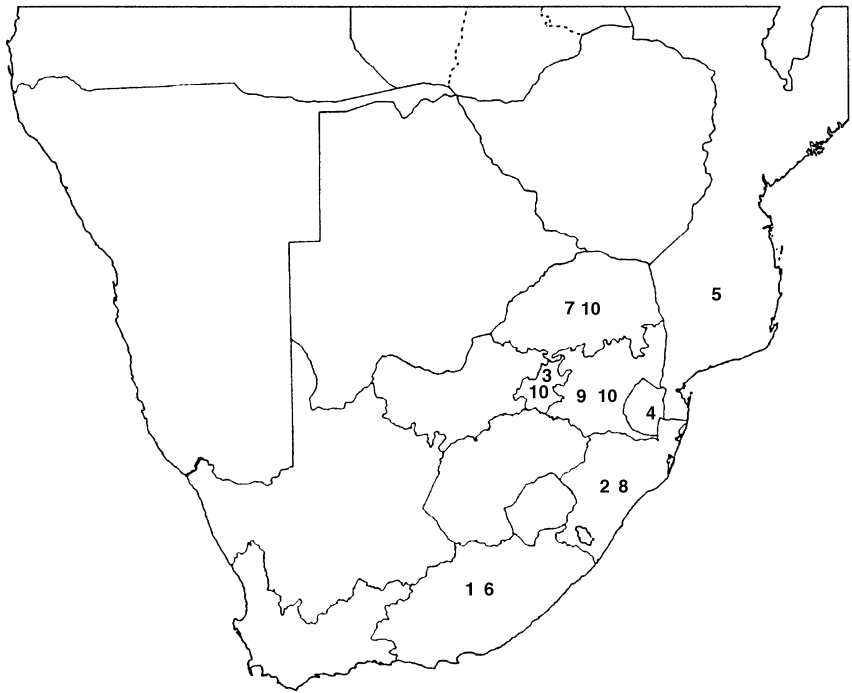
Acanthodon monticoloides Hewitt, 1919b: 67, 76.
Idiops monticoloides: Roewer, 1942: 153.

Type locality: male holotype (TM), Pigg's Peak (25.58S; 31.14E) Swaziland.

Distribution: Swaziland.

5. *Segregara mossambicus* (Hewitt, 1919b) comb. nov.

Acanthodon mossambicus Hewitt, 1919b: 72.
Idiops mossambicus: Roewer, 1942: 153.



- Mozambique**
5. *S. mossambicus*
- Swaziland**
4. *S. monticoloides*
- Northern Province**
7. *S. paucispinulosus*
10. *S. transvaalensis*
- Gauteng**
3. *S. monticola*
10. *S. transvaalensis*
- Mpumalanga**
9. *S. sylvestris*
10. *S. transvaalensis*
- KwaZulu-Natal**
2. *S. grandis*
8. *S. pectinipalpis*
- Eastern Cape**
1. *S. abrahami*
6. *S. ochreolus*

Fig. 46. Distribution of *Segregara* species in Southern Africa.

Type locality: male holotype (TM), Magude, Mozambique.

Distribution: Mozambique.

6. *Segregara ochreolus* (Pocock, 1902a) comb. nov.

Acanthodon ochreolum Pocock, 1902a: 9; Hewitt, 1915c: 306; 1920: 76.

Idiops ochreolus: Roewer, 1942: 153.

Type locality: male holotype (NM), Jansenville, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Jansenville).

7. *Segregara paucispinulosus* (Hewitt, 1915b) comb. nov.

Acanthodon transvaalensis paucispinulosus Hewitt, 1915b: 98.

Acanthodon paucispinulosus: Hewitt, 1916b: 186.

Segregara transvaalensis paucispinulosus: Roewer, 1942: 156.

Type locality: two female syntypes and juveniles (TM), Gravelotte (23.57S; 30.36E), near Leydsdorp, Northern Province.

Distribution: South Africa (*Northern Province*: Gravelotte).

8. *Segregara pectinipalpis* (Purcell, 1903c) comb. nov.

Acanthodon pectinipalpis Purcell, 1903c: 87; Hewitt, 1919b: 76.

Ctenolophus pectinipalpis: Purcell, 1904: 119; Roewer, 1942: 150.

Type locality: three male syntypes (SAM 9938),

Zululand, KwaZulu-Natal (no exact locality).

Distribution: South Africa (*KwaZulu-Natal*: Zululand).

9. *Segregara sylvestris* (Hewitt, 1925) comb. nov.

Acanthodon sylvestris Hewitt, 1925: 282.

Idiops sylvestris: Roewer, 1942: 154.

Type locality: male holotype (AM), Woodbush (23.49S; 29.54E), Mpumalanga.

Distribution: South Africa (*Mpumalanga*: Woodbush).

10. *Segregara transvaalensis* (Hewitt, 1913a)

Ctenolophus transvaalensis Hewitt, 1913a: 412.

Acanthodon transvaalensis: Hewitt, 1916b: 182; Van Dam & Roberts, 1917: 222.

Segregara transvaalensis: Tucker, 1917: 134; Roewer, 1942: 156.

Type locality: female holotype (AM), Newington (Soutpansberg district). (Note: according to an atlas 'Newington' is close to Pilgrim's Rest (24.51S; 31.25E) in Mpumalanga Province.)

Distribution: South Africa (*Gauteng*: various localities in the Pretoria district: Mayville, Rietfontein, Witfontein, Skinner's Court between Lyttelton junction and Irene, Rietfontein, Schoemansrust, Roodeplaat, Rosslyn and Zeekoegat); *Northern Province*: Mosdene, Nylsvley Nature Reserve; *Mpumalanga*: Newington, Pan station Middelburg, Steynsdorp, Carolina district).



Front-eyed trapdoor spider
(Idiopidae: *Gorgyrella schreineri* minor).

FAMILY MICROSTIGMATIDAE

forest-floor mygalomorphs

(figs 47–50)

The Microstigmatidae are a small family represented by two subfamilies, the Micromygalinae and Microstigmatinae. They are known from Panama, South America and Africa. Only the latter subfamily is known from Southern Africa, represented by one genus and six species.

Genus recorded from Southern Africa

Microstigmatinae: *Microstigmata* Strand, 1932.

Diagnostic characters

Microstigmatids are small to medium-sized (4–13 mm) spiders with the following synapomorphies:

- small, oval booklung openings (fig. 48e),
- body covered with blunt-tipped or clavate setae (fig. 48b),
- thoracic region elevated behind fovea,
- domed apical segment of posterior spinnerets.

Descriptive characters

- **carapace:** glabrous, covered with blunt-tipped or clavate setae (fig. 48b); thoracic region as high as cephalic region; fovea straight to slightly recurved, depressed (fig. 47);
- **sternum:** 1–3 pairs of small marginal sternal sigilla;
- **eyes:** eight; closely grouped in two rows on a tubercle (fig. 48c);
- **chelicerae:** furrow with 6–13 teeth on promargin; rastellum absent;
- **mouthparts:** labium domed, with cuspules; serrula present or reduced;
- **legs:** three claws; paired claw with two rows of teeth, originating near dorsal surface; tarsal organ protruding above dorsal surface of tarsus; tibia I with prolateral spur (fig. 49a);
- **abdomen:** oval; covered with erect, blunt-tipped setae (fig. 48a);
- **spinnerets:** four; posterior spinnerets short;
- **genitalia:** female spermathecae simple or branched, chitinous walls with numerous pores (fig. 49g); male palp with bulb pyriform, conductor absent; third

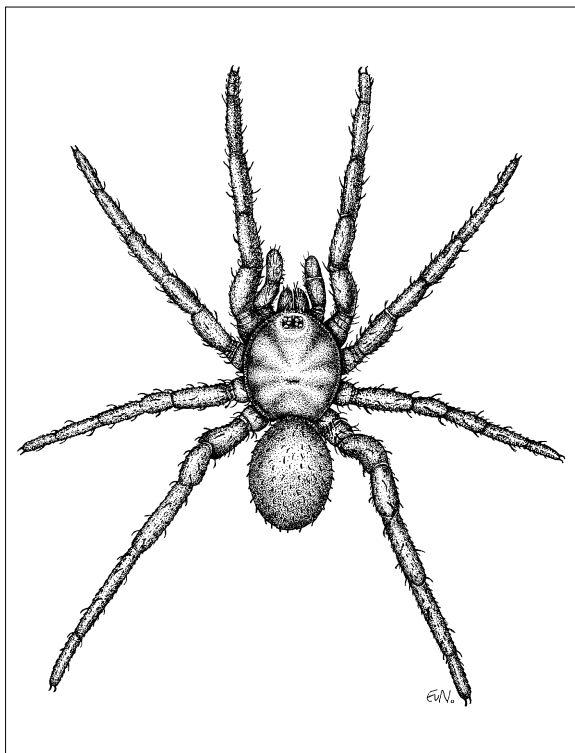


Fig. 47. Microstigmatidae — *Microstigmata* sp.

haematodocha not evident (fig. 49c);

- **body size:** 4–13 mm;
- **colour:** various shades of brown.

Higher classification

The Microstigmatidae falls in the microorder Tuberculotae and is the sister group of the Mecicobothriidae. Together they form the Mecicobothrioidina, sharing the following characters: loss of tarsal spines, absence of palpal conductor, and sloping thorax (Raven, 1985).

Natural history

The microstigmatids are the only mygalomorphs that wander about freely — they do not

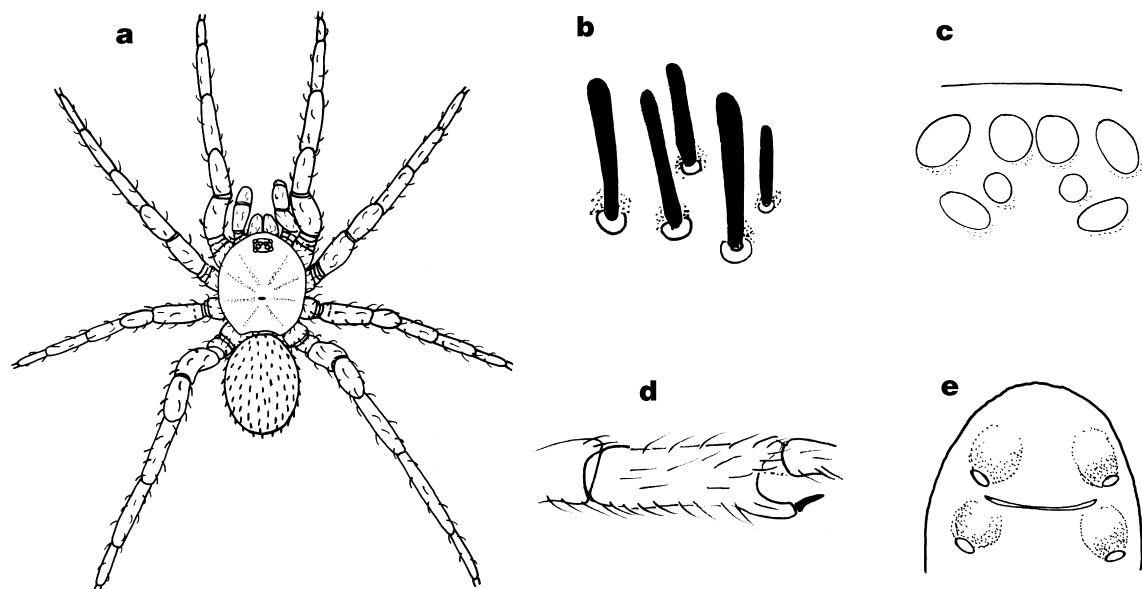


Fig. 48. Microstigmatidae — *Microstigmata* sp. **a:** female, dorsal view; **b:** clavate setae; **c:** eye pattern; **d:** tibia I of male; **e:** booklung openings.

live in burrows. They make minimal use of silk and are found under stones and logs, and in damp, decomposing logs.

Subfamily MICROSTIGMATINAE Roewer, 1942

Microstigmatinae Roewer, 1942: 194.

Diagnostic characters

Body bears digitiform cuticular setae (fig. 48a); tarsal organ without concentric ridges and protruding above dorsal surface of tarsus; two rows of dorso-laterally-originating teeth on tarsal claws; anterior lateral spinnerets absent; palpal conductor absent; smooth trichobothrial bases; body size 4–13 mm.

Taxonomic notes

Raven & Platnick (1981) removed the Microstigmatinae from the Dipluridae and elevated it to familial rank. The genus *Microstigmata* was revised by Griswold (1985a), who provided a key to the species.

Distribution

Microstigmatinae are represented by three genera: *Pseudonemesia* (South America), *Ministigmata* (Brazil) and *Microstigmata*, an exclusively African genus with six species.

Genus *MICROSTIGMATA* Strand, 1932

Microstigma Hewitt, 1916b: 206; 1925: 286; Lawrence, 1938: 459; Bonnet, 1957: 2906.

Microstigmata Strand, 1932: 142 (*nomen novum* for *Microstigma* Hewitt, preoccupied in the Odonata); Roewer, 1942: 194; Raven & Platnick, 1981: 15; Griswold, 1985a: 15; Raven, 1985: 69.

Type species: *Microstigma geophilum* Hewitt, 1916b.

Diagnostic characters

Body glabrous but with attenuated, blunt-tipped or clavate setae (fig. 48b); thoracic fovea straight to slightly recurved and depressed (fig. 48a); eight eyes closely grouped on low tubercle (fig. 48c); rastellum absent; cheliceral furrow with 6–13 teeth along promargin; fang long and slender; labium domed, usually with cuspules; endites with cuspules numbering from 15 to more than 55; abdomen with erect, blunt-tipped bristles at least at apex; small spiders, body size 4–13 mm.

Natural history

The *Microstigmata* are the only mygalomorphs in Southern Africa that do not excavate or live in burrows. They are found under stones, logs, and in damp, decomposing logs, and make minimal use of silk (Griswold, 1985a). According to Lawrence (1952), they may be restricted to microhabitats with high humidity and even temperatures. They are found in undergrowth and the litter-layer of indigenous forests and closed-

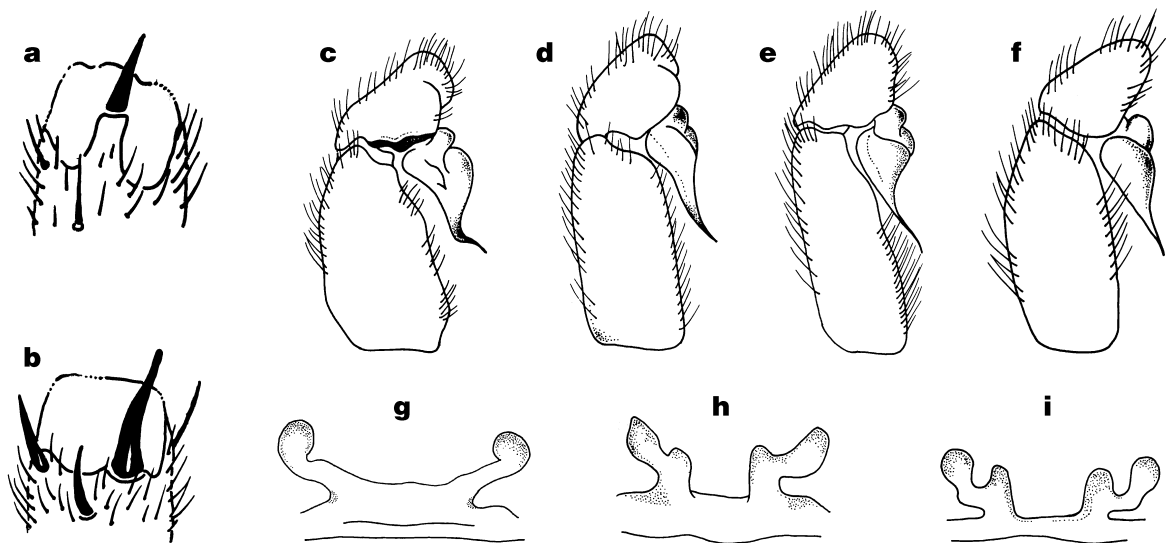


Fig. 49. Microstigmatidae — *Microstigmata* spp.; **a**: tibia I with apical apophysis equal in size to the clasping spur; **b**: tibia I without apophysis; **c**: embolus, *M. geophila*; **d–f**: embolus of (**d**) *M. amatola*, (**e**) *M. longipes* and (**f**) *M. zuluense*; **g–i**: spermathecae of (**g**) *M. lawrencei*, (**h**) *M. geophila* and (**i**) *M. amatola*. (After Griswold, 1985a.)

canopy bush. Their cuticles are usually encrusted with detritus clinging to the setae on their bodies.

Van der Merwe (1994) collected numerous specimens from pit traps during a survey of different habitats in the Ngome State Forest.

Distribution

Microstigmata are known from eastern South Africa, from near the border with Mozambique at 27°S to Grahamstown in the Eastern Cape Province at 33° extending from sea-level to the Drakensberg escarpment (1500 m) (fig. 50).

KEY TO THE SOUTHERN AFRICAN SPECIES OF MICROSTIGMATA
(after Griswold, 1985a)

1. Males 2

— Females 6

2. Tibia I with apical apophysis equal to or greater than length of clasping spur (fig. 49a) · 3

— Tibia I without apical apophysis or, if present, shorter than length of clasping spur (fig. 49b) 5

3. Endites with <15 cuspules grouped near margin of labium; cheliceral fang furrow with fewer than eight, widely spaced teeth on promargin *M. ukhahlamba*

— Endites with >25 cuspules spread across proximal margin; cheliceral fang furrow with 10 or more teeth on promargin, usually variable in size 4

4. Length of palpal tibia less than 1.8 times width; embolus sharply bent (fig. 49c) *M. geophila*

— Length of palpal tibia more than 1.9 times width; embolus short and slender (fig. 49d) *M. amatola*

5. Cheliceral fang furrow with 10 or more widely spaced teeth; length of palpal tibia greater than 2.2 times width; embolus long and slender (fig. 49e) *M. longipes*

— Cheliceral fang furrow with seven or fewer widely spaced teeth; length of palpal tibia less than two times width; embolus short and slender (fig. 49f) *M. zuluense*

6. Cheliceral fang furrow with large and small teeth; endites with 30 or more cuspules; sclerotized portion of genitalia visible through epigynal cuticle; spermathecal duct long, unbranched or with small mesal diverticulum (fig. 49g) 7

— Cheliceral fang furrow with series of equal-sized teeth; endites with fewer than 30 cuspules; genitalia not visible through epigynal cuticle; spermathecal duct short, usually branched 9

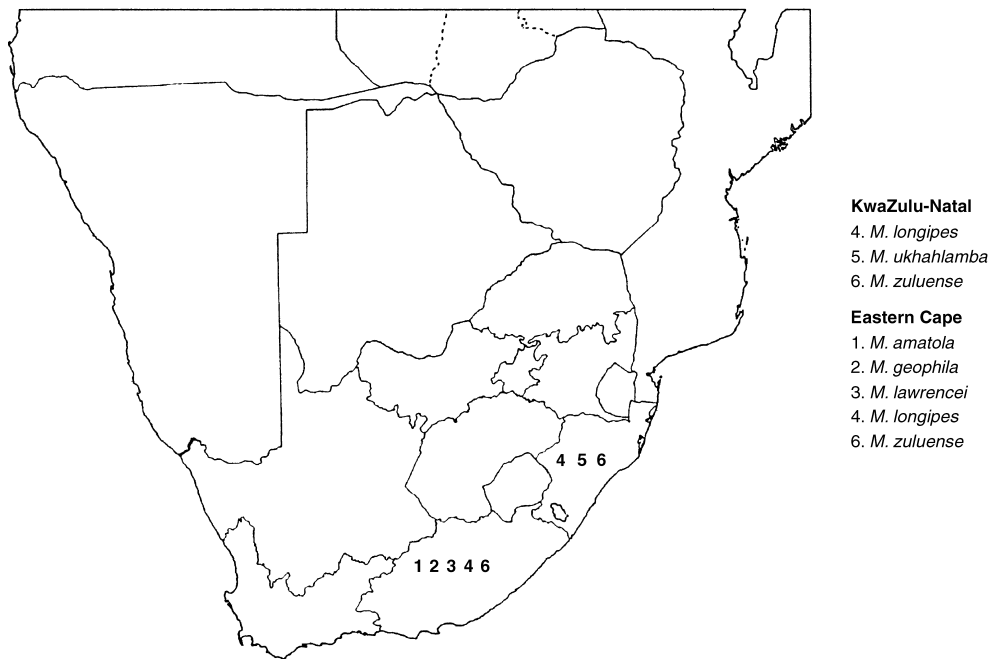


Fig. 50. Distribution of *Microstigmata* species in Southern Africa.

7. Endites with 55 cuspules; spermathecal ducts long, straight and unbranched (fig. 49g).....	<i>M. lawrencei</i>
— Endites with fewer than 50 cuspules; spermathecal ducts sharply bent, usually with short inner diverticulum	8
8. Opening of spermathecal duct into vulval chamber surrounded by chitinous ring (fig. 49h)	<i>M. geophila</i>
— Opening of spermathecal duct into vulval chamber without chitinous ring (fig. 49i)	<i>M. amatola</i>
9. Cheliceral fang furrow with 10 or more closely spaced teeth; base of femora with attenuated bristles on ventral surface; abdomen without clavate bulbous setae	<i>M. longipes</i>
— Cheliceral fang furrow with seven or fewer widely spaced teeth; base of femora with blunt-tipped or clavate bulbous bristles on ventral surface; abdomen with clavate bulbous setae	10
10. Sternum and coxa with clavate bulbous bristles; dorsum of abdomen with pale markings on dark grey background	<i>M. zuluense</i>
— Sternum and coxa with blunt-tipped	

bristles; dorsum of abdomen with dark-brown markings on brown background

M. ukhahlamba

Species recorded from Southern Africa

1. *Microstigmata amatola* Griswold, 1985a
Microstigmata amatola Griswold, 1985a: 30; Platnick, 1989: 74.
Type locality: male holotype (SAM 2709) and numerous male and female paratypes (AM & AMNH), Amatola Mountains near Hogsback (32.35S; 27.5E), Eastern Cape Province.
Distribution: South Africa (*Eastern Cape*: Hogsback (Amatola mountains), Stutterheim (Kologha Forest)).

2. *Microstigmata geophila* (Hewitt, 1916b)
Microstigma geophilum Hewitt, 1916b: 207; 1925: 286.
Microstigmata geophila: Strand, 1932: 142; Griswold, 1985a: 33; Platnick, 1989: 74.
Type locality: female lectotype and seven female paralectotypes (AM), Grahamstown (34.8S; 19.2E), Eastern Cape Province.
Distribution: South Africa (*Eastern Cape*: Grahams-town).

3. *Microstigmata lawrencei* Griswold, 1985a
Microstigmata lawrencei Griswold, 1985a: 29; Platnick, 1989: 74.
Type locality: female holotype and female paratype

(NM 8099); Kambi Forest (31.35S; 28.48E), Transkei, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Kambi Forest, between Franklin and Riverside).

4. *Microstigmata longipes* (Lawrence, 1938)

Microstigma longipes Lawrence, 1938: 463; Bonnet, 1957: 2906.

Microstigmata longipes: Roewer, 1942: 194; Raven & Platnick, 1981: 15; Griswold, 1985a: 17; Platnick, 1989: 74.

Type locality: male lectotype and female paralectotype (NM 123), Umkomaas Valley, near Bulwer (29.48S; 29.46E), KwaZulu-Natal.

Distribution: South Africa (*Eastern Cape*: Port St Johns at Umzimvubu; *KwaZulu-Natal*: Kokstad at Ingali Forest, Port Shepstone, Pietermaritzburg at Umkomaas River, Empangeni, Fort Nottingham, Drakensberg, Bulwer, Richmond, Umhlali, Shooters Hill, Table Mountain, Karkloof Forest, New Hanover, Kranskop, Estcourt, Champagne Castle, Ngoye Forest, Gwaliweni Forest, Oudeni Forest, Ngome State Forest).

5. *Microstigmata ukhahlamba* Griswold, 1985a

Microstigmata ukhahlamba Griswold, 1985a: 29; Platnick, 1989: 74.

Type locality: male holotype and two female paratypes (TM 14679), Drakensberg Mountain, Cathedral Peak (28.57S; 29.12E), Ndumeni Forest, KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Ndumeni Forest, Cathedral Peak forest station 75 km WSW Estcourt).

6. *Microstigmata zuluense* (Lawrence, 1938)

Microstigma zuluense Lawrence, 1938: 461; Bonnet, 1957: 2906.

Microstigmata zuluense: Roewer, 1942: 194; Raven & Platnick, 1981: 15; Griswold, 1985a: 22; Platnick, 1989: 74.

Type locality: male lectotype and male and female paralectotypes (NM 1389), Nkandla Forest (28.38S; 31.6E), Zululand, KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Nkandla Forest; *Eastern Cape*: Umzimvubu (Port St Johns)).

FAMILY MIGIDAE

tree and banded-legged trapdoor spiders

(figs 51–56)

The family Migidae is represented by nine genera and 79 species. Their distribution is restricted to countries on the Gondwanian continents of the southern hemisphere (Africa, Madagascar, Australia, Argentina and Chile). Raven (1985) recognized three subfamilies: Miginae, Paramiginae and Calathotarsinae. Of the three subfamilies, Miginae and Paramiginae are known from Southern Africa, represented by two genera and 26 species.

Genera recorded from Southern Africa

Miginae: *Poecilomigas* Simon, 1903.

Paramiginae: *Moggridgea* O. P.-Cambridge, 1875.

Diagnostic characters

Migids are small to medium-sized (5–25 mm) mygalomorph spiders characterized by the following synapomorphies:

- two distinct, longitudinal keels on outer surface of cheliceral fang (fig. 52b),
- chelicerae short with fangs directed obliquely (plagiognathy) (fig. 52c).

Descriptive characters

- **carapace:** fovea recurved (fig. 51), straight or T-shaped; cephalic region smooth without hair; thoracic region arched or lower than fovea;
- **sternum:** with one pair of sigilla (fig. 52f);
- **eyes:** in two rows, occupying almost half the head width (fig. 52d);
- **chelicerae:** short, fangs directed obliquely (fig. 52c); outer surface of cheliceral fang with two distinct, longitudinal keels (fig. 52b); fang furrow usually with two rows of teeth; rastellum absent;
- **mouthparts:** cuspules present on labium and endites of females, absent or present in males; serrula absent;
- **legs:** three claws; metatarsi I and II with four or more pairs of strong setae (fig. 52g);
- **abdomen:** oval;
- **spinnerets:** apical segment of posterior spinnerets domed;

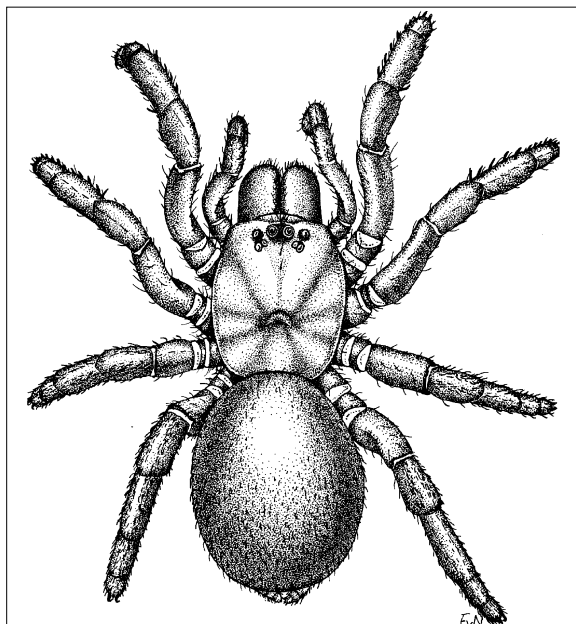


Fig. 51. Migidae — *Moggridgea* sp.

- **genitalia:** female spermathecae paired, simple, unbranched (fig. 53c); male palp with distal haematodocha of bulbus small, distal sclerite conical (fig. 53d);
- **body size:** 5–25 mm;
- **colour:** various shades of brown to black; legs and/or abdomen frequently with patterns.

Higher classification

Platnick & Shadab (1976) indicated that the vertical inclination of the chelicerae is autapomorphic for the migids. However, Raven (1985) proposed that the two low keels on the outer surface of the cheliceral fangs are a better character to use as they occur in all migids. He listed the Migidae in the microorder Fornicephalae in the Ctenizoidina along with the Actinopodidae and Ctenizidae. It is placed in the superfamily Migoidea with the Actinopodidae as sister group. According to Goloboff & Platnick (1987), these two families are the most apomorphic representatives of the microorder Fornicephalae.

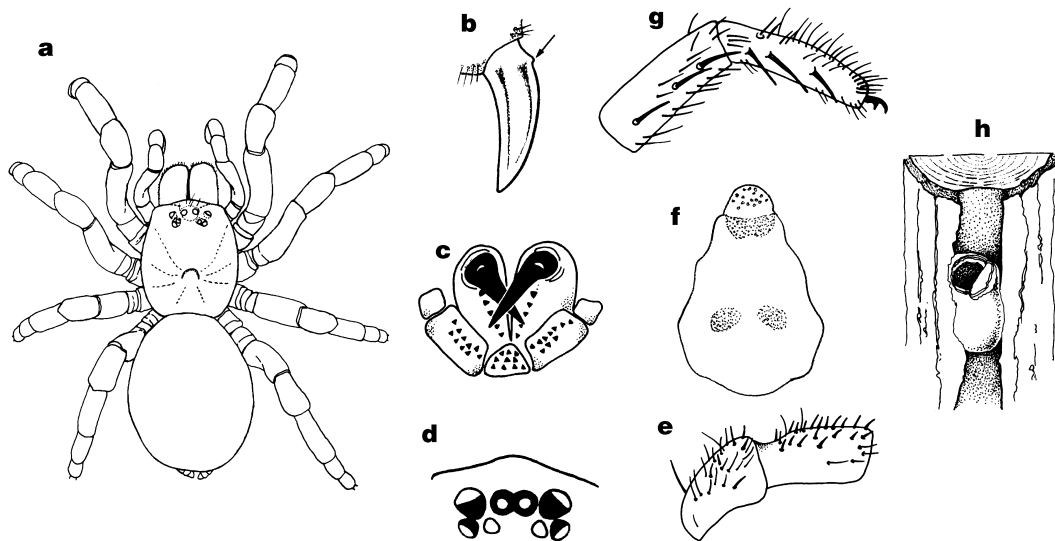


Fig. 52. Migidae. **a:** female, dorsal view; **b:** chelicera with keels; **c:** chelicerae, ventral view; **d:** eye pattern; **e:** tibia III with dorsal excavation; **f:** sternum; **g:** leg I with pairs of strong setae; **h:** retreat on tree trunk.

Natural history

Migids construct tubular silk retreats either in burrows in the ground or as aerial retreats on tree trunks (fig. 53h) or rocks. Most species close the entrance with a trapdoor. The aerial retreats are usually shorter than the terrestrial burrows and the outside surfaces are well camouflaged and strengthened with plant material (bark, lichen or moss). According to Coyle (1986), aerial retreats may be involved in prey detection as they provide a greater prey-sensing area. The placement of retreats on tree trunks may also increase prey capture, as trunks are sites of high prey density. Migids have no rastellums and Simon (1903c) suggested that the dorsolateral keels of the fangs may be used to enlarge cavities in bark. Griswold (1987a) reported that the spinules on the patella often show wear and suggested that they could be used in the digging process.

KEY TO THE SOUTHERN AFRICA SUBFAMILIES OF MIGIDAE

1. Small medial tooth present on outer fang (fig. 52b); tibia III without excavation dorsally **Miginae**
- Outer fang without small medial tooth; tibia III with dorsal excavation (fig. 52e) **Paramiginae**

Subfamily MIGINAE Simon, 1889

Miginae Simon, 1889b: 178; Raven, 1985: 144.

Diagnostic characters

Thorax as high as cephalic region; small basal tooth present between keels of outer surface of fangs (fig. 52b); metatarsi and tarsi III and IV with dense scopulae; femora with dorsal stout spines; tibia III without dorsal excavation.

Distribution

The Miginae are represented by two genera, *Poecilomigas* from South Africa and *Migas* from Australia and South America.

Genus *POECILOMIGAS* Simon, 1903

banded-legged trapdoor spiders

Poecilomigas Simon, 1903a: 23; Roewer, 1942: 192; Bonnet, 1958: 3736; Griswold, 1987b: 484.

Caedmon O. P.-Cambridge, 1903: 143; Roewer, 1942: 193.

Type species: *Moggridgea abrahami* O. P.-Cambridge, 1889.

Diagnostic characters

Carapace smooth in females to weakly wrinkled in males; basal tooth present on fang (fig. 52b); tibia III cylindrical without a dorsobasal depression; patellae without ventral, erect, lamellate setae; body size 6–22 mm.

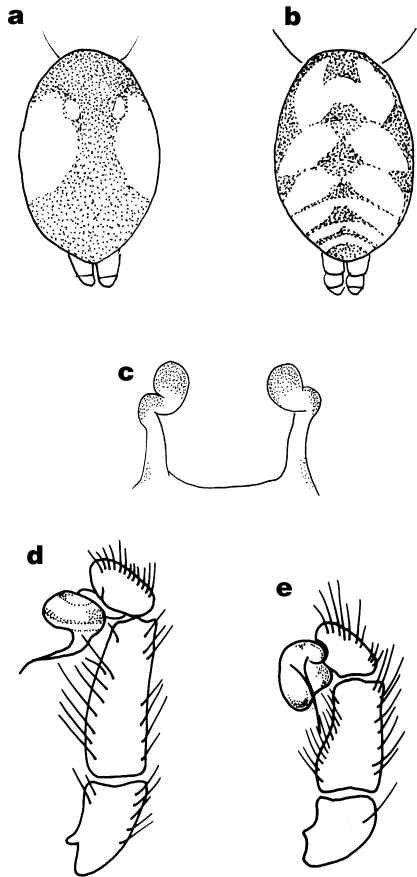


Fig. 53. Migidae — *Poecilomigas* spp. **a:** abdomen, *P. abrahami*; **b:** abdomen, *P. elegans*; **c:** spermathecae, *P. abrahami*; **d:** palp, *P. abrahami*; **e:** palp, *P. elegans*. (After Griswold, 1987b.)

Natural history

Only the habits of *Poecilomigas abrahami* are known. It appears to be exclusively arboreal and the retreats are made on the trunks of various tree species. Most trees appear to be suitable as long as the bark is soft and parts can be picked off for camouflage. The retreats are situated in depressions in the tree; they are wider than long and about twice the length of the occupant, usually vertically orientated. The retreats have wafer-type doors at both ends. The doors are oval, wider than long and the upper door is slightly larger than the lower one. When disturbed the spiders drop through the lower door to escape. They seem to be quite mobile and larger spiders have been seen to evict smaller ones to

occupy the burrows. The retreats are difficult to locate in the field as they are very well camouflaged with bits of material from the surrounding substrate.

Distribution

Three species are known from Africa, two of which occur in Southern Africa (fig. 54). They occur in the moister regions of eastern Africa (Griswold, 1987b), from Algoa Bay and the Uitenhage basin in South Africa to Tanzania in East Africa. They are more commonly found in forest and woodland areas.

KEY TO THE SOUTHERN AFRICA SPECIES OF POECILOMIGAS

(after Griswold, 1987b)

- 1. Dorsum of abdomen with broad, dark band; middle of sides pale (fig. 53a); spermathecae with distal curve, length greater than diameter (fig. 53c); palpal tibia slender, embolus length less than 1.5 times bulb length (fig. 53d) . . . *P. abrahami*
- Only males known; dorsum of abdomen pale with anteromedian dark diamond and chevron pattern (fig. 53b); palpal tibia relatively stout, embolus elongate, length greater than width, 1.8 times bulb length (fig. 53e) *P. elegans*

Species recorded from Southern Africa

1. *Poecilomigas abrahami* (O. P.-Cambridge, 1889)

Moggridgea abrahami O. P.-Cambridge, 1889: 41.
Migas abrahami Simon, 1892a: 82.
Poecilomigas abrahami: Purcell, 1903c: 72; Hewitt, 1915b: 91; Bonnet, 1958: 3737; Griswold, 1987b: 486; Platnick, 1989: 73.
Caedmon abrahami: O. P.-Cambridge, 1903: 144; Roewer, 1942: 193.
Moggridgea tidmarshi Lenz, 1889: 578.
Moggridgea stauntoni Pocock, 1902b: 319.
Moggridgea abrahami stauntoni Hewitt, 1915d: 125.
Moggridgea pulchripes Simon, 1903a: 23; Roewer, 1942: 192; Bonnet, 1958: 3737

Type locality: female holotype (UMO-Oxford) Grahamstown (33.19S; 26.22E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Grahams-town, King William's Town (Pirie Forest, Isidenge State Forest), East London, Alexandra State Forest, Port St Johns; *Northern Cape*: Fraserburg; *KwaZulu-Natal*: Durban, Port Edward, Krantzklloof, Kloof,

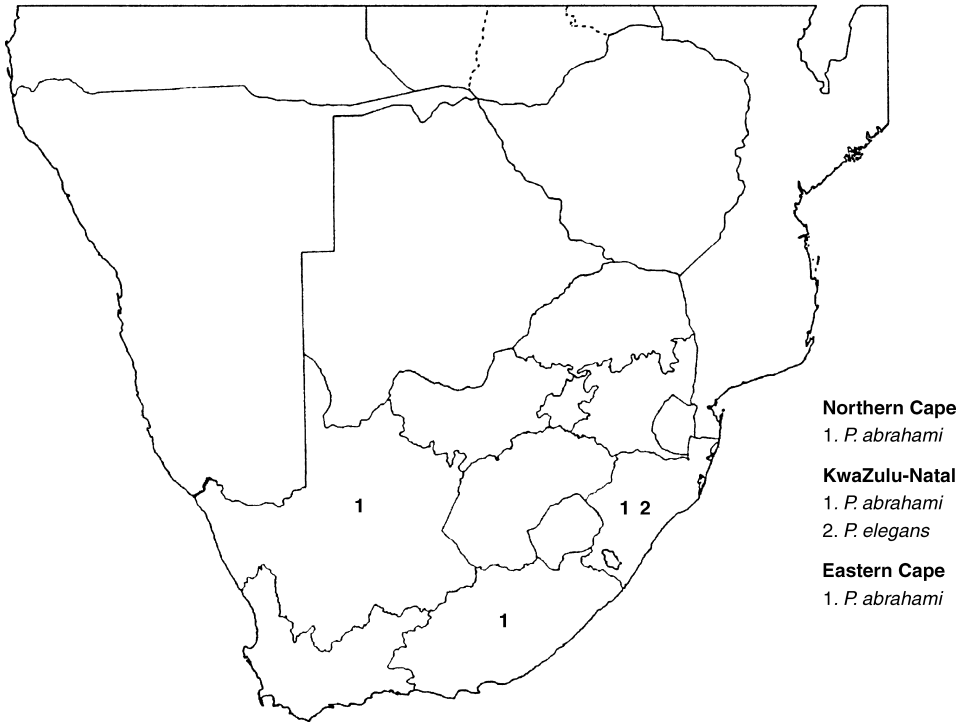


Fig. 54. Distribution of *Poecilomigas* species in Southern Africa.

Pietermaritzburg, Umgeni Valley Nature Reserve, Kar-kloof Forest, Nkandla Forest, Howick, Hluhluwe Nature Reserve, Eshowe (at Dlinza Forest), Sordwana Bay).

2. *Poecilomigas elegans* Griswold, 1987b

Poecilomigas elegans Griswold, 1987b: 493.

Type locality: male holotype (NM 3366), one juvenile female, Eshowe (28.54S; 31.28E), Zululand, KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Eshowe).

Subfamily PARAMIGINAE Petrunkevitch, 1939
 Myrtalae Simon, 1892a: 84 (unavailable through homonymy of type genus).
 Paramiginae Petrunkevitch, 1939: 154; Raven, 1985: 182.

Diagnostic characters

Outer fang without small medial tooth; tibia III with some excavation dorsally (fig. 52e).

Distribution

The Paramiginae include four genera, three of which are known from Madagascar and one, *Moggridgea*, from South Africa.

Genus *MOGGRIDGEA* O. P.-Cambridge, 1875

African tree trapdoor spiders

Moggridgea O. P.-Cambridge, 1875: 319; Roewer, 1942: 191; Griswold, 1987a: 9.
 Type species: *Moggridgea dyeri* O. P.-Cambridge, 1875.

Diagnostic characters

Carapace smooth in females and wrinkled to striate in males; a group of erect, lamellate setae present beneath patellae I, II and IV (rarely III); a group of stout, elongated setae present beneath femur II; preening comb present on metatarsus IV; medium-sized: 5–25 mm in length and strongly sexually dimorphic.

Natural history

Moggridgea are trapdoor spiders that close the entrance to their retreat with a hinged trapdoor. According to Griswold (1987a), two basic types of retreats are found:

A bag or sac-like retreat, oval to pear-shaped, constructed of tough silk, in a niche or crevice where little excavation is required. Several species construct their retreats in crevices in rocks. Sac-like retreats of, e.g., *M. paucispina*

have been found in crevices in rocks at Wonderboompoort, near Pretoria (Van Dam & Roberts, 1917). The sacs are short (length 32 mm), rather flattened and pouch-like (width 17 mm) with 9.75-mm thick lids. The spiders cling tightly to the lids when disturbed. Claw and tooth marks are present centrally on the underside of the lids. Hewitt (1913b) reported that *M. crudeni* is a rock-frequenting species found in retreats in earth-filled cracks and crevices in the Alicedale region. The retreats are often wedged in very narrow spaces and the lids are mostly oval. Lawrence (1927b) reported that *M. purpurea* was collected from a burrow made in a cleft almost 1 m from the base of a vertical wall of a limestone rock. Other *Moggridgea* species are found in trees. The retreats of *M. microps* have been found in a wooded ravine at Malelane in Mpumalanga Province. They are usually made in hollows or crevices in the bark, and cleverly covered with bits of bark (Van Dam & Roberts, 1917).

True silk-lined tubular burrows are usually terrestrial, the burrow being longer than wide in, for example, *M. mordax*. The entrance and the trapdoor are slightly raised above the surface and resemble a piece of loose earth. The lid is D-shaped and rather thick and flat, becoming thinner at the margin. The hinge is wide, occupying almost the entire truncated margin. The upper side is covered with earth and generally overgrown with a black fungus, while the underside is white with a pair of conspicuous pits (Purcell, 1903c). Mating occurs in the burrow. The egg sac of *M. mordax* consists of a compressed oval sack, rounded on one side and truncated on the other. The female suspends the egg sac above the bottom of the burrow with silk threads, the truncated side uppermost. The young remain with the female for some time.

Distribution

Moggridgea are known from Africa and Australia. Thirty-one species are known from Africa and the surrounding islands and 24 from Southern Africa. They are found in a variety of habitats from cool-temperate to tropical wet forest, and from bushveld and open savanna to true desert areas (fig. 56).

KEY TO THE SOUTHERN AFRICAN SPECIES OF MOGGRIDGEA

(after Griswold, 1987a)

1. Females 2
- Males 24
2. Pair of setae present anterior to fovea 3
- Setae absent 5
3. Fovea almost straight, central part absent; carapace dark with a pale area around fovea; carapace raised, thoracic declivity steep (fig. 55a) (Leydsdorp) · · *M. albimaculata*
- Fovea recurved, entire; carapace colouration not contrasting; carapace lower with thoracic declivity gentle 4
4. Coxa I with 15 or more thorns (Eastern Cape) *M. dyeri*
- Coxa I without thorns or with fewer than 10 (Botswana) *M. whytei*
5. Coxae lacking thorns 16
- At least coxa I with thorns 6
6. Thorns present on coxae I–III 7
- Thorns absent from coxa I 9
7. Small pointed tubercles present laterad of fovea (fig. 55b); tibia III with well-developed, shiny basal depression (fig. 52e) (Gravelotte) *M. breyeri*
- Sides of carapace smooth, tubercles absent; tibia III with weakly developed, shiny basal depression, or absent 8
8. Carapace with bold contrasting markings radiating from fovea; clypeus usually angular; patella IV with broad band of setae with numerous spinules at base; coxae I and II with >50 thorns; coxa III usually with >30 thorns (Alicedale) · · · *M. pseudocrudeni*
- Carapace without bold contrasting markings; clypeus usually rounded; patella IV with narrow band of setae and few spinules at base; coxa I with <50 thorns; coxa III usually with <30 thorns (Alicedale) *M. crudeni*
9. Coxa II with thorns 10
- Coxa II lacking thorns 14
10. Fovea without posterior longitudinal groove; spermathecae relatively short, widely separated at base (fig. 55d) 11

- Fovea usually with posterior longitudinal groove (fig. 55c); spermathecae long, slender, usually sinuate with narrow base . . . 12
- 11. Sternal sigilla narrow (fig. 55e); coxae II and III each with >12 thorns; diameter ALE:AME >2.2 (Montagu) *M. mordax*
- Sternal sigilla almost round (fig. 53f); coxae II and III each with >100 thorns; diameter ALE:AME >1.8 (Zimbabwe, Soutpansberg) *M. pyimi*
- 12. Abdomen with chevron pattern breaking up into spots posteriorly and laterally (Grahamstown) *M. rupicoloides*
- Abdomen uniformly dark dorsally or with faint, pale chevrons on dark background . . . 13
- 13. Ocular area broad; posterior eye row re-curved; patella IV with anterolateral band of slender setae; coxae II and III each with <25 thorns (Alicedale) *M. rupicola*
- Ocular area narrow; posterior eye row nearly straight; patella IV with band of setae with at least 20 spinules at base; coxae II and III each with >50 thorns (Alicedale) *M. terrestris*
- 14. Spermathecae bottle-shaped, diameter at base nearly twice that of distal bulb; carapace and legs yellow-brown, chelicerae, clypeus and ocular area red-brown, abdomen dark grey, sides with white spots (Piketberg) *M. ampullata*
- Spermathecae cylindrical or increasing in diameter distally 15
- 15. Palp coxa with narrow band of cuspules; distance between sternal sigilla less than half their length (fig. 55g); spermathecae slender, sinuous (fig. 55h) (Clanwilliam) *M. leipoldti*
- Palp coxa with broad band of cuspules; distance between sternal sigilla greater than half their length; spermathecae straight and usually stout (fig. 55i) (Stellenbosch) *M. terricola*
- 16. Sternal sigilla narrow-oval; length:width >2.1 17
- Sternal sigilla nearly round, pear-shaped or irregular; length:width <2.0 19
- 17. Carapace and legs pale yellow-white with strong contrasting dark markings extending from fovea to ocular area and clypeus; femora with dorsal longitudinal bands on distal half (Namibia) *M. pallida*
- Carapace and legs yellow-brown to dark, without contrasting markings 18
- 18. Venter of legs I and II with thin, curved setae, longer than height of segment; femur II with undifferentiated setae; fovea broad width/length >4.0; sclerotized region of spermathecae a narrow band at base (Namibia) *M. purpurea*
- Venter of legs I and II without slender, curved setae; femur II with stout setae; fovea strongly recurved, width:length <2.2; spermathecae sclerotized for half of length (Karoo) *M. peringueyi*
- 19. Coxa I with cuspules (Table Mountain) *M. teresae*
- Coxa I without cuspules 20
- 20. Femur I ventrally with rows of thorn-like setae; sigilla pear-shaped or irregular (Cape Peninsula) *M. quercina*
- Femur I ventrally with slender setae; sigilla almost round 21
- 21. Preening comb consists of two setae; ocular width area:caput >0.65 (Pretoria) *M. paucispina*
- Preening comb consists of three or more setae; ocular width area:caput <0.63 22
- 22. Spermathecae short and broad (fig. 55j), angled towards centre (Knysna) *M. intermedia*
- Spermathecae slender 23
- 23. Spermathecae widely separated at base (fig. 55k), length:base width <0.85 (Stellenbosch) *M. loistata*
- Spermathecal base narrow, width less than length of spermathecae (widespread in eastern parts of South Africa) *M. microps*
- 24. At least one coxa with thorns 25
- Coxae without thorns, only setal patches may be present 27
- 25. Coxae I–III with thorns, at least 30 on each (Alicedale) *M. pseudocrudeni*
- Coxa I without thorns, coxa III with <20 26
- 26. Coxae II and III with thorns; posterior median eyes smaller than posterior lateral eyes (Alicedale) *M. rupicola*

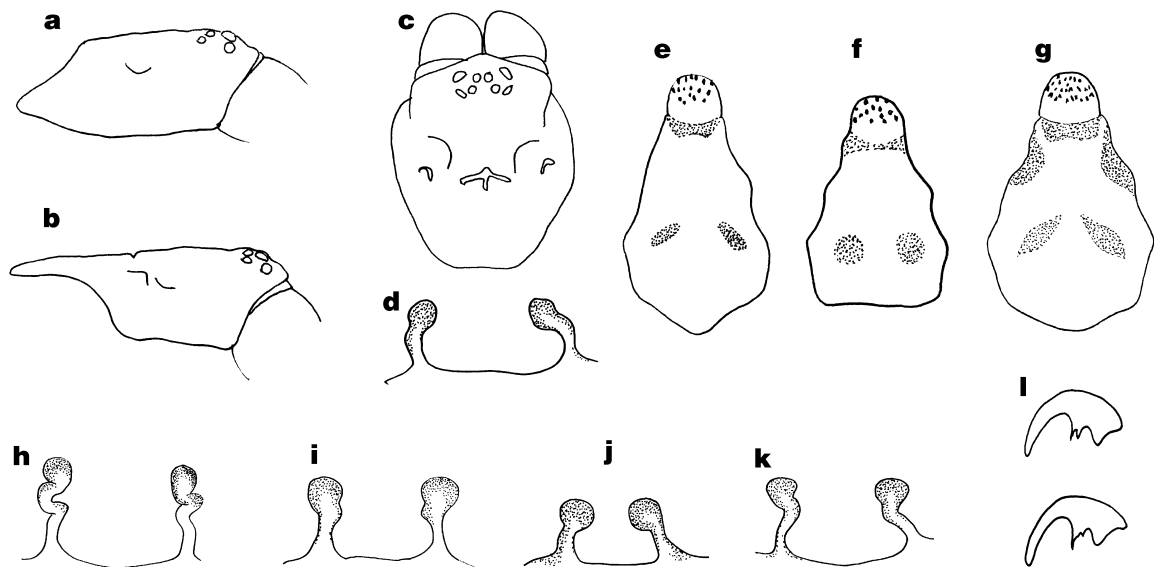


Fig. 55. Migidae — *Moggridgea* spp. **a:** raised carapace of *M. albimaculata*, lateral view; **b:** carapace with tubercles, *M. breyeri*; **c:** fovea with median groove, *M. breyeri*; **d:** spermathecae, *M. mordax*; **e:** sternum, *M. mordax*; **f:** sternum, *M. pyimi*; **g:** sternum, *M. leipoldti*; **h:** spermathecae, *M. leipoldti*; **i:** spermathecae, *M. terricola*; **j:** spermathecae, *M. intermedia*; **k:** spermathecae of *M. loistata*; **l:** paired tarsal claw of *M. peringueyi*. (After Griswold 1987a.)

—	Coxae II without thorns; posterior median eyes larger than posterior lateral eyes (Stellenbosch)	<i>M. terricola</i>
27.	Femur I evenly rounded, not carinate	28
—	Femur I carinate ventrally	29
28.	Labium and endites with cuspules; legs I and II with scopula on tarsi extending onto apex of metatarsi (Table Mountain)	<i>M. teresa</i>
—	Cuspules absent; scopulae restricted to tarsi	29
29.	Paired claws on tarsi I and II with basal tooth serrate (fig. 55l)	<i>M. peringueyi</i>
—	Paired claws on tarsi I and II with basal tooth simple or bifid	30
30.	Sternal sigilla narrow-oval; carapace low, height at fovea equal to caput height (Namibia)	<i>M. eremicola</i>
—	Sternal sigilla oval to round; caput raised, height at least 1.5 times height of fovea (Knysna)	<i>M. intermedia</i>

- Species recorded from Southern Africa**
- 1. *Moggridgea albimaculata* Hewitt, 1925**
Moggridgea albimaculata Hewitt, 1925: 277; Roewer, 1942: 191; Griswold, 1987a: 16; Platnick, 1989: 72.
Type locality: female lectotype and female paralectotype (TM 2795), Sekororo, 35 m SSW of Leydsdorp (23.59S; 30.31E), Northern Province.
Distribution: South Africa (Northern Province: Leydsdorp (Sekororo)).
- 2. *Moggridgea ampullata* Griswold, 1987a**
Moggridgea ampullata Griswold, 1987a: 19; Platnick, 1989: 72.
Type locality: female holotype (AM), three female paratypes (AM & TM), Piketberg (32.55S; 18.45E), Western Cape Province.
Distribution: South Africa (Western Cape: Piketberg).
- 3. *Moggridgea breyeri* Hewitt, 1915b**
Moggridgea breyeri Hewitt, 1915b: 89; Griswold, 1987a: 24; Platnick, 1989: 72.
Moggridgea beyeri (lapsus) Roewer, 1942: 191.
Type locality: female lectotype and female paralectotypes (TM 2802), Gravelotte (23.57S; 30.36E), Northern Province.
Distribution: South Africa (Northern Province: Gravelotte, Mamoranga at Letaba River).

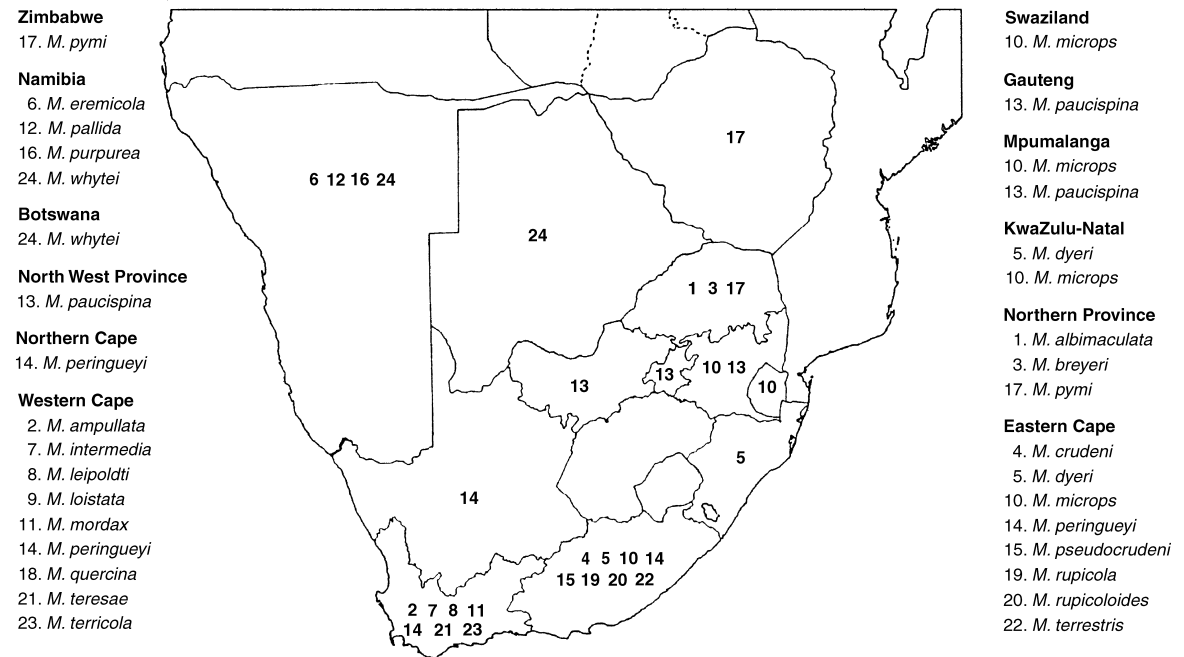


Fig. 56. Distribution of *Moggridgea* species in Southern Africa.

4. *Moggridgea crudeni* Hewitt, 1913b

Moggridgea crudeni Hewitt, 1913b: 47; Roewer, 1942: 191; Bonnet, 1957: 2972; Griswold, 1987a: 28; Platnick, 1989: 72.
Moggridgea crudeni transversa Hewitt, 1919a: 213.

Type locality: female lectotype and three female paralectotypes (AM), Alicedale (33.19S; 26.05E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Alicedale, Woodfields Krantz near Zuurberg Hotel (*M. crudeni transversa* type locality), Port Alfred).

5. *Moggridgea dyeri* O. P.-Cambridge, 1875

Moggridgea dyeri O. P.-Cambridge, 1875: 319; Roewer, 1942: 191; Bonnet, 1957: 2972; Griswold, 1987a: 31; Platnick, 1989: 72.

Type locality: female holotype (UMO), Uitenhage (33.47S; 25.28E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Uitenhage; Port Elizabeth, Dunbrody, Sidbury, Port Alfred, Alice, Fort Brown at Andries Vosloo Kudu Reserve, Somerset East, Redhouse, Grahamstown; KwaZulu-Natal: Harding).

6. *Moggridgea eremicola* Griswold, 1987a

Moggridgea eremicola Griswold, 1987a: 34; Platnick, 1989: 72; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: male holotype (SMW), Dome Gorge, Rössing Mine (22.31S; 14.52E), Namibia.

Distribution: Namibia.

7. *Moggridgea intermedia* Hewitt, 1913a

Moggridgea intermedia Hewitt, 1913a: 433; Roewer, 1942: 191; Bonnet, 1957: 2972; Griswold, 1987a: 36; Platnick, 1989: 72.

Type locality: female holotype (TM 2803), Knysna (33.57S; 23.10E), Western Cape Province.

Distribution: South Africa (Western Cape: Knysna, Krantzhoek at Harkerville State Forest, Diepwalle Forest Station).

8. *Moggridgea leipoldti* Purcell, 1903c

Moggridgea leipoldti Purcell, 1903c: 71; Bonnet, 1957: 2972; Griswold, 1987a: 41; Platnick, 1989: 72.
Moggridgea leipoldti (lapsus) Roewer, 1942: 191.

Type locality: female holotype (SAM 3619), Clanwilliam (Van Rhynsdorp road) (32.13S; 18.59E), Western Cape Province.

Distribution: South Africa (Western Cape: Clanwilliam).

9. *Moggridgea loistata* Griswold, 1987a

Moggridgea loistata Griswold, 1987a: 43; Platnick, 1989: 72.

Type locality: female holotype and female paratypes (AM), Houwhoek (34.11S; 19.05E), Western Cape Province.

Distribution: South Africa (Western Cape: Houwhoek, Stellenbosch).

10. *Moggridgea microps* Hewitt, 1915a

Moggridgea microps Hewitt, 1915a: 90; Roewer, 1942: 191; Bonnet, 1957: 2972; Griswold, 1987a: 45; Platnick, 1989: 72.
Moggridgea quercina, Tucker, 1917: 79 (not *M. quercina* Simon, 1903).

Type locality: female holotype (TM 2789), Malelane (25.29S; 31.31E), Mpumalanga Province.

Distribution: South Africa (Mpumalanga: Malelane,

Louw's Creek, Barberton; *KwaZulu-Natal*: Eshowe, Durban, Port Shepstone, St Lucia; *Eastern Cape*: East London, Port St Johns and Swaziland (Wyldelesdale).

11. *Moggridgea mordax* Purcell, 1903c

Moggridgea mordax Purcell, 1903c: 69; Roewer, 1942: 191; Bonnet, 1957: 2972; Griswold, 1987a: 48; Platnick, 1989: 72.

Type locality: female lectotype (SAM 12405) and female paralectotype, Hot Springs, Montagu (33.47S; 20.07E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Montagu, St Helena Bay).

12. *Moggridgea pallida* Hewitt, 1914a

Moggridgea pallida Hewitt, 1914a: 150; Roewer, 1942: 191; Bonnet, 1957: 2973; Griswold, 1987a: 56; Platnick, 1989: 72; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (TM 2793), Kraikluft, Namaqualand, Namibia

Distribution: Namibia.

13. *Moggridgea paucispina* Hewitt, 1916b

Moggridgea paucispina Hewitt, 1916b: 205; Roewer, 1942: 191; Bonnet, 1957: 2973; Griswold, 1987a: 59; Platnick, 1989: 72.

Type locality: female holotype (TM 2797), Wonderboompoort (25.45S; 28.12E), Pretoria, Gauteng Province.

Distribution: South Africa (*Gauteng*: Wonderboompoort, near Pretoria; *North West Province*: Rustenburg at Wolhuterskop, Silkaatsnek; *Northern Province*: Soutpansberg, Leydsdorp; *Mpumalanga*: Barberton).

14. *Moggridgea peringueyi* Simon, 1903a

Moggridgea peringueyi Simon, 1903a: 23; Bonnet, 1957: 2973; Griswold, 1987a: 62; Platnick, 1989: 72.
Moggridgea coegensis Purcell, 1903d: 71; Bonnet, 1957: 2972; Griswold, 1987a: 62 (synonym).
Moggridgea nigra Purcell, 1904: 115; Bonnet, 1957: 2973; Griswold, 1987a: 62 (synonym).
Moggridgea latus Tucker, 1917: 81; Griswold, 1987a: 62 (synonym).
Moggridgea lata; Tucker, 1920: 486; Roewer, 1942: 191; Bonnet, 1957: 2972.

Type locality: female holotype (MNHN 19274), Matjiesfontein (33.14S; 20.35E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Matjiesfontein, Beaufort West, Keurboom, Oudtshoorn, Worcester, Caledon, Houwhoek, Ashton, Swartzkop; *Northern Cape*: Victoria West, De Aar; *Eastern Cape*: Rooispruit, Cradock, Peddie, Fort Brown, Dunbrody, Kirkwood, Uitenhage, Port Elizabeth, Port Alfred, Alicedale, Swartberg, Redhouse, Blaauwkrantz).

15. *Moggridgea pseudocrudeni* Hewitt, 1919a

Moggridgea pseudocrudeni Hewitt, 1919a: 210; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 68; Platnick, 1989: 72.

Type locality: female lectotype (AM 2473) and six female paralectotypes, Alicedale (33.19S; 26.05E),

Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alicedale, Bushman's River (Dassie Klip), Committeesdrift, Wydgeleë).

16. *Moggridgea purpurea* Lawrence, 1927b

Moggridgea purpurea Lawrence, 1927b: 219; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 72; Platnick, 1989: 72; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (SAM 6686), Cameis, Namibia.

Distribution: Namibia (Kaokoveld and Ovamboland in the northern interior).

17. *Moggridgea pyimi* Hewitt, 1914b

Moggridgea pyimi Hewitt, 1914b: 15; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 75; Platnick, 1989: 73.
Moggridgea chirindaensis Benoit, 1962: 279; Brignoli, 1983: 121; Griswold, 1987a: 75 (synonym).

Type locality: female holotype (AM), 50 miles S Umtali (Melssetter Road), Zimbabwe.

Distribution: Zimbabwe (Umtali, Mount Selinda (Chirinda Forest) and South Africa (*Northern Province*: Soutpansberg (Wyllie's Poort)).

18. *Moggridgea quercina* Simon, 1903a

Moggridgea quercina Simon, 1903a: 22; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 77; Platnick, 1989: 73.
Caedmon congener O. P.-Cambridge, 1903: 146; Roewer, 1942: 193; Griswold, 1987a: 77 (synonym).
Poecilomigas congener, Bonnet, 1956: 919; 1958: 3737.
Caedmon dubia O. P.-Cambridge, 1903: 147; Roewer, 1942: 193; Griswold, 1987a: 77 (synonym).
Poecilomigas dubius, Bonnet, 1956: 919; 1958: 3737.
Caedmon thoracica O. P.-Cambridge, 1903: 144; Roewer, 1942: 193; Griswold, 1987a: 77 (synonym).
Poecilomigas thoracicus Bonnet, 1956: 919; 1958: 3737.

Type locality: female lectotype (MNHN 15254) and female paralectotype, Cape Town (Cape of Good Hope) (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Cape Town, Muizenberg, Simon's Town, Lion's Hill, Franschhoek).

19. *Moggridgea rupicola* Hewitt, 1913c

Moggridgea rupicola Hewitt, 1913c: 462; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 82; Platnick, 1989: 73.

Type locality: female lectotype and female paralectotypes (AM), Alicedale (33.19S; 26.05E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alicedale, Grahamstown, Bushman's River at Dassie Klip, Avontuur).

20. *Moggridgea rupicoloides* Hewitt, 1914b

Moggridgea rupicoloides Hewitt, 1914b: 14; Roewer, 1942: 192; Griswold, 1987a: 87; Platnick, 1989: 73.
Moggridgea rupicoloides (lapsus) Bonnet, 1957: 2973.

Type locality: female lectotype and eight female paralectotypes (AM), Grahamstown (33.19S; 26.22E)

Distribution: South Africa (*Eastern Cape*: Grahams-town, Uitenhage, Somerset East).

21. *Moggridgea teresae* Griswold, 1987a

Moggridgea teresae Griswold, 1987a: 93; Platnick, 1989: 73.

Type locality: female holotype (NM 3336), male paratype and 14 female paratypes, slopes of Table Mountain, Cape Town, Western Cape Province. Skeleton Gorge forest, Kirstenbosch Botanical Gardens, Cape Province.

Distribution: South Africa (*Western Cape*: slopes of Table Mountain (Skeleton Gorge forest), Kirstenbosch Botanical Gardens, Fernwood, Bat's Cave).

22. *Moggridgea terrestris* Hewitt, 1914b

Moggridgea terrestris Hewitt, 1914b: 13; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 98; Platnick, 1989: 73.

Type locality: female holotype (AM), Alicedale (33.19S; 26.05E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alicedale).

23. *Moggridgea terricola* Simon, 1903a

Moggridgea terricola Simon, 1903a: 22; Tucker, 1917: 82; Roewer, 1942: 192; Bonnet, 1957: 2973; Griswold, 1987a: 100; Platnick, 1989: 73.

Caedmon affinis O. P.-Cambridge, 1903: 145; Griswold, 1987a: 100 (synonym).

Poecilomigas affinis, Bonnet, 1956: 919; 1958: 3737.

Type locality: female lectotype (MNHN 15537) and three female paralectotypes, Stellenbosch (33.56S; 18.51E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Stellenbosch, Swellendam, Cape Town, Signal Hill).

24. *Moggridgea whytei* Pocock, 1897

Moggridgea whytei Pocock, 1897: 733; Roewer, 1942: 192; Bonnet, 1957: 2974; Griswold, 1987a: 106; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: female holotype (BMNH), Nyika Plateau, Malawi.

Distribution: Malawi, Botswana, Namibia, Democratic Republic of Congo and Zambia.



Female tree trapdoor spider
(Migidae: *Moggridgea breyeri*).

FAMILY NEMESIIDAE

wishbone trapdoor spiders

(figs 57–65)

The Nemesiidae are known from Africa, Madagascar, Australia, New Zealand, Burma, India, Malaysia, southern Europe, North (California) and South America, where it is represented by 26 genera and about 300 species. The family comprises six subfamilies, two of which, the Anaminae and Bemmerinae, are known from Southern Africa. The family is represented by five genera and 49 species.

Genera recorded from Southern Africa

Anaminae: *Entypesa* Simon, 1902; *Hermacha* Simon, 1889; *Lepthercus* Purcell, 1902;

Bemmerinae: *Pionothele* Purcell, 1902; *Spiroctenus* Simon, 1889.

Diagnostic characters

Nemesiids are medium-sized to large (13–30 mm) spiders with the following synapomorphies:

- broad biserially dentate tarsal claws,
- palpal claw of female with teeth on promargin.

Descriptive characters

- **carapace:** low, with cephalic region slightly arched; fovea short, more or less straight or procurved (fig. 57); clypeus narrow; hirsute;
- **sternum:** with some marginal sigilla (fig. 61b);
- **eyes:** eight in two rows about twice as wide as long; on a well-defined eye tubercle (fig. 58b);
- **chelicerae:** porrect; rastellum absent or when present consisting of weak spines on a low mound (fig. 58d); fangs long; cheliceral groove with teeth on promargin;
- **mouthparts:** labium wider than long with numerous (*Spiroctenus*) to no (Anaminae) cuspules; endites with cuspules; serrula present or absent;
- **legs:** three claws; paired claws broad with two rows of teeth (fig. 58e), S-shaped in males (fig. 58f) (Bemmerinae); leg formula 4123; tarsi with scopulae; tibiae of male with spur on low mound (fig. 58i);
- **female palp:** claw with row of teeth on promargin;
- **abdomen:** oval; hirsute;

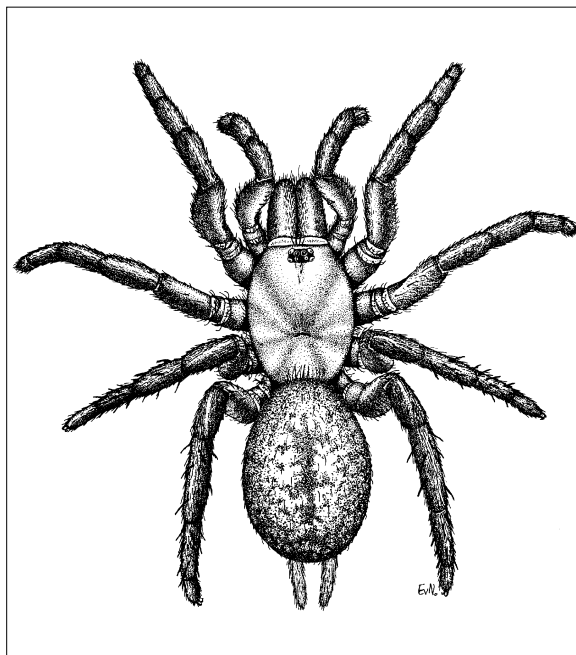


Fig. 57. Nemesiidae — *Spiroctenus* sp.

- **spinnerets:** long but differ from Dipluridae in that the median spinnerets are situated closer together; apical segment of posterior spinnerets digitiform (fig. 58g);
- **genitalia:** female spermathecae entire or bilobed; male palp with cymbium short, bilobate, spinose; bulbus pyriform with short embolus, conductor absent; bulbus with broadly-flanged embolus, or tapering (Anaminae) (fig. 58h);
- **body size:** 13–30 mm;
- **colour:** various shades of brown; variegated dark patterns.

Higher classification

The Nemesiidae of Simon (1892b) was elevated to family rank by Raven (1985) and included genera previously placed in the Ctenizidae. He considered three characters (biserially dentate and broad, paired tarsal claw and teeth on promargin of female palp) to support the monophyly of the Nemesiidae. Goloboff (1995) suggested that the

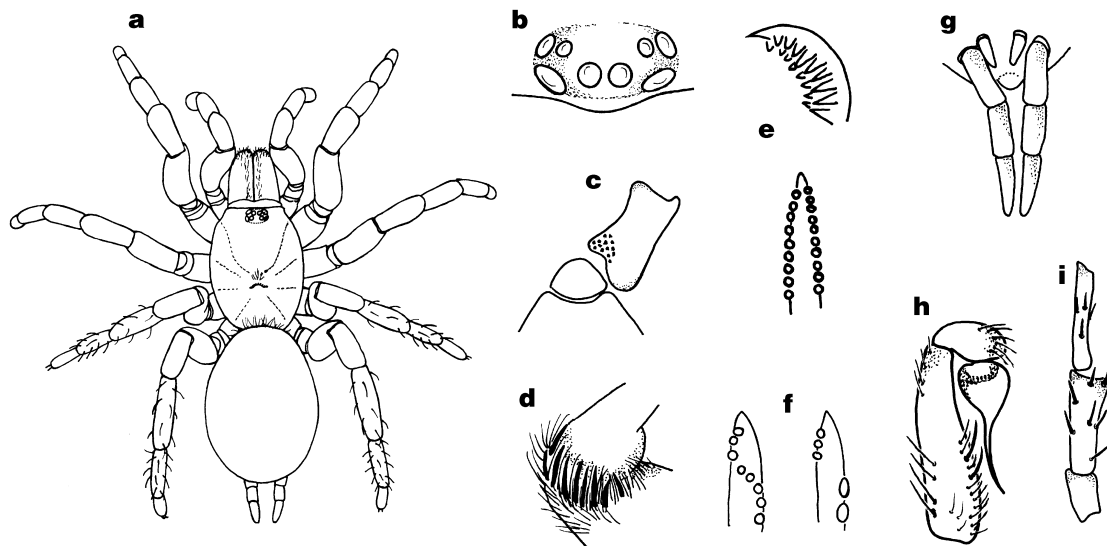


Fig. 58. Nemesiidae. **a:** female, dorsal view; **b:** eye pattern; **c:** mouthparts; **d:** rastellum, *Hermacha*; **e:** two rows of teeth on tarsal claws; **f:** S-shaped row of teeth on tarsal claw; **g:** spinnerets, *Hermacha*; **h:** male palp, *Hermacha*; **i:** leg I of male.

three characters constitute a single character, the biserially dentate tarsal claw, and found this character to be present also in the Dipluridae (Diplurinae), Microstigmatidae, Theraphosoidina and Rastelloidina. Goloboff (1995) reassessed relationships within Nemesiidae and considered it a paraphyletic group. However, current data do not allow a redelimitation at the familial level. The Nemesiidae are at present considered the sister group of the Theraphosoidina and these groups form the Crassitarsae and share scopulate tarsi and an edentate third tarsal claw (Raven, 1985).

KEY TO THE SOUTHERN AFRICAN SUBFAMILIES OF NEMESIIDAE

1. Both males and females with two rows of teeth on paired tarsal claws (fig. 58e); endites with strongly produced heel with cuspules (fig. 58c); scopulae entire on tarsus II of female **Anaminae**
- Males with S-shaped row of teeth on paired claws (fig. 58f) and females with two rows (fig. 58e); endites rectangular without heel but with numerous cuspules; scopulae on tarsus II of female reduced or absent **Bemmerinae**

Natural history

Little is known about the behaviour of the nemesiids. Most species seem to live in silk-lined burrows that vary in shape from a simple, deep burrow, to a Y-shaped burrow (fig. 60a), to burrows with side passages or chambers (fig. 60b) made under rocks. Some members of the Anaminae make silk-lined tubes under or on the sides of rocks, or are found in silk webbing. Only a few species are known to cover the entrance to the burrow with a lid. In most instances the entrances are without lids and flush with the soil surface or rimmed with sticks and grass to form a turret.

Subfamily ANAMINAE Simon, 1889

Anaminae Simon, 1889b: 178; Raven, 1985: 82.

Diagnostic characters

Endites with strongly produced heel, with cuspules extending posteriorly (fig. 58c); labium without cuspules; rastellum usually absent, if present comprises only a few strong setae; sternum with small marginal posterior sigilla; males and females with two rows of teeth on paired claws (fig. 58e); scopula entire on tarsus II of female; scopulae present on tarsi I–III of male; tibia I of male with mating spur consisting of a strong spine or cuticular spur (fig. 58i); apical segment of posterior spinnerets digitiform (fig. 58g); palpal bulb with broad embolus, flanged or tapering (fig. 58h).

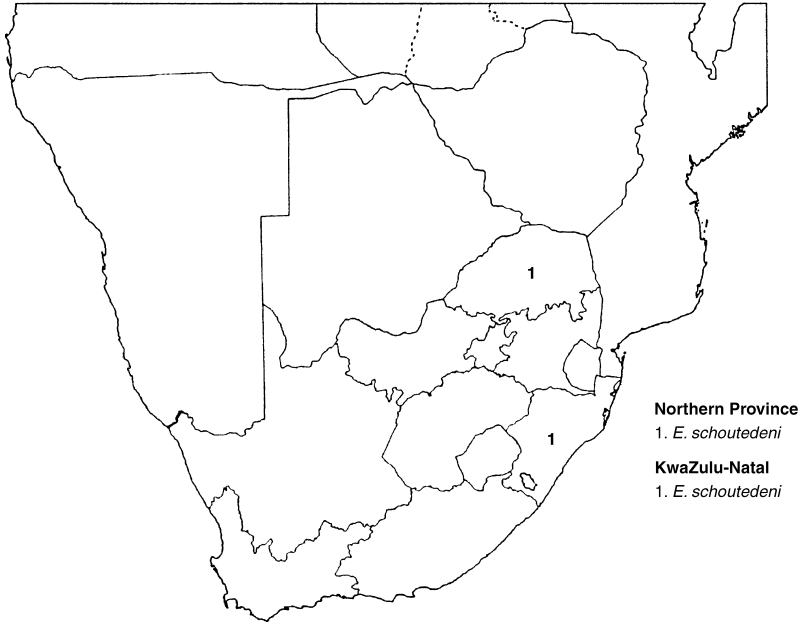


Fig. 59. Distribution of *Entypesa schoutedeni* in Southern Africa.

Distribution

The subfamily is represented by 10 genera, with *Entypesa*, *Hermacha* and *Lepthercus* recorded from Southern Africa.

**KEY TO THE SOUTHERN AFRICAN
GENERA OF THE ANAMINAE**

1. Tarsi I–IV with strong scopulae; preening comb present or absent 2

— Tarsus I with thin scopulae, tarsi II–IV with scopulae weak or absent; preening comb present; tarsi of male palp elongate · · *Entypesa*

2. Metatarsal preening comb present; tibia I of male with one strong spine (fig. 58i) *Hermacha*

— Metatarsal preening comb absent; tibia I of male with cuticular spur (fig. 61c) · *Lepthercus*

Genus ENTYPESA Simon, 1902

Entypesa Simon, 1902a: 599; Platnick, 1989: 85; Roewer, 1942: 198.

Pseudohermacha Strand, 1907a: 549; 1907b: 184; Roewer, 1942: 183; Raven, 1985: 86 (synonym).

Type species: *Entypesa nebulosa* Simon, 1902a.

Diagnostic characters

Eye tubercle and cephalic region low; fovea short

and straight; endites rectangular with anterior lobe and heel rounded, with numerous cuspules on inner corner; labium without cuspules; chelicerae without rastellum; serrula on anterior lobe; small marginal posterior sternal sigilla; metatarsal preening comb present on legs II and IV; scopulae very thin and divided on tarsus I, weak or absent on tarsi II–IV; distal tibia I of male with one large spine on low spur; apical segment of posterior spinnerets elongate; male palpal tarsi elongate; male palpal bulb pyriform with long, tapering embolus (fig. 61e); body size 13–25 mm.

Taxonomic notes

Entypesa was transferred from the Dipluridae and is considered a senior synonym of *Pseudohermacha* Strand, 1907 (Raven, 1985). It differs from *Hermacha* in that the scopulae are very thin on tarsus I and weak or absent on tarsi II–IV, and that the tarsi of the male palp are long (Raven, 1985).

Natural history

Specimens have been collected from under the bark of trees, in transparent webbing. They construct sheet-like burrows under stones with several entrances radiating from a central part.



Fig. 60. Nemesiid burrows. **a:** Y-shaped burrow; **b:** burrow underneath rocks.

Distribution

Entypesa is an Afrotropical genus with one species known from South Africa and two species from Madagascar (fig. 59).

Species recorded from Southern Africa

1. *Entypesa schoutedeni* Benoit, 1965e

Entypesa schoutedeni Benoit, 1965e: 261; Brignoli, 1983: 124; Raven, 1983: 552; 1985: 86; Platnick, 1989: 85.

Type locality: male holotype, female allotype (MRAC 127.593), Soutpansberg, Northern Province.

Distribution: South Africa (KwaZulu-Natal: Warburton; Northern Province: Soutpansberg).

Genus *HERMACHA* Simon, 1889

Hermacha Simon, 1889c: 408; Roewer, 1942: 177; Raven, 1985: 85; Platnick, 1989: 85.

Brachytheliscus Pocock, 1902b: 317; Hewitt, 1915d: 125; Roewer, 142: 171; Raven, 1985: 85 (synonym).

Damarchodes Simon, 1903b: 43; Tucker, 1917: 112; Roewer, 1942: 171; Raven, 1985: 85 (synonym).

Hermachola Hewitt, 1915c: 314; Roewer, 142: 178; Raven, 1985: 85 (synonym).

Type species: *Hermacha caudata* Simon, 1888c.

Diagnostic characters

Fovea short, more or less straight (fig. 58a); eye tubercle raised and well defined (fig. 58b); clypeus narrow; endites rectangular, anterior lobe rounded with numerous cusps on inner corner; labium without cusps; anterior surface of chelicerae with numerous short, stout bristles; strong scopulae on

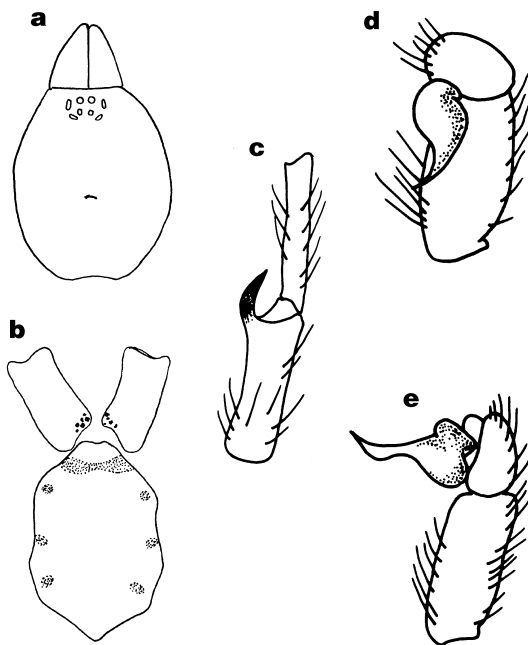


Fig. 61. Nemesiidae. **a–d:** *Lepthercus* sp., **a:** carapace, dorsal view; **b:** sternum; **c:** leg I of male; **d:** male palp; **e:** *Entypesa* sp. male palp.

tarsi I–III; female with broad, dense scopula over entire surface of metatarsi I and II; paired tarsal claws biserially dentate; male lacking a cuticular spur on tibia I bearing only a slender or moderately stout spine on a low mound (fig. 58i); rastellum if present usually consisting of stout setiform spines (fig. 58d); metatarsal preening comb present; posterior spinnerets very long, digitiform or triangular (fig. 58g); colour varies from medium-brown to reddish yellow to dark brown; body size 13–30 mm.

Taxonomic notes

Hermacha was transferred by Raven (1985) from Ctenizidae and considered a senior synonym of *Damarchodes* Simon, 1903, *Hermachola* Hewitt, 1915, and *Brachytheliscus* Pocock, 1902 (Raven, 1985). *Hermacha* differs from *Entypesa* in that tarsi I–III bear strong scopulae and from *Lepthercus* by lacking a cuticular spur on tibia I of the male.

Natural history

Hermacha bicolor constructs vertical burrows in the ground, similar to those of the Ctenizidae, but the burrows do not have lids. Most burrows

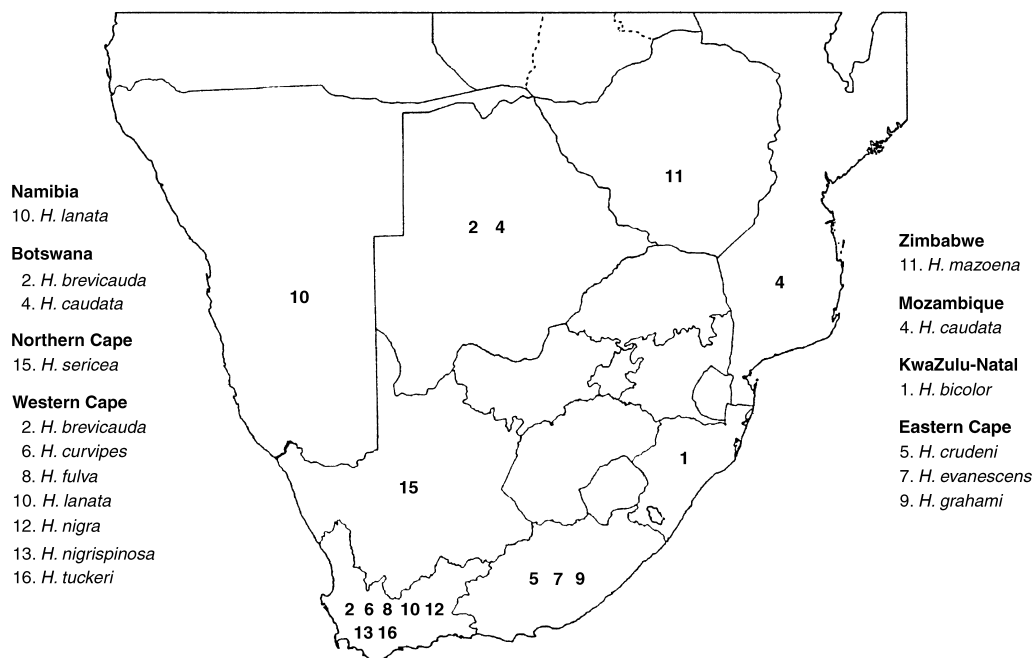


Fig. 62. Distribution of *Hermacha* species in Southern Africa.

recorded were made in the soil, and were well lined with silk, the openings flush with the surface (Hewitt, 1915d). The burrow of *H. evanescens* is deep and vertical, the rim flush with its surroundings. The burrows are made in ground that becomes very hard when dry (Purcell, 1903a). In *H. brevicauda* similar burrows are made but the rim of the burrow is decorated with sticks (Purcell, 1903c).

Distribution

Hermacha are known from Southern Africa and Brazil. Sixteen species have been recorded from Southern Africa, mostly from South Africa (fig. 62).

Species recorded from Southern Africa

1. *Hermacha bicolor* (Pocock, 1897)

Brachythele bicolor Pocock, 1897: 735; 1898a: 199.
Brachytheliscus bicolor: Pocock, 1902b: 317.
Hermacha bicolor: Hewitt, 1915d: 125 (male); Tucker, 1917: 105; Roewer, 1942: 177.

Type locality: female holotype (BMNH), Durban (29.57S; 30.59E), KwaZulu-Natal Province.

Distribution: South Africa (*KwaZulu-Natal*: Durban, Lower Umkomaas, Port Shepstone, Stella Bush near Durban, Howick, Umbilo).

2. *Hermacha brevicauda* Purcell, 1903c

Hermacha brevicauda Purcell, 1903c: 98; Roewer, 1942: 177.

Type locality: two male syntypes (SAM 4453, 8898), foot of Devil's Peak, Cape Town (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Cape Town).

3. *Hermacha capensis* (Ausserer, 1871)

Brachythele capensis Ausserer, 1871: 175; Purcell, 1902b: 373; Roewer, 1942: 197.
Hermacha capensis: Benoit, 1964b: 417; Platnick, 1993: 93.

Type locality: female holotype, Cape Province (no exact locality).

Distribution: South Africa.

4. *Hermacha caudata* Simon, 1889c

Hermacha caudata Simon, 1889c: 408; 1892a: 114; Roewer, 1942: 177.

Type locality: male holotype, Delagoa Bay (26.00S; 32.40E), Mozambique.

Distribution: Mozambique.

5. *Hermacha crudeni* Hewitt, 1913c

Hermacha crudeni Hewitt, 1913c: 466; Roewer, 1942: 177.

Type locality: female holotype (AM), Alicedale (33.19S; 26.5E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alicedale).

6. *Hermacha curvipes* Purcell, 1902b

Hermacha curvipes Purcell, 1902b: 377; Tucker, 1917: 106 (female); Roewer, 1942: 177.

Type locality: two male syntypes (SAM 6011), moun-

tainside at Simon's Town (34.51S; 18.28E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Simon's Town, Cape Peninsula, St James, Platteklip Ravine, Newlands (Skeleton Ravine), Wynberg Hill).

7. *Hermacha evanescens* Purcell, 1903c

Hermacha evanescens Purcell, 1903c: 99; Roewer, 1942: 177.

Type locality: series of female syntypes (SAM 11833), Hanover (31.5S; 24.27E) and Vlagkop, north of Hanover, Eierfontein west of Hanover and Poortjesfontein north of Hanover, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Hanover).

8. *Hermacha fulva* Tucker, 1917

Hermacha fulvus Tucker, 1917: 109.

Hermacha fulva: Roewer, 1942: 177.

Type locality: male holotype (SAM 150,406), Caledon (34.13S; 19.25E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Caledon).

9. *Hermacha grahami* (Hewitt, 1915c)

Hermachola grahami Hewitt, 1915c: 314; Roewer, 1942: 178.

Hermacha grahami: Raven, 1985: 85.

Type locality: male holotype (AM), Grahamstown (33.19S; 26.22E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Grahamstown).

10. *Hermacha lanata* Purcell, 1902b

Hermacha lanata Purcell, 1902b: 373; Roewer, 1942: 177; Griffin & Dippenaar-Schoeman, 1991: 156.

Type locality: male holotype (SAM 3639), Bokkeveld east of Pakhuisberg, Clanwilliam (32.10S; 18.52E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Clanwilliam) and Namibia.

11. *Hermacha mazoena* Hewitt, 1915c

Hermacha mazoena Hewitt, 1915c: 312; Roewer, 1942: 178.

Type locality: female holotype (BMNH), Mazoe, Mashonaland, Zimbabwe.

Distribution: Zimbabwe.

12. *Hermacha nigra* Tucker, 1917

Hermacha nigra Tucker, 1917: 110; Roewer, 1942: 178.

Type locality: two female syntypes (SAM 13,899), Bergvliet Flats, Cape Peninsula (34.21S; 18.30E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Bergvliet Flats).

13. *Hermacha nigrispinosa* Tucker, 1917

Hermacha nigrispinosus Tucker, 1917: 112.

Hermacha nigrispinosa: Roewer, 1942: 178.

Type locality: series of male and female syntypes (SAM

B2593), Sneeugat Valley, Great Winterhoek Mt, Tulbagh (33.18S; 19.9E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Tulbagh).

14. *Hermacha nigromarginata* Strand, 1907a

Hermacha nigromarginata Strand, 1907a: 548; 1907c: 181; Roewer, 1942: 178.

Type locality: male holotype (Lübeck Museum), Cape Province (no exact locality).

Distribution: South Africa.

15. *Hermacha sericea* Purcell, 1902b

Hermacha sericea Purcell, 1902b: 375; Roewer, 1942: 178.

Type locality: male holotype (SAM 3666), Van Rhyn's Dorp (31.36S; 18.45E) and the western part of Calvinia, Northern Cape Province.

Distribution: South Africa (*Northern Cape*: Van Rhyn's Dorp, Calvinia).

16. *Hermacha tuckeri* Raven, 1985

Hermacha purcelli, Tucker, 1917: 112; Roewer, 1942: 178
Raven, 1985: 161 (junior homonym, replaced by *Hermacha tuckeri*).

Hermacha tuckeri: Raven, 1985: 161; Platnick, 1989: 85.

Type locality: one female and two male syntypes (SAM 2670, 12395), Ashton (33.50S; 20.05E), Robertson, Western Cape Province.

Distribution: South Africa (*Western Cape*: Ashton).

Genus *LEPTHERCUS* Purcell, 1902

Lepthercus Purcell, 1902b: 379; Roewer, 1942: 178; Platnick, 1989: 87.

Type species: *Lepthercus dregei* Purcell, 1902b.

Diagnostic characters

Fovea short, more or less straight (fig. 61a); eye area twice as wide as long; eye tubercle raised and well defined; clypeus narrow; endites rectangular with anterior lobe rounded, with numerous cuspules on inner corner (fig. 61b); labium without cuspules; rastellum absent or with slender, moderately stout setae; preening comb absent; tibia I of male with a long, spine-tipped, spur-like apical tubercle (fig. 61c); palpal bulb pyriform with embolus fairly long, tapering and twisted (fig. 61d); posterior spinnerets long and slender; colour dull-brown with long, silky, pale-brown setae; body size 16–20 mm.

Taxonomic notes

Lepthercus was transferred from the Ctenizidae by Raven (1985). According to Hewitt (1917a), the females of this genus are difficult to distinguish from *Hermacha* as they present few structural characters that can be used, apart from the absence of a preening comb.

Natural history

No information on their behaviour has been published. Based on collection records they seem to live on the soil surface under stones and rocks. They make silk retreats running along the underside of rocks, with transparent webbing in the vicinity, or shallow silk-lined burrows under rocks. The more active males are often found in pit traps and swimming pools.

Distribution

The genus includes two species known only from South Africa (fig. 63). There are undescribed species from Gauteng in the NCA.

Species recorded from Southern Africa

1. *Lepthercus dregei* Purcell, 1902b

Lepthercus dregei Purcell, 1902b: 379; Simon, 1903c: 907; Roewer, 1942: 178; Raven, 1985: 86; Platnick, 1989: 87.

Type locality: male holotype (SAM 5692), Doornnek in the Zuurbergen, Alexandria (33.35S; 26.10E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Alexandria).

2. *Lepthercus rattrayi* Hewitt, 1917a

Lepthercus rattrayi Hewitt, 1917a: 699; Roewer, 1942: 179.

Type locality: male holotype and female paratypes, East London (33.01S; 27.58E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: East London).

Subfamily BEMMERINAE Simon, 1903

Bemmereae Simon, 1903c: 885.
Bemmerinae, Raven, 1985: 90.

Diagnostic characters

Teeth on paired claws of male in S-shaped row (fig. 58f), of female in two rows (fig. 58e); labium with numerous, few or no cuspules; posterior sternal sigilla submarginal or central; preening comb present or absent; apical posterior spinnerets domed or digitiform.

Taxonomic note

Raven (1985) listed four genera in this subfamily, with *Spiroctenus* and *Pionothele* recorded from Southern Africa. According to Raven (1985), however, the inclusion of *Pionothele* in this subfamily is tentative until more material can be examined.

KEY TO THE SOUTHERN AFRICAN GENERA OF BEMMERINAE

1. Labium and endites with numerous cuspules; eye tubercle low; rastellum with close-set, stout setae *Spiroctenus*
- Labium without cuspules, endites with few; eye tubercle raised and well defined; rastellum absent *Pionothele*

Natural history

Members of this subfamily construct a variety of burrow types, varying from simple burrows to burrows with side chambers and tunnels. The entrances can be closed with a trapdoor or left open, and may be with or without a turret.

Distribution

The subfamily is known from India, Burma, Malaysia, Sumatra and Africa.

Genus *PIONOTHELE* Purcell, 1902

Pionothele Purcell, 1902b: 380; Raven, 1985: 183.
Type species: *Pionothele straminea* Purcell, 1902b.

Diagnostic characters

Fovea short, more or less straight; eye tubercle raised and well defined, close to clypeal edge; eyes wide, more than twice as wide as long; labium without cuspules; endites with few cuspules; rastellum moderately stout setae; preening comb absent; third claw on anterior legs reduced; paired claws long, biseriate; posterior spinnerets stout, apical segment of posterior spinnerets domed; tarsus IV spined; leg I in male medially swollen, tibia slender; colour a pale mahogany-brown with darker radiating stripes; body size 13.5 mm.

Taxonomic note

Pionothele, a monotypic genus, was transferred to this subfamily from the Ctenizidae by Raven (1985).

Natural history

Nothing is known about the natural history of this genus.

Distribution

Known only from the type locality of *P. straminea* in South Africa (fig. 63).

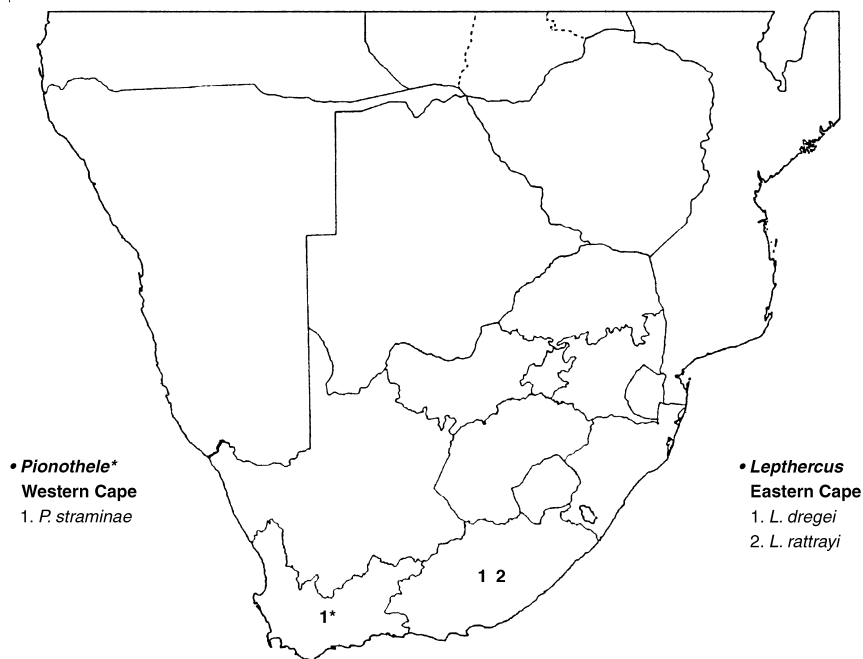


Fig. 63. Distribution of *Lepthercus* and *Pionothele** species in Southern Africa.

Species recorded from Southern Africa

1. *Pionothele straminea* Purcell, 1902b

Pionothele straminea Purcell, 1902b: 381; Simon, 1903c: 907; Tucker, 1917: 117; Roewer, 1942: 183; Raven, 1985: 93; Platnick, 1989: 89.

Type locality: male holotype (SAM 3586), Rondegat, 5 miles S of Clanwilliam (32.11S; 18.53E), Western Cape Province.

Distribution: South Africa (Western Cape: Clanwilliam).

Genus *SPIROCTENUS* Simon, 1889

Spiroctenus Simon, 1889c: 409; Roewer, 1942: 186; Raven, 1985: 182; Platnick, 1989: 91.

Hermachastes Pocock, 1900a: 319; Purcell, 1904: 121 (synonym); Roewer, 1942: 164; Raven, 1985: 154.

Bessia Pocock, 1900a: 320; Hewitt, 1916a: 221 (synonym); Roewer, 1942: 165; Raven, 1985: 150.

Bemmeris Simon, 1903b: 42; Purcell, 1904: 121 (synonym); Roewer, 1942: 165; Raven, 1985: 150.

Ctenonemus Simon, 1903b: 43; Purcell, 1904: 121 (synonym); Roewer, 1942: 165; Raven, 1985: 151.

Type species: *Spiroctenus personatus* Simon, 1889c.

Diagnostic characters

Fovea deep, more or less straight (procurved in *Spiroctenus armatus*); eye tubercles low; labium with few to numerous cuspules; endites with numerous cuspules (fig. 64b); rastellum consists of short spines or dense stout bristles (fig. 64d); paired claw in male with S-shaped row of teeth (fig. 58f); preening comb present on metatarsi III–IV; apical segment of

posterior spinnerets domed; male palp (fig. 64e) colour varies from dark brown to dark reddish to olive-yellow to slate-blue; legs of males longer than in females; body size 12–21 mm.

Taxonomic notes

Spiroctenus was transferred from the Ctenizidae to the Nemesiidae by Raven (1985). It is considered a senior synonym of *Hermachastes* Pocock, *Bessia* Pocock, *Bemmeris* Simon and *Ctenonemus* Simon by Raven (1985).

Natural history

Juveniles and females of *Spiroctenus* construct a variety of burrow shapes, varying from a plain burrow to burrows with side chambers. Males of different species are frequently found under stones or in shallow burrows (Purcell, 1903c). The entrances of the burrows are either closed with a trapdoor or decorated with a turret. Published information on burrow structure of some *Spiroctenus* species is summarized in Table 6.

Distribution

Spiroctenus is an African genus represented by 29 species known only from Southern Africa (fig. 65).

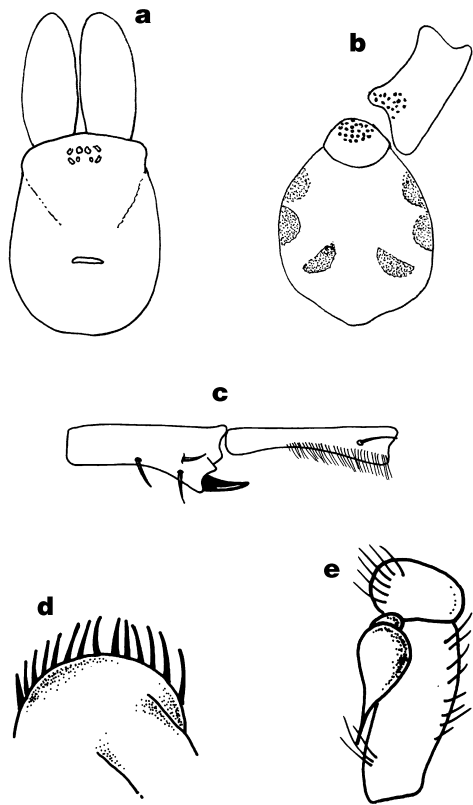


Fig. 64. Nemesiidae — *Spiroctenus* spp. **a:** carapace, dorsal view; **b:** sternum; **c:** leg I of male; **d:** rastellum; **e:** male palp.

Species recorded from Southern Africa

- 1. *Spiroctenus armatus* Hewitt, 1913c**
Spiroctenus armatus Hewitt, 1913c: 466; 1915c: 305; Roewer, 1942: 184.
Type locality: male holotype (AM), west bank of Kowie River, 4.5 km from Port Alfred (25.46S; 27.46E), Eastern Cape Province.
Distribution: South Africa (Eastern Cape: Port Alfred).
- 2. *Spiroctenus broomi* Tucker, 1917**
Spiroctenus broomi Tucker, 1917: 101; Roewer, 1942: 184.
Type locality: 10 female syntypes (SAM 13843), Stellenbosch (33.56S; 18.51E), Western Cape Province.
Distribution: South Africa (Western Cape: Stellenbosch).
- 3. *Spiroctenus cambierae* (Purcell, 1902b)**
Hermachastes cambierae Purcell, 1902b: 369.
Spiroctenus cambierae, Simon, 1903d: 107; Tucker, 1917: 98; Roewer, 1942: 184.
Type locality: male and female syntypes (SAM 8345), Houwhoek, Caledon (34.13S; 19.25E), Western Cape Province, South Africa.
Distribution: South Africa (Western Cape: Caledon).
- 4. *Spiroctenus coeruleus* Lawrence, 1952**
Spiroctenus coeruleus Lawrence, 1952: 184; Brignoli, 1983: 116.
Type locality: two male syntypes (NM 5626), Pietermaritzburg (29.38S; 30.28E), KwaZulu-Natal.
Distribution: South Africa (KwaZulu-Natal: Pietermaritzburg).

Table 6. Burrows of some *Spiroctenus* species in Southern Africa.

SPECIES	BURROW SHAPE	ENTRANCE	REFERENCES
<i>S. armatus</i>	simple	trapdoor present	Hewitt, 1913c
<i>S. broomi</i>	looping passage connecting a lower portion with upper passage	low turret around entrance	Tucker, 1917
<i>S. collinus</i>	simple	turret around entrance	Tucker, 1917
<i>S. flavopunctatus</i>	burrow found in rotten tree trunk	no information	Purcell, 1903c
<i>S. londinensis</i>	sloping downwards for a distance of 18–20 cm with the lower portion free of silk	lid of burrow with hinge down middle of lid forming two distinct halves	Cruden, 1916; Hewitt, 1919b
<i>S. minor</i>	sloping downwards for a distance of 18–20 cm with the lower portion free of silk	similar to <i>S. londinensis</i> but heavier	Cruden, 1916; Hewitt, 1913c
<i>S. pallidipes</i>	vertical, enlarging at a depth of 18–23 cm into a chamber with an additional small chamber in upper wall used as a retreat when in danger	without a lid or turret	Purcell, 1904
<i>S. pectiniger</i>	burrow sloping for about 4 cm, leading into a larger chamber (23 cm in diameter) out of which two side burrows lead: a deep vertical burrow and a short, more horizontal burrow	entrance with silk lining slightly spread out on soil surface, flush with surface or forming a very short turret, covered by bits of plant material.	Purcell, 1904
<i>S. schreineri</i>	about 10–15 cm deep, fairly straight but in some with a forked entrance or blind chamber	turret decorated with sticks and grass seeds	Purcell, 1903c
<i>S. validus</i>	Y-shaped	turret around entrance	Tucker, 1917

Free State22. *S. pilosus***Northern Cape**26. *S. schreineri***Western Cape**

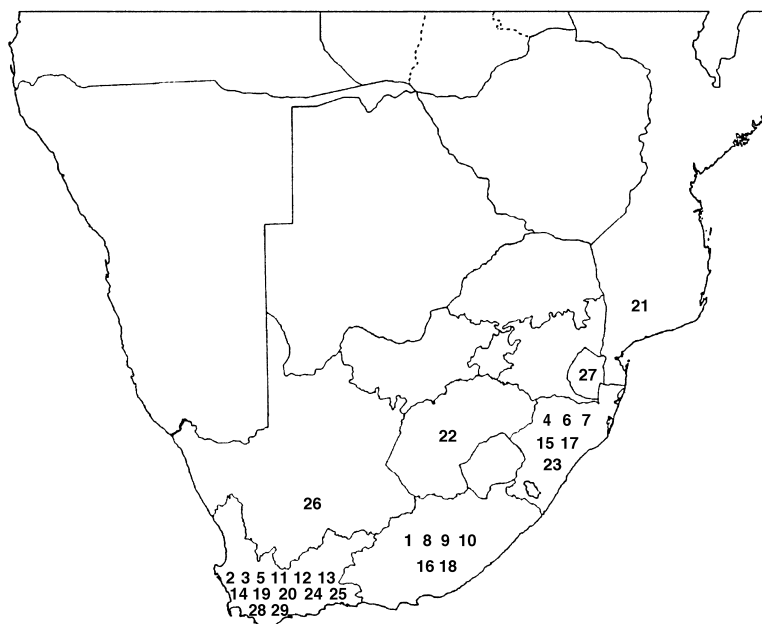
2. *S. broomi*
 3. *S. cambierae*
 5. *S. collinus*
 11. *S. gooldi*
 12. *S. inermis*
 13. *S. latus*
 14. *S. lightfooti*
 19. *S. pallidipes*
 20. *S. pectiniger*
 24. *S. purcelli*
 25. *S. sagittarius*
 28. *S. tricalcaratus*
 29. *S. validus*

Mozambique21. *S. personatus***Swaziland**27. *S. spinipalpis***KwaZulu-Natal**

4. *S. coeruleus*
 6. *S. curvipes*
 7. *S. exilis*
 15. *S. lignicolus*
 17. *S. marleyi*
 23. *S. punctatus*

Eastern Cape

1. *S. armatus*
 8. *S. flavopunctatus*
 9. *S. fossorius*
 10. *S. fuliginosus*
 16. *S. londonensis*
 18. *S. minor*

**Fig. 65.** Distribution of *Spiroctenus* species in Southern Africa.**5. *Spiroctenus collinus* (Pocock, 1900a)**

Hermachastes collinus Pocock, 1900a: 319; 1902a: 15; Purcell, 1902b: 364.

Spiroctenus collinus: Simon, 1903c: 907; Tucker, 1917: 99; Roewer, 1942: 184.

Bemmeris pardalina: Simon, 1903c: 896, 907; Purcell, 1904: 121 (synonym).

Type locality: male and female syntypes (BMNH), Table Mountain, Cape Town (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Cape Peninsula, Tulbagh (Sneeugat Valley, SAM B2583), Tulbagh Road Station).

6. *Spiroctenus curvipes* Hewitt, 1919b

Spiroctenus curvipes Hewitt, 1919b: 88; Roewer, 1942: 184.

Type locality: male and female syntypes (NM), Klipspruit, Utrecht (27.40S; 30.20E), KwaZulu-Natal.

Distribution: South Africa (*KwaZulu-Natal*: Utrecht).

7. *Spiroctenus exilis* Lawrence, 1938

Spiroctenus exilis Lawrence, 1938: 458; Roewer, 1942: 184.

Type locality: male holotype (NM), Port Edward (31.03S; 30.14E), KwaZulu-Natal Province.

Distribution: South Africa (*KwaZulu-Natal*: Port Edward).

8. *Spiroctenus flavopunctatus* (Purcell, 1903c)

Hermachastes flavopunctatus Purcell, 1903c: 97.

Spiroctenus flavopunctatus: Roewer, 1942: 184.

Type locality: two female syntypes (SAM 12695), Hogsback (32.35S; 27.05E), Amatola mountain range, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Hogsback).

9. *Spiroctenus fossorius* (Pocock, 1900a)

Bessia fossoria Pocock 1900a: 320.

Bessia fossorina: Simon, 1903c: 896; Hewitt, 1915c: 304; Roewer, 1942: 164.

Spiroctenus fossorina: Raven, 1985: 92.

Spiroctenus fossorius: Platnick, 1989: 91.

Type locality: female holotype (BMNH), Port Elizabeth (33.58S; 25.36E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Port Elizabeth).

10. *Spiroctenus fuliginosus* (Pocock, 1902a)

Hermachastes fuliginosus Pocock, 1902a: 14.

Spiroctenus fuliginosus: Roewer, 1942: 185.

Type locality: female holotype, Brakkloof (33.12S; 26.40E), Grahamstown, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Grahamstown).

11. *Spiroctenus gooldi* (Purcell, 1903c)

Hermachastes gooldi Purcell, 1903c: 95.

Spiroctenus gooldi Tucker, 1917: 104; Roewer, 1942: 185.

Type locality: three male syntypes (SAM 11706), Stompneus, St Helena Bay, Malmesbury (33.28S; 18.43E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Malmesbury, St James, Cape Peninsula).

12. *Spiroctenus inermis* (Purcell, 1902b)

Hermachastes inermis Purcell, 1902b: 370.

Spiroctenus inermis: Simon, 1903c: 907; Roewer, 1942: 185.

Type locality: male syntypes (SAM 3664), Pakhuisberg, Clanwilliam (32.11S; 18.53E), Western Cape Province.

Distribution: South Africa (*Western Cape*: Clanwilliam, Onder Bokkeveld, Nieuwoudtville, Oorlogskloof, Calvinia).

13. *Spiroctenus latus* Purcell, 1904*Spiroctenus latus* Purcell, 1904: 125; Roewer, 1942: 185.**Type locality:** male holotype (SAM), Wellington (33.39S; 18.58E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Wellington).**14. *Spiroctenus lightfooti* (Purcell, 1902b)***Hermachastes lightfooti* Purcell, 1902b: 363.*Spiroctenus lightfooti*: Roewer, 1942: 185.**Type locality:** male and female syntypes (SAM 657, 8543, 8550), Signal Hill, Cape Town (33.56S; 18.28E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Cape Town, Brandvlei, Worcester).**15. *Spiroctenus lignicolus* Lawrence, 1937***Spiroctenus lignicolus* Lawrence, 1937: 216; Roewer, 1942: 185.**Type locality:** female syntypes (NM), Hluhluwe Game Reserve (28.02S; 32.17E), KwaZulu-Natal Province.**Distribution:** South Africa (*KwaZulu-Natal*: Hluhluwe Game Reserve).**16. *Spiroctenus londinensis* Hewitt, 1919b***Spiroctenus londinensis* Hewitt, 1919b: 86; Roewer, 1942: 185.**Type locality:** male and female syntypes (TM), East London (33.01S; 25.55E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: East London).**17. *Spiroctenus marleyi* Hewitt, 1919b***Spiroctenus marleyi* Hewitt, 1919b: 83; Roewer, 1942: 185.**Type locality:** male holotype (AM), Eshowe (28.53S; 31.28E), KwaZulu-Natal.**Distribution:** South Africa (*KwaZulu-Natal*: Eshowe).**18. *Spiroctenus minor* (Hewitt, 1913c) comb. nov.***Bessia minor* Hewitt, 1913c: 469; 1915c: 302 (male); Roewer, 1942: 165.**Type locality:** female syntypes (AM), Alicedale (33.19S; 26.05E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Alicedale).**19. *Spiroctenus pallidipes* Purcell, 1904***Spiroctenus pallidipes* Purcell, 1904: 121; Roewer, 1942: 185.**Type locality:** male and female syntypes (SAM), Touws River Station, Worcester (33.39S; 19.26E), Western Cape.**Distribution:** South Africa (*Western Cape*: Worcester).**20. *Spiroctenus pectiniger* (Simon, 1903b) comb. nov.***Ctenonemus pectiniger* Simon, 1903b: 42; 1903c: 896; Roewer, 1942: 165.*Hermachastes pectiniger*: Purcell, 1904: 121.**Type locality:** female from South Africa (no exact locality).**Distribution:** South Africa (*Western Cape*: Matjiesfontein).**21. *Spiroctenus personatus* Simon, 1889c***Spiroctenus personatus* Simon, 1889c: 409; 1892a: 115; Roewer, 1942: 185; Raven, 1985: 92; Platnick, 1989: 91.**Type locality:** male syntypes from Delagoa Bay (26.00S; 32.40E), Mozambique.**Distribution:** Mozambique.**22. *Spiroctenus pilosus* Tucker, 1917***Spiroctenus pilosus* Tucker, 1917: 102; Roewer, 1942: 185.**Type locality:** female holotype (SAM B250), Smithfield (30.13S; 26.32E), Free State Province.**Distribution:** South Africa (*Free State*: Smithfield).**23. *Spiroctenus punctatus* Hewitt, 1916a***Spiroctenus punctatus* Hewitt, 1916a: 222; Roewer, 1942: 185.**Type locality:** female syntype (DM), Zululand, South Africa.**Distribution:** South Africa (*KwaZulu-Natal*: Zululand).**24. *Spiroctenus purcelli* Tucker, 1917***Spiroctenus purcelli* Tucker, 1917: 105; Roewer, 1942: 185.**Type locality:** male holotype (SAM 150.472), Simonstown (34.51S; 18.28E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Simon's Town).**25. *Spiroctenus sagittarius* (Purcell, 1902b)***Hermachastes sagittarius* Purcell, 1902b: 365.*Spiroctenus sagittarius*, Simon, 1903c: 907; Roewer, 1942: 185.**Type locality:** male holotype (SAM 7814), Brandvlei, Worcester (33.39S; 19.26E), Western Cape Province.**Distribution:** South Africa (*Western Cape*: Worcester).**26. *Spiroctenus schreineri* (Purcell, 1903c)***Hermachastes schreineri* Purcell, 1903c: 96.*Spiroctenus schreineri*: Tucker, 1917: 132; Roewer, 1942: 185.**Type locality:** series of female syntypes (SAM 11793), Hanover (31.05S; 24.27E), and the farms Poortjiesfontein and Vlagkop, Northern Cape Province.**Distribution:** South Africa (*Northern Cape*: Hanover).**27. *Spiroctenus spinipalpis* Hewitt, 1919b***Spiroctenus spinipalpis* Hewitt, 1919b: 84; Roewer, 1942: 185.**Type locality:** male holotype (TM), Ruby Creek, Swaziland.**Distribution:** Swaziland.**28. *Spiroctenus tricalcaratus* (Purcell, 1903c)***Hermachastes tricalcaratus* Purcell, 1903c: 94.*Spiroctenus tricalcaratus*: Roewer, 1942: 185.**Type locality:** eight male syntypes (SAM 11724), Steenberg's Cove, Stompneus and Soldaten Post all at St Helena Bay (32.16S; 18.08E), Malmesbury, Western Cape Province.**Distribution:** South Africa (*Western Cape*: St Helena Bay).**29. *Spiroctenus validus* (Purcell, 1902b)***Hermachastes validus* Purcell, 1902b: 367.*Spiroctenus validus*: Simon, 1903c: 907; Tucker, 1917: 132; Roewer, 1942: 185.**Type locality:** male and female syntypes (SAM 150.408), Bonnie Vale farm near Bushman's Drift on Breede River near Ashton (33.50S; 20.05E), Swellendam, Western Cape Province.**Distribution:** South Africa (*Western Cape*: Ashton, Swellendam).

FAMILY THERAPHOSIDAE

baboon spiders

(figs 66–74)

The Theraphosidae is the most diverse family of mygalomorph spiders, represented by 86 genera and about 612 species with a pantropical distribution. Theraphosids are known from the Oriental, Indian, Australian and Afrotropical Regions, and South and Central America. Of the eight recognized subfamilies, only one, the Harpactirinae, represented by five genera and 47 species, occurs in Southern Africa.

Genera recorded from Southern Africa

Harpactirinae: *Brachionopus* Pocock, 1897; *Ceratogyrus* Pocock, 1897; *Harpactira* Ausserer, 1871; *Harpactirella* Purcell, 1902; *Pterinochilus* Pocock, 1897.

Diagnostic characters

Theraphosids are medium-sized to very large (13–90 mm) mygalomorph spiders with the following synapomorphies:

- two tarsal claws with well-developed scopulae and claw tufts (fig. 67d),
- tarsi with >6 clavate trichobothria dorsally (fig. 7c),
- labium and endites with dense cuspsules.

Descriptive characters

- **carapace:** clypeus wide; fovea short, varies from straight to procurved (fig. 67a), in *Ceratogyrus* with distinct horn or depression (fig. 69e–k); carapace hirsute;
- **sternum:** with moderately small, oval, marginal to sub-central posterior sigilla;
- **eyes:** two rows on distinct eye tubercle (fig. 67b);
- **chelicerae:** outer surface of chelicerae hirsute or with dense scopulae (fig. 71b) (except *Harpactirella* and *Brachionopus* (fig. 71e); rastellum absent or weakly developed;
- **mouthparts:** anterior lobe of endites well developed into conical process; labium and endites with dense cuspsules;
- **legs:** two claws with thick tarsal scopulae (fig. 67d)

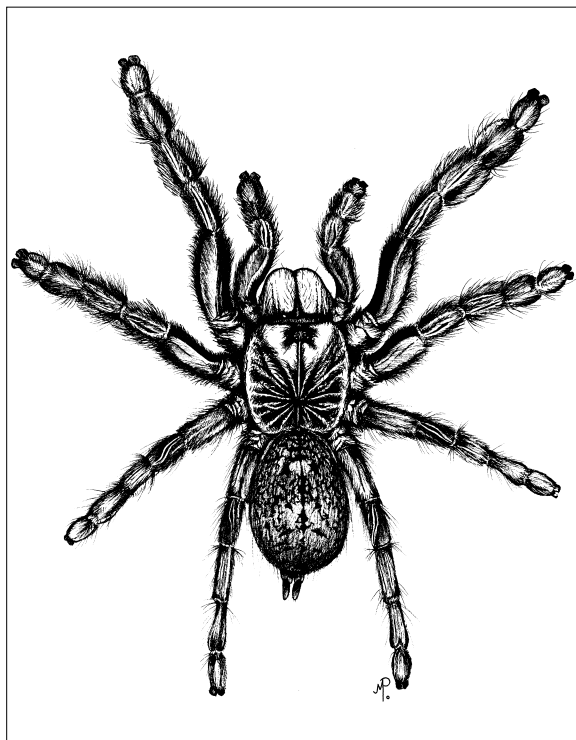


Fig. 66. Theraphosidae — *Pterinochilus junodi*.

and claw tufts; paired tarsal claws with only one row of teeth each; >6 clavate trichobothria dorsally on tarsi; tibia I in males usually with mating spur (except *Brachionopus*);

- **abdomen:** oval; hirsute;
- **spinnerets:** apical segment of posterior spinnerets long and digitiform (fig. 7d);
- **genitalia:** female spermathecae paired, simple, unbranched; male palp with small second haematocha and coniform distal sclerite; subtegulum large; embolus broad and acuminate;
- **body size:** 13–90 mm;
- **colour:** various hues of brown that vary from yellowish to dark; abdomen with variegated markings.

Higher classification

The monophyly of the theraphosids is supported by their well-developed claw tufts and scopulae

on the tarsi. The Theraphosidae and Paratropididae belong to the Theraphosoidea and, with the Barychelidae as sister group, they all belong to the Theraphosoidina on the basis of their claw tufts and the tibial mating spur on the male's front leg (Raven, 1985; Gallon, 2001).

Taxonomic notes

Raven (1985) recognized seven subfamilies. Smith (1990a) discussed some of the African and the Middle East species of Theraphosidae and provided keys and distribution maps. Only one genus, *Ceratogyrus*, has been revised (De Wet & Dippenaar-Schoeman, 1991), with new species added by Gallon (2001).

Natural history

Theraphosids, in Africa are known as baboon spiders, and occur in a variety of habitats. They are commonly found in dry *Acacia* scrubland, grassland or savanna woodland. In arid regions their burrows are usually deep to provide protection from high temperatures (Smith, 1990a). Most baboon spiders are ground-living and construct relatively permanent silk-lined burrows or retreats under stones and rocks. The silk lining usually extends beyond the entrance to form a silk rim that may assist in prey detection and that sometimes incorporates pieces of plant material. Theraphosids are predominantly nocturnal sit-and-wait hunters and most species await the approach of prey in the entrance of their burrows. Prey is usually captured at or near the entrance. They presumably rely on sensory detection systems like trichobothria to detect air currents generated by moving prey, or soil and silk vibration detectors such as slit sensilla or club-shaped trichobothria. The tarsal claws of the first and second legs and palp maintain contact with the silk encircling the burrow entrance. From this position, prey is ambushed and dragged into the burrow. The spiders hide during the day in the burrows and the entrances are frequently silked over during daylight hours with a thin, transparent cover. When they wander beyond the burrow entrance, draglines of silk are usually laid down. Theraphosids may live up to 25 years (Baerg & Peck, 1970) and take about 10 years to mature.

Some theraphosids are known to deliver painful bites. Perret (1974a,b) and Perret & Freyvogel (1973) reported on the venom of a *Pterinochilus* species from Tanzania. Another member of the Harpactirinae known to deliver a painful bite is *Harpactirella lightfooti* that occurs in the southern parts of South Africa. Finlayson (1939) and Prins & Leroux (1986) reported on bite symptoms, and Smithers (1939) provided some biological data on this species.

Conservation

Owing to the demand for these spiders as pets, they are classified as Commercially Threatened in terms of the IUCN system. In February 1987, three genera, *Ceratogyrus*, *Harpactira* and *Pterinochilus*, were added to Schedule VII of the Provincial Nature Conservation Ordinance of 1983 of the Transvaal as Protected Invertebrate Animals. This restriction is still in place in most South African provinces.

Subfamily HARPACTIRINAE Pocock, 1897
Harpactirinae Pocock, 1897: 744; Raven, 1985: 117; Smith, 1990a: 62.

Diagnostic characters

Chelicerae and trochanter of palp with characteristic plumose pad of setae used as stridulating organ (fig. 69b,c) (absent in *Harpactirella* and *Brachionopus*); clypeus wide (fig. 67a); legs usually with spines on tibia and metatarsi III and IV; femur IV sometimes with scopula on retrolateral face; males with simple, long and tapering tibial mating spurs (absent in *Brachionopus*); palpal bulb with embolus long and tapering, sometimes with simple keels.

KEY TO THE SOUTHERN AFRICAN
GENERA OF HARPACTIRINAE

- 1. Foveal groove U-shaped or takes the form of a horn (fig. 69d–k) or mound . . ***Ceratogyrus***
- Fovea a transverse groove without tubercle or mound, more or less straight 2
- 2. Chelicerae with plumose pad of setae on side (fig. 71b) and plumose setae on prolateral face of male palp's trochanter (fig. 71c) sometimes extending to proximal part of femur 3

- Setae absent 4
- 3. Pad of plumose setae present between chelicerae when viewed from above (less distinct in juveniles); chelicerae with a row of stout setae below plumose pad on side, with a group of stout plumose setae on palpal coxa (fig. 71b,c) **Harpactira**
- Plumose setae absent from inside of chelicerae; stout setae beneath plumose pad on outside absent **Pterinochilus**
- 4. Tibial spur present in male; tarsal segment IV (sometimes III) divided by a thin band of setae **Harpactirella**
- Tibial spur absent; tarsal segments of legs II–IV divided by thin setae **Brachionopus**

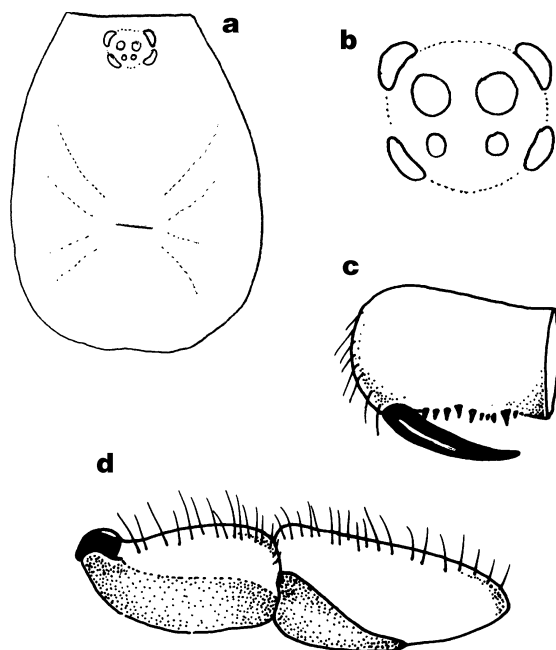


Fig. 67. *Brachionopus* spp. **a:** carapace, dorsal view; **b:** eye pattern; **c:** chelicera without setae, lateral view; **d:** scopula on tarsus and metatarsus I.

Distribution

Members of the Harpactirinae are distributed throughout the southern and southeastern parts of Africa. From Southern Africa five genera are known, represented by 47 species.

Genus *BRACHIONOPUS* Pocock, 1897

Brachionopus Pocock, 1897: 740; Purcell, 1903c: 105; Roewer, 1942: 219; Raven, 1985: 117; Smith, 1990a: 66.
Type species: *Brachionopus robustus* Pocock, 1897.

Diagnostic characters

Chelicerae without plumose pad on outside (fig. 67c); clypeus equal to half the length of eye tubercle (fig. 67a); cuspules on labium sometimes reduced or absent; legs short and robust; scopulae not broader than segment; scopulae on tarsus I entire, tarsi II–IV divided by setae; male lacks tibial spur; scopulae on metatarsi III and IV not very dense, extending to middle of segment (fig. 67d); basal segment of posterior spinnerets as long as other two segments; colour varies from golden-brown to greenish black, abdomen mottled or decorated with a median line and chevrons; body size 13.5–21.5 mm.

Taxonomic notes

Pocock (1897) originally placed *Brachionopus* in the Barychelidae because it lacks stridulating organs on the chelicerae. Raven (1985) transferred it to the Theraphosidae in the subfamily Harpactirinae based on the presence of a

wide clypeus. Charpentier (1993) did not accept this placement, but did not suggest an alternative relationship. Platnick (1998) followed the placement of Raven (1985). *Brachionopus* resembles *Harpactirella* very closely but differs in the absence of a tibial mating spur in the male.

Natural history

Very few documented records exist of their behaviour. Locality data for specimens in the National Collection of Arachnida in Pretoria indicate that *Brachionopus* species are frequently found in tubular, silk-lined burrows made under rocks or logs, sometimes with light webbing at the entrance. The males wander around and are readily collected in pit traps. On several occasions, specimens were collected in areas infested with harvester termites.

Distribution

Brachionopus is an Afrotropical genus known

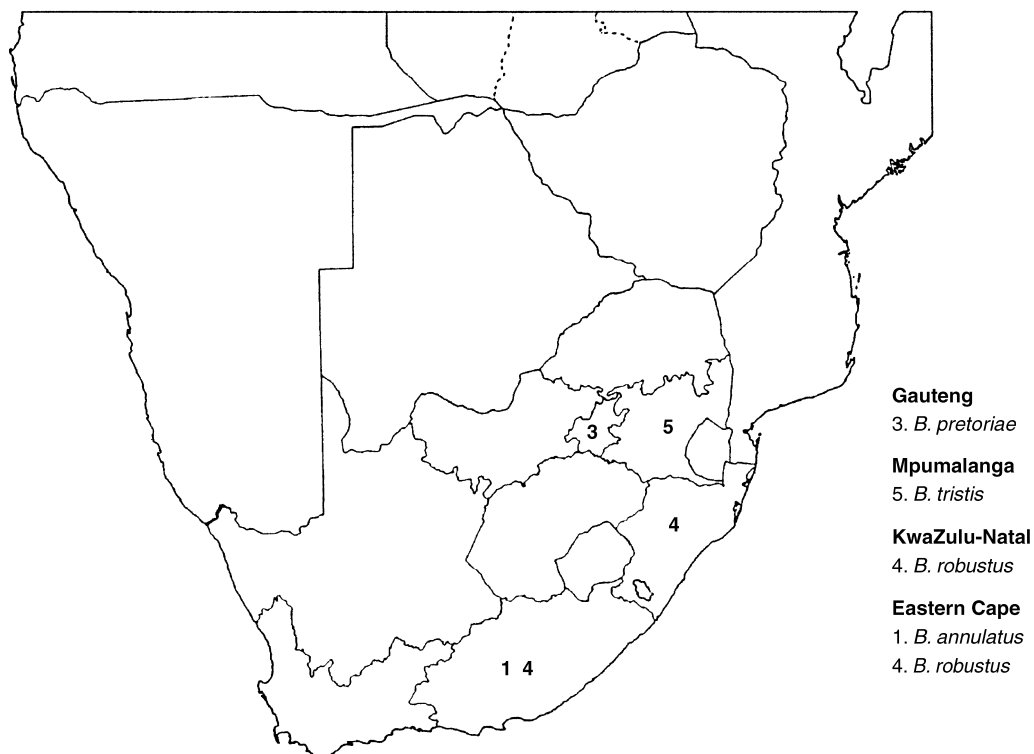


Fig. 68. Distribution of *Brachionopus* species in Southern Africa.

only from the southeastern regions of South Africa where it is represented by five species (fig. 68).

List of species from Southern Africa

1. *Brachionopus annulatus* Purcell, 1903c (Kentani orange lesser baboon spider)

Brachionopus annulatus Purcell, 1903c: 105; Roewer, 1942: 219; Smith, 1990a: 67.

Type locality: female holotype (SAM 12752), Kentani (32.30S; 28.21E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Kentani).

2. *Brachionopus leptopelmiformis* Strand, 1907a

Brachionopus leptopelmiformis Strand, 1907a: 552; 1907c: 204; Roewer, 1942: 219.

Type locality: male holotype (Lübeck Museum) from Cape Province (no exact locality).

Distribution: South Africa (known only from type locality).

3. *Brachionopus pretoriae* Purcell, 1904 (Pretoria lesser baboon spider)

Brachionopus pretoriae Purcell, 1904: 126; Roewer, 1942: 219.

Type locality: female holotype (TM 12769) from Pretoria (25.35S; 28.11E), Gauteng Province.

Distribution: South Africa (Gauteng: Pretoria).

4. *Brachionopus robustus* Pocock, 1897 (East London golden brown lesser baboon spider)

Brachionopus robustus Pocock, 1897: 740; Simon, 1903c: 915; Hewitt, 1921: 6; Roewer, 1942: 219; Smith, 1990a: 67; Platnick, 1993: 102.

Type locality: female holotype (BMNH 91-8-2), East London (33.01S; 27.58E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: East London; KwaZulu-Natal: Hluhluwe Game Reserve).

5. *Brachionopus tristis* Purcell, 1903c (Barberton yellow lesser baboon spider)

Brachionopus tristis Purcell, 1903c: 105; Roewer, 1942: 219; Smith, 1990a: 67.

Type locality: female holotype (SAM 4197), Barberton (25.48S; 31.03E), Mpumalanga Province.

Distribution: South Africa (Mpumalanga: Barberton).

Genus *CERATOGYRUS* Pocock, 1897 horned baboon spiders

Ceratogyrus Pocock, 1897: 754; Roewer, 1942: 268; Platnick, 1989: 97; Smith, 1990a: 67; 1990b: 11; De Wet & Dippenaar-Schoeman, 1991: 43.

Coelogenium Purcell, 1902a: 338; Smith, 1990a: 75; Gallon, 2001: 2 (synonym).

Type species: *Ceratogyrus darlingi* Pocock, 1897.

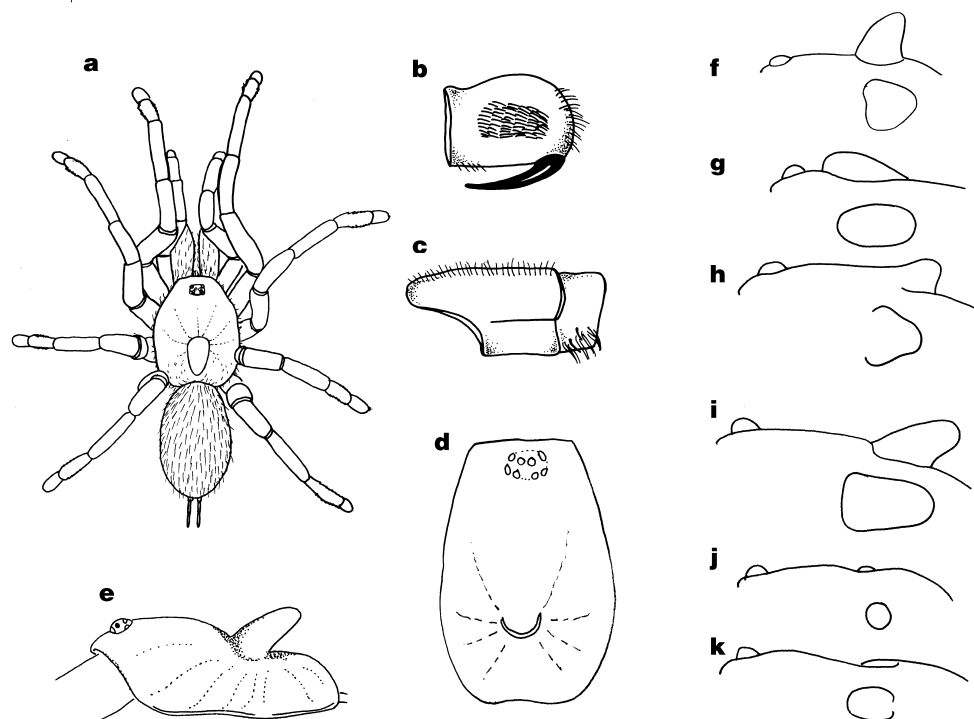


Fig. 69. Theraphosidae — *Ceratogyrus* spp. **a**: female, dorsal view; **b**: chelicera with pad of plumose setae; **c**: coxa; **d**: carapace, dorsal view; **e**: carapace, lateral view; **f–h**: carapace in lateral view, and dorsal view of foveal tubercle, of (**f**) *C. darlingi*, (**g**) *C. brachycephalus*, (**h**) *C. dolichocephalus*, (**i**) *C. bechuanicus*, (**j**) *C. marshalli* and (**k**) *C. sanderi*. (f–k after De Wet & Dippenaar-Schoeman, 1991.)

Diagnostic characters

Foveal tubercle a low dome or prominent horn (fig. 67e–i), varies in shape between species; eyes grouped in a small rectangle on a compact tubercle; rastellum absent; dense scopulae on side of chelicerae (fig. 69b); labium and endites with cuspules; tarsal and metatarsal scopulae ventrally on all legs; abdomen oval, usually with fishbone pattern dorsally, uniformly setose; male with mating spur on tibia I; body size 30–55 mm.

Natural history

Females and juveniles live in silk-lined burrows usually made in areas sparsely covered with grass (De Wet & Dippenaar-Schoeman, 1991). Openings of burrows are frequently found in open areas near grass tufts. The diameter of the entrance varies from 20–25 mm and the depth from 240–400 mm. Some burrows are wider at the surface but taper inwards. Most burrows are J-shaped, but shapes vary depending on pre-existing cavities.

Storage places are sometimes found as hollowed-out portions of the side wall near the bottom of the burrow. Eggs are usually laid in summer and are deposited in an egg sac at the bottom of the burrow. The number of eggs varies between 30 and 44 and can hatch within two weeks. The males wander around in search of a mate. Kirk (1990) reported on the breeding of *Ceratogyrus bechuanicus* and De Wet & Schoonbee (1991) on the occurrence and conservation status of some *Ceratogyrus* spp. in South Africa.

Distribution

Ceratogyrus is known by 10 species from southern Sudan and Malawi and seven from Southern Africa (fig. 70).

Gallon (2001) described *C. ezendami* from Mozambique, but without exact locality data. This species is not included in the following key:

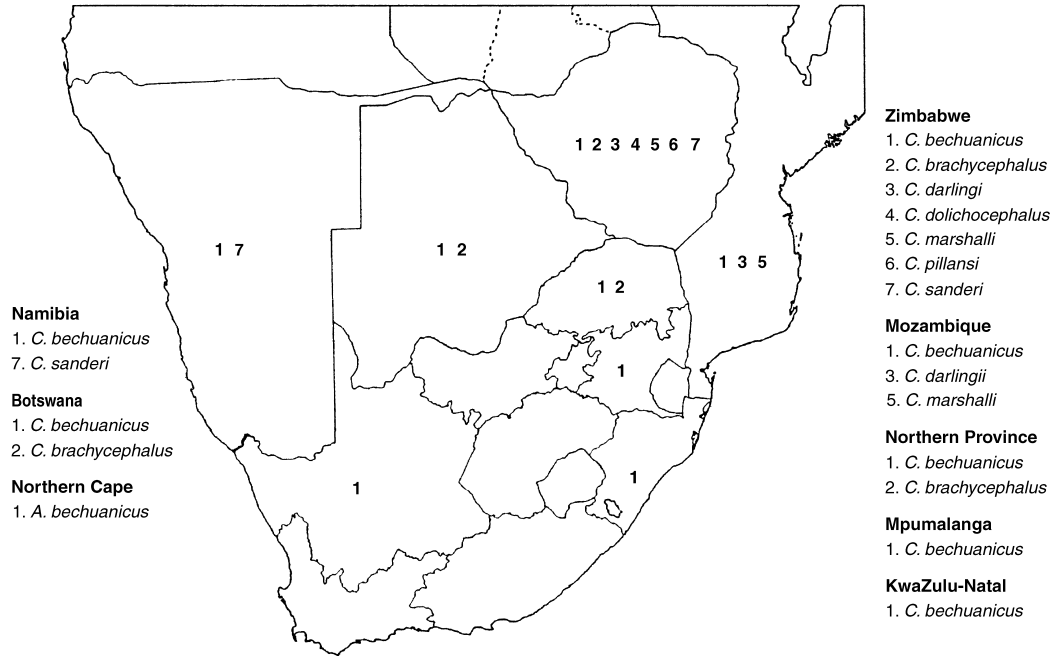


Fig. 70. Distribution of *Ceratogyrus* species in Southern Africa.

KEY TO THE SOUTHERN AFRICAN SPECIES OF CERATOGYRUS

(after De Wet & Dippenaar-Schoeman, 1991; Gallon, 2001)

1. Fovea procurved without distinct, laterally visible protuberance (fig. 69d) *C. pillansi*
- Fovea with low-set plug or clearly visible tubercle 2
2. Foveal tubercle distinct and clearly visible laterally (fig. 67g) 3
- Foveal tubercle a low dome, round or rectangular, not clearly visible laterally (fig. 69j) 6
3. Foveal tubercle a posteriad extension of the cephalic region (fig. 69h) *C. dolichocephalus*
- Foveal tubercle not a posteriad extension of the cephalic region, inclined posteriad or anteriad, or vertical 4
4. Foveal tubercle inclined posteriad (fig. 69i) 5
- Foveal tubercle inclined anteriad (fig. 69g) *C. brachycephalus*
5. Foveal tubercle broad and obtuse, strongly inclined posteriad, anterior slope of tubercle slightly curved (fig. 69i) *C. bechuanicus*

- Foveal tubercle slender, straight, conical, inclined posteriad, anterior slope of tubercle not curved (fig. 69f) *C. darlingi*
- 6. Foveal tubercle a low round dome (fig. 69j) *C. marshalli*
- Foveal tubercle a low rectangular dome (fig. 69k) *C. sanderi*

Species recorded from Southern Africa

1. *Ceratogyrus bechuanicus* Purcell, 1902a (starbust horned baboon spider)

Ceratogyrus bechuanicus Purcell, 1902a: 339; Lawrence, 1927a: 2; 1927b: 219; 1936: 145; Roewer, 1942: 268; Smith, 1987: 128; 1990a: 70; De Wet & Dippenaar-Schoeman, 1991: 45; Griffin & Dippenaar-Schoeman, 1991: 156; Platnick, 1998: 155; Schmidt, 1993: 61; Platnick, 1989: 98; Platnick, 1993: 102; Fitzpatrick, 2001: 177.

Ceratogyrus schultzei Purcell, 1908: 213; Roewer, 1942: 268; Smith, 1987: 129; 1990a: 75; De Wet & Dippenaar-Schoeman, 1991: 45 (*C. schultzei* synonym).

Type locality: two male syntypes (badly damaged) (SAM 4539), Mochudi, Botswana.

Distribution: Botswana, Central Namibia, Zimbabwe (widespread), Mozambique and South Africa (Northern Province: widespread in the province; Mpumalanga:

northeastern parts; *Northern Cape*: northern parts; *KwaZulu-Natal*: Mkuzi).

2. *Ceratogyrus brachycephalus* Hewitt, 1919b

(rhino-horned baboon spider)

Ceratogyrus brachycephalus Hewitt, 1919b: 103; Lessert, 1936: 208; Roewer, 1942: 268; Smith, 1987: 128; 1990a: 70; De Wet & Dippenaar-Schoeman, 1991: 50; Platnick, 1993: 102; Fitzpatrick, 2001: 177.

Type locality: lectotype female and paralectotype females (TM 2992, 2993, 2994), Njelele River, Soutpansberg, Northern Province, South Africa.

Distribution: South Africa (*Northern Province*: Soutpansberg, Messina, Maasstroom), central Botswana and Zimbabwe (widespread).

3. *Ceratogyrus darlingi* Pocock, 1897

(East African horned baboon spider)

Ceratogyrus darlingi Pocock, 1897: 754; Purcell, 1902a: 340; Roewer, 1942: 268; Smith, 1987: 128; 1990a: 71; De Wet & Dippenaar-Schoeman, 1991: 54; Schmidt, 1993: 116; Platnick, 1989: 98; 1993: 102; 1998: 155; Fitzpatrick, 2001: 177.

Type locality: female lectotype (BMNH 1897/4/6/3/5), Enkeldoorn (= Chivu), Zimbabwe.

Distribution: central and eastern Zimbabwe, and Mozambique.

4. *Ceratogyrus dolichocephalus* Hewitt, 1919b

Ceratogyrus dolichocephalus Hewitt, 1919b: 104; Roewer, 1942: 268; Smith, 1987: 128; 1990a: 73; De Wet & Dippenaar-Schoeman, 1991: 57; Platnick, 1993: 102; Fitzpatrick, 2001: 177; Gallon, 2001: 4.

Type locality: female lectotype and female paralectotype (TM 2990, 2991), Victoria, Zimbabwe.

Distribution: Zimbabwe (widespread).

5. *Ceratogyrus marshalli* Pocock, 1897

(Zimbabwe lesser horned baboon spider)

Ceratogyrus marshalli Pocock, 1897: 754; Purcell, 1902a: 340; Roewer, 1942: 268; Smith 1987: 128; 1990a: 74; De Wet & Dippenaar-Schoeman, 1991: 60; 1988: 128; Platnick, 1993: 102; Fitzpatrick, 2001: 177.

Ceratogyrus cornuatus De Wet & Dippenaar-Schoeman, 1991: 52; Gallon, 2001: 10 (synonym).

Type locality: male lectotype (BMNH 1897/4/6/1), Salisbury (= Harare), Zimbabwe.

Distribution: northeastern Zimbabwe and Mozambique.

6. *Ceratogyrus pillansi* (Purcell, 1902a)

(Zimbabwe golden half-moon baboon spider)

Coelogenium pillansi Purcell, 1902a: 338; Simon, 1903c: 948; Roewer, 1942: 268; Smith, 1990a: 75; Schmidt, 1993: 116; Platnick, 1993: 102; 1998: 156; Fitzpatrick, 2001: 177.

Ceratogyrus pillansi, Gallon, 2001: 15.

Type locality: female holotype (NM 5749), Balla Balla, Zimbabwe.

Distribution: southern Zimbabwe and Malawi.

7. *Ceratogyrus sanderi* Strand, 1906a

(Windhoek lesser horned baboon spider)

Ceratogyrus sanderi Strand, 1906a: 23; 1907c: 239; Roewer, 1942: 268; Smith, 1987: 129; 1990: 75; De Wet & Dippenaar-Schoeman, 1991: 62; Griffin & Dippenaar-Schoeman, 1991: 156; Platnick, 1993: 102; Fitzpatrick, 1994: 121; 2001: 177.

Type locality: male holotype (destroyed, Museum Stuttgart) from Windhoek, Namibia (redescription De Wet & Dippenaar-Schoeman, 1991).

Distribution: Namibia and northwestern Zimbabwe.

Genus *HARPACTIRA* Ausserer, 1871

common baboon spiders

Harpactira Ausserer, 1871: 204; Pocock, 1897: 749; Purcell, 1902a: 333; Roewer, 1942: 269; Smith, 1990a: 79.

Type species: *Harpactira atra* (Latreille, 1832).

Diagnostic characters

Pad of plumose setae present between chelicerae when viewed from above (less distinct in juveniles); pad of plumose setae also present on side of chelicerae, with a row of long, stout setae below with corresponding group of stout, plumose setae on palpal coxa (fig. 71b,c); fovea transverse (fig. 71a); clypeus wide; male tibial mating spur a single mound with a spine; male palp with embolus tapering; colour varies from mouse-brown to greenish black to golden-yellow; carapace frequently with radiating bands and pale border; abdomen frequently with median line and bands; darker ventrally; body size 26–64 mm.

Natural history

Harpactira species are usually found in fynbos, *Acacia* scrub, grassland and dry forest. Some species such as *H. guttata* Strand have been recorded from tropical coastal rainforests (Smith, 1990a). *Harpactira* is an opportunistic burrower that frequently excavates retreats beneath rocks and logs, or uses old animal burrows, as does *H. atra*. The latter is commonly found on the Cape Peninsula and Robben Island where it lives in silk-lined burrows made under stones, frequently on hillsides or on open sandy plains. These spiders use old mole burrows if no stones are available (Purcell, 1902a). *Harpactira baviana*, another common species, makes similar shallow excavations under stones, but some live in burrows a few centimetres deep (Purcell, 1903b). A female of *H. curator* was collected from under stones in a large web-lined chamber

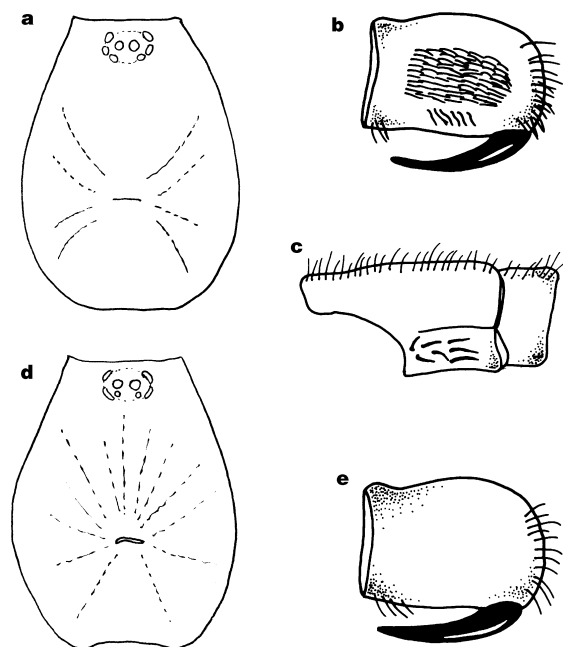


Fig. 71. Theraphosidae. **a-c:** *Harpactira* sp. **a:** carapace, dorsal view; **b:** chelicera with pad of plumose setae and row of stout setae; **c:** stout setae on palpal coxa; **d-e:** *Harpactirella* sp. **d:** carapace, dorsal view; **e:** chelicera without pad of plumose setae.

with two exit passages (Pocock, 1898). Paulsen (1999a) reported on prey items.

Distribution

Harpactira species seem to be mainly restricted to South Africa, with only *H. namaquensis* found in Namibia and *H. tigrina* in Zimbabwe (fig. 72). Pocock (1897) mentioned a record of *H. tigrina* from Somaliland. However, Smith (1990a), who has collected widely in Africa, considers this a dubious record.

Species recorded from Southern Africa

1. *Harpactira atra* (Latreille, 1832)

(Common pigmy Cape baboon spider)

Mygale atra Latreille, 1832: 70.

Mygale funebra Walckenaer, 1837: 226.

Mygale (Eurypelma) coracina C.L. Koch, 1842: 37; Pocock, 1897: 749 (synonym).

Mygale funebria C.L. Koch, 1842: 81.

Harpactira atra Ausserer, 1871: 204; Simon, 1892a: 150; Pocock, 1897: 749; Purcell, 1902a: 320; Simon, 1903c: 948; Strand, 1907c: 227; Roewer, 1942: 269; Smith, 1990a: 80; Schmidt, 1993: 118; Platnick, 1993: 107.

Type locality: female (BMNH 01-3-10-3) and male

(BMNH 48-63) syntypes from Cape Province (no exact locality).

Distribution: South Africa (Western Cape: common throughout the Cape Peninsula, Houtbay, Simon's Town, Worcester, Darling, Gordons Bay, Robben Island; KwaZulu-Natal: Zululand (possibly introduced)).

2. *Harpactira baviana* Purcell, 1903b

(Hanover olive starburst baboon spider)

Harpactira baviana Purcell, 1903b: 30; Roewer, 1942: 269; Smith, 1990a: 81; Platnick, 1993: 107.

Type locality: two male (SAM 9562, 9959) and two female syntypes (SAM 9960, 10048), Hanover (31.04S; 27.27E), farms Poortjesfontein and Krapfontein, Northern Cape Province.

Distribution: South Africa (Northern Cape: Hanover).

3. *Harpactira cafreriana* (Walckenaer, 1837)

(Cape orange baboon spider)

Mygale cafreriana Walckenaer, 1837: 225; C.L. Koch, 1842: 80.

Mygale villosa, Walckenaer, 1837: 226.

Harpactira cafreriana, Purcell, 1902a: 325; Roewer, 1942: 269; Smith, 1990a: 81.

Type locality: male and female syntypes (SAM), 'Caffraria, South Africa'.

Note: exact locality not established as there were various areas referred to as 'Caffraria' in the Eastern Cape Province, 'Natal' and 'Transvaal'.

Distribution: South Africa (Caffraria; Western Cape: Cape Peninsula, Jonkershoek near Stellenbosch, Caledon, Knysna, Swellendam, Slanghoek near Worcester).

4. *Harpactira chrysogaster* Pocock, 1897

(banded red-rumped baboon spider)

Harpactira chrysogaster Pocock, 1897: 750; Roewer, 1942: 269; Smith, 1990a: 81; Platnick, 1993: 107.

Type locality: male (badly damaged, BMNH 44-139) and female syntypes (BMNH 98-1-9-1), South Africa (no exact locality).

Distribution: South Africa (Western Cape: Stellenbosch).

5. *Harpactira curator* Pocock, 1898a

(Malvern starburst baboon spider)

Harpactira curator Pocock, 1898a: 199; Roewer, 1942: 269; Smith, 1990a: 83; Platnick, 1993: 107.

Type locality: male (lost) and female syntypes (BMNH 1897-11-4-14-15), Malvern near Durban (29.53S; 30.56E), KwaZulu-Natal.

Distribution: South Africa (KwaZulu-Natal: Malvern and Stamford Hill near Durban).

6. *Harpactira curvipes* Pocock, 1897

(Kleinpoort mouse baboon spider)

Harpactira curvipes Pocock, 1897: 750; Roewer, 1942: 269; Smith, 1990a: 84.

Type locality: juvenile female holotype (BMNH 54-22),

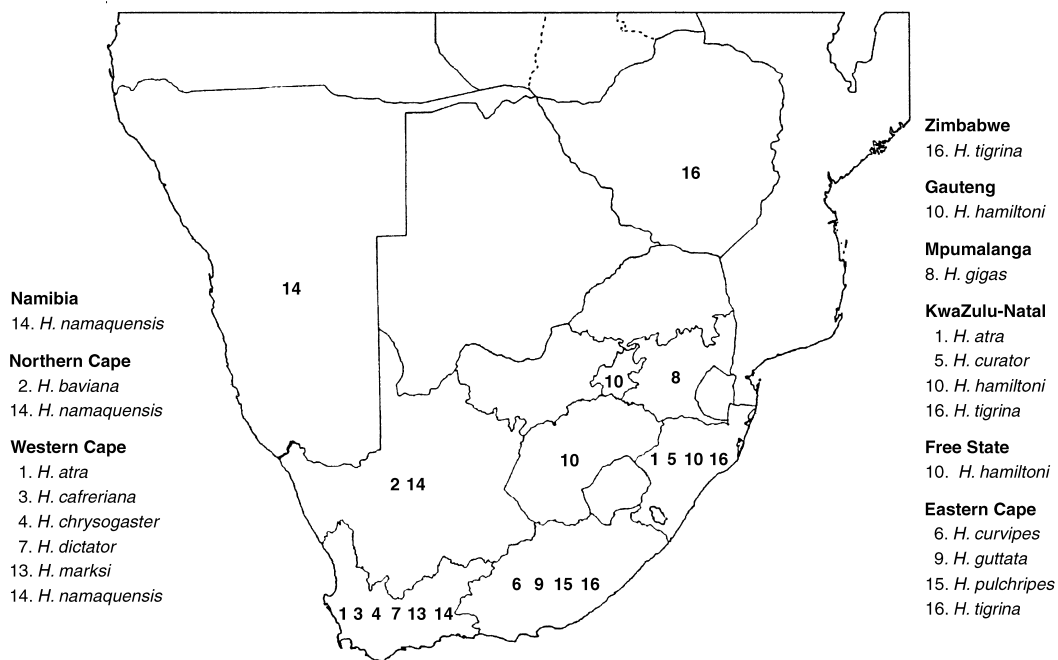


Fig. 72. Distribution of *Harpactira* species in Southern Africa.

'Natal' (no exact locality).

Note: Smith (1991) designated a paralectotype female (BMNH 96-7-24-2) labelled by Pocock as *H. curvipes* from Kleinpoort. 'Kleinpoort' could not be located in KwaZulu-Natal, only a Kleinpoort (33.09S; 26.51E) in the Eastern Cape Province.

Distribution: South Africa (KwaZulu-Natal? and Kleinpoort).

7. *Harpactira dictator* Purcell, 1902b

(*Swellendam baboon spider*)

Harpactira dictator Purcell, 1902b: 331; Roewer, 1942: 269; Smith, 1990a: 84.

Type locality: male and female syntypes (SAM 8838), Bonnie Vale Farm close to Bushman's Drift, a few miles from Ashton (33.50S; 20.05E), Swellendam, Western Cape Province.

Note: Smith (1991) suggested that *H. dictator* could be a junior synonym of *H. lineata* Pocock, 1897.

Distribution: South Africa (*Western Cape*: Bonnie Vale Farm and Zandvliet Farms near Ashton, Avontuur near Storms Vlei (Swellendam), Slanghoek Rabiesberg near Nuy River Station, Worcester, Ladismith).

8. *Harpactira gigas* Pocock, 1898b

(*Transvaal banded baboon spider*)

Harpactira gigas Pocock, 1898b: 316; Strand, 1907c: 230; Purcell, 1902a: 324; Roewer, 1942: 269; Smith, 1990a: 85.

Type locality: female holotype (BMNH 1898-5-7-23), Barberton (25.47S; 31.03E), Mpumalanga Province.

Note: Smith (1990) doubts the Cape record of this species by Strand (1907).

Distribution: South Africa (*Mpumalanga*: Barberton).

9. *Harpactira guttata* Strand, 1907a

Harpactira guttata Strand, 1907a: 553; 1907c: 230; Roewer, 1942: 269; Smith, 1990a: 85.

Type locality: female type (Lübeck Museum), East London (33.01S; 27.55E), Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: East London, Grahamstown).

10. *Harpactira hamiltoni* Pocock, 1902b

(*golden starbust baboon spider*)

Harpactira hamiltoni Pocock, 1902b: 316; Roewer, 1942: 269; Smith, 1990a: 85; Platnick, 1993: 107.

Type locality: male and female syntypes (BMNH), Vredefort (27.01S; 27.22E) Road, Free State Province.

Distribution: South Africa (*Free State*: Vredefort; *Gauteng*: Johannesburg; *KwaZulu-Natal*: Durban).

11. *Harpactira lineata* Pocock, 1897

(*white starbust baboon spider*)

Harpactira lineata Pocock, 1897: 749; Roewer, 1942: 269; Smith, 1990a: 86.

Note: according to Smith (1990) possibly a senior synonym of *H. dictator*.

Type locality: female holotype (BMNH 44-6), South Africa (no exact locality).

Distribution: South Africa.

12. *Harpactira lyrata* (Simon, 1892a)

Planadecta lyrata Simon, 1892a: 169.

Harpactira lyrata, Simon, 1903c: 948; Roewer, 1942: 269; Smith, 1990a: 87.

Type locality: female holotype (MNHP), South Africa (no exact locality).

Distribution: South Africa.

13. *Harpactira marksi* Purcell, 1902a

(*Elizabethfontein baboon spider*)

Harpactira marksi Purcell, 1902a: 322; Roewer, 1942: 269.

Type locality: male and female syntypes (SAM 2161), Gutverwacht Mission Station (32.54S; 18.46E), Piketberg district, Western Cape Province.

Distribution: South Africa (*Western Cape*: Piketberg, Elizabethfontein (Clanwilliam), Kogmans Kloof near Ashton (Robertson)).

14. *Harpactira namaquensis* Purcell, 1902a

(*Okiep baboon spider*)

Harpactira namaquensis Purcell, 1902a: 324; 1908: 212; Strand, 1917: 164; Roewer, 1942: 270; Smith, 1990a: 87; Griffin & Dippenaar-Schoeman, 1991: 157.

Type locality: male and female syntypes (SAM 3975), Okiep (29.35S; 17.52E), Kradfontein, Concor dia,(29.31S;17.59E) and Steinkopf (29.15S; 17.44E) in Namaqualand, Northern Cape Province.

Distribution: South Africa (*Northern Cape*: Okiep, Kradfontein, Concordia, Steinkopf, Van Wyksvlei; *West-ern Cape*: Touws River) and Namibia (Lüderitz Bay, Kuibis).

15. *Harpactira pulchripes* Pocock, 1901a

(*yellow-collared baboon spider*)

Harpactira pulchripes Pocock, 1901a: 287; Roewer, 1942: 270; Smith, 1990a: 87.

Type locality: female holotype and four juveniles (BMNH 01-3-5-32-33), Brakkloof (33.12S; 26.40E, Grahamstown, Eastern Cape Province.

Distribution: South Africa (*Eastern Cape*: Grahams-town, Jansenville).

16. *Harpactira tigrina* Ausserer, 1875

(*common yellow-banded baboon spider*)

Harpactira tigrina Ausserer, 1875: 185; Pocock, 1897: 748; Purcell, 1902a: 328; Strand, 1907c: 233; Roewer, 1942: 270; Smith, 1990a: 88; Platnick, 1993: 107; Fitzpatrick, 2001: 177.

Type locality: female holotype, (BMNH 97-11-10-26), South Africa (no exact locality).

Distribution: South Africa (*Eastern Cape*: Port Elizabeth, Kleinpoort, East London, Bizana (Pondoland), King William's Town, Port Alfred, Grahamstown, Dunbrody Umtata, Bedford district; *KwaZulu-Natal*: Eastern Zululand) and Zimbabwe (Matabeleland).

Note: Pocock (1897) listed a specimen from Somalia

but according to Smith (1990a) it could have been incorrectly labelled.

Genus *HARPACTIRELLA* Purcell, 1902

lesser baboon spiders

Harpactirella Purcell, 1902a: 340; 1903c: 101; Smith, 1990b: 89.

Ashantia Strand, 1908: 770; Raven, 1985: 149 (synonym).

Luphocemus Denis, 1960: 186; Benoit, 1965f: 297 (synonym).

Type species: *Harpactirella treleaveni* Purcell, 1902a.

Diagnostic characters

Fovea transverse (fig. 71d); lacks plumose pad on outside of chelicerae (fig. 71e); clypeus wide; number of cuspules on labium varies from 12–50; male with single tibial spur, capped long spine; scopulae on metatarsus IV usually confined to distal fourth; scopulae on tarsus IV divided by a thin band of setae; spines limited to base of tibia and metatarsi of all legs; serrula absent; body size 13–35 mm.

Taxonomic note

Raven (1985) transferred *Harpactirella* from the Barychelidae to the Theraphosidae. This was accepted by Platnick (1998) but not by Charpentier (1993). Raven (1985) considered *Harpactirella* a senior synonym of *Ashantia* Strand, 1908 (West Africa), and *Luphocemus* Denis, 1960 (Morocco).

Natural history

Harpactirella are commonly found in dry grass-land, *Acacia* veld and forests, and their burrows range from silk-lined excavations under rocks to branched burrows. A female of *H. domicola* was collected from a cylindrical burrow (17–18 cm deep) that widened towards the opening. The burrow was strongly inclined and descended towards the hinge side at an angle of nearly 45°. The burrow was covered with a flat lid, oval in outline, but broadly truncated at the hinge. The underside of the lid was lined with white silk but the upper part was rough and covered with clay. The lid broadly overlapped the edge of the burrow (Purcell, 1903c).

Harpactirella treleaveni has been collected from silk-lined chambers under stones, similar to those of *Harpactira* species (Purcell, 1902a).

Toxicity

Harpactira lightfooti is the only species of baboon spider from Southern Africa that is of

medical importance. These spiders are fairly aggressive and people sometimes get bitten (G. Müller, pers. comm.). Finlayson (1939) was the first to report on the neurotoxic effect of the venom. Bites in humans result in a burning pain at the site of the bite. After about two hours, victims start to vomit and show distinct signs of shock, turning pale and walking with difficulty.

Distribution

According to Raven (1985), *Harpactirella* is distributed throughout Southern Africa (fig. 73), West Africa and southern Morocco.

Species recorded from Southern Africa

1. *Harpactirella domicola* Purcell, 1903c

(yellow-legged lesser baboon spider)

Harpactirella domicola Purcell, 1903c: 103; Roewer, 1942: 221; Smith, 1990a: 90.

Type locality: female holotype (SAM 8897), Bonnie Vale farm at Bushman's Drift, on the Breede River near Ashton (33.50S; 20.05E), Swellendam, Western Cape Province.

Distribution: South Africa (Western Cape: Swellendam).

2. *Harpactirella flavipilosa* Lawrence, 1936

(Botswana lesser baboon spider)

Harpactirella flavipilosa Lawrence, 1936: 145; Roewer, 1942: 221; Griffin & Dippenaar-Schoeman, 1991: 157.

Type locality: female holotype (TM), Kabulabula (17.50S; 24.58E) Botswana.

Distribution: Botswana, Namibia, South Africa (North West Province: Brits; Gauteng: Roodepoort).

3. *Harpactirella helenae* Purcell, 1903c

(St Helena Bay lesser baboon spider)

Harpactirella helenae Purcell, 1903c: 101; Roewer, 1942: 221; Smith, 1990a: 90.

Type locality: male and female syntypes (SAM 11710), Stompneus, St Helena Bay (32.45S; 18.05E), Malmesbury, Western Cape Province.

Distribution: South Africa (Western Cape: Malmesbury).

4. *Harpactirella karrooica* Purcell, 1902a

(Prince Albert sunburst lesser baboon spider)

Harpactirella karrooica Purcell, 1902a: 344; Roewer, 1942: 221; Smith, 1990a: 90.

Type locality: male and female syntypes (SAM 3432), Prince Albert (33.13S; 22.02E), Western Cape Province.

Distribution: South Africa (Western Cape: Prince Albert).

5. *Harpactirella lapidaria* Purcell, 1908

(Steinkopf mustard lesser baboon spider)

Harpactirella lapidaria Purcell, 1908: 214; Roewer, 1942: 221;

Griffin & Dippenaar-Schoeman, 1991: 157; Smith, 1990a: 90; Platnick, 1993: 108.

Type locality: male and female syntypes (SAM), Steinkopf (29.15S; 17.44E), Little Namaqualand, Northern Cape Province.

Distribution: South Africa (Northern Cape: Steinkopf).

6. *Harpactirella leleupi* Benoit, 1965f

(Zimbabwe lesser chestnut baboon spider)

Harpactirella leleupi Benoit, 1965f: 298; Brignoli, 1983: 130; Smith, 1990a: 91; Fitzpatrick, 2001: 177.

Type locality: female holotype (MRAC 116-016), Melsetter, 64 km N Chipinga, Zimbabwe.

Distribution: Zimbabwe.

7. *Harpactirella lightfooti* Purcell, 1902a

(Lightfoot's lesser baboon spider)

Harpactirella lightfooti Purcell, 1902a: 346; 1903c: 104; Roewer, 1942: 221; Smith, 1990a: 91.

Note: Purcell (1903c) suggested that *H. longipes* could be the male of *H. lightfooti*.

Type locality: female holotype (SAM 3219), Paarl, Cape Town (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (Western Cape: Cape Town, Paarl).

8. *Harpactirella longipes* Purcell, 1902a

(Clanwilliam lesser baboon spider)

Harpactirella longipes Purcell, 1902a: 343; 1908: 215; Roewer, 1942: 221; Smith, 1990a: 91; Platnick, 1993: 108.

Type locality: male holotype (SAM 3567), Clanwilliam (32.13S; 18.59E), Western Cape Province.

Distribution: South Africa (Western Cape: Olyvenbosch near Bergvlei River and Onder Bergvlei (both Clanwilliam district), Porterville (Piketberg), Boschklouf Waterfall in Cedarberg Mountains).

9. *Harpactirella magna* Purcell, 1903c

(Dunbrody golden yellow lesser baboon spider)

Harpactirella magna Purcell, 1903c: 102; Roewer, 1942: 221; Smith, 1990a: 91.

Type locality: female holotype (SAM 5248), Dunbrody (33.28S; 25.33E), Sundays River, Uitenhage, Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Dunbrody).

10. *Harpactirella schwarzi* Purcell, 1904

(Willowmore lesser baboon spider)

Harpactirella schwarzi Purcell, 1904: 126; Roewer, 1942: 221.

Type locality: female holotype (SAM 12928), Vleikuil, north of Blyde Berg, Willowmore (33.18S; 23.30E), Eastern Cape Province.

Distribution: South Africa (Eastern Cape: Willowmore).

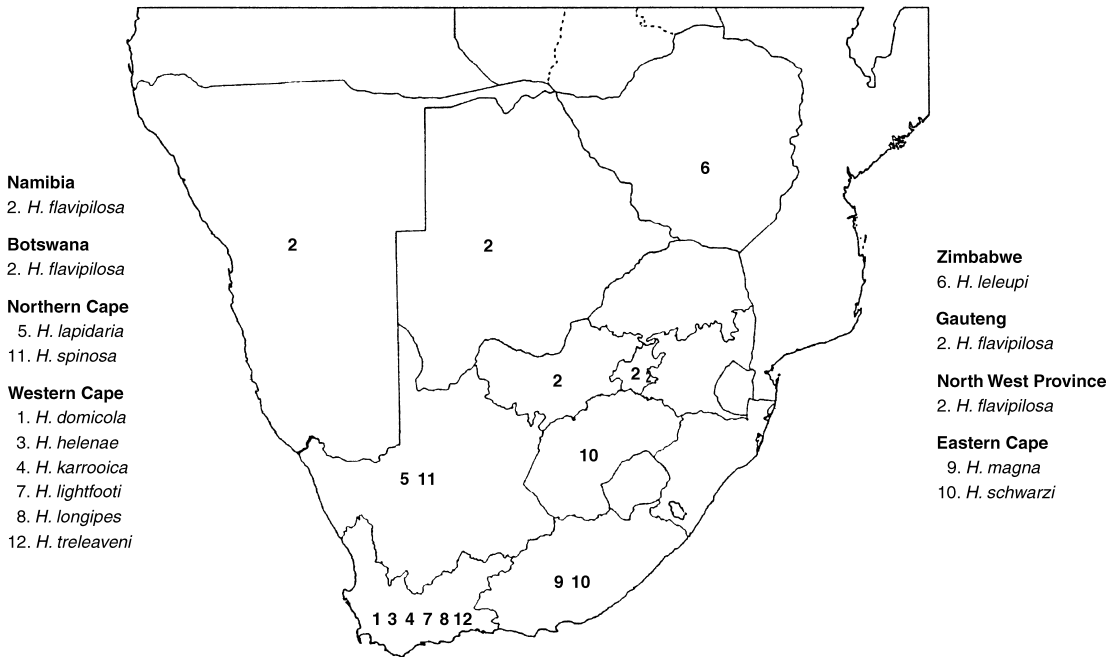


Fig. 73. Distribution of *Harpactirella* species in Southern Africa.

11. *Harpactirella spinosa* Purcell, 1908

(Steinkopf brown lesser baboon spider)

Harpactirella schwarzi Purcell, 1908: 214; Roewer, 1942: 221; Griffin & Dippenaar-Schoeman, 1991: 157; Smith, 1990a: 91; Platnick, 1993: 108.

Type locality: male and female syntypes (Schultz collection), Steinkopf (29.15S; 17.44E), Little Namaqualand, Northern Cape Province.

Distribution: South Africa (Northern Cape: Steinkopf, Kamaggas).

12. *Harpactirella treleaveni* Purcell, 1902a

(Table Mountain golden lesser baboon spider)

Harpactirella treleaveni Purcell, 1902a: 341; Roewer, 1942: 221; Smith, 1990a: 92.

Type locality: male and female syntypes (SAM 4496), Table Mountain on Cape Town side (33.56S; 18.28E), Western Cape Province.

Distribution: South Africa (Western Cape: Cape Town, Cape Peninsula, Signal Hill, Devils Peak, Camps Bay).

Genus *PTERINCHILUS* Pocock, 1897

Pterinochilus Pocock, 1897: 752; Berland, 1914: 49; Laurent, 1946: 316; Roewer, 1942: 270; Smith, 1990a: 92. *Idiothele* Hewitt, 1919b: 101; Roewer, 1942: 221; Raven, 1985: 154 (synonym).

Type species: *Pterinochilus vorax* Pocock, 1897.

Diagnostic characters

Plumose setae absent from inside of chelicerae

and coxae of pedipalp; large plumose pad on side of chelicerae but stout setae beneath plumose pad absent; clypeus wide; spines on tibiae and metatarsi III and IV; single tibial spur in male (except *P. junodi*); embolus long, sometimes with a simple keel; body size 10–55 mm.

Taxonomic note

Idiothele is considered a junior synonym of *Pterinochilus* by Raven (1985). The genus is closely related to *Harpactira* but differs by both lacking setae on the inside of the chelicerae and stridulatory setae. The genus is currently being revised (Gallon, in press).

Natural history

Pterinochilus species inhabit silk-lined chambers beneath rocks or logs with a single entrance that is heavily lined with silk, or in burrows made beside rocks. Trapdoors were reported for the first two species described in the genus *Idiothele*. The retreat of *P. nigrofulvus* was described as tubular with a well-developed trapdoor. The door is very large and thin, becoming very delicate at the margin. It is semi-circular or D-shaped (Hewitt, 1919b). Similar retreats have been reported for *P. pluridentatum*, with the hinge of the lid 20 mm

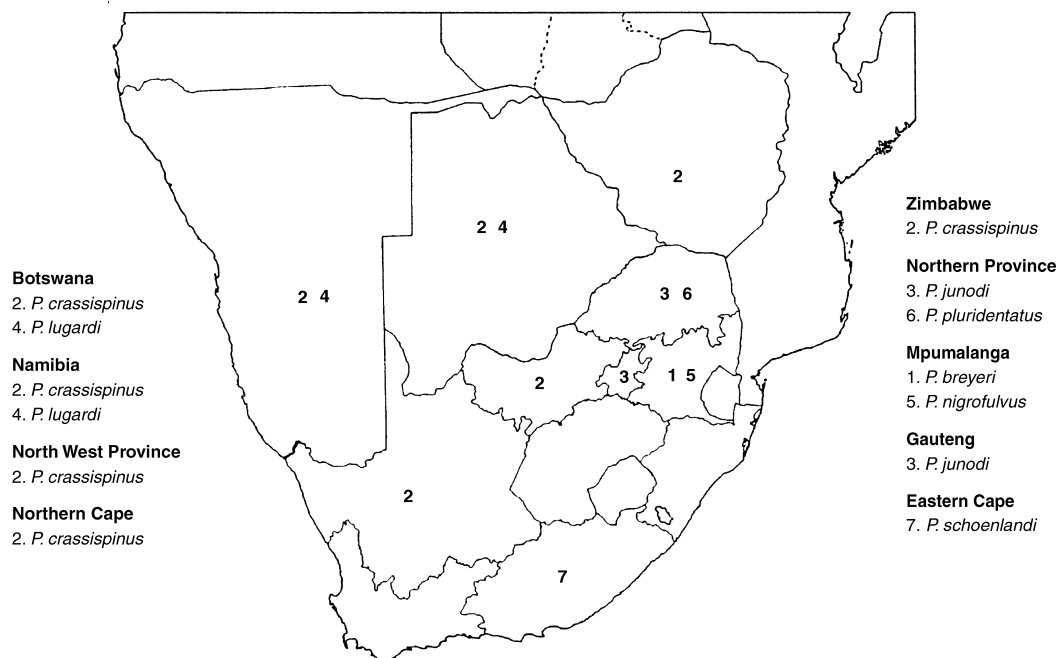


Fig. 74. Distribution of *Pterinochilus* species in Southern Africa.

in length (Hewitt, 1919b). However, Smith (1990a) reported that *P. nigrofulvus* only excavates chambers under rocks. The burrow of *P. junodi* is deep, with a short, vertical shaft leading into the ground, without a trapdoor. The burrows are frequently made in hard clayey soil and are common in *Acacia* scrubland (Paulsen, 1998).

Pterinochilus junodi are able to produce two egg sacs per year, one after the other, after mating once. The male of *P. junodi* is small compared to the female (Paulsen, 1998).

Toxicity

Perret (1974a) reported on a *Pterinochilus* species from Tanzania that produces a neurotoxic venom that tested highly potent on mice and guineapigs. However, in man the effect is less severe. Reports on the venom of this species have been published by Perret (1974b) and Perret & Freyvogel (1973).

Distribution

The genus is represented by 24 species, seven of which are known from Southern Africa (fig. 74). It is widely distributed in the eastern parts of Africa, from South Africa to Ethiopia.

Species recorded from Southern Africa

1. *Pterinochilus breyeri* Hewitt, 1919b

(Malelane golden-brown baboon spider)

Pterinochilus breyeri Hewitt, 1919b: 102; Roewer, 1942: 270; Smith, 1990a: 94.

Type locality: female holotype (TM), Malelane (25.29S; 31.31E), Mpumalanga Province.

Distribution: South Africa (Mpumalanga: Malelane, Hectorspruit, Gollel).

2. *Pterinochilus crassispinus* Purcell, 1902a (Common Zimbabwe autumn-gold baboon spider)

Pterinochilus crassispina Purcell, 1902a: 335; Strand, 1917: 165; Roewer, 1942: 270; Smith, 1989: 13; 1990a: 94; Griffin & Dippenaar-Schoeman, 1991: 157; Fitzpatrick, 2001: 177.

Pterinochilus crassispinus Platnick, 1993: 113.

Type locality: male holotype (TM 6252), Matopo district, Matabeleland, Zimbabwe.

Distribution: Zimbabwe, Namibia (Okahandja), Botswana and South Africa (North West Province: Vryburg; Northern Cape: Barkly West).

3. *Pterinochilus junodi* Simon, 1904a

(Soutpansberg starburst baboon spider)

Pterinochilus junodi Simon, 1904a: 65; Roewer, 1942: 270.

Type locality: female holotype (MNHN), Shilouvane, 30 km from Leydsdorp (24.08S; 29.19E), Northern Province.

Distribution: South Africa (Northern Province: Leydsdorp; Gauteng: Hammanskraal).

4. *Pterinochilus lugardi* Pocock, 1900a*(Botswana baboon spider)**Pterinochilus lugardi* Pocock, 1900a: 318; Roewer, 1942: 271; Smith, 1990a: 96; Griffin & Dippenaar-Schoeman, 1991: 157; Platnick, 1993: 114.**Type locality:** male holotype (BMNH1899-3-10-1), Kwebe Hills near Lake Ngami, Botswana.**Distribution:** Botswana and Namibia.**Note:** Erroneously reported from South Africa by Roewer (1942) and Smith (1990a).**5. *Pterinochilus nigrofulvus* Pocock, 1898b***(Transvaal golden baboon spider)**Pterinochilus nigrofulvus* Pocock, 1898b: 317; Smith, 1990a: 100; Platnick, 1993: 114.*Idiothele nigrofulvus*, Hewitt, 1919b: 100; Roewer, 1942: 221.**Note:** Hewitt (1919) suggested that *P. crassispinus* could be a junior synonym of this species.**Type locality:** male and female syntypes (BMNH 98-5-7-24a / 89-5-7-24), Barberton, Mpumalanga Province.**Distribution:** South Africa (*Mpumalanga*: Barberton, Hectorspruit, Malelane).**6. *Pterinochilus pluridentatus* (Hewitt, 1919b) comb. nov.***Idiothele pluridentatum* Hewitt, 1919b: 101.*Idiothele pluridentata*, Roewer, 1942: 221.**Type locality:** female holotype, Nuanetzi River (24.27; 31.59E), Soutpansberg district, Northern Province.**Distribution:** South Africa (*Northern Province*: Soutpansberg district).**7. *Pterinochilus schoenlandi* Pocock, 1900a**
*(Grahamstown baboon spider)**Pterinochilus schönlandi* Pocock, 1900a: 318; Roewer, 1942: 271; Smith, 1990a: 101; Platnick, 1993: 114.**Type locality:** male holotype (BMNH 99-7-24-37), paratype (BMNH01-3-13-108), Grahamstown (33.19S; 26.32E), Eastern Cape Province.**Distribution:** South Africa (*Eastern Cape*: Grahams-town).

Female baboon spider
(Theraphosidae: *Pterinochilus nigrofulvus*).

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APPENDIX I

Alphabetical list of Southern African Mygalomorphae families, genera and species

1. FAMILY ATYPIDAE Thorell, 1870

Genus *Calommata* Lucas, 1837

1. *C. simoni* Pocock, 1903

2. FAMILY BARYCHELIDAE Simon, 1889

Genus *Cyphonisia* Simon, 1889

1. *C. arcturus* (Tucker, 1917)
2. *C. macequece* (Tucker, 1920)
3. *C. selindanus* (Benoit, 1965)

Genus *Pisenor* Simon, 1889

1. *P. notius* Simon, 1889

Genus *Sipalolasma* Simon, 1892

1. *S. humicola* (Benoit, 1965)

3. FAMILY CTENIZIDAE Thorell, 1887

Genus *Stasimopus* Simon, 1892

1. *S. artifex* Pocock, 1902
2. *S. astutus* Pocock, 1902
3. *S. bimaculatus* Purcell, 1903
4. *S. brevipalpis* Purcell, 1903
5. *S. caffrus* (C.L. Koch, 1842)
6. *S. castaneus* Purcell, 1903
7. *S. coronatus* Hewitt, 1915
8. *S. dreyeri* Hewitt, 1915
9. *S. erythrognathus* Purcell, 1903
10. *S. fordii* Hewitt, 1927
11. *S. gigas* Hewitt, 1915
- 12a. *S. insculptus* Pocock, 1901
- 12b. *S. insculptus peddiensis* Hewitt, 1917
13. *S. kentanicus* Purcell, 1903
14. *S. kolbei* Purcell, 1903
15. *S. leipoldti* Purcell, 1902
16. *S. longipalpis* Hewitt, 1917
17. *S. maraisi* Hewitt, 1914
18. *S. minor* Hewitt, 1915
19. *S. nanus* Tucker, 1917
20. *S. nigellus* Pocock, 1902
21. *S. obscurus* Purcell, 1908
22. *S. oculatus* Pocock, 1897
23. *S. palpiger* Pocock, 1902
24. *S. patersonae* Hewitt, 1913
25. *S. poweri* Hewitt, 1915
26. *S. purcelli* Tucker, 1917
27. *S. quadratimaculatus* Purcell, 1903
28. *S. qumbu* Hewitt, 1913
29. *S. robertsi* Hewitt, 1910
30. *S. rufidens* (Ausserer, 1871)
31. *S. schoenlandi* Pocock, 1900
32. *S. schreineri* Purcell, 1903
33. *S. schultzei* Purcell, 1908
34. *S. spiniceps* Hewitt, 1917
35. *S. spinosus* (Hewitt, 1914)

36. *S. steynsburgensis* Hewitt, 1915

37. *S. suffusus* Hewitt, 1916

38. *S. tysoni* Hewitt, 1919

- 39a. *S. umtalicus* Purcell, 1903

- 39b. *S. umtalicus rangeri* Hewitt, 1927

44. *S. unispinosus* Purcell, 1903

4. FAMILY CYRTAUCHENIIDAE Simon, 1892

Genus *Ancylotrypa* Simon, 1889

1. *A. barbertoni* (Hewitt, 1913)
2. *A. bicornuta* Strand, 1906
3. *A. brevicornis* (Hewitt, 1919)
4. *A. brevipalpis* (Hewitt, 1916)
5. *A. breyeri* (Hewitt, 1919)
6. *A. bulcocki* (Hewitt, 1916)
7. *A. coloniae* (Pocock, 1902)
8. *A. cornuta* Purcell, 1904
9. *A. crudeni* (Hewitt, 1915)
10. *A. dentata* (Purcell, 1903)
11. *A. dreyeri* (Hewitt, 1915)
12. *A. elongata* Purcell, 1908
13. *A. flavidofusula* (Hewitt, 1915)
14. *A. granulata* (Hewitt, 1935)
15. *A. lateralis* (Purcell, 1902)
16. *A. magnisigillata* (Hewitt, 1914)
17. *A. namaquensis* (Purcell, 1908)
18. *A. nigriceps* (Purcell, 1902)
19. *A. nuda* (Hewitt, 1916)
20. *A. onelli* (Purcell, 1902)
21. *A. pallidipes* (Purcell, 1904)
22. *A. parva* (Hewitt, 1916)
23. *A. pretoriae* (Hewitt, 1913)
24. *A. pusilla* Purcell, 1903
25. *A. rufescens* (Hewitt, 1916)
26. *A. schultzei* (Purcell, 1908)
27. *A. sororum* (Hewitt, 1916)
28. *A. spinosa* Simon, 1889
29. *A. tookei* (Hewitt, 1919)
30. *A. vryheidensis* (Hewitt, 1915)
31. *A. zebra* (Simon, 1892)
32. *A. zuluensis* (Lawrence, 1937)

Genus *Homostola* Simon, 1892

1. *H. abernethyi* (Purcell, 1903)
2. *H. pardalina* (Hewitt, 1913)
3. *H. reticulata* (Purcell, 1902)
4. *H. vulpecula* Simon, 1892
5. *H. zebrina* Purcell, 1902

5. FAMILY DIPLURIDAE Simon, 1889

Genus *Allothele* Tucker, 1920

1. *A. australis* (Purcell, 1903)
2. *A. caffer* (Pocock, 1902)
3. *A. malawi* Coyle, 1984

4. *A. tereitis* Tucker, 1920

Genus: *Thelechoris* Karsch, 1881

1. *T. striatipes* (Simon, 1889)

6. FAMILY IDIOPIDAE Simon, 1892

Genus *Ctenolophus* Purcell, 1904

1. *C. cregoei* (Purcell, 1902)

2. *C. fenoulheti* Hewitt, 1913

3. *C. kolbei* (Purcell, 1902)

4. *C. oomi* Hewitt, 1913

5. *C. spiricola* (Purcell, 1903)

Genus *Galeosoma* Purcell, 1903

1a. *G. coronatum* Hewitt, 1915

1b. *G. coronatum sphaeroideum* Hewitt, 1919

2. *G. hirsutum* Hewitt, 1916

3. *G. mossambicum* Hewitt, 1919

4. *G. pallidum* Hewitt, 1915

5. *G. pilosum* Hewitt, 1916

6. *G. planiscutatum* Hewitt, 1919

7. *G. pluripunctatum* Hewitt, 1919

8a. *G. robertsi* Hewitt, 1916

8b. *G. robertsi crinitum* Hewitt, 1919

9. *G. schreineri* Hewitt, 1913

10. *G. scutatum* Purcell, 1903

11a. *G. vandami* Hewitt, 1915

11b. *G. vandami circumjunctum* Hewitt, 1919

12. *G. vernayi* Hewitt, 1935

Genus *Gorgyrella* Purcell, 1902

1. *G. hirschhorni* (Hewitt, 1919)

2. *G. inermis* Tucker, 1917

3. *G. namaquensis* Purcell, 1902

4a. *G. schreineri* Purcell, 1903, comb. nov.

4a. *G. schreineri minor* (Hewitt, 1916) comb. nov.

Genus *Heligmomerus* Simon, 1892

1. *H. astutus* (Hewitt, 1915)

2. *H. caffer* Purcell, 1903

3. *H. deserti* Pocock, 1901

Genus *Idiops* Perty, 1833

1. *I. arnoldi* Hewitt, 1914

2. *I. bersebaensis* Strand, 1917

3. *I. briodae* (Schenkel, 1937)

4. *I. castaneus* Hewitt, 1913

5. *I. crudeni* (Hewitt, 1914)

6. *I. damarensis* Hewitt, 1934

7. *I. flaveolus* (Pocock, 1901)

8. *I. fryi* (Purcell, 1903)

9. *I. gerhardti* Hewitt, 1913

10. *I. gracillipes* (Hewitt, 1919)

11a. *I. gunningi* Hewitt, 1913

11b. *I. gunningi elongatus* Hewitt, 1915

12. *I. hamiltoni* (Pocock, 1902)

13. *I. hepburni* (Hewitt, 1919)

14. *I. hirsutus* (Hewitt, 1919)

15. *I. kentanicus* (Purcell, 1930)

16. *I. mafeae* Lawrence, 1927

17. *I. microps* (Hewitt, 1913)

18. *I. nigropilosus* (Hewitt, 1919)

19. *I. palapyi* Tucker, 1917

20. *I. pallidipes* Purcell, 1908

21. *I. parvus* Hewitt, 1915

22. *I. pretoriae* (Pocock, 1898)

23. *I. pulcher* Hewitt, 1914

24. *I. pulloides* Hewitt, 1919

25. *I. pullus* Tucker, 1917

26. *I. pungwensis* Purcell, 1904

27. *I. spiriferus* Roewer, 1942

28. *I. striatipes* Purcell, 1908

29. *I. thorelli* O. P.-Cambridge, 1870

30. *I. vandami* (Hewitt, 1925)

31. *I. versicolor* (Purcell, 1903)

Genus *Segregara* Tucker, 1917

1. *S. abrahami* (Hewitt, 1913)

2. *S. grandis* (Hewitt, 1915)

3. *S. monticola* (Hewitt, 1916) comb. nov.

4. *S. monticoloides* (Hewitt, 1919) comb. nov.

5. *S. mossambicus* (Hewitt, 1919) comb. nov.

6. *S. ochreolus* (Pocock, 1902) comb. nov.

7. *S. paucispinulosus* (Hewitt, 1915)

8. *S. pectinipalpis* (Purcell, 1903) comb. nov.

9. *S. sylvestris* (Hewitt, 1925) comb. nov.

10. *S. transvaalensis* (Hewitt, 1915)

7. FAMILY MICROSTIGMATIDAE Roewer, 1942

Genus *Microstigmata* Strand, 1932

1. *M. amatola* Griswold, 1985

2. *M. geophila* (Hewitt, 1916)

3. *M. lawrencei* Griswold, 1985

4. *M. longipes* (Lawrence, 1938)

5. *M. ukhahlamba* Griswold, 1985

6. *M. zuluense* (Lawrence, 1938)

8. FAMILY MIGIDAE Simon, 1892

Genus *Moggridgea* O. P.-Cambridge, 1875

1. *M. albimaculata* Hewitt, 1925

2. *M. ampullata* Griswold, 1987

3. *M. breyeri* Hewitt, 1915

4. *M. crudeni* Hewitt, 1913

5. *M. dyeri* O. P.-Cambridge, 1875

6. *M. eremicola* Griswold, 1987

7. *M. intermedia* Hewitt, 1913

8. *M. leipoldti* Purcell, 1903

9. *M. loistata* Griswold, 1987

10. *M. microps* Hewitt, 1915

11. *M. mordax* Purcell, 1903

12. *M. pallida* Hewitt, 1914

13. *M. paucispina* Hewitt, 1916

14. *M. peringueyi* Simon, 1903

15. *M. pseudocrudeni* Hewitt, 1919

16. *M. purpurea* Lawrence, 1928

17. *M. pyimi* Hewitt, 1914

18. *M. quercina* Simon, 1903

19. *M. rupicola* Hewitt, 1913

20. *M. rupicoloides* Hewitt, 1914

21. *M. teresae* Griswold, 1987

22. *M. terrestris* Hewitt, 1914

23. *M. terricola* Simon, 1903

24. *M. whytei* Pocock, 1897

Genus *Poecilomigas* Simon, 1903

1. *P. abrahami* (O. P.-Cambridge, 1889)
2. *P. elegans* Griswold, 1987

9. FAMILY NEMESIIDAE Simon, 1892**Genus *Entypesa* Simon, 1902**

1. *E. schoutedeni* Benoit, 1965

Genus *Hermacha* Simon, 1889

1. *H. bicolor* (Pocock, 1897)
2. *H. brevicauda* Purcell, 1903
3. *H. capensis* (Ausserer, 1871)
4. *H. caudata* Simon, 1889
5. *H. cruden* Hewitt, 1913
6. *H. curvipes* Purcell, 1902
7. *H. evanescens* Purcell, 1903
8. *H. fulva* Tucker, 1917
9. *H. grahami* (Hewitt, 1915)
10. *H. lanata* Purcell, 1902
11. *H. mazoena* Hewitt, 1915
12. *H. nigra* Tucker, 1917
13. *H. nigripinosa* Tucker, 1917
14. *H. nigromarginata* Strand, 1907
15. *H. sericea* Purcell, 1902
16. *H. tuckeri* Raven, 1985

Genus *Lepthercus* Purcell, 1902

1. *L. dregei* Purcell, 1902
2. *L. rattrayi* Hewitt, 1917

Genus *Pionothele* Purcell, 1902

1. *P. straminea* Purcell, 1902

Genus *Spiroctenus* Simon, 1889

1. *S. armatus* Hewitt, 1913
2. *S. broomi* Tucker, 1917
3. *S. cambierae* (Purcell, 1902)
4. *S. coeruleus* Lawrence, 1952
5. *S. collinus* (Pocock, 1900)
6. *S. curvipes* Hewitt, 1919
7. *S. exilis* Lawrence, 1938
8. *S. flavopunctatus* (Purcell, 1903)
9. *S. fossorius* (Pocock, 1900)
10. *S. fuliginosus* (Pocock, 1902)
11. *S. gooldi* (Purcell, 1903)
12. *S. inermis* (Purcell, 1903)
13. *S. latus* Purcell, 1904
14. *S. lightfooti* (Purcell, 1902)
15. *S. lignicolus* Lawrence, 1937
16. *S. londinensis* Hewitt, 1919
17. *S. marleyi* Hewitt, 1919
18. *S. minor* (Hewitt, 1913) comb. nov.
19. *S. pallidipes* Purcell, 1904
20. *S. pectiniger* (Simon, 1903) comb. nov.
21. *S. personatus* Simon, 1888
22. *S. pilosus* Tucker, 1917
23. *S. punctatus* Hewitt, 1916
24. *S. purcelli* Tucker, 1917
25. *S. sagittarius* (Purcell, 1902)
26. *S. schreineri* (Purcell, 1903)
27. *S. spinipalpis* Hewitt, 1919

28. *S. tricalcaratus* (Purcell, 1903)

29. *S. validus* (Purcell, 1902)

10. FAMILY THERAPHOSIDAE Thorell, 1870**Genus *Brachionopus* Pocock, 1897**

1. *B. annulatus* Purcell, 1903
2. *B. leptopelmiformis* Strand, 1907
3. *B. pretoriae* Purcell, 1904
4. *B. robustus* Pocock, 1897
5. *B. tristis* Purcell, 1903

Genus *Ceratogyrus* Pocock, 1897

1. *C. bechuanicus* Purcell, 1902
2. *C. brachycephalus* Hewitt, 1919
3. *C. darlingi* Pocock, 1897
4. *C. dolichocephalus* Hewitt, 1919
5. *C. marshalli* Pocock, 1897
6. *C. pillansi* (Purcell, 1902)
7. *C. sanderi* Strand, 1906

Genus *Harpactira* Ausserer, 1871

1. *H. atra* (Latreille, 1832)
2. *H. baviana* Purcell, 1903
3. *H. cafreriana* (Walckenaer, 1837)
4. *H. chrysogaster* Pocock, 1897
5. *H. curator* Pocock, 1898
6. *H. curvipes* Pocock, 1897
7. *H. dictator* Purcell, 1902
8. *H. gigas* Pocock, 1898
9. *H. guttata* Strand, 1907
10. *H. hamiltoni* Pocock, 1902
11. *H. lineata* Pocock, 1897
12. *H. lyrata* (Simon, 1892)
13. *H. marksii* Purcell, 1902b
14. *H. namaquensis* Purcell, 1902
15. *H. pulchripes* Pocock, 1901
16. *H. tigrina* Ausserer, 1875

Genus *Harpactirella* Purcell, 1902

1. *H. domicola* Purcell, 1903
2. *H. flavipilosa* Lawrence, 1936
3. *H. helenae* Purcell, 1903
4. *H. karrooica* Purcell, 1902
5. *H. lapidaria* Purcell, 1908
6. *H. leleupi* Benoit, 1965
7. *H. lightfooti* Purcell, 1902
8. *H. longipes* Purcell, 1902
9. *H. magna* Purcell, 1903
10. *H. schwarzi* Purcell, 1904
11. *H. spinosa* Purcell, 1908
12. *H. treleaveni* Purcell, 1902

Genus *Pterinochilus* Pocock, 1897

1. *P. breyeri* Hewitt, 1919
2. *P. crassispinus* Purcell, 1902
3. *P. junodi* Simon, 1904
4. *P. lugardi* Pocock, 1900
5. *P. nigrofulvus* Pocock, 1898
6. *P. pluridentatus* (Hewitt, 1919)
7. *P. schoenlandi* Pocock, 1900

APPENDIX II

Alphabetical list of generic synonyms

<i>Acanthodon</i> Guérin, 1839	= <i>Idiops</i> Perty, 1833 (Idiopidae)
<i>Bemmeris</i> Simon, 1903	= <i>Spiroctenus</i> Simon, 1889 (Nemesiidae)
<i>Bessia</i> Pocock, 1900	= <i>Spiroctenus</i> Simon, 1889 (Nemesiidae)
<i>Brachytheliscus</i> Pocock, 1902	= <i>Hermacha</i> Simon, 1889 (Nemesiidae)
<i>Caedmon</i> O. P.-Cambridge, 1903	= <i>Poecilomigas</i> Simon, 1903 (Migidae)
<i>Coelogenium</i> Purcell, 1902	= <i>Ceratogyrus</i> Pocock, 1897 (Theraphosidae)
<i>Ctenonemus</i> Simon, 1903	= <i>Spiroctenus</i> Simon, 1889 (Nemesiidae)
<i>Cyclopelma</i> Benoit, 1965	= <i>Sipalolasma</i> Simon, 1892 (Barychelidae)
<i>Damarchodes</i> Simon, 1903	= <i>Hermacha</i> Simon, 1889 (Nemesiidae)
<i>Diplothele</i> O.P.-Cambridge, 1890	= <i>Pisenor</i> Simon, 1899 (Barychelidae)
<i>Hermachastes</i> Pocock, 1900	= <i>Spiroctenus</i> Simon, 1889 (Nemesiidae)
<i>Hermachola</i> Hewitt, 1915	= <i>Hermacha</i> Simon, 1889 (Nemesiidae)
<i>Idiothele</i> Hewitt, 1919	= <i>Pterinochilus</i> Pocock, 1897 (Theraphosidae)
<i>Paromostola</i> Purcell, 1903	= <i>Homostola</i> Simon, 1892 (Cyrtaucheniidae)
<i>Pelmatorycter</i> Pocock, 1902	= <i>Ancylotrypha</i> Simon, 1888 (Cyrtaucheniidae)
<i>Pisenorina</i> Benoit, 1966	= <i>Cyphonisia</i> Simon, 1889 (Barychelidae)
<i>Pisenorodes</i> Pocock, 1898	= <i>Cyphonisia</i> Simon, 1889 (Barychelidae)
<i>Pseudohermacha</i> Strand, 1907	= <i>Entypesa</i> Simon, 1902 (Nemesiidae)
<i>Stictogaster</i> Purcell, 1902	= <i>Homostola</i> Simon, 1892 (Cyrtaucheniidae)
<i>Titanidiops</i> Simon, 1903	= <i>Idiops</i> Perty, 1833 (Idiopidae)
<i>Urothele</i> Tullgren, 1910	= <i>Pisenor</i> Simon, 1889 (Barychelidae)

APPENDIX III

List of abbreviations

AM – Albany Museum, Grahamstown
AMNH – American Museum of Natural History, New York
BMNH – The Natural History Museum, London
DM – Durban Natural History Museum, South Africa
MRAC – Koninklijk Museum voor Midden-Afrika, Tervuren
NCA – National Collection of Arachnida, ARC-Plant Protection Research Institute, Pretoria
NM – Natal Museum, Pietermaritzburg
NMB – Natural History Museum, Bulawayo, Zimbabwe
NMW – Natural History Museum, Wiesbaden, Germany
SAM – South African Museum, Cape Town
TM – Northern Flagship Institution: Transvaal Museum, Pretoria
UMO – University Museum, Oxford, UK

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The CD-ROM contains the following elements:

- 1) The Identification Manual in Adobe Acrobat 4.0 format.
- 2) A collection of 41 hyperlinked colour images of trapdoor and baboon spiders in a PhotoGallery.
- 3) Adobe Acrobat Reader v. 4.0.

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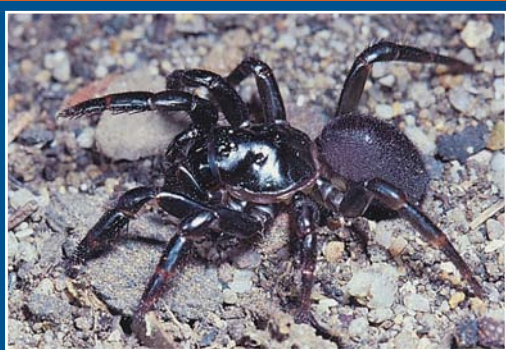
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The Mygalomorphae is a suborder that includes some of the largest and most fascinating spiders — the baboon and trapdoor spiders. Within the Araneae they are regarded as more primitive, with the oldest fossil records dating back to the Triassic Period. They are long-lived animals and can reach up to 20 years in captivity. Most mygalomorph families are terrestrial and live in silk-lined retreats, either burrows of various shapes made in the soil or sac-like chambers made under rocks or on tree trunks. The entrances to these retreats can be left open or closed off with a trapdoor made of silk and soil particles. Most species are nocturnal and hide in their retreats during the day. At night they wait in the entrance of their retreats for passing prey, or they wander around in search of a meal. They prey on a variety of insects and small animals and form an important part of the ecological food chain.

Southern Africa has a rich mygalomorph fauna of 10 families, 28 genera and 281 species. This manual provides the reader with an overview of the suborder Mygalomorphae in Southern Africa. The morphology, systematics and natural history of the families, subfamilies and genera are described, and illustrated keys are provided. Distribution records of the 281 species are listed and illustrated in maps.

Manuals such as this one are very important tools for determining the richness and diversity of the Southern African spider fauna. They provide information that is of fundamental importance in the formulation of conservation measures, especially for the Mygalomorphae which can easily be exploited.



The author of this book, Dr Ansie Dippenaar-Schoeman, is a professional arachnologist and head of the Spider Research Centre in the Biosystematics Division of the Plant Protection Research Institute, Agricultural Research Council, Pretoria, South Africa. She has devoted her entire career, spanning more than 30 years, to the study of Afrotropical spiders, both from a biological and a taxonomic perspective.

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